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Preface to the Second Volume

In accordance with the intention announced in the preface to the first volume, I am now able to place the second volume of my *Textbook of Algebra* before the public. The plan laid down there has been carried out in its essential points. In the applications I have tried to select problems that had already acquired an independent interest in other fields, such as geometry or the theory of functions, and that at the same time bring the principal points of algebraic theory before the reader in as many different ways as possible.

The application of the theory of algebraic numbers has been carried through as far as the theory of cyclotomic numbers. If life and strength endure, I hope in a continuation of my work to present the further applications to the domain of elliptic functions, which are contained only in part in my book *Elliptic Functions and Algebraic Numbers*.

During the preparation and printing of the second volume as well, I have had beside me the help and counsel of the friends whom I already named in the preface to the first edition. But the first volume has by now also won for itself many new friends who have advanced my work by suggestions and advice. To them all I express my thanks here, and add the request that they may continue to preserve their interest in the work.

Strasbourg, July 1896.
The Author.

Part I. General Group Theory

§1. Definition of Groups

In the first volume, in connection with permutations, we became acquainted with the concept of a group and made important algebraic applications of it. Our next task must therefore be to formulate more generally this concept, so extraordinarily important in all of modern mathematics, and to become familiar with the laws that govern it. We place the following definition at the head of the discussion.

A system P of objects (elements) of any kind whatsoever becomes a *group* when the following assumptions are satisfied.

1. A rule is given according to which, from a first and a second element of the system, a completely determined third element of the same system is derived.

Symbolically, if a is the first, b the second, and c the third element, one writes

$$ab = c, \qquad c = ab,$$

and says that c is *composed* from a and b, while a and b are the *components* of c.

For this composition one does not in general assume the commutative law; that is, ab may differ from ba. On the other hand, one does assume

2. the *associative law*,

that is, whenever a, b, c are any three elements of P, one has

$$(ab)c = a(bc).$$

From this, by complete induction, it follows that one always arrives at the same result when, in any finite sequence of elements of P, a, b, c, d, $\ldots$, one first composes two neighboring elements, then again two neighboring elements, and so on until the whole sequence has been reduced to a single element, denoted by $abcd\ldots$. Thus, for example,

$$\begin{aligned} abcd &= (ab)cd = [(ab)c]d = (ab)(cd) \\ &= a(bc)d = [a(bc)]d = a[(bc)d] \\ &= ab(cd) = (ab)(cd) = a[b(cd)]. \end{aligned}$$

§2. Intersection of two groups

Cosets do not themselves have the defining properties of a group. For, in order that the composition of two elements of Q_1 should again yield an element of Q_1, one would need, for example,

$$a_1b\,a_2b = a_3b,$$

and hence

$$a_1ba_2 = a_3, \qquad b = a_1^{-1}a_3a_2^{-1}.$$

Thus b would, contrary to the assumption, have to belong to Q. The designation *coset* is therefore to be understood only in an inexact or analogical sense.

Two divisors Q, Q' of a group P always have the element 1 in common. They may, however, have further elements in common, and these common elements form a group, which we denote by R. For if a and b belong both to Q and to Q', then by the group property of Q and Q' the same is true of the product ab; therefore R is a group. This common group is called the greatest common divisor of Q and Q', or, with a geometric turn of phrase, the *intersection* of Q and Q'.

§3. Normal subgroups

If Q is a divisor of P that is identical with all of its conjugate divisors, then Q is called a *normal subgroup* of P.

The group consisting only of the single element 1 is a normal subgroup of every group. We obtain another normal subgroup by taking the greatest common divisor R of all divisors conjugate to some divisor Q of P.

Indeed, it has already been proved that R, as the intersection of several divisors of P, is itself a group. Now let

$$Q,\ Q',\ Q'', \dots$$

be the divisors conjugate to Q, and let b be any element of P. Then the system of groups

$$b^{-1}Qb,\ b^{-1}Q'b,\ b^{-1}Q''b, \dots$$

is not different from the original system. If R is the intersection of the first system, then $b^{-1}Rb$ is the intersection of the second system, and consequently $R = b^{-1}Rb$; that is, R is a normal subgroup of P.

If N is a normal subgroup of a group P and b is any element of P, then

$$b^{-1}Nb = N, \qquad \text{or equivalently} \qquad Nb = bN.$$

If P has order n, N has order ν, and index μ, so that $n = \mu\nu$, then we may choose the μ elements $1, b_1, \dots, b_{\mu-1}$ in such a way that the decomposition of P into cosets is

$$P = N + Nb_1 + Nb_2 + \cdots + Nb_{\mu-1} = N + b_1N + b_2N + \cdots + b_{\mu-1}N.$$

§4. Composition of subsets

Let A and B be any two collections of elements of a group P (whether groups or not). By the symbolic product AB we understand the set of all elements obtained by composing each element a of A with each element b of B.

The two equations

$$AA = A, \qquad AB = BA,$$

where A is fixed and B is an arbitrary subset of P, express that A is a normal subgroup of P. For from the first relation A is itself a divisor of P, and from the second it follows for every element $b \in P$ that

$$b^{-1}Ab = A,$$

that is, A is normal in P.

If A and B are two divisors of P, then one of the two equalities

$$AB = BA, \qquad \text{or equivalently} \qquad BA = AB,$$

is the necessary and sufficient condition for the product set AB to be a group.

Because of the defining property of normal subgroups, N commutes with every b, and therefore

$$b_\nu N = N b_\nu,$$

so that

$$Nb_\nu \cdot Nb_\mu = NN \cdot b_\nu b_\mu.$$

Since $b_\nu b_\mu$ lies in P, the set $Nb_\nu b_\mu$ coincides with one of the cosets occurring in the decomposition above; and because $NN = N$, it follows that if we write $Nb_\nu b_\mu = M$, then

$$NM = M.$$

Thus property (1) in the definition of a group has been verified for the composition of cosets. That the associative law also holds was already noted. It remains only to verify the analogue of cancellation.

§5. Multiple isomorphism

These quotient groups are related to one another in the same way as isomorphic groups in the ordinary sense, except that to each element A there may correspond not just one element a, but several. This gives occasion to extend the notion of isomorphism by the following definition.

A group P with elements $a, b, c, \ldots$ is said to be *multiply isomorphic* to a group Q with elements $A, B, C, \ldots$ if the two groups can be put into correspondence in such a way that to each of the elements $A, B, C, \ldots$ there correspond one or more of the elements $a, b, c, \ldots$, while every element of Q corresponds to one and only one element of P, and such that whenever a corresponds to A and b corresponds to B, then ab corresponds to AB.

In this way one proves first that to each of the elements of the quotient group there corresponds an entire coset of the original group. Following H"older, one writes such a quotient group in the form P/N. The order of the complementary group is equal to the index of the group N.

We mention in conclusion a theorem about normal subgroups. If N is a divisor of P, and N' is a divisor of N which is at the same time a normal subgroup of P, then N' is also a normal subgroup of N. This is immediate, because for every element $b \in P$ one has

$$b^{-1}N'b = N',$$

and every element of N is also an element of P.

The converse, however, is not valid in general: if N is a normal subgroup of P and N' is a normal subgroup of N, it need not follow that N' is a normal subgroup of P.

§6. Composition series and the theorem of C. Jordan

A group P is called *simple* if it has no normal subgroup other than itself and the identity; it is called *composite* if other normal subgroups are present.

A normal subgroup which is not contained properly in any larger proper normal subgroup of P is called a *maximal normal subgroup* of P. If P_1 is a maximal normal subgroup of P, P_2 is a maximal normal subgroup of P_1, and so on, then one obtains a chain

$$P,\ P_1,\ P_2,\ \ldots,\ P_\nu = 1,$$

in which each subgroup is a maximal normal subgroup of the preceding one. Such a chain is called a *composition series*; the corresponding indices are its indices of composition.

The important theorem expressing the uniqueness of the composition factors was established by C. Jordan.

Theorem (Jordan). *However a composition series of a group P may be chosen, the sequence of its indices is always the same up to order.*

The proof rests on the quotient groups introduced above. Suppose Q and Q_1 are normal subgroups of P, and that in addition Q_1 is a divisor (hence, by the previous section, a normal subgroup) of Q. Then the quotient group Q/Q_1 is a normal subgroup of P/Q_1, and the index of Q/Q_1 in P/Q_1 is equal to the index of Q in P.

This reduction is the crucial device in the proof that the composition factors are independent of the chosen composition series, apart from their order.

§7. Further results on composition series

We derive first several important consequences of Jordan's theorem.

Let P be a group and Q a normal subgroup of P, and suppose that P and Q have composition factors

$$A_1, A_2, \ldots, A_\lambda \qquad \text{and} \qquad C_1, C_2, \ldots, C_\mu.$$

Then the quotient group P/Q has composition factors

$$A_1, A_2, \ldots, A_\lambda,\ C_\mu^{-1}, \ldots, C_2^{-1}, C_1^{-1}.$$

For every composition series of P/Q can be completed to one of P.

Corollary 1.1. If P, Q, R are three groups, each of which is a normal subgroup of the preceding one, and if the index of R in P is a power of a prime number, then the index of R in Q and the index of Q in P are powers of the same prime.

§8. Metacyclic groups

A group P is called *metacyclic* if it possesses a composition series all of whose factors are cyclic.

This definition includes all cyclic groups, and also all groups whose order is a power of a prime number. For every group of prime-power order is, by Sylow's theorem, solvable, hence metacyclic.

Metacyclic groups play an important role in the theory of algebraic equations, since the equations whose groups are metacyclic are solvable by radicals.

Appendix: foundations of group theory

The preceding paragraphs contain the foundations of general group theory as they are used in modern algebra. We have treated the definition of a group, the theory of divisors and cosets, normal subgroups and their properties, many-valued isomorphism, and composition series together with Jordan's theorem.

These concepts are of fundamental importance for the further development of algebra. Upon them is based the theory of abelian groups, which is presented in the following paragraphs of the second section.

Second Section. Abelian groups

§9. Representation of Abelian groups by a basis

We have already called those groups in which the commutative law holds for composition *commutative* or *Abelian* groups. For these groups, which are of particular importance in applications, the laws are much simpler than in general groups, as we now see.

Let S be an Abelian group of order n. We shall show that the elements of S can be represented in the form

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}, \tag{1}$$

where $A_1, \ldots, A_\nu$ are suitable elements of the group and the exponents $\alpha_1, \ldots, \alpha_\nu$ run through complete residue systems modulo positive integers $a_1, \ldots, a_\nu$, the orders of $A_1, \ldots, A_\nu$. Such a system $A_1, \ldots, A_\nu$ is called a *basis* of the Abelian group.

One sees immediately that every element of the form (1) actually occurs: for example, one obtains A_i by taking $\alpha_i = 1$ and $\alpha_k = 0$ for $k \neq i$.

It remains to prove that every element occurs equally often. Suppose

$$1 = A_1^{h_1} A_2^{h_2} \cdots A_n^{h_n} \tag{2}$$

is a representation of the identity. Then the value of θ in (1) is unchanged if one replaces $\alpha_1, \ldots, \alpha_n$ by $\alpha_1 + h_1, \ldots, \alpha_n + h_n$. Hence each element θ is represented at least as many times by (1) as the identity is represented by (2).

Conversely, if two exponent systems $\alpha_1, \ldots, \alpha_n$ and $\alpha_1', \ldots, \alpha_n'$ yield the same element θ, then

$$1 = A_1^{\alpha_1 - \alpha_1'} A_2^{\alpha_2 - \alpha_2'} \cdots A_n^{\alpha_n - \alpha_n'}. \tag{3}$$

Thus the difference of the two exponent systems gives a representation of the identity. Therefore the number of representations of θ can be no larger than the number of representations of 1. If we denote that number by h, then the total number of possible exponent choices in (1) is $a_1 a_2 \cdots a_n$, while n is the number of distinct group elements. Hence

$$nh = a_1 a_2 \cdots a_n. \tag{4}$$

From formula (4) one obtains:

Proposition 2. If r is a prime dividing the order n of the group S, then S contains an element of order r.

Indeed, one of the orders $a_1, \ldots, a_n$ must be divisible by r. If, say, $a_1 = rk$, then A_1^k has order r.

Proposition 3. If $A, B, \ldots$ are arbitrary elements of S with orders $a, b, \ldots$, then the product $AB \cdots$ again lies in S, and its order divides the least common multiple m of $a, b, \ldots$.

For

$$(AB \cdots)^m = A^m B^m \cdots = 1,$$

so m is a multiple of the order of $AB \cdots$.

Proposition 4. Suppose the order n of the group is factored as

$$n = ab$$

with a and b relatively prime. Then there exist exactly a elements A whose order divides a, and exactly b elements B whose order divides b, and every element of the group is represented uniquely in the form

$$\theta = AB. \tag{5}$$

To see this, let $\mathcal{A}$ be the set of elements whose order divides a. By Proposition 3 this is a subgroup of S. Similarly the set $\mathcal{B}$ of elements whose order divides b is a subgroup.

Let a' be the order of $\mathcal{A}$, and b' the order of $\mathcal{B}$. Then a' is relatively prime to b: if a prime r divided a', then by Proposition 2 the subgroup $\mathcal{A}$ would contain an element of order r, and since every element of $\mathcal{A}$ has order dividing a, that prime r would divide a, not b. Similarly b' is relatively prime to a.

Now choose integers x, y such that

$$ax + by = 1. \tag{6}$$

For any element $\theta \in S$ we have

$$\theta = \theta^{ax}\theta^{by}. \tag{7}$$

Since $(\theta^{ax})^b = 1$, the element θ^{ax} lies in $\mathcal{A}$, and similarly $\theta^{by} \in \mathcal{B}$. Therefore every element has a representation

$$\theta = AB. \tag{8}$$

This representation is unique. If $AB = A'B'$, then by raising to the suitable powers $by = 1 - ax$ and $ax = 1 - by$ one obtains $A = A'$ and hence $B = B'$. The number of products AB is therefore $a'b'$, and since every element of S occurs once, we have

$$n = ab = a'b'.$$

Because a is relatively prime to b' and b is relatively prime to a', it follows that

$$a = a', \qquad b = b'.$$

This proves Proposition 4 in full.

If the two subgroups $\mathcal{A}, \mathcal{B}$ are each represented by bases, then formula (8) shows that S itself is represented by a basis, namely by adjoining the basis elements of $\mathcal{A}$ and $\mathcal{B}$.

Repeating this decomposition whenever a and b can themselves be split into relatively prime factors, we reduce matters to the case where the order of the group is a power of a single prime:

$$n = p^{\mu}. \tag{9}$$

Thus it remains to prove that every Abelian group of prime-power order can be represented by a basis. Let A be an element of maximal order a. Then a is a power of p, and the order of every element of S divides a. Hence for every $\theta \in S$,

$$\theta^a = 1. \tag{10}$$

The elements

$$1, A, A^2, \ldots, A^{a-1} \tag{11}$$

are all distinct and form a subgroup of S. If they already exhaust the group, then S is represented by the one-element basis A.

Otherwise, for every $\theta \in S$ there exists a least positive integer b such that

$$\theta^b = A^t \tag{12}$$

for some integer t. This b divides a, hence is a power of p, and it also divides the order of θ. From (12) one finds that t must be divisible by b: writing $a = qb + b'$ with $0 \leq b' < b$, one checks that if $b' \neq 0$ then $\theta^{b'}$ would already be a power of A, contradicting the minimality of b.

Writing $a/b = k$ and setting

$$B = \theta A^{-tk}, \tag{13}$$

we obtain

$$B^b = 1, \tag{14}$$

and b is the least exponent with this property. Thus the elements

$$A^{\alpha_1} B^{\alpha_2}, \qquad \alpha_1 = 0, 1, \ldots, a-1, \quad \alpha_2 = 0, 1, \ldots, b-1, \tag{15}$$

are all distinct and form a subgroup S' represented by the basis A, B. If $S' = S$, we are done; otherwise the construction continues inductively.

Proceeding by induction, one obtains a subgroup $S_{\nu-1}$ with basis

$$S_{\nu-1} = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_{\nu-1}^{\alpha_{\nu-1}}, \tag{16}$$

such that:

1. the orders satisfy
$$a_1 \geq a_2 \geq \cdots \geq a_{\nu-1},$$
2. for every element $\theta \in S$, the power $\theta^{a_{\nu-1}}$ lies in $S_{\nu-1}$, and the corresponding exponents are all divisible by $a_{\nu-1}$.

If $S_{\nu-1} \neq S$, choose $\theta \in S$ so that the least positive exponent a_ν with $\theta^{a_\nu} \in S_{\nu-1}$ is as large as possible. Then $a_\nu \leq a_{\nu-1}$, and a_ν is again a power of p. Writing

$$\theta^{a_\nu} = A_1^{h_1 a_\nu} A_2^{h_2 a_\nu} \cdots A_{\nu-1}^{h_{\nu-1} a_\nu}, \tag{17}$$

define

$$A_\nu = \theta A_1^{-h_1} A_2^{-h_2} \cdots A_{\nu-1}^{-h_{\nu-1}}. \tag{18}$$

Then

$$A_\nu^{a_\nu} = 1, \tag{19}$$

and a_ν is the smallest such exponent. Consequently, the elements

$$A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}, \qquad 0 \leq \alpha_i < a_i, \tag{20}$$

are all distinct and form a larger subgroup S_ν, again satisfying the inductive conditions.

Continuing this process, one eventually obtains the whole group S represented by a basis. This proves the theorem that every finite Abelian group can be represented by a basis.

It follows at once that every Abelian group is metacyclic. Indeed, if

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu},$$

and if p is a prime dividing the order a_1 of A_1, then the elements

$$\theta' = A_1^{p\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}$$

form a subgroup of index p, represented by the basis $A_1^p, A_2, \ldots, A_\nu$. Repeating this procedure shows that S admits a chain of subgroups of prime index, hence is metacyclic.

§10. The invariants of Abelian groups

The proof just given for the existence of a basis also provides a method for finding one in which the orders of the basis elements are prime powers. Nevertheless, the same Abelian group can admit different bases.

For example, let A and B have relatively prime orders a and b. Then they determine a two-element basis with group

$$A^\alpha B^\beta, \qquad \alpha = 0, 1, \ldots, a-1, \quad \beta = 0, 1, \ldots, b-1. \tag{1}$$

But the same group can also be represented by the one-element basis AB, namely in the form

$$(AB)^s, \qquad s = 0, 1, \ldots, ab-1. \tag{2}$$

Indeed, given arbitrary α, β, one can determine s modulo ab so that

$$s \equiv \alpha \pmod{a}, \qquad s \equiv \beta \pmod{b},$$

and then (2) reproduces (1).

Even so, there is a precise sense in which the structure of the basis is completely determined by the group itself.

Theorem 0.2. Suppose

$$A_1, A_2, \ldots, A_\nu$$

with orders

$$a_1, a_2, \ldots, a_\nu$$

and

$$B_1, B_2, \dots, B_\mu$$

with orders

$$b_1, b_2, \dots, b_\mu$$

are two bases of the same Abelian group S of order n. Let p be a prime dividing n, and write

$$a_i = p^{\alpha_i}\bar{a}_i, \qquad b_j = p^{\beta_j}\bar{b}_j,$$

where $\bar{a}_i, \bar{b}_j$ are not divisible by p. Then the prime powers p^{α_i} greater than 1 occur among the p^{β_j}, and conversely.

To prove this, order the basis elements so that

$$a_1 \leq a_2 \leq \cdots \leq a_\nu, \qquad b_1 \leq b_2 \leq \cdots \leq b_\mu. \tag{3}$$

Let m be the least common multiple of $a_1, \dots, a_\nu$. Since the $\bar{a}_i$ are not divisible by p, the number m is also not divisible by p.

For an arbitrary element

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}, \tag{4}$$

one then has

$$\theta^{p^{\alpha_\nu} m} = 1.$$

Applying this to the basis elements $B_1, \dots, B_\mu$, one sees that every b_j divides $p^{\alpha_\nu} m$. Hence p^{β_μ} divides p^{α_ν}, and the least common multiple of the b_j is again m. Reversing the rôles of the two bases, one obtains the opposite divisibility, and therefore

$$p^{\alpha_\nu} = p^{\beta_\mu}. \tag{5}$$

Assume inductively that

$$p^{\alpha_1} = p^{\beta_1}, \quad p^{\alpha_2} = p^{\beta_2}, \quad \dots, \quad p^{\alpha_{s-1}} = p^{\beta_{s-1}} \tag{6}$$

have already been proved. Then for any element θ,

$$\theta^{p^{\alpha_s} m} = A_1^{\alpha_1 p^{\alpha_s} m} A_2^{\alpha_2 p^{\alpha_s} m} \cdots A_\nu^{\alpha_\nu p^{\alpha_s} m}. \tag{7}$$

Counting the number of distinct elements of this form yields

$$q = \frac{p^{\alpha_1}}{p^{\alpha_s}} \cdot \frac{p^{\alpha_2}}{p^{\alpha_s}} \cdots \frac{p^{\alpha_{s-1}}}{p^{\alpha_s}}. \tag{9}$$

If one now expresses the same elements in the B-basis and assumes

$$p^{\alpha_s} > p^{\beta_s}, \tag{10}$$

then the counting argument forces p^{β_s} to divide p^{α_s}, which contradicts (10) unless

$$p^{\alpha_s} = p^{\beta_s}. \tag{14}$$

This proves Theorem 0.2.

The prime powers occurring in the orders of a basis are therefore independent of the choice of basis. We call them the *invariants* of the Abelian group. Their product is equal to the order n of the group.

By §9 we can choose a basis whose element orders are prime powers. Hence those orders are exactly the invariants and are completely determined by the group.

That the invariants completely determine the group as well follows from the next theorem.

Theorem 0.3. Two Abelian groups with the same invariants are isomorphic, and isomorphic Abelian groups have the same invariants.

Indeed, if

$$\theta = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}$$

are the elements of one group S, and

$$\theta' = B_1^{\alpha_1} B_2^{\alpha_2} \cdots B_\nu^{\alpha_\nu}$$

those of another group S', with the same exponent ranges modulo the same basis orders $a_1, \dots, a_\nu$, then we obtain an isomorphism by letting θ correspond to θ' whenever the exponent systems are identical.

Conversely, if two groups are isomorphic, then the images of a basis of one group form a basis of the other, and therefore the invariants must agree in both.

§11. Group characters

A principal problem in the theory of Abelian groups is to determine all divisors of an Abelian group. The solution of this problem is substantially facilitated by introducing the concept of *group characters*, which is also important in many other respects.

Let S be an Abelian group of order n, whose elements we denote by A, and let

$$A_1, A_2, \dots, A_\nu \tag{1}$$

be the elements of a basis of S, of orders

$$a_1, a_2, \dots, a_\nu.$$

Then every element A is represented, and represented uniquely, in the form

$$A = A_1^{\alpha_1} A_2^{\alpha_2} \cdots A_\nu^{\alpha_\nu}, \tag{2}$$

where

$$0 \le \alpha_1 < a_1, \qquad 0 \le \alpha_2 < a_2, \qquad \dots, \qquad 0 \le \alpha_\nu < a_\nu. \tag{3}$$

The exponents $\alpha_1, \alpha_2, \ldots, \alpha_\nu$, or any integers congruent to them modulo $a_1, a_2, \ldots, a_\nu$, are called the *indices* of the element A. Then one has:

1. The indices of a product AA' are obtained by adding the corresponding indices of the two factors.

If we assign to each element A some numerical value, we may call this assignment a function of A. Such a function $\chi(A)$ will be called a *group character* if for every two elements $A, A' \in S$ it satisfies

$$\chi(AA') = \chi(A)\chi(A'). \tag{4}$$

Hence it also satisfies the more general multiplicative law

$$\chi(A'A''A'''\cdots) = \chi(A')\chi(A'')\chi(A''')\cdots .$$

Two such functions χ and χ_1 are regarded as distinct if there exists at least one element A for which $\chi(A) \neq \chi_1(A)$. We now determine these characters more closely.

Setting $A' = 1$ in (4), one obtains

$$\chi(A) = \chi(A)\chi(1). \tag{5}$$

Excluding once and for all the degenerate case in which all $\chi(A) = 0$, one gets:

$$\chi(1) = 1. \tag{6}$$

2. Every character takes the value 1 on the identity element.

Setting $A' = A$ in (4), we obtain

$$\chi(A^2) = [\chi(A)]^2,$$

and by complete induction, for every exponent h,

$$\chi(A^h) = [\chi(A)]^h. \tag{7}$$

If a denotes the order of the element A, then with $h = a$ and using (6) one gets

$$[\chi(A)]^a = 1. \tag{8}$$

3. The values of a character $\chi(A)$ are roots of unity whose orders divide the order of A.

We can now determine all characters easily. Set

$$\chi(A_1) = \omega_1, \qquad \chi(A_2) = \omega_2, \qquad \ldots, \qquad \chi(A_\nu) = \omega_\nu. \tag{9}$$

Then

$$\omega_1^{a_1} = 1, \qquad \omega_2^{a_2} = 1, \qquad \ldots, \qquad \omega_\nu^{a_\nu} = 1, \tag{10}$$

and from (2) and (4),

$$\chi(A) = \omega_1^{\alpha_1}\omega_2^{\alpha_2}\cdots\omega_\nu^{\alpha_\nu}. \tag{11}$$

Conversely, whenever $\omega_1, \omega_2, \ldots, \omega_\nu$ is any solution of (10), formula (11) defines a function of A that satisfies (4), and hence is a character of the group.

Each of the equations (10) has $a_1, a_2, \ldots, a_\nu$ roots, and if all root choices are combined with one another, one obtains

$$a_1 a_2 \cdots a_\nu = n$$

possible systems. These lead to n distinct characters $\chi(A)$. Indeed, if another choice

$$\omega_1', \omega_2', \ldots, \omega_\nu'$$

yields the same function for all A, then taking $\alpha_1 = 1$, $\alpha_2 = \cdots = \alpha_\nu = 0$ shows $\omega_1 = \omega_1'$, and similarly for the others.

Thus:

4. There exist exactly n distinct characters of an Abelian group of order n, all given by formula (11).

Let $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_\nu$ be primitive roots of the equations (10). Then we may write

$$\omega_1 = \varepsilon_1^{\beta_1}, \qquad \omega_2 = \varepsilon_2^{\beta_2}, \qquad \ldots, \qquad \omega_\nu = \varepsilon_\nu^{\beta_\nu}, \tag{12}$$

where $\beta_1, \ldots, \beta_\nu$ are taken from complete residue systems modulo $a_1, \ldots, a_\nu$. Then all n characters are given by

$$\chi(A) = \varepsilon_1^{\alpha_1\beta_1}\varepsilon_2^{\alpha_2\beta_2}\cdots\varepsilon_\nu^{\alpha_\nu\beta_\nu}. \tag{13}$$

Among these occurs the character obtained by setting all $\beta_i = 0$; it has value 1 on every element A, and we call it the *unit character*.

The n characters of S can themselves be combined into an Abelian group, indeed into one isomorphic to S.

For by (13), each character is determined by a system of integers

$$\beta_1, \beta_2, \ldots, \beta_\nu$$

taken modulo $a_1, \ldots, a_\nu$, and this same system is the index system of a definite element $B \in S$, namely

$$B = A_1^{\beta_1} A_2^{\beta_2} \cdots A_\nu^{\beta_\nu}. \tag{14}$$

Thus every element $B \in S$ corresponds to a definite character, which we denote by $\chi_B(A)$, or simply χ_B when the variable A is omitted.

If B' is a second element of S, with indices $\beta_1', \beta_2', \ldots, \beta_\nu'$, then its corresponding character $\chi_{B'}$ is defined likewise. If now $\chi_B\chi_{B'}$ denotes the character given on each A by

$$\chi_B\chi_{B'}(A) = \varepsilon_1^{\alpha_1(\beta_1+\beta_1')}\varepsilon_2^{\alpha_2(\beta_2+\beta_2')}\cdots\varepsilon_\nu^{\alpha_\nu(\beta_\nu+\beta_\nu')}, \tag{15}$$

then the characters are thereby combined into a group isomorphic to S. The identity element of this character group is the unit character.

Accordingly, the character group can itself be represented by a basis, obtained by setting

$$\chi_1(A) = \varepsilon_1^{\alpha_1}, \qquad \chi_2(A) = \varepsilon_2^{\alpha_2}, \qquad \dots, \qquad \chi_\nu(A) = \varepsilon_\nu^{\alpha_\nu}. \tag{16}$$

Then every character has the form

$$\chi_B = \chi_1^{\beta_1}\chi_2^{\beta_2}\cdots\chi_\nu^{\beta_\nu}, \tag{17}$$

and if $B, B' \in S$, then

$$\chi_B\chi_{B'} = \chi_{BB'}. \tag{18}$$

If χ_1, χ_2 are any two characters and $\chi_1\chi_2$ their composite, then for every A,

$$\chi_1(A)\chi_2(A) = (\chi_1\chi_2)(A). \tag{19}$$

We may summarize this as:

5. The characters of a group S can be combined into a group isomorphic to S.

One has also, immediately from definition (13),

$$\chi_B(A) = \chi_A(B). \tag{20}$$

Finally there is the following theorem.

6. If A runs through all elements of the group S, then for a fixed character χ,

$$\sum_A \chi(A) = \begin{cases} n, & \text{if } \chi \text{ is the unit character,} \\ 0, & \text{otherwise.} \end{cases} \tag{21}$$

Likewise, if the element A is held fixed and χ runs through all characters, then

$$\sum_\chi \chi(A) = \begin{cases} n, & \text{if } A = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{22}$$

When χ is the unit character, or A is the identity, these formulas are evident, since each of the n summands is then equal to 1. If χ is not the unit character, there exists $B \in S$ such that $\chi(B) \neq 1$. Writing

$$\sum_A \chi(A) = \sigma,$$

and multiplying by $\chi(B)$, one obtains

$$\sum_A \chi(AB) = \sigma\chi(B).$$

But as A runs through the whole group, so does AB, hence the left side is also σ. Therefore

$$\sigma[1 - \chi(B)] = 0,$$

and since $\chi(B) \neq 1$, we get $\sigma = 0$, proving (21). Formula (22) follows similarly, and also immediately from (20).

§12. Divisors of an Abelian group. Reciprocal groups

A divisor T of an Abelian group S is, as already remarked above, always a normal subgroup. Hence by §4 there exists a complementary group

$$S/T,$$

whose elements are the cosets of T, namely

$$T, \quad TA', \quad TA'', \quad \ldots, \tag{1}$$

where $A', A'', \ldots$ are certain elements of S. If t is the order of T, then the number of such cosets, that is, the order of the group S/T, is equal to the index of T in S, namely

$$\frac{n}{t}.$$

This group S/T is itself Abelian, because

$$TA' \cdot TA'' = TA'A'', \qquad TA'' \cdot TA' = TA''A',$$

and these are equal.

Let us now determine the characters of the group S/T. Denote its elements by

$$T_0, T_1, T_2, \ldots, T_j, \tag{2}$$

and let one of its characters be written $\xi(T_i)$.

From this function $\xi(T)$ we can derive a function $\xi(A)$ on the elements of S by setting

$$\xi(A_i) = \xi(T_i) \tag{3}$$

whenever A_i is any element belonging to the coset T_i. For the elements of the group T itself one then has $\xi(A) = 1$.

But this function $\xi(A)$ is among the characters of S. For if $A_i \in T_i$ and $A_k \in T_k$, then $A_i A_k \in T_i T_k$, and hence

$$\xi(A_i)\xi(A_k) = \xi(T_i)\xi(T_k) = \xi(T_i T_k) = \xi(A_i A_k). \tag{4}$$

This is precisely the defining property of a character of S.

Conversely, suppose one of the characters $\chi = \xi$ of S takes the same value on every element of T. Since the identity belongs to T, this common value can only be 1. If now A_i is any element of the coset T_i, and A runs through the whole group T, then AA_i runs through the coset T_i. Since $\xi(A) = 1$, one has

$$\xi(AA_i) = \xi(A_i). \tag{5}$$

That is, $\xi(A)$ has one and the same value on all elements of a fixed coset T_i, and may therefore be regarded as a function of T_i, denoted $\xi(T_i)$.

Since whenever $A_i \in T_i$ and $A_k \in T_k$, the product $A_i A_k \in T_i T_k$, it follows that

$$\xi(T_i)\xi(T_k) = \xi(T_i T_k). \tag{6}$$

So $\xi(T_i)$ is indeed a character of the quotient group S/T.

Therefore there are exactly j such characters $\xi(A)$, namely those that take the value 1 on every element of T. These j characters form, under composition of characters, a group; for if $\xi_1(A) = 1$ and $\xi_2(A) = 1$ on T, then also $\xi_1\xi_2(A) = 1$. Since these functions ξ can equally well be viewed as the characters of the quotient group S/T, it follows from §11, theorem 5, that the group of the ξ is isomorphic to S/T.

Hence:

7. If an Abelian group S has a divisor T of order t and index j, then among the characters of S there are exactly j, and no more, that take the value 1 on all elements of T; and for every element of S not contained in T, at least one of these characters takes a value different from 1. These characters form a group isomorphic to S/T.

Since every character χ_B corresponds to a definite element $B \in S$, if χ_B runs through the group of characters belonging to T, then B runs through a group of order j, isomorphic to the group of the χ_B, and therefore also to S/T. We denote this group by U and call it the *group reciprocal to* T.

Again, one should note that in general the notion of reciprocal group depends on the chosen basis and the chosen roots of unity ε, so it does not belong intrinsically to the group S and the subgroup T alone.

The reciprocal group of T is characterized by the following theorem.

8. Let A run through the elements of a divisor $T \subset S$. Then the elements $B \in S$ satisfying

$$\chi_B(A) = 1 \tag{7}$$

for every $A \in T$ are precisely the elements of the group U reciprocal to T, whose order is equal to the index of T.

By §11, (20), the group T is in turn the reciprocal group to U; the relation between the two groups is therefore mutual.

One also has the following theorem.

9. If T' is a divisor of T, then conversely the reciprocal group U of T is a divisor of the group U' reciprocal to T'.

For every element $A' \in T'$ also lies in T. Hence, if B runs through U, then $\chi_B(A') = 1$, and therefore B must occur in the group U'.

If one selects from the totality of characters χ an arbitrary finite number

$$\xi_1, \xi_2, \ldots, \xi_k,$$

whether or not they form a group, then all elements A satisfying the k conditions

$$\xi_1(A) = 1, \qquad \xi_2(A) = 1, \qquad \ldots, \qquad \xi_k(A) = 1 \tag{8}$$

form a group, because from $\xi_i(A) = 1$ and $\xi_i(A') = 1$ it follows that $\xi_i(AA') = 1$.

If the characters $\xi_1, \ldots, \xi_k$ do not already form a group, then there follow from equations (8) additional equations

$$\xi_{k+1}(A) = 1, \quad \ldots$$

so that $\xi_1, \ldots, \xi_k$ may be completed to a group.

10. It follows from theorem 7 that in this way one can obtain all possible divisors of S.

This batch stops at the general construction of reciprocal groups; the illustrative examples that follow in Weber are left for the next batch.

§4. Cyclic equations

163. Reduction of Abelian equations to cyclic equations

We have already seen above that in a transitive Abelian group, besides the identity, no permutation occurs that leaves any symbol fixed. From this one obtains another theorem.

Assume that a permutation π of such a group is decomposed into its cycles, and let the cycle with the fewest elements be an r-cycle. Then π^r leaves the elements of this cycle fixed and therefore must be the identity permutation. But from $\pi^r = 1$ it follows that, if other cycles are present, they too must have r elements. Thus:

Theorem. A permutation of a transitive Abelian group contains only cycles of the same length.

If m is the degree of the group, then the number r of elements in one cycle must divide m, and if $m = rs$, then s is the number of r-cycles of which π consists.

Now let P be the group of an Abelian equation $F(x) = 0$, and choose any non-identity permutation π in P. Decompose it into cycles

$$\pi = \gamma_1 \gamma_2 \cdots \gamma_s, \tag{1}$$

where each cycle γ refers to r roots of the equation $F(x) = 0$. Arrange these roots so that

$$\gamma_1 = (\alpha, \alpha_1, \alpha_2, \ldots, \alpha_{r-1}),$$

$$\gamma_2 = (\beta, \beta_1, \beta_2, \ldots, \beta_{r-1}),$$

$$\ldots,$$

$$\gamma_s = (\delta, \delta_1, \delta_2, \ldots, \delta_{r-1}). \tag{2}$$

If π_i is any permutation of P, then because of commutativity one has

$$\pi^{-1}\pi_i\pi = \pi_i, \qquad \text{that is,} \qquad \pi_i\pi = \pi\pi_i. \tag{3}$$

By the theorem on the formation of the permutations $\pi^{-1}\pi_i\pi$ (§154, no. 6), it follows that γ_1 cannot be changed when the permutations π are carried out. But since γ_1 is completely determined inside the cycles of π, apart from its ordering and its initial element, it follows that under any permutation π_i in P the elements belonging to one cycle γ cannot be separated from one another; they can only be cyclically permuted among themselves, and the cycles can moreover be permuted among one another.

Hence, if $s > 1$, the group is imprimitive.

A rational function of the arguments $\alpha, \alpha_1, \ldots, \alpha_{r-1}$ that does not change its value when the α are subjected to the cyclic permutation γ_1 and its repetitions is called a *cyclic function* of the α. If we denote by $\omega, \omega_1, \ldots, \omega_{s-1}$ the conjugate values of ω, setting for instance

$$\omega = \psi(\alpha, \alpha_1, \ldots, \alpha_{r-1}),$$

$$\omega_1 = \psi(\beta, \beta_1, \ldots, \beta_{r-1}), \qquad \ldots, \tag{4}$$

then these quantities are, by §158, the roots of an irreducible equation of degree s,

$$\Phi(\omega) = 0, \tag{5}$$

whose group is obtained by determining the permutations induced on the quantities (4) by the permutations coming from P.

But the permutation of the quantities (4) induced by $\pi_i\pi_j$ is obtained by composing the permutations induced separately by π_i and π_j; hence the group of permutations of (4) is itself commutative.

Therefore $\Phi(\omega) = 0$ is an Abelian equation of degree s. Its first factor $F(x, \omega)$ has the roots $\alpha, \alpha_1, \ldots, \alpha_{r-1}$, and the group of the equation

$$F(x, \omega) = 0$$

consists only of the permutations belonging to the period of the cyclic permutation γ_1.

We shall call an equation whose group consists of a single cycle and its repetitions a *cyclic equation*. Thus cyclic equations are the simplest special case of Abelian equations.

We have therefore proved that the solution of every Abelian equation is reduced to the solution of an Abelian equation of lower degree together with the solution of a collection of cyclic equations.

One can apply the same theorem again to the auxiliary equation of degree s, and continue in this way until that auxiliary equation is reduced to degree one.

Thus we obtain the result:

Theorem. The solution of an Abelian equation can always be reduced to the solution of a sequence of cyclic equations whose degrees divide the degree of the given equation.

It is not necessary to include irreducibility in the definition of cyclic equations. We may therefore formulate the definition in general as follows.

An equation of degree m, $F(x) = 0$, with m distinct roots is called a *cyclic equation* over the field K if its roots $\alpha, \alpha_1, \alpha_2, \ldots, \alpha_{m-1}$ are not rational, but can be arranged in such a way that the cyclic functions of the roots are rational over K.

If therefore

$$\pi = (\alpha, \alpha_1, \alpha_2, \ldots, \alpha_{m-1}), \tag{6}$$

then every function belonging to K and invariant under the permutations of the cyclic group

$$C = 1, \pi, \pi^2, \pi^3, \ldots, \pi^{m-1} \tag{7}$$

must be rational.

The Galois group of a cyclic equation is either the whole period C, in which case the equation is irreducible, or else it is a subgroup of C; in the latter case, if e and f are two integer divisors of m with $m = ef$, then this group consists of the permutations

$$C^e = 1, \pi^e, \pi^{2e}, \ldots, \pi^{(f-1)e}, \tag{8}$$

and the cyclic equation $F(x) = 0$ splits into e factors of degree f.

For let

$$t = \psi(\alpha, \alpha^e, \alpha^{2e}, \ldots, \alpha^{(f-1)e})$$

be a function belonging to the group C, and assume it is equal to a quantity in K, which we denote by a.

Then on the rational equation

$$F(x) = F(x, a)F(x, a_1) \cdots F(x, a_{e-1})$$

no permutation not contained in C can act, and consequently the group of the equation can contain no substitutions other than those occurring in C.

If now π itself occurs in the group of the equation, then the group is identical with C, and since C is transitive, the equation is irreducible.

If, however, $\pi^{e'}$ is the smallest power of π occurring in the group of the equation, then $C^{e'}$ is the group. If $e' < m$, then $C^{e'}$ is intransitive, and the equation is reducible.

The same reduction can also be applied to cyclic equations whose degree is not prime. If namely $m = ef$ is any factorization of m into two factors, then the permutation π^e splits into e cycles γ, each with f elements:

$$\gamma = (\alpha, \alpha^e, \alpha^{2e}, \ldots, \alpha^{(f-1)e}). \tag{9}$$

Now determine a function ψ belonging to the first of these cycles, and set

$$\begin{aligned}
\eta = \eta_0 &= \psi(\alpha, \alpha^e, \alpha^{2e}, \ldots, \alpha^{(f-1)e}), \\
\eta_1 &= \psi(\alpha_1, \alpha^{e+1}, \alpha^{2e+1}, \ldots, \alpha^{(f-1)e+1}), \\
&\ldots, \\
\eta_{e-1} &= \psi(\alpha_{e-1}, \alpha_{2e-1}, \alpha_{3e-1}, \ldots, \alpha_{m-1}).
\end{aligned} \tag{10}$$

These quantities are all distinct, yet under application of the permutation π they pass cyclically into one another. Hence their cyclic functions, and therefore also their symmetric functions, are rational. So they are the roots of a cyclic equation of degree e, while $F(x)$ splits into e factors of degree f,

$$F(x) = F(x, \eta)F(x, \eta_1) \cdots F(x, \eta_{e-1}),$$

each of which gives a cyclic equation of degree f for the roots of one of the cycles γ.

Indeed, if for example

$$F_1 = (x - \alpha)(x - \alpha^e) \cdots (x - \alpha^{(f-1)e}),$$

then this function admits the permutations in the period of γ. It also admits the permutations of the group that becomes the Galois group of our equation after adjoining η, a group that can consist only of powers of $\gamma_0, \gamma_1, \ldots, \gamma_{e-1}$.

Hence, by Lagrange's theorem (§155, equation (2)), F_1 is rationally expressible in terms of η, so $F_1 = F(x, \eta)$. But the equation

$$F(x, \eta) = 0$$

is again cyclic over $K(\eta)$, since the cyclic functions of its roots belong to that field.

Like all Abelian equations, cyclic equations have the property that every root is rationally expressible in terms of every other. Here these expressions can be arranged in the following cyclic manner. If $\alpha, \alpha_1, \alpha_2, \ldots, \alpha_{m-1}$ are the roots of the cyclic equation $F(x) = 0$, then the entire function of degree $m - 1$ in x

$$\Psi_m\left(x + \frac{F'(x)}{\Psi'_m(x)}\right) = \Psi_m$$

is unchanged by the permutation π and therefore belongs to K. If in it we set

$$x = \alpha, \alpha_1, \ldots, \alpha_{m-1}$$

and introduce the symbol θ, then it follows that

$$\alpha_1 = \theta(\alpha), \qquad \alpha_2 = \theta(\alpha_1), \qquad \ldots, \qquad \alpha_{m-1} = \theta(\alpha_{m-2}), \qquad \alpha = \theta(\alpha_{m-1}), \tag{11}$$

and this holds whether $F(x)$ is reducible or irreducible, provided only that $F(x)$ and $F'(x)$ have no common factor.

The solution of cyclic equations of degree m has thus been made dependent on the solution of cyclic equations whose degrees are the prime factors of m. If for example m is a power of 2, then the solution is carried out by a sequence of square roots. In this way Gauss first treated the cyclotomic equations.

Finally we add the remark, important for what follows, that for prime degree the notions of normal equation and cyclic equation coincide. Certainly a cyclic equation of prime degree is irreducible and hence a normal equation.

Conversely, if the degree n of a normal equation, which is also the degree of its group, is prime, and if π is any non-identity permutation of this group, then the degree of π, which must divide n, is equal to n; hence

$$\pi^n = 1, \qquad \pi, \pi^2, \dots, \pi^{n-1}$$

constitute the group of the equation, and the equation is cyclic.

§5. Lagrange resolvents

164. The method of solution for cyclic equations

The method by which we shall now learn to solve cyclic equations applies uniformly to prime degrees and to composite degrees. One uses for this purpose certain expressions known under the name of *Lagrange resolvents*, which are of great use in all investigations concerning the algebraic solution of equations.

Let $F(x) = 0$ be an equation with roots

$$\alpha, \alpha_1, \alpha_2, \dots, \alpha_{m-1}.$$

Let ε be any mth root of unity, and introduce the notation

$$(\varepsilon, \alpha) = \alpha + \varepsilon\alpha_1 + \varepsilon^2\alpha_2 + \cdots + \varepsilon^{m-1}\alpha_{m-1}. \tag{1}$$

These sums are what one calls the *Lagrange resolvents.* If these functions are known for all mth roots of unity ε, then the equation itself is solved; for by §133, equation (6),

$$\sum_{\varepsilon} \varepsilon^{-k} = m \quad \text{or} \quad 0, \tag{2}$$

according as k is divisible by m or not, and hence

$$m\alpha_k = \sum_{\varepsilon} (\varepsilon, \alpha)\varepsilon^{-k}, \tag{3}$$

where the sums extend over all mth roots of unity ε.

The other roots can be expressed in the same way:

$$m\alpha_k = \sum_{\varepsilon} \varepsilon^{-k}(\varepsilon, \alpha), \tag{4}$$

so that everything is indeed reduced to knowledge of (ε, α) and the functions formed from it.

The sums (2) and (3) can also be written in another fashion, since all mth roots of unity are powers of a primitive one.

If therefore ε denotes a fixed primitive mth root of unity and h runs through a complete system of residues modulo m, say $0, 1, \ldots, m-1$, then equations (2), (3) may be written as

$$m\alpha = \sum_{h} (\varepsilon^h, \alpha), \qquad m\alpha_k = \sum_{h} \varepsilon^{-hk}(\varepsilon^h, \alpha). \tag{5}$$

We first investigate these resolvents as functions of m independent variables α, and for simpler notation we adopt the convention that $\alpha_h = \alpha_k$ whenever $h \equiv k \pmod{m}$.

Then the entire system of variables α is obtained by letting the index h run through a complete residue system modulo m.

For these resolvents the following statements hold.

1. If one applies to the indices of the α the cyclic permutation

$$\pi = (0, 1, \ldots, m-1),$$

then (ε, α) passes into $\varepsilon^{-1}(\varepsilon, \alpha)$, and under the permutation π^k it passes into $\varepsilon^{-k}(\varepsilon, \alpha)$.

This follows immediately from the defining formula (1).

Now let ν be any positive exponent and form $(\varepsilon, \alpha)^\nu$. Expand this power by the polynomial theorem, set $\varepsilon^m = 1$, and order the result by powers of ε. One obtains an expression of the form

$$(\varepsilon, \alpha)^\nu = A_0^{(\nu)} + \varepsilon A_1^{(\nu)} + \varepsilon^2 A_2^{(\nu)} + \cdots + \varepsilon^{m-1} A_{m-1}^{(\nu)}, \tag{5'}$$

where the $A_h^{(\nu)}$ are forms of degree ν with integer coefficients in the variables α, but are independent of ε. Again we stipulate that $A_h^{(\nu)} = A_k^{(\nu)}$ whenever $h \equiv k \pmod{m}$.

From this we prove the following statement.

2. If one applies the cyclic permutation π to the indices of the α, then the indices of the coefficients in $(\varepsilon, \alpha)^\nu$ undergo the cyclic permutation π^ν; that is, $A_h^{(\nu)}$ passes into $A_{h+\nu}^{(\nu)}$.

To prove this, note that formula $(5')$ remains valid for every mth root of unity ε, including 1. Using (2) one then obtains

$$mA_k^{(\nu)} = \sum_{\varepsilon} \varepsilon^{-k}(\varepsilon, \alpha)^\nu. \tag{6}$$

If we apply the permutation π on the right-hand side, then by statement 1 the term $(\varepsilon, \alpha)^\nu$ is multiplied by $\varepsilon^{-\nu}$, so the whole sum becomes

$$\sum_{\varepsilon} \varepsilon^{-(k+\nu)}(\varepsilon, \alpha)^\nu = mA^{(\nu)}_{k+\nu},$$

which is exactly the assertion.

3. Thus, when the permutation π is applied to the indices of the α, it induces on the indices of the coefficients of the product ordered by powers of ε the permutation determined by the total exponent-weight.

This general principle will be used in the further computation of the Lagrange resolvents.

This batch stops at the opening structural properties of the Lagrange resolvents.

§37. Linear substitutions and their composition

One of the most effective means for forming groups, and at the same time one that leads to many applications, is given by linear substitutions and their composition. We have already encountered such linear substitutions several times, for example in the second section of the first volume on the occasion of the multiplication law for determinants, and then again in the eleventh section in connection with equivalent numbers.

By a *linear substitution* in n variables we understand a system of equations by which one system of n variables $y_1, y_2, \ldots, y_n$ is expressed linearly in terms of another system $x_1, x_2, \ldots, x_n$. According to the number of variables we distinguish unary, binary, ternary, quaternary substitutions, and in general we shall call the number of variables the *dimension* of the substitution. For the time being we restrict attention to homogeneous substitutions, so that, if the coefficients are denoted by $a_i^{(k)}$, the substitution is represented by the system

$$y_\kappa = \sum_{i=1}^{n} a_i^{(\kappa)} x_i \qquad (\kappa = 1, 2, \ldots, n). \tag{1}$$

Very often the variables themselves are not important, so that such a substitution is sufficiently characterized by its coefficients $a_i^{(\kappa)}$. One sometimes uses a single letter, say A, to denote a substitution, setting

$$A = \begin{pmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} \end{pmatrix}. \tag{2}$$

We shall also write briefly $A = (a_i^{(\kappa)})$, and denote the determinant of the substitution coefficients by $|A|$. This determinant is always assumed to be nonzero. We also write the system symbolically as

$$(y_1, y_2, \ldots, y_n) = A(x_1, x_2, \ldots, x_n), \tag{3}$$

or, more briefly,

$$(y) = A(x). \tag{4}$$

The substitution

$$y_1 = x_1, \quad y_2 = x_2, \quad \ldots, \quad y_n = x_n \tag{5}$$

or

$$J = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \tag{6}$$

is called the *identity substitution*.

A substitution of the form

$$y_1 = \mu_1 x_1, \quad y_2 = \mu_2 x_2, \quad \ldots, \quad y_n = \mu_n x_n \tag{7}$$

or

$$M = \begin{pmatrix} \mu_1 & 0 & \cdots & 0 \\ 0 & \mu_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mu_n \end{pmatrix} \tag{8}$$

will be called a *multiplicative substitution*, or simply a *multiplication*; the elements $\mu_1, \mu_2, \ldots, \mu_n$ are called the *multipliers*.

Now suppose that we have a second linear substitution of the same dimension n, introducing a new system of variables z:

$$z_i = \sum_{h=1}^{n} b_h^{(i)} y_h, \qquad (z) = B(y). \tag{9}$$

If we substitute for y the values from (1), then we obtain the z expressed in terms of the x by means of a new linear substitution E, which we denote by

$$z = E(x) = BA(x), \tag{10}$$

and the substitution $E = BA$ is said to be *composed* from B and A. One must, however, distinguish between AB and BA. Two substitutions A, B having the special property $AB = BA$ are called *interchangeable* or *commutative*.

The coefficients of E are obtained by inserting the expressions (1) into (9):

$$z_\kappa = \sum_h e_h^{(\kappa)} x_h, \qquad e_h^{(\kappa)} = \sum_{i=1}^{n} b_i^{(\kappa)} a_h^{(i)}. \tag{11}$$

These formulas are exactly the same as those used earlier for the multiplication of determinants. Thus we may express the composition law of substitutions as follows.

1. To form the substitution BA composed from B and A, one proceeds exactly as if the determinants $|B|$ and $|A|$ were to be multiplied by the determinant multiplication rule. In order to obtain the elements of one row, one multiplies the elements of a row of the first factor by the corresponding elements of the columns of the second factor and then adds.

From this rule there follows immediately:

2. The determinant of a composed substitution is equal to the product of the determinants of its components.

To compose three substitutions C, B, A of the same dimension, we insert the expressions (10) for z into a further linear substitution

$$(u) = C(z). \tag{12}$$

Then the variables u are expressed in terms of the x by

$$(u) = CBA(x). \tag{13}$$

Since it is evidently immaterial whether one first introduces (10) into (12), or first expresses z by y and then y by x, one obtains for this composition the associative law

$$C(BA) = (CB)A = CBA. \tag{14}$$

The same may of course be verified directly from (11). Thus:

3. For the composition of linear substitutions, the associative law holds, but in general not the commutative law.

A multiplication in which all multipliers are equal,

$$N = \begin{pmatrix} \nu & 0 & \cdots & 0 \\ 0 & \nu & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \nu \end{pmatrix},$$

will be called a *similarity substitution.* Two substitutions that are obtained from one another by composition with a similarity substitution, such as A and AN or A and NA, are called *similar.* From the composition law one sees immediately:

4. A similarity substitution is commutative with every substitution A of the same dimension; two similarity substitutions, composed with one another, again yield a similarity substitution; and two substitutions that are each similar to a third are also similar to one another.

Equally immediate is the statement:

5. Composition with the identity substitution J leaves every substitution unchanged.

The substitution J therefore plays the role of a unit in composition and may be omitted wherever it occurs composed with other substitutions.

We have also the following theorem.

6. To every substitution A there exists one and only one inverse substitution A^{-1} satisfying

$$AA^{-1} = A^{-1}A = J. \tag{15}$$

This last statement follows from the fundamental formulas of determinant theory if one determines the elements $\alpha_h^{(\kappa)}$ of A^{-1} from the linear equations

$$\sum_i a_i^{(h)} \alpha_\kappa^{(i)} = \begin{cases} 0, & h \neq \kappa, \\ 1, & h = \kappa, \end{cases} \tag{16}$$

from which, if $A_\kappa^{(h)}$ denotes the minor of $|A|$, one obtains

$$|A|\alpha_h^{(\kappa)} = A_\kappa^{(h)}. \tag{17}$$

These in turn imply the system

$$\sum_i \alpha_i^{(h)} a_\kappa^{(i)} = \begin{cases} 0, & h \neq \kappa, \\ 1, & h = \kappa. \end{cases} \tag{18}$$

The inverse substitution of A is nothing other than the solution of the system of equations defining the direct substitution A, so that from

$$(y) = A(x)$$

there follows

$$(x) = A^{-1}(y). \tag{19}$$

For the composition of inverse substitutions one obtains from $ABB^{-1}A^{-1} = J$ the formula

$$(AB)^{-1} = B^{-1}A^{-1}. \tag{20}$$

If A, B, C are three substitutions, then composition with A^{-1} shows from either of the equations

$$AB = AC, \qquad BA = CA$$

that necessarily $B = C$. Hence the characteristic properties required for a group are satisfied by the composition of substitutions, and thus:

The totality of all substitutions of a fixed dimension forms an (infinite) group.

Inverse substitutions

Composition with the identity substitution J changes no substitution. Thus, in the composition law, J is regarded as the unit element, and may be omitted whenever it occurs as a factor composed with other substitutions.

We now have the following theorem.

Theorem. For every substitution A there exists a unique inverse substitution A^{-1} satisfying

$$AA^{-1} = A^{-1}A = J. \tag{1}$$

This follows from the basic formulas of determinant theory. If the entries of A^{-1} are denoted by $\alpha_h^{(k)}$, then they are determined by the linear equations obtained from the requirement that the product with A be the identity. Since the determinant of A is assumed non-zero, these equations have a unique solution. The inverse substitution is therefore uniquely determined.

For inverses of composites one obtains, from

$$ABB^{-1}A^{-1} = J,$$

the formula

$$(AB)^{-1} = B^{-1}A^{-1}. \tag{2}$$

If A, B, C are three substitutions, then from either of the equations

$$AB = AC, \qquad BA = CA,$$

composition with A^{-1} gives $B = C$. Thus the characteristic properties required for a group are satisfied under composition. The totality of all substitutions of a fixed dimension is therefore an infinite group. Moreover, if one selects from this totality any set closed under composition, then that set itself forms a group, finite or infinite according to the case.

Transformed substitutions

Let L be a fixed substitution. From any substitution A of the same dimension one obtains another substitution

$$A' = L^{-1}AL, \tag{3}$$

which is called the transform of A by L.

Putting

$$(x) = L(x'), \qquad (y) = L(y'), \tag{4}$$

one obtains, from $(y) = A(x)$,

$$(y') = A'(x'). \tag{5}$$

Consequently the passage to the transformed substitution is equivalent to applying the same change of variables, namely L^{-1}, to both systems of variables.

If

$$A' = L^{-1}AL, \qquad B' = L^{-1}BL,$$

then the laws of composition give

$$A'B' = L^{-1}ABL.$$

Hence:

Theorem. As A runs through the substitutions of a group, and L is kept fixed, the transformed substitutions $L^{-1}AL$ run through the substitutions of an isomorphic group.

Transposed substitutions

Inverse substitutions must be distinguished from transposed substitutions. From a substitution A one obtains a definite new substitution, denoted here by A_t, by turning rows into columns and columns into rows; equivalently, in Weber's notation, one interchanges the upper and lower indices of the coefficients of A.

If in the coefficient formula for the product $C = BA$ one interchanges upper and lower indices and simultaneously interchanges the letters a and b, one obtains the coefficients of the transpose of the product. Thus, when the factors in the product are transposed, their order is reversed:

$$(BA)_t = A_t B_t. \tag{6}$$

The operation of transposition is therefore formally similar to taking inverses, in that an order reversal occurs. Combining both operations gives

$$(A^{-1})_t = (A_t)^{-1}, \tag{7}$$

and, where no ambiguity is possible, this may be written simply as A_t^{-1}.

Linear fractional substitutions in one variable

We now apply the preceding general results to some special cases, beginning with substitutions of one variable. These substitutions are the same as the linear fractional functions

$$y = \frac{\alpha x + \beta}{\gamma x + \delta}, \tag{8}$$

for which Weber introduces the abbreviation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \tag{9}$$

Their composition is governed by

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} = \begin{pmatrix} \alpha\alpha' + \beta\gamma' & \alpha\beta' + \beta\delta' \\ \gamma\alpha' + \delta\gamma' & \gamma\beta' + \delta\delta' \end{pmatrix}. \tag{10}$$

The coefficients $\alpha, \beta, \gamma, \delta$ are arbitrary numbers subject only to the condition that

$$D = \alpha\delta - \beta\gamma \neq 0.$$

The identity substitution is

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{11}$$

and the inverse of (9) is

$$\begin{pmatrix} \dfrac{\delta}{D} & \dfrac{-\beta}{D} \\ \dfrac{-\gamma}{D} & \dfrac{\alpha}{D} \end{pmatrix}. \tag{12}$$

The substitution transposed to (9) is obtained by interchanging the diagonal entries and changing the signs of the off-diagonal entries:

$$\begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}. \tag{13}$$

Equivalently, this is obtained by replacing

$$\alpha, \beta, \gamma, \delta \quad \text{by} \quad \delta, -\gamma, -\beta, \alpha.$$

One verifies immediately that

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} = \begin{pmatrix} D & 0 \\ 0 & D \end{pmatrix}. \tag{14}$$

This both confirms the preceding rule for transposes and shows that every such substitution commutes with its transposed substitution.

Permutation groups as linear substitutions

Let $x_1, x_2, \dots, x_n$ be a system of n variables, and let

$$\alpha_1, \alpha_2, \dots, \alpha_n$$

be some ordering of the numbers $1, 2, \dots, n$. The equations

$$x_1 = x'_{\alpha_1}, \quad x_2 = x'_{\alpha_2}, \quad \dots, \quad x_n = x'_{\alpha_n} \tag{15}$$

define a linear substitution of dimension n,

$$A = \begin{pmatrix} a_1^{(1)} & \cdots & a_n^{(1)} \\ \cdots & \cdots & \cdots \\ a_1^{(n)} & \cdots & a_n^{(n)} \end{pmatrix}, \qquad (x) = A(x'), \tag{16}$$

in which every row and every column contains exactly one non-zero coefficient, and that coefficient is 1. Hence $|A| = \pm 1$. If, after carrying out the substitution, one writes x_i again in place of x'_i, the result is just the permutation

$$\begin{pmatrix} 1, 2, \dots, n \\ \alpha_1, \alpha_2, \dots, \alpha_n \end{pmatrix}$$

of the indices.

Let B be another such substitution,

$$(x') = B(x''), \tag{17}$$

or, more explicitly,

$$x'_1 = x''_{\beta_1}, \quad x'_2 = x''_{\beta_2}, \quad \dots, \quad x'_n = x''_{\beta_n}. \tag{18}$$

Then composition gives

$$x_1 = x''_{\beta_{\alpha_1}}, \quad x_2 = x''_{\beta_{\alpha_2}}, \quad \dots, \quad x_n = x''_{\beta_{\alpha_n}}, \tag{19}$$

which is abbreviated by

$$(x) = AB(x''). \tag{20}$$

In other words,

$$\begin{pmatrix} 1, 2, \dots, n \\ \alpha_1, \alpha_2, \dots, \alpha_n \end{pmatrix} \begin{pmatrix} 1, 2, \dots, n \\ \beta_1, \beta_2, \dots, \beta_n \end{pmatrix} = \begin{pmatrix} 1, 2, \dots, n \\ \beta_{\alpha_1}, \beta_{\alpha_2}, \dots, \beta_{\alpha_n} \end{pmatrix}. \tag{21}$$

Thus, when the substitutions A and B are regarded as permutations of the indices, their product as substitutions agrees with their composite as permutations.

Therefore permutation groups on n letters are simply a special case of finite groups of linear substitutions of dimension n. Even permutations correspond to substitutions of determinant $+1$, and odd permutations to substitutions of determinant -1.

This connection between permutations and linear substitutions makes it possible to derive from every permutation group a group of linear substitutions, and conversely to treat certain questions about permutation groups by means of the theory of linear substitutions.

§40. Groups of binary linear substitutions with real coefficients

For the binary substitutions

$$y_1 = \alpha x_1 + \beta x_2, \qquad y_2 = \gamma x_1 + \delta x_2, \tag{22}$$

the determinant is

$$D = \alpha\delta - \beta\gamma.$$

We shall first suppose that the coefficients $\alpha, \beta, \gamma, \delta$ are real, and ask under what conditions a group of such substitutions can be finite.

The characteristic equation of the substitution is

$$\begin{vmatrix} \alpha - \lambda & \beta \\ \gamma & \delta - \lambda \end{vmatrix} = \lambda^2 - (\alpha + \delta)\lambda + D = 0. \tag{23}$$

Its roots are

$$\lambda = \frac{\alpha + \delta \pm \sqrt{(\alpha + \delta)^2 - 4D}}{2}. \tag{24}$$

If

$$(\alpha + \delta)^2 - 4D > 0,$$

then the roots are real and distinct. In that case the substitution can be brought, by a similar substitution, into the form

$$y_1 = \lambda_1 x_1, \qquad y_2 = \lambda_2 x_2. \tag{25}$$

The n-fold iterate is then

$$y_1 = \lambda_1^n x_1, \qquad y_2 = \lambda_2^n x_2.$$

For some positive power to become the identity, λ_1 and λ_2 must be roots of unity. Since they are real, this forces

$$\lambda_1, \lambda_2 \in \{1, -1\}.$$

If

$$(\alpha + \delta)^2 - 4D = 0,$$

then the two characteristic roots coincide: $\lambda_1 = \lambda_2 = \lambda$. The substitution can then either be reduced to the diagonal form above, or to the Jordan form

$$y_1 = \lambda x_1, \qquad y_2 = x_1 + \lambda x_2. \tag{26}$$

In the latter case the n-fold iterate contains the term

$$y_2 = n\lambda^{n-1} x_1 + \lambda^n x_2,$$

and therefore cannot be the identity for any positive n, unless the nilpotent part vanishes. Thus a finite group contains no nontrivial substitution of this Jordan type.

It remains to consider the case

$$(\alpha + \delta)^2 - 4D < 0.$$

Then the two characteristic roots are conjugate complex roots. A real substitution with such roots is, over the real field, similar to a rotation-dilatation. If the substitution has finite order, its characteristic roots must again be roots of unity; hence the quotient of the two roots is a root of unity and the corresponding projective transformation is periodic.

Consequently, a real binary linear substitution which belongs to a finite group is characterized by the fact that, after multiplication by a scalar if necessary, it can be interpreted as a rotation through an angle commensurable with 2π, or as one of the exceptional involutory cases in which the characteristic roots are 1 and -1.

This gives a first reduction of the problem of finite real binary substitution groups to the problem of finite rotation groups and their associated projective actions.

§41. Invariants and covariants of groups of linear substitutions

The theory of linear substitutions is connected in the closest way with the theory of algebraic forms. By a *form* in the variables

$$x_1, x_2, \ldots, x_n$$

we mean an integral homogeneous function of these variables. A form

$$f(x_1, \ldots, x_n)$$

of degree p is transformed by a linear substitution A into another form

$$f'(x_1', \ldots, x_n')$$

of the same degree in the new variables. If $f' = f$, that is, if the form is transformed into itself by the substitution, then f is called an *invariant* of A.

More generally, suppose that f is a function not only of the variables x, but also of other variables $u, v, \ldots$. If f is transformed into itself by applying A to the x-variables and at the same time applying corresponding substitutions to the variables $u, v, \ldots$, then f is called a *covariant.*

We now consider a group S of linear substitutions in the variables

$$x_1, \ldots, x_n,$$

and seek those forms which are simultaneously invariants, or covariants, for all substitutions of the group.

Let

$$F_1, F_2, \ldots, F_r$$

be a system of such invariants which are rationally independent, meaning that no identical rational relation holds among them, while every other invariant can be rationally expressed through them. Such a system is called a *basis* of the invariant system of the group S.

The most general invariant has the form

$$F = A_1F_1 + A_2F_2 + \cdots + A_rF_r, \tag{27}$$

where $A_1, A_2, \ldots, A_r$ are arbitrary constants, or more generally quantities lying in the appropriate coefficient field.

From the general theory of algebraic forms one obtains the following fundamental facts.

I. There always exists a finite system of invariants from which all the remaining invariants may be generated rationally.

II. If the group is finite, the invariant system may be chosen so that it reflects the finite averaging process over the substitutions of the group.

III. The corresponding covariants are obtained in the same manner once the simultaneous transformation law of the auxiliary variables is fixed.

These facts allow the invariant theory of a finite linear substitution group to be treated algebraically: the full system of invariant forms is replaced by a finite basis and the rational relations among its elements.

0.1 The extended notion of invariant

Relative invariants are in essence a special case of a more general notion, which, in a manner similar to that of relative invariants, can be used for reducing the form problem.

We look for systems of homogeneous forms of the same degree in the variables

$$x_1, x_2, \ldots, x_n,$$

namely

$$X_1, X_2, \ldots, X_m, \tag{28}$$

with the property that, when the substitutions of the group S are applied, the functions $X_1, \ldots, X_m$ undergo a system Σ of linear substitutions. These substitutions Σ then themselves form a group; this group is isomorphic to S, either simply or with several stages of identification. If $m = 1$, one recovers the notion of a relative invariant.

We now seek all substitutions of S which leave each of the functions (28) individually unchanged. These substitutions form a group, which we denote by T, and we shall show that T is a normal divisor of S. Let s be a substitution in S which induces the substitution σ on the X's. Then s^{-1} induces σ^{-1}. If τ is a substitution in T, then τ leaves all the X's fixed. Therefore $s^{-1}\tau s$ induces

$$\sigma^{-1}\sigma = 1$$

on the X's, and hence also leaves all the X's fixed. Thus $s^{-1}\tau s$ belongs to T, and T is a normal divisor of S.

Let μ and ν be the orders of S and T, and write

$$\mu = e\nu.$$

Then e is the index of T and at the same time the order of the group Σ.

Every absolute invariant of the group T, that is, every function of x which admits the substitutions of T, can be expressed rationally over Ω in terms of the X's.

The proof is the same argument that has already been used several times. Let ϑ be a function of the X's which assumes the e distinct values

$$\vartheta, \vartheta_1, \ldots, \vartheta_{e-1},$$

and put

$$\Phi(t) = (t-\vartheta)(t-\vartheta_1)\cdots(t-\vartheta_{e-1}). \tag{29}$$

Then $\Phi(t)$ is a rational function of the variable t over Ω.

Let r be an absolute invariant of T. It assumes at most e different values corresponding to $\vartheta, \vartheta_1, \ldots, \vartheta_{e-1}$, say

$$r, r_1, \ldots, r_{e-1},$$

where repetitions may occur. Then the function

$$\frac{\Phi(t)r}{t-\vartheta} + \frac{\Phi(t)r_1}{t-\vartheta_1} + \cdots + \frac{\Phi(t)r_{e-1}}{t-\vartheta_{e-1}} = F(t)$$

belongs to Ω, and for $t = \vartheta$ one obtains

$$r = \frac{F(\vartheta)}{\Phi'(\vartheta)}.$$

Since $\Phi'(\vartheta) \neq 0$, this expresses r rationally in ϑ. Hence r belongs to the field

$$\Omega(X_1, X_2, \ldots, X_m).$$

The invariants of T are also invariants of S. After the form problem for the group Σ has been solved, the functions $X_1, \ldots, X_m$ are known, and therefore the invariants of T are known as well. Thus the form problem for S is reduced to the successive solution of the two form problems belonging to the groups Σ and T, both of lower degree than the original form problem for S. A form problem which can be decomposed in this way into two lower-degree form problems will be called *imprimitive*. Since such a reduction requires the existence of a normal divisor T of S distinct from both S and the identity, it follows that, if S is simple, the corresponding form problem is necessarily *primitive*.

Conversely, if S has a normal divisor of index e, then the form problem is imprimitive. To obtain a system of functions $X_1, \ldots, X_m$, one takes an invariant ϑ of T which assumes e different values under S, and one may take these values

$$\vartheta, \vartheta_1, \ldots, \vartheta_{e-1}$$

as the functions $X_1, \ldots, X_m$. The group Σ is then the permutation group induced by S on these ϑ's. In the sense of the preceding section, however, one wants m to be as small as possible, and a useful reduction of the problem is achieved only when $m < n$.

0.2 Normal forms

Before turning to special applications, we need one more general observation. We have already seen that from any group S of linear substitutions one obtains infinitely many isomorphic groups by transforming with an arbitrary substitution L of the same dimension, namely by forming

$$LSL^{-1}. \tag{30}$$

This is equivalent to replacing the variables x by variables y through

$$(x) = L(y). \tag{31}$$

This transformation can be used to put the group S into a simpler normal form; then the invariants also acquire definite normal forms, which in many cases may be quite simple.

For the normal form, let the resolvent of the form problem be written

$$\Phi(\vartheta, A_1, A_2, \ldots) = 0, \tag{32}$$

where $A_1, A_2, \ldots$ denote invariants of the group S. The variables x_i can then be expressed rationally in terms of ϑ and the A_i:

$$x_i = \varphi_i(\vartheta, A_1, A_2, \ldots). \tag{33}$$

Once the function ϑ has been fixed, these expressions take a definite form for the normal form of the group.

It may also happen that the invariants are not given in the normal form, but in some other form, which we shall call the *general form*. One then has to determine the substitution L which transforms the general form into a previously recognized possible normal form. This task is no other than the form problem itself for the normal form.

Assume that the invariants in the general form, considered as functions of y, are $B_1, B_2, \ldots$, so that the substitution (31) produces the identities

$$A_1 = B_1, \qquad A_2 = B_2, \qquad \ldots. \tag{34}$$

The task is to determine the substitution L making these equations identities, when the forms of the functions A_i and B_i are known. This is solved once one assumes solved equation (32) for a suitably chosen special system of values of the A_i.

Indeed, differentiate (32) and (33):

$$\begin{aligned} 0 &= \Phi'(\vartheta)d\vartheta + \sum_i \Phi'(A_i)dA_i, \\ dx_i &= \varphi_i'(\vartheta)d\vartheta + \sum_i \varphi_i'(A_i)dA_i. \end{aligned}$$

Eliminating $d\vartheta$ gives

$$dx_i = \sum_i \frac{\varphi_i'(A_i)\Phi'(\vartheta) - \Phi'(A_i)\varphi_i'(\vartheta)}{\Phi'(\vartheta)}\, dA_i. \tag{35}$$

In consequence of (34), one has

$$dA_i = \sum_k \frac{\partial B_i}{\partial y_k}\, dy_k. \tag{36}$$

Comparison with the differentials of the substitution $x = L(y)$ then determines the coefficients of L as rational expressions once a special value system for the y's has been chosen, subject only to the condition that $\Phi'(\vartheta)$ not vanish. Thus the determination of the normalizing substitution has been reduced to the same resolvent which solves the form problem.

0.3 Ternary orthogonal substitutions

We now have to pick out, from the totality of linear substitutions, narrower groups, and finally finite groups. Such narrower groups are obtained by imposing the condition that certain homogeneous functions of the variables should remain invariant. Instead of treating this in full generality, we pass immediately to the most important special case. We restrict ourselves to ternary substitutions and require that a quadratic form of non-vanishing determinant remain invariant. Since every such quadratic form can be transformed linearly into a sum of squares, no generality is lost by taking the form to be the sum of squares. Such substitutions are called *orthogonal*.

Thus

$$(y_1, y_2, y_3) = A(x_1, x_2, x_3) \tag{37}$$

is orthogonal if its coefficients are chosen so that the identity

$$y_1^2 + y_2^2 + y_3^2 = x_1^2 + x_2^2 + x_3^2 \tag{38}$$

holds. Substituting the expressions (37) into (38) and equating corresponding coefficients yields six relations among the nine coefficients of A.

If

$$A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}, \tag{39}$$

these relations are

$$\begin{aligned} a_1^2 + b_1^2 + c_1^2 &= 1, \quad a_2a_3 + b_2b_3 + c_2c_3 = 0, \\ a_2^2 + b_2^2 + c_2^2 &= 1, \quad a_3a_1 + b_3b_1 + c_3c_1 = 0, \\ a_3^2 + b_3^2 + c_3^2 &= 1, \quad a_1a_2 + b_1b_2 + c_1c_2 = 0. \end{aligned} \tag{40}$$

They are the familiar relations of analytic geometry. From them, using the multiplication theorem for determinants, one obtains $|A|^2 = 1$, and therefore $|A| = \pm 1$.

Equations (40) also show that the inverse substitution of A is identical with the transpose:

$$A^{-1} = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}.$$

This property could itself serve as the definition of orthogonal substitutions.

All orthogonal substitutions form a group. Inside it lies the narrower group distinguished by

$$|A| = +1, \tag{41}$$

which Weber calls the group of *proper orthogonal substitutions.*

The substitution (37) may be interpreted as the passage from one rectangular coordinate system to another with the same origin. If the determinant condition (41) holds, then the first coordinate system can be brought into coincidence with the second by a rotation about a fixed axis through a definite angle.

To see this, set

$$D = \begin{vmatrix} a_1 - 1 & a_2 & a_3 \\ b_1 & b_2 - 1 & b_3 \\ c_1 & c_2 & c_3 - 1 \end{vmatrix}.$$

From the orthogonality relations one obtains $|A|D = -D$; hence, when $|A| = +1$, $D = 0$. Therefore one can determine λ, μ, ν from

$$\begin{aligned} \lambda &= a_1\lambda + a_2\mu + a_3\nu, \\ \mu &= b_1\lambda + b_2\mu + b_3\nu, \\ \nu &= c_1\lambda + c_2\mu + c_3\nu, \\ \lambda^2 + \mu^2 + \nu^2 &= 1. \end{aligned} \tag{42}$$

These numbers determine the direction of a straight line making with the axes y_1, y_2, y_3 the same angles as with x_1, x_2, x_3; this line is the axis of rotation. In the case $|A| = -1$ this conclusion no longer applies.

There is also another interpretation: (x) and (y) may be regarded as the coordinates of two different points, referred to the same coordinate system. If x describes a line, surface, or body, then y describes a congruent figure; when $|A| = -1$, it is the mirror image.

The group of proper orthogonal substitutions is equivalent to the group obtained from the possible positions of a rigid body rotating about a fixed point. If a position E is chosen as the identity, every other position A is reached by a rotation about a certain axis through a certain angle. If B is another position, then the compound position AB is the position obtained by carrying out the rotation which leads to A, but starting from the position B rather than from the identity. This is exactly the composition law for the corresponding orthogonal substitutions.

By composing the proper orthogonal substitutions with one improper substitution, for example the reflection

$$(x_1, x_2, x_3) = (y_1, y_2, -y_3),$$

one obtains all orthogonal substitutions.

The finite groups contained in the group of proper orthogonal substitutions are of special interest. Geometrically, such a finite group of degree g corresponds to a finite set of positions of a body. If one imagines the body fixed simultaneously in all these positions and unites the whole into a new rigid body, one obtains a figure that can cover itself in g different positions. The clearest cases occur when the body is bounded by planes: regular pyramids, double pyramids, and the regular solids.

A regular pyramid with g sides covers itself in g positions by rotations about its principal axis. A regular double pyramid has additional half-turn symmetries about axes in the equatorial plane. The tetrahedron has 12 such positions, the cube and octahedron have 24, and the dodecahedron and icosahedron have 60. In each case the number is equivalently the product of the number of vertices with the number of edges or faces meeting at a vertex, or the product of the number of faces with the number of edges or vertices of one face, or twice the number of edges.

0.4 Linear fractional substitutions

The group of ternary orthogonal substitutions is isomorphic to the group of linear fractional substitutions of one variable, or equivalently to the binary substitutions of ratios.

Let

$$(y_1, y_2, y_3) = A(x_1, x_2, x_3), \tag{43}$$

$$y_1^2 + y_2^2 + y_3^2 = x_1^2 + x_2^2 + x_3^2. \tag{44}$$

Apply the fixed, non-orthogonal substitution

$$L = \begin{pmatrix} i & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & i/\sqrt{2} & -i/\sqrt{2} \end{pmatrix}, \quad L^{-1} = \begin{pmatrix} -i & 0 & 0 \\ 0 & 1/\sqrt{2} & -i/\sqrt{2} \\ 0 & 1/\sqrt{2} & i/\sqrt{2} \end{pmatrix}, \tag{45}$$

where $i = \sqrt{-1}$. Set

$$(y_1, y_2, y_3) = L(y_1', y_2', y_3'), \qquad (x_1, x_2, x_3) = L(x_1', x_2', x_3'), \tag{46}$$

$$(y_1', y_2', y_3') = A'(x_1', x_2', x_3'), \tag{47}$$

where

$$A' = L^{-1} A L. \tag{48}$$

Then (44) becomes

$$-y_1'^2 + 2y_2'y_3' = -x_1'^2 + 2x_2'x_3'. \tag{49}$$

Put

$$x_1' = \sqrt{2}\xi_1\xi_2, \qquad x_2' = \xi_1^2, \qquad x_3' = \xi_2^2. \tag{50}$$

Then $x_1'^2 - 2x_2'x_3'$ vanishes identically, and the images y_1', y_2', y_3' become binary quadratic forms in ξ_1, ξ_2 satisfying

$$y_1'^2 - 2y_2'y_3' = 0. \tag{51}$$

It follows that $y_2'y_3'$ is a perfect square. The forms y_2' and y_3' cannot have a common factor; otherwise the determinant of A' would vanish. Hence y_2' and y_3' must themselves be squares of linear forms. Write

$$\eta_1 = \alpha\xi_1 + \beta\xi_2, \qquad \eta_2 = \gamma\xi_1 + \delta\xi_2. \tag{52}$$

Then one may put

$$y_1' = \sqrt{2}\eta_1\eta_2, \qquad y_2' = \eta_1^2, \qquad y_3' = \eta_2^2. \tag{53}$$

Expanding and reverting from the ξ's to x_1', x_2', x_3', one obtains

$$A' = \begin{pmatrix} \alpha\delta + \beta\gamma & \sqrt{2}\alpha\gamma & \sqrt{2}\beta\delta \\ \sqrt{2}\alpha\beta & \alpha^2 & \beta^2 \\ \sqrt{2}\gamma\delta & \gamma^2 & \delta^2 \end{pmatrix}. \tag{54}$$

The condition (49) requires

$$\alpha\delta - \beta\gamma = \pm 1. \tag{55}$$

The plus sign corresponds to proper orthogonal substitutions and the minus sign to improper ones.

Thus A' is determined by the binary substitution

$$(\eta_1, \eta_2) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\xi_1, \xi_2), \qquad \alpha\delta - \beta\gamma = \pm 1, \tag{56}$$

or, in terms of ratios $\eta = \eta_1/\eta_2$, $\xi = \xi_1/\xi_2$, by the linear fractional substitution

$$\eta = \frac{\alpha\xi + \beta}{\gamma\xi + \delta}. \tag{57}$$

The four coefficients $\alpha, \beta, \gamma, \delta$ determine the same ternary substitution after simultaneous multiplication by -1. Thus the ternary substitution corresponds not to a single binary matrix but to a pair of opposite binary matrices, or equivalently to the induced fractional-linear transformation.

Moreover, if A'', B'' are two such binary substitutions and A', B' the associated ternary substitutions, then the ternary substitution associated with $A''B''$ is $A'B'$. Hence the corresponding groups are isomorphic.

0.5 Reality conditions

We must still answer one question. Up to now no attention has been paid to the reality of the coefficients. If a geometric application is intended, the orthogonal substitution A must be real. We must therefore determine the conditions on $\alpha, \beta, \gamma, \delta$ which make the coefficients of A real.

Using the expression $A = LA'L^{-1}$ one obtains

$$A = \begin{pmatrix} \alpha\delta + \beta\gamma & i(\alpha\gamma + \beta\delta) & \alpha\gamma - \beta\delta \\ -i(\alpha\beta + \gamma\delta) & \dfrac{\alpha^2 + \beta^2 + \gamma^2 + \delta^2}{2} & \dfrac{-\alpha^2 + \beta^2 - \gamma^2 + \delta^2}{2} \\ \alpha\beta - \gamma\delta & i\dfrac{\alpha^2 - \beta^2 - \gamma^2 + \delta^2}{2} & \dfrac{\alpha^2 - \beta^2 + \gamma^2 - \delta^2}{2} \end{pmatrix}.$$

The entries of this matrix are to be real and

$$\alpha\delta - \beta\gamma = \pm 1. \tag{58}$$

The result is the following necessary and sufficient condition. The orthogonal substitution is real precisely when α and $\pm\delta$, and β and $\mp\gamma$, form two conjugate-imaginary pairs; the upper signs hold for proper orthogonal substitutions, and the lower signs for improper ones.

0.6 Finite groups of linear fractional substitutions and their poles

If all finite groups of linear fractional substitutions are known, then all finite groups of proper ternary orthogonal substitutions and all finite groups of binary linear substitutions of determinant 1 are thereby known as well.

Let G be a finite group of degree n whose elements are

$$\Theta_0(x) = x, \quad \Theta_1(x), \quad \dots, \quad \Theta_{n-1}(x), \tag{59}$$

where each Θ denotes a linear fractional function

$$\Theta(x) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \tag{60}$$

By transforming with an arbitrary linear substitution L, one obtains an isomorphic group

$$1, \quad L\Theta_1 L^{-1}, \quad \dots, \quad L\Theta_{n-1}L^{-1}.$$

The classification is considered complete once one representative from each such transformed family is found.

Since G is finite, every element Θ has a finite order: there is a least positive integer t such that $\Theta^t = 1$, and t divides n. For every non-identity substitution we consider the values of x fixed by the substitution:

$$x = \Theta(x). \tag{61}$$

These fixed values are called the *poles* of Θ. Inserting (60), one obtains the quadratic equation

$$\gamma x^2 + (\delta - \alpha)x - \beta = 0. \tag{62}$$

The roots coincide only in the exceptional case

$$(\delta - \alpha)^2 + 4\beta\gamma = 0, \tag{63}$$

or equivalently

$$(\alpha + \delta)^2 = 4\Delta, \qquad \Delta = \alpha\delta - \beta\gamma. \tag{64}$$

0.7 The possible types of groups

Let a be a pole of the group G, and suppose that there are exactly $\nu - 1$ non-identity elements

$$\Theta_1, \Theta_2, \dots, \Theta_{\nu-1}$$

of G such that

$$a = \Theta_1(a) = \Theta_2(a) = \cdots = \Theta_{\nu-1}(a). \tag{65}$$

Such a pole is called a ν-fold pole. The elements

$$1, \Theta_1, \Theta_2, \dots, \Theta_{\nu-1} \tag{66}$$

form a subgroup Q of G, because if both Θ_1 and Θ_2 fix a, then so does their product. Hence ν divides n:

$$n = \nu\mu. \tag{67}$$

This group Q is cyclic. After transforming a to ∞, the substitutions fixing it take the form

$$x, \quad \varepsilon_1 x + c_1, \quad \varepsilon_2 x + c_2, \quad \ldots,$$

and the composition laws show that the ε_i are the ν-th roots of unity. No two of them can be equal without making the corresponding substitutions identical. Hence the group is generated by an element whose multiplier is a primitive ν-th root of unity:

$$Q = \{1, \Theta, \Theta^2, \ldots, \Theta^{\nu-1}\}. \tag{68}$$

Thus:

Theorem. *A ν-fold pole determines a cyclic subgroup Q of degree ν contained in G.*

Theorem. *The two poles of any substitution Θ of the group G are poles of the same multiplicity, and they are the only poles of the corresponding cyclic group Q.*

Choose representatives $\Psi_1, \ldots, \Psi_{\mu-1}$ for the non-trivial right cosets of Q, so that

$$G = Q + \Psi_1 Q + \cdots + \Psi_{\mu-1} Q. \tag{69}$$

Put

$$a_1 = \Psi_1(a), \quad \ldots, \quad a_{\mu-1} = \Psi_{\mu-1}(a). \tag{70}$$

Then

$$a, a_1, \ldots, a_{\mu-1}$$

are distinct and are all ν-fold poles. They form a system of conjugate ν-fold poles of G.

Every substitution χ of G permutes these poles. Hence two systems of conjugate poles are either identical or disjoint. Therefore all poles of G can be arranged into systems of conjugate poles.

Counting every ν-fold pole with multiplicity $\nu - 1$, the total number of poles is $2n - 2$. Thus

$$2n - 2 = \mu(\nu - 1) + \mu'(\nu' - 1) + \mu''(\nu'' - 1) + \cdots . \tag{71}$$

If h is the number of systems of conjugate poles, and $n = \mu\nu = \mu'\nu' = \cdots$, this becomes

$$2n - 2 = nh - \mu - \mu' - \mu'' - \cdots . \tag{72}$$

Since every ν is at least 2, one obtains

$$2 \le h \le 4 - \frac{4}{n}.$$

Therefore h is only 2 or 3.

If $h = 2$, then $\mu = \mu' = 1$ and $\nu = \nu' = n$, giving the cyclic case. If $h = 3$, one obtains exactly the following possibilities:

I. **Cyclic group.**

$$\nu = \nu' = n, \qquad \mu = \mu' = 1.$$

II. **Dihedral group.**

$$\nu = \nu' = 2, \qquad \nu'' = m, \qquad n = 2m, \qquad \mu = \mu' = m, \qquad \mu'' = 2.$$

III. **Tetrahedral group.**

$$\nu = 2, \qquad \nu' = 3, \qquad \nu'' = 3, \qquad n = 12.$$

IV. **Octahedral group.**

$$\nu = 2, \qquad \nu' = 3, \qquad \nu'' = 4, \qquad n = 24.$$

V. **Icosahedral group.**

$$\nu = 2, \qquad \nu' = 3, \qquad \nu'' = 5, \qquad n = 60.$$

These exhaust the possibilities. It remains only to show that groups of all five types actually exist.

0.8 Basic forms

To every system of conjugate poles of the group G there corresponds an invariant binary form. If the poles are

$$a, a_1, \dots, a_{\mu-1},$$

then the corresponding form has roots at precisely these poles; in homogeneous variables it is

$$f(x_1, x_2) = \prod_j (x_1 - a_j x_2), \tag{73}$$

up to a constant factor. Under a substitution of the group the roots are merely permuted, so f is transformed into itself up to a constant factor.

Thus:

Theorem. *Every system of conjugate poles of G gives an invariant form whose degree is the number of poles in the system and whose roots are exactly those poles.*

These invariant forms will be called the *basic forms* of the group.

If $F(x_1, x_2)$ is any invariant form of G and shares a root with a basic form f, then F shares all the roots of f; hence F is divisible by f.

Consequently, if an invariant form F has degree less than the order of G, it must be, up to a constant factor, a product of powers of the basic forms. Equivalently: an invariant form of degree smaller than the order of the group which is not such a product must vanish identically.

0.9 Cyclic and dihedral groups

We now apply the preceding principles to construct the polyhedral groups. For a cyclic group of degree n, there are only two systems of conjugate poles, each consisting of a single $(n-1)$-fold pole. These may be taken to be 0 and ∞, giving the linear basic forms

$$f_1 = x_1, \qquad f_2 = x_2. \tag{74}$$

All substitutions are multiplicative, and their multipliers are n-th roots of unity. Thus, for a primitive n-th root ε, the group is

$$C_n = \left\{ \begin{pmatrix} \varepsilon^r & 0 \\ 0 & 1 \end{pmatrix} : r = 0, 1, \dots, n-1 \right\}, \tag{75}$$

or, with determinant 1,

$$C_n = \left\{ \begin{pmatrix} \varepsilon^{r/2} & 0 \\ 0 & \varepsilon^{-r/2} \end{pmatrix} : r = 0, 1, \dots, n-1 \right\}. \tag{76}$$

For the dihedral group D_m of degree $2m$, there are two systems of m conjugate two-fold poles and one system of two conjugate m-fold poles. Taking the latter to be 0 and ∞, the quadratic basic form is

$$f_1 = x_1 x_2.$$

The group consists of

$$\begin{gathered} x, \varepsilon x, \varepsilon^2 x, \dots, \varepsilon^{m-1} x, \\ \frac{1}{x}, \frac{\varepsilon}{x}, \frac{\varepsilon^2}{x}, \dots, \frac{\varepsilon^{m-1}}{x}, \end{gathered}$$

where ε is a primitive m-th root of unity. In matrix form,

$$D_m = \begin{pmatrix} \varepsilon^r & 0 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & \varepsilon^r \\ 1 & 0 \end{pmatrix}, \qquad r = 0, 1, \dots, m-1. \tag{77}$$

With determinant 1, this may be written

$$\begin{pmatrix} \varepsilon^{r/2} & 0 \\ 0 & \varepsilon^{-r/2} \end{pmatrix}, \qquad \begin{pmatrix} 0 & i\varepsilon^{r/2} \\ i\varepsilon^{-r/2} & 0 \end{pmatrix}. \tag{78}$$

The remaining poles are obtained from

$$x = \frac{\varepsilon^h}{x},$$

and give the two basic forms

$$f_2 = x_1^m + x_2^m, \qquad f_3 = x_1^m - x_2^m. \tag{79}$$

0.10 The tetrahedral group

For the tetrahedral group there is a system of six conjugate two-fold poles. We may assume that the group contains

$$\Theta(x) = -x, \qquad \psi(x) = \frac{1}{x},$$

and hence also

$$\psi_1(x) = \psi\Theta(x) = -\frac{1}{x}.$$

The poles of these substitutions are $0, \infty, \pm 1, \pm i$. Therefore the equation of the two-fold poles is $x(x^4 - 1) = 0$, and the first basic form of degree 6 is

$$f(x_1, x_2) = x_1 x_2 (x_1^4 - x_2^4). \tag{80}$$

There must also be a substitution χ carrying -1 to 0 and $+1$ to ∞, and hence it has the form

$$\chi = \lambda \frac{x+1}{x-1}. \tag{81}$$

Requiring it to permute the six poles forces $\lambda = \pm i$. Choosing the upper sign, one obtains

$$\chi(x) = i\frac{x+1}{x-1}.$$

Then

$$\chi^2(x) = \frac{x+i}{x-i}, \qquad \chi^3(x) = x,$$

and the twelve substitutions of the tetrahedral group are

$$\begin{array}{cccc} 1, & \Theta, & \psi, & \Theta\psi, \\ \chi, & \Theta\chi, & \psi\chi, & \Theta\psi\chi, \\ \chi^2, & \Theta\chi^2, & \psi\chi^2, & \Theta\psi\chi^2. \end{array} \tag{82}$$

Equivalently they are

$$\pm x, \quad \pm\frac{1}{x}, \quad \pm i\frac{x+1}{x-1}, \quad \pm i\frac{x-1}{x+1}, \quad \pm\frac{x+i}{x-i}, \quad \pm\frac{x-i}{x+i}. \tag{83}$$

They satisfy the defining relations listed in the source, and therefore form the tetrahedral group generated by Θ, ψ, χ.

After a change of variables, the remaining basic forms can be written

$$\begin{aligned} \Phi_1 &= x_1^4 + 2\sqrt{-3}\, x_1^2 x_2^2 + x_2^4, \\ \Phi_2 &= x_1^4 - 2\sqrt{-3}\, x_1^2 x_2^2 + x_2^4. \end{aligned} \tag{84}$$

Eliminating x_1, x_2 from these formulas gives

$$12\sqrt{-3}\, f^2 = \Phi_1^3 - \Phi_2^3. \tag{85}$$

The determinant-1 representatives of Θ, ψ, χ are

$$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \qquad \begin{pmatrix} \frac{1-i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & \frac{1+i}{2} \end{pmatrix}. \tag{86}$$

They satisfy the reality conditions above, so the corresponding ternary orthogonal group is real.

The Hessian of the basic form f is again an invariant:

$$\begin{aligned} H &= -\frac{1}{25}(f''(x_1,x_1)f''(x_2,x_2) - f''(x_1,x_2)^2) \\ &= (x_1^4 + x_2^4)^2 + 12x_1^4x_2^4, \end{aligned}$$

and is the product of the two functions Φ_1 and Φ_2.

0.11 The octahedral group

For the octahedral group, the general classification gives systems of conjugate poles of multiplicities $2, 3, 4$ and group order 24. Weber constructs it by adjoining to the tetrahedral group a further substitution that enlarges the order from 12 to 24.

Let the tetrahedral group be generated by substitutions Θ, ψ, χ satisfying

$$\begin{aligned} \Theta^2 = \psi^2 = 1, \qquad \chi^3 = 1, \\ \chi\Theta = \Theta\psi\chi, \qquad \chi\psi = \Theta\chi, \\ \chi^2\Theta = \psi\chi^2, \qquad \chi^2\psi = \Theta\psi\chi^2. \end{aligned} \tag{87}$$

One obtains a group of order 24 with elements of the form

$$\chi^\lambda\omega^\mu\Theta^\nu,$$

where χ has order 3, ω has order 2, and Θ has order 4, subject to the relations displayed in Weber's construction. The resulting group is isomorphic to the octahedral group.

The remaining basic forms of degrees 8 and 12 are obtained from the tetrahedral basic form by covariant operations. One obtains

$$W = x_1^8 + 14x_1^4x_2^4 + x_2^8, \tag{88}$$

and from the functional determinant of f and W,

$$K = x_1^{12} - 33x_1^8x_2^4 - 33x_1^4x_2^8 + x_2^{12}. \tag{89}$$

0.12 The icosahedral group

For the icosahedral group there is a system of conjugate five-fold poles. We may therefore assume that the group contains

$$\Theta(x) = \varepsilon x, \qquad \psi(x) = -\frac{1}{x}, \tag{90}$$

where ε is a primitive fifth root of unity. The basic form of degree 12 associated with the five-fold poles must then have the form

$$f(x_1,x_2) = x_1x_2(x_1^{10} + mx_1^5x_2^5 - x_2^{10}), \tag{91}$$

and the problem is to determine the constant m. The two other basic forms have degrees 20 and 30.

The constant m can be determined by forming a covariant of degree lower than the order 60 of the group which cannot be expressed as a product of powers of the three basic forms of degrees $12, 20, 30$; by the preceding divisibility theorem, such a covariant must vanish identically. Weber describes this through polar and Hessian operations on the binary form f.

The source chunk becomes unstable at this point and conflates part of the tetrahedral Hessian computation with the beginning of the icosahedral calculation. I therefore stop this batch at the clean mathematical transition: the icosahedral group is reduced to the determination of the degree-12 basic form (91) and its degree-20 and degree-30 companions.

§42. The absolute invariant and the ground field

Let

$$F_1, F_2, \ldots, F_r$$

be a basis of the invariant system J, as determined above, and let F be any other nonconstant invariant of S. The forms $A_1, A_2, \ldots, A_r$ may then be chosen so that

$$F = A_1F_1 + A_2F_2 + \cdots + A_rF_r. \tag{92}$$

Apply all substitutions of the group S to this identity. By assumption,

$$F, F_1, F_2, \ldots, F_r$$

remain unchanged; the coefficients $A_1, A_2, A_3, \ldots$, however, may be transformed into

$$A_1^{(i)}, A_2^{(i)}, A_3^{(i)}, \ldots.$$

If one adds all identities obtained in this way and divides by the order s of the group S, then one obtains

$$F = \overline{A}_1F_1 + \overline{A}_2F_2 + \cdots + \overline{A}_rF_r, \tag{93}$$

where now $\overline{A}_1, \overline{A}_2, \ldots$ are themselves invariants of the group S. By the defining property of the basis, these averaged coefficients are rationally expressible through $F_1, \ldots, F_r$.

Hence:

Theorem. *Every invariant F can be represented as a rational function of the basis invariants $F_1, \ldots, F_r$.*

Let

$$\vartheta_1, \vartheta_2, \ldots, \vartheta_h$$

be the coefficients of the general substitution of S, and let Ω be the rational field generated by these quantities. Every invariant F is then a function of the variables x and of the quantities ϑ. If the x's are treated as independent variables and the ϑ's as parameters, the totality of all invariants so obtained forms a field. We shall call it the *field of invariants* of the group S.

The absolute invariants are precisely those functions that remain unchanged under all substitutions of the group and depend only on the orbit of the point under the group action. Thus they furnish the rational functions on the quotient of the space of variables by the group.

§43. The form problem

The problem of determining the values of

$$x_1, x_2, \dots, x_n$$

which give a prescribed value F_0 to a given invariant F, when the coefficients of the group are assumed known, is called the *form problem* of the group S.

Let $\Omega(t)$ denote a general function of the field Ω which is carried into itself by the substitutions of S. The roots of the equation

$$\Omega(t) = \omega,$$

where ω is any fixed value, are all distinct and are transformed into one another by the substitutions of S.

The solution of the form problem is closely connected with the solution of the equation

$$\Omega(t) = \omega.$$

Indeed, if ϑ is a root of this equation and one sets

$$x_i = \vartheta_i(\vartheta), \tag{94}$$

for suitable functions ϑ_i, then one obtains values of the x_i satisfying all invariant relations.

If the invariant J can be represented as a rational function of ϑ, then this function cannot change when any of the substitutions

$$(\vartheta, \vartheta_1), \quad (\vartheta, \vartheta_2), \quad \dots$$

is applied. Consequently it is rationally expressible through the coefficients of $\Omega(t)$. Whether this representation is possible depends on the nature of the group.

If S' is a subgroup of S of index j, and no larger subgroup has the same invariance property, then ϑ satisfies an equation of degree j. This equation is to be regarded as a *resolvent* of the form problem. Every function which admits exactly the substitutions of S' satisfies such a resolvent.

Thus the form problem transforms invariant theory into an equation problem: the passage from a group to its subgroups corresponds to the passage from a general equation to its resolvents.

§44. Galois groups and resolvents

The theory of groups of linear substitutions is closely related to the theory of algebraic equations. Let

$$t^n + a_1 t^{n-1} + a_2 t^{n-2} + \cdots + a_n = 0 \tag{95}$$

be an equation of degree n, with roots

$$\alpha_1, \alpha_2, \ldots, \alpha_n.$$

The totality of all permutations of these roots which leave unchanged every rational relation among them forms the *Galois group* of the equation.

A function

$$F(\alpha_1, \ldots, \alpha_n)$$

of the roots that is unchanged by all permutations of the Galois group is rationally expressible through the coefficients

$$a_1, \ldots, a_n$$

of the equation.

Solving the equation is equivalent to determining a function ϑ of the roots that is left fixed by no nonidentity permutation of the Galois group. Such a function is called a *primitive function* of the field.

The Galois group may always be regarded as a group of linear substitutions: replace each permutation of the roots by the corresponding substitution of the variables

$$x_1, \ldots, x_n,$$

as described in §39. In this way the group of the equation becomes a concrete linear substitution group, and the methods of invariant theory become available for the study of algebraic equations.

Resolvents arise by passing to subgroups. If a function of the roots is invariant under a subgroup S' but not under the whole group S, then the conjugates of that function under S are indexed by the cosets of S'. Their elementary symmetric functions are invariant under S, hence rational functions of the coefficients of the original equation. The function therefore satisfies an auxiliary equation whose degree is the index $[S : S']$.

§45. Metacyclic groups

A group S is called *metacyclic* if it has a distinguished chain of subgroups

$$S,\ S_1,\ S_2,\ \ldots,\ S_\nu = 1, \tag{96}$$

such that each S_i is a normal subgroup of S_{i-1} and each quotient group

$$S_{i-1}/S_i$$

is cyclic.

Such a group can be built by solving a chain of equations, each of which is cyclic. The solvability of an algebraic equation by radicals is therefore expressed group-theoretically by the metacyclic nature of its Galois group, or equivalently by the existence of a suitable chain with cyclic factors.

For the symmetric group on n letters, metacyclicity occurs only for $n \leq 4$. Likewise the alternating group on n letters is metacyclic only for $n \leq 4$. This is the group-theoretic obstruction behind the impossibility, in general, of solving equations of degree at least five by radicals.

§46. Abstract and concrete groups

Until now the exposition of group theory has been tied to definite operations: substitutions, linear transformations, and similar concrete processes. But the group concept may also be detached from any particular representation and defined purely formally.

By an *abstract group* one understands a system of elements

$$A, B, C, \ldots$$

for which a composition law, or multiplication, is defined and satisfies the following rules.

1. From any two elements A, B, their product is a uniquely determined element $C = AB$.
2. The associative law holds:
$$(AB)C = A(BC).$$
3. There is an *identity element* J such that
$$JA = AJ = A$$
for every element A.
4. For every element A there is an inverse element A^{-1} such that
$$AA^{-1} = A^{-1}A = J.$$

Two abstract groups are called *isomorphic* if their elements can be named in such a way that the multiplication rule is the same in both groups. This definition separates the intrinsic structure of the group from the particular objects on which the group happens to act.

The point of abstraction is not to discard concrete examples, but to recognize when two different systems of operations have the same internal law of composition. Permutation groups, linear substitution groups, rotation groups, and Galois groups can then be compared by their multiplication tables rather than by their external form.

§47. The alternating group and its invariants

The alternating group on n letters consists of all even permutations and has index 2 in the symmetric group. The function

$$\Delta = \prod_{i<k}(x_i - x_k) \tag{97}$$

is an invariant of the alternating group. More precisely, it is an invariant which changes sign under every odd permutation.

The square

$$\Delta^2$$

is therefore an absolute invariant of the symmetric group. It is a symmetric function of the x_i, and is represented by the discriminant of the corresponding equation.

The most general invariant of the alternating group can be written in the form

$$F = F_1 + \Delta F_2, \tag{98}$$

where F_1 and F_2 are symmetric functions, that is, invariants of the full symmetric group.

For $n \geq 5$, the alternating group is simple: it has no normal subgroup other than the identity and itself. This fact is one of the central structural reasons why the general equation of degree at least five cannot be solved by radicals.

§48. Orthogonal substitutions

A linear substitution is called *orthogonal* if it transforms the quadratic form

$$x_1^2 + x_2^2 + \cdots + x_n^2 \tag{99}$$

into itself. For a substitution A with coefficients $a_i^{(k)}$, this means that the identity

$$\sum_k \left(\sum_i a_i^{(k)} x_i \right)^2 = \sum_i x_i^2 \tag{100}$$

holds in the variables x.

Comparing coefficients gives the relations

$$\sum_k a_i^{(k)} a_j^{(k)} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \tag{101}$$

For example, if

$$A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix},$$

then the orthogonality relations are

$$a_1^2 + b_1^2 + c_1^2 = 1, \qquad a_1 a_2 + b_1 b_2 + c_1 c_2 = 0, \tag{102}$$
$$a_2^2 + b_2^2 + c_2^2 = 1, \qquad a_1 a_3 + b_1 b_3 + c_1 c_3 = 0, \tag{103}$$
$$a_3^2 + b_3^2 + c_3^2 = 1, \qquad a_2 a_3 + b_2 b_3 + c_2 c_3 = 0. \tag{104}$$

These equations say exactly that the row vectors of A form an orthonormal system.

From these relations one obtains

$$|A|^2 = 1,$$

and hence

$$|A| = \pm 1.$$

The orthogonal substitutions with determinant $+1$ are those of the *first kind*, or rotations. Those with determinant -1 are those of the *second kind*, or reflections.

The totality of all orthogonal substitutions in n variables forms a group, the *orthogonal group*. One may regard either the positions of the body or the rotations themselves as the elements of the group.

§49. Linear fractional substitutions

We shall now show that the group of ternary orthogonal substitutions is isomorphic to the group of linear fractional substitutions in one variable, or equivalently to the group of binary substitutions acting on ratios.

Write a ternary orthogonal substitution as

$$(y_1, y_2, y_3) = A(x_1, x_2, x_3), \tag{105}$$

so that

$$y_1^2 + y_2^2 + y_3^2 = x_1^2 + x_2^2 + x_3^2. \tag{106}$$

We first apply a fixed, nonorthogonal change of variables of the form

$$x_1 = \xi_1 + \xi_2, \qquad y_1 = \eta_1 + \eta_2, \tag{107}$$

$$x_2 = \xi_1 - \xi_2, \qquad y_2 = \eta_1 - \eta_2, \tag{108}$$

$$x_3 = \xi_3, \qquad y_3 = \eta_3. \tag{109}$$

Under this transformation the quadratic condition is converted into an equivalent condition in the new variables.

Now introduce binary variables by setting, in effect,

$$x_1 = \sqrt{2}\,\xi_1\xi_3, \qquad x_2 = \sqrt{2}\,\xi_2\xi_3.$$

Then the corresponding quadratic relation vanishes identically. The transformed variables y_1, y_2, y_3 become binary quadratic forms in ξ_1, ξ_2, and these forms must satisfy the transformed quadratic equation identically.

The product relation then forces the relevant quadratic forms to decompose into linear factors. Since y_2 and y_3 cannot have a common factor without making the determinant of the transformation vanish, the factors are arranged in such a way that the ternary orthogonal substitution is represented by a binary linear substitution, considered projectively.

Thus ternary rotations are represented by fractional linear transformations. In modern language, this is the classical passage from the binary linear group, modulo scalar multiplication, to the rotation group in three dimensions.

§51. Finite groups of linear fractional substitutions; poles of groups

From the preceding developments it follows that, once all finite groups of linear fractional substitutions are known, one immediately obtains all finite groups of proper ternary orthogonal substitutions and all finite groups of binary linear substitutions with determinant 1.

Let S be a finite group of linear fractional substitutions in a variable x. Write its substitutions in canonical form

$$x' = \frac{\alpha x + \beta}{\gamma x + \delta}, \tag{110}$$

with determinant condition

$$\alpha\delta - \beta\gamma = 1.$$

A nonidentity substitution of this form generally has two fixed points, or *poles*, obtained from the quadratic equation

$$\gamma x^2 + (\delta - \alpha)x - \beta = 0. \tag{111}$$

Only in the exceptional case $\gamma = 0$ and $\alpha = \delta$, that is, for a translation $x' = x + \beta$, does the equation reduce to degree one; then there is only one improper pole, situated at infinity.

The pole structure is the key to the classification of finite fractional linear groups. A finite group acts on the sphere of projective values, and the exceptional points with nontrivial stabilizer determine the group.

§52. The cyclic and dihedral groups

The simplest finite groups are the *cyclic groups.* A cyclic group of degree n is generated by the substitution

$$x' = \varepsilon x, \tag{112}$$

where ε is a primitive n-th root of unity. The n substitutions of the group are

$$x,\ \varepsilon x,\ \varepsilon^2 x,\ \ldots,\ \varepsilon^{n-1} x. \tag{113}$$

The two poles are 0 and ∞, and both are fixed by every substitution of the group.

A *dihedral group* of degree $2n$ contains, besides these substitutions, the n substitutions

$$x' = \frac{\varepsilon^k}{x}, \qquad k = 0, 1, \ldots, n-1. \tag{114}$$

These interchange the two poles 0 and ∞. Thus the group consists of the substitutions

$$x' = \varepsilon^k x, \qquad x' = \frac{\varepsilon^k}{x}, \qquad k = 0, 1, \ldots, n-1. \tag{115}$$

The element ε must run through a full system of n-th roots of unity. Hence, if the ε occurring in the generator is primitive, the whole cyclic subgroup is obtained from its powers, and the dihedral group is obtained by adjoining the inversion $x \mapsto 1/x$, together with its products with the rotations.

§58. The Icosahedral Group

Since in the icosahedral group there is only one system of conjugate fivefold poles, we may, by the preceding reduction, assume that the group contains the two substitutions

$$\Theta(x) = \varepsilon x, \qquad \psi(x) = -\frac{1}{x}, \tag{116}$$

where ε is a primitive fifth root of unity. We choose here, as we are free to do, the second substitution in the form $-1 : x$, because the formulae thereby become simpler.

The basic form of degree twelve belonging to the system of fivefold poles must, since it admits the substitutions (116), have the form

$$f(x_1, x_2) = x_1 x_2 (x_1^{10} + m x_1^5 x_2^5 - x_2^{10}), \tag{117}$$

and it remains to determine the constant factor m. The two other basic forms have degrees twenty and thirty. We can determine m from the theorem of §54, if we can construct from f a covariant of degree less than sixty which cannot be represented as a product of powers of three forms of degrees twelve, twenty, and thirty. Such a covariant must then vanish identically.

A covariant of degree sixteen of the form f is easily obtained by taking, in the sense of the first volume, the fourth polar of the binary form $f(x_1, x_2)$:

$$12 \cdot 11 \cdot 10 \cdot 9\, P_4(x, \xi) = u_0 \xi_1^4 + 4u_1 \xi_1^3 \xi_2 + 6u_2 \xi_1^2 \xi_2^2 + 4u_3 \xi_1 \xi_2^3 + u_4 \xi_2^4, \tag{118}$$

where the u_i are obtained from the fourth partial derivatives of f. From this fourth polar one forms, as in the theory of binary quartics, its first invariant,

$$u_0 u_4 - 4u_1 u_3 + 3u_2^2. \tag{119}$$

Since a form of degree sixteen cannot be a product of forms of degrees twelve, twenty, and thirty, this covariant must vanish identically.

For the form (117) one has, up to the common numerical factor coming from the fourth differentiation,

$$\begin{aligned}
u_0 &= 11 \cdot 10 \cdot 9 \cdot 8\, x_1^7 x_2 + 6 \cdot 5 \cdot 4 \cdot 3\, m x_1^2 x_2^6, \\
u_1 &= 4 \cdot 11 \cdot 10 \cdot 9\, x_1^6 x_2^2 + 4 \cdot 6 \cdot 5 \cdot 4\, m x_1 x_2^7, \\
u_2 &= 6 \cdot 5 \cdot m\, x_1^5 x_2^5, \\
u_3 &= -4 \cdot 11 \cdot 10 \cdot 9\, x_1^2 x_2^6 + 4 \cdot 6 \cdot 5 \cdot 4\, m x_1^7 x_2, \\
u_4 &= -11 \cdot 10 \cdot 9 \cdot 8\, x_1 x_2^7 + 6 \cdot 5 \cdot 4 \cdot 3\, m x_1^6 x_2^2.
\end{aligned}$$

Substitution in (119), and comparison of the coefficient of the appropriate middle term, gives

$$m = \pm 11.$$

Both signs are admissible. The one choice is carried into the other by replacing x_1 by $-x_1$, or equivalently by an allowable transformation of the group. We therefore put

$$f(x_1, x_2) = x_1 x_2 (x_1^{10} + 11 x_1^5 x_2^5 - x_2^{10}). \tag{120}$$

The ten remaining fivefold poles are consequently obtained by solving

$$x^{10} + 11x^5 - 1 = 0.$$

Writing

$$\omega = -\frac{11 + 5\sqrt{5}}{2}, \qquad \omega' = -\frac{11 - 5\sqrt{5}}{2}, \tag{121}$$

so that

$$\omega^2 + \omega = 1, \qquad \omega + \omega' = -1, \qquad \omega\omega' = -1, \tag{122}$$

these fivefold poles, besides 0 and ∞, may be written in the form

$$x = \varepsilon^\lambda \omega, \qquad x = \varepsilon^\lambda \omega', \qquad \lambda = 0, 1, 2, 3, 4. \tag{123}$$

Here $\varepsilon + \varepsilon^{-1} = 2\cos(2\pi/5) = (-1 + \sqrt{5})/2$ and $\varepsilon^2 + \varepsilon^{-2} = 2\cos(4\pi/5) = (-1 - \sqrt{5})/2$, so that the notation is compatible with the elementary identities just displayed.

We now apply the theorem of §53. If a, a' are the two poles of a substitution Θ, and if b is a pole conjugate with a, then there is a substitution χ of the group such that $b = \chi(a)$, and such that $b' = \chi(a')$ is the second pole of $\chi^{-1}\Theta\chi$. We may take $a = \infty$, $a' = 0$, and $b = \omega$. Then b' is a still undetermined pole ξ among those in (123), and χ has the form

$$\chi(x) = \frac{\omega\alpha x + \beta}{\alpha x + \beta\omega^{-1}}, \qquad \alpha\beta(\omega^{-1} - \omega) = 1. \tag{124}$$

For every exponent h not divisible by five, the substitution Θ^h may be used. The poles of $\chi^{-1}\Theta^h\chi$ must be ω and ξ. From the corresponding quadratic equation for the fixed points one obtains

$$\frac{\beta}{\alpha} = \pm\omega\sqrt{-\xi\omega^{-1}},$$

and the condition (124) then determines the remaining scalar factor. In Weber's normalization this reduces to the simple form

$$\chi(x) = \frac{\omega x + \beta}{x - \omega^{-1}}, \tag{125}$$

where β is still to be determined and, from the determinant condition, has the value $-\omega\xi$.

If χ is known, the whole icosahedral group can be generated. Let r, s run through the residues $0, 1, 2, 3, 4$. The group then contains the substitutions

$$\Theta^r, \qquad \psi\Theta^r, \qquad \Theta^r\chi\Theta^s, \qquad \Theta^r\chi\psi\Theta^s. \tag{126}$$

There are sixty of them. Written as matrices of fractional-linear substitutions, they are

$$\Theta^r = \begin{pmatrix} \varepsilon^r & 0 \\ 0 & 1 \end{pmatrix}, \tag{127}$$

$$\psi\Theta^r = \begin{pmatrix} 0 & 1 \\ -\varepsilon^r & 0 \end{pmatrix}, \tag{128}$$

$$\Theta^r\chi\Theta^s = \begin{pmatrix} \varepsilon^r\omega & \varepsilon^{r-s}\beta \\ 1 & -\varepsilon^{-s}\omega \end{pmatrix}, \tag{129}$$

$$\Theta^r\chi\psi\Theta^s = \begin{pmatrix} -\varepsilon^{r+s}\beta & \varepsilon^r\omega \\ \varepsilon^s\omega & 1 \end{pmatrix} = \begin{pmatrix} -\varepsilon^r\beta\omega^{-1} & \varepsilon^{r-s} \\ 1 & \varepsilon^{-s}\omega^{-1} \end{pmatrix}. \tag{130}$$

Since $-\omega^2$ is no power of ε, no two of these matrices represent the same substitution.

The icosahedral group has, as we have seen, twelve fivefold poles. Any two of these are the poles of four substitutions of fifth order; together with the identity these form a cycle of five elements. Thus there are twenty-four elements of fifth order. Similarly, the twenty threefold poles lead to twenty substitutions of third order, and the thirty twofold poles, paired two by two, lead to fifteen substitutions of second order. Together with the identity this gives

$$24 + 20 + 15 + 1 = 60.$$

The determination of β in (125) is obtained by considering the orders of the substitutions (129) and (130). For an arbitrary linear substitution

$$S = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

the second and third iterates are

$$S^2 = \begin{pmatrix} a^2 + bc & b(a+d) \\ c(a+d) & bc + d^2 \end{pmatrix},$$

$$S^3 = \begin{pmatrix} a^3 + 2abc + bcd & b(a^2 + bc + ad + d^2) \\ c(a^2 + bc + ad + d^2) & d^3 + abc + 2bcd \end{pmatrix}.$$

Disregarding the cases where b or c vanishes, which do not occur here, one obtains the necessary and sufficient conditions

$$a + d = 0 \tag{131}$$

for a substitution of second order, and

$$a^2 + bc + ad + d^2 = 0 \tag{132}$$

for one of third order.

The substitutions (127), apart from the identity, have fifth order. The five substitutions (128) have second order. Among the substitutions (129), exactly the five of the form $\Theta^r \chi \Theta^{-r}$ have second order. Hence five among the substitutions (130) must also have second order. From (131) this is possible only if β is a power of ε.

The substitutions of third order must now be distributed among the forms (129) and (130). The condition that (129) have third order is

$$\omega^2(\varepsilon^r + \varepsilon^{-s} + \varepsilon^{-(r+s)}) = 1, \tag{133}$$

and for (130) the condition is

$$\omega^2 = \beta - \varepsilon^{-(r+s)} - \beta^2 \varepsilon^{r+s}. \tag{134}$$

Either one of these conditions can be satisfied only when

$$\beta = 1.$$

Indeed, in (133) the left side is real only when β is real; since β is a power of ε, this forces $\beta = 1$. In (134) one obtains the same conclusion by passing to conjugate-imaginary quantities; the expression can remain invariant under conjugation only if $\beta = \beta^{-1}$, hence $\beta = 1$ among fifth roots of unity.

With $\beta = 1$, five of the substitutions (130), namely those of the form $\Theta^r \chi\psi\Theta^{-r}$, have second order. The substitutions (129) have third order precisely when

$$r + s \equiv \pm 2 \pmod 5, \tag{135}$$

and the substitutions (130) have third order precisely when

$$r + s \equiv \pm 1 \pmod 5. \tag{136}$$

Thus exactly ten substitutions of second order and twenty of third order occur in the forms (129) and (130); the remaining cases are of fifth order.

Summarizing, χ must have the form

$$\chi(x) = \frac{\omega x + 1}{x - \omega^{-1}}. \tag{137}$$

Among the sixty substitutions

$$\Theta^r, \quad \psi\Theta^r, \quad \Theta^r\chi\Theta^s, \quad \Theta^r\chi\psi\Theta^s, \tag{138}$$

there occur, besides the identity,

1. fifteen substitutions of second order:

$$\psi\Theta^r, \qquad \Theta^r\chi\Theta^{-r}, \qquad \Theta^r\chi\psi\Theta^{-r} \quad (r = 0, 1, 2, 3, 4);$$

2. twenty substitutions of third order:

$$\Theta^r\chi\Theta^s \quad (r + s \equiv \pm 2), \qquad \Theta^r\chi\psi\Theta^s \quad (r + s \equiv \pm 1);$$

3. twenty-four substitutions of fifth order:

$$\Theta^r \ (r \neq 0), \quad \Theta^r\chi\Theta^s \ (r + s \equiv \pm 1), \quad \Theta^r\chi\psi\Theta^s \ (r + s \equiv \pm 2).$$

In explicit fractional-linear form these substitutions include

$$\varepsilon^r x, \qquad -\frac{1}{\varepsilon^r x}, \qquad \frac{\varepsilon^r \omega x + \varepsilon^{r-s}}{x - \varepsilon^{-s}\omega}, \qquad \frac{-\varepsilon^r\omega^{-1}x + \varepsilon^{r-s}}{x + \varepsilon^{-s}\omega^{-1}}. \tag{139}$$

It remains only to verify that the totality of substitutions (138) is indeed a group. This follows from the composition laws, which we now record. First, directly from the meaning of Θ, ψ, χ,

$$\Theta^5 = 1, \qquad \psi^2 = 1, \qquad \psi\Theta = \Theta^{-1}\psi, \qquad \chi\psi = \psi\chi. \tag{140}$$

For an arbitrary exponent λ, use

$$\omega = \varepsilon + \varepsilon^{-1}, \qquad \omega' = \varepsilon^2 + \varepsilon^{-2}, \qquad \omega\omega' = -1. \tag{141}$$

Then, putting first $\lambda = 1$ and then $\lambda = 2$, Weber obtains

$$\chi\Theta^{\pm 1}\chi = \Theta^{\pm 1}\chi\psi\Theta^{\pm 1}, \qquad \chi\Theta^{\pm 2}\chi = \Theta^{\pm 2}\chi\Theta^{\pm 2}. \tag{142}$$

Together with (140), these formulae allow any product of two substitutions of the form (138) to be reduced again to one of the same forms. Thus the group property is proved.

From the first formula in (142) one also derives

$$\psi = \Theta\chi^{-1}\Theta^{-1}\chi^{-1}\Theta,$$

so that ψ may be derived from Θ and χ. The substitutions Θ and χ alone may therefore be taken as generators of the icosahedral group.

To write χ with determinant one, we use

$$-\omega^{-2} - 1 = -\omega^{-1} = (\varepsilon - \varepsilon^{-1})^2.$$

Replacing ω by $\varepsilon + \varepsilon^{-1}$, this gives

$$\chi = \frac{1}{\varepsilon - \varepsilon^{-1}} \begin{pmatrix} \omega & 1 \\ 1 & -\omega^{-1} \end{pmatrix} = \begin{pmatrix} \dfrac{\omega}{\varepsilon - \varepsilon^{-1}} & \dfrac{1}{\varepsilon - \varepsilon^{-1}} \\ \dfrac{1}{\varepsilon - \varepsilon^{-1}} & -\dfrac{\omega^{-1}}{\varepsilon - \varepsilon^{-1}} \end{pmatrix}. \tag{143}$$

§59. Divisors of the Icosahedral Group

The divisors, or subgroups, of the icosahedral group must be sought among the lower polyhedral groups. First there are cyclic groups C_n, each generated by one of the icosahedral substitutions. Counting them immediately from the preceding arrangement according to the orders of the substitutions gives

$$15 \text{ groups } C_2, \qquad 10 \text{ groups } C_3, \qquad 6 \text{ groups } C_5.$$

There are also dihedral groups D_2, D_3, D_5 in the icosahedral group. Representatives may be written in the form

$$\begin{aligned} &D_2 : 1,\ \chi\psi,\ \Theta^{-1}\chi\psi\Theta^{-1},\ \Theta\chi\psi\Theta, \\ &D_3 : \Theta^r\chi\Theta^s \quad (r+s \equiv \pm 2 \pmod 5), \\ &D_5 : \Theta^r,\ \psi\Theta^r, \qquad r = 0,1,2,3,4. \end{aligned}$$

The number of such dihedral groups is found from the fact that a dihedral group is completely determined by its cyclic subgroup. For the four-group D_2 one must remember that the same group is obtained whether one begins with ψ, $\chi\psi$, or $\Theta\chi\psi\Theta$; hence the directly obtained number is to be divided by three. Thus the numbers of groups D_2, D_3, D_5 are respectively

$$5, \qquad 10, \qquad 6.$$

Of special importance are the tetrahedral groups contained in the icosahedral group. We determine one of them, denoted by Q. A tetrahedral group must contain a four-group D_2; we take the one generated by ψ and $\chi\psi$. In addition, Q must contain a substitution of third order. Such a substitution can be taken in one of the two forms

$$\Theta^r\chi\Theta^s, \qquad \Theta^r\chi\psi\Theta^s.$$

Both choices lead to the same result; choose

$$\varphi = \Theta^r\chi\Theta^s, \qquad r+s \equiv \pm 2 \pmod 5. \tag{144}$$

Neither of the numbers r, s may be zero; otherwise either $\chi\varphi$ or $\varphi\chi$ would be a power of Θ, hence of fifth order, while no element of fifth order can occur in Q.

Using the composition laws of §58 one obtains

$$\chi\varphi = \Theta^r\chi\psi\Theta^{r+s} \quad (r \neq \pm 1), \qquad \chi\varphi = \Theta^{-r}\chi\Theta^{-r+s} \quad (r \neq \pm 2),$$

with the exclusions required by the same reason just given. The four possible cases left are

$$r = \pm 1, \quad s = \pm 2, \qquad \text{or} \qquad r = \pm 2, \quad s = \pm 1 \pmod 5.$$

If one of the four corresponding substitutions of third order belongs to Q, the other three do also. Put

$$\varphi = \Theta\chi\Theta. \tag{145}$$

Then from the relations of §58 one obtains

$$\varphi^2 = \Theta^{-1}\chi\Theta^{-1}, \qquad \varphi\chi = \Theta^{-1}\chi\psi\Theta^{-2}, \qquad \chi\varphi = \Theta^2\chi\psi\Theta, \tag{146}$$

and, by adjoining the compositions with ψ, one sees that the same group is obtained if one begins with a substitution of the second form.

The whole group Q is therefore completely determined by the chosen four-group D_2. It contains the elements

$$1, \quad \psi, \quad \chi\psi, \quad \Theta^{-1}\chi\psi\Theta^{-1}, \quad \varphi, \quad \varphi\psi, \quad \varphi\chi, \quad \varphi\chi\psi, \quad \varphi^2, \quad \varphi^2\psi, \quad \varphi^2\chi, \quad \varphi^2\chi\psi. \tag{147}$$

Equivalently, Q is generated by φ, ψ, χ and is represented in the form of a tetrahedral group. The multiplication rules following from (145) are

$$\psi\varphi^{-1} = \chi\varphi^{-1}\psi, \qquad \chi\varphi = \varphi\chi\psi, \qquad \psi\chi = \chi\psi, \qquad \chi^2 = \psi^2 = 1. \tag{148}$$

They are the defining composition laws of the tetrahedral group.

Since the tetrahedral group contains five four-groups D_2, the icosahedral group contains five tetrahedral subgroups. They are obtained from Q by transforming with the powers of Θ:

$$\Theta^{-r}Q\Theta^r, \qquad r = 0, 1, 2, 3, 4.$$

No two of these groups have, besides the identity, a common substitution. Indeed, Q and $\Theta^{-1}Q\Theta$ have in common, besides the identity, only two displayed elements, and neither of these occurs in $Q\Theta^{-1}$.

It follows from the theorem on permutation representations by cosets that the icosahedral group is isomorphic to a permutation group of degree sixty on five letters. The representation is obtained by adjoining to the five right cosets

$$Q, \quad Q\Theta, \quad Q\Theta^2, \quad Q\Theta^3, \quad Q\Theta^4$$

each element δ of the icosahedral group, forming

$$Q\delta, \quad Q\Theta\delta, \quad Q\Theta^2\delta, \quad Q\Theta^3\delta, \quad Q\Theta^4\delta,$$

and examining the permutation so produced. For the generating substitutions $\delta = \Theta, \chi$, one obtains respectively the permutations

$$(Q, Q\Theta, Q\Theta^2, Q\Theta^3, Q\Theta^4) \mapsto (Q\Theta, Q\Theta^2, Q\Theta^3, Q\Theta^4, Q),$$

and

$$(Q, Q\Theta, Q\Theta^2, Q\Theta^3, Q\Theta^4) \mapsto (Q, Q\Theta^2, Q\Theta^4, Q\Theta, Q\Theta^3).$$

The second computation follows from the formulae

$$\Theta\chi = \varphi\Theta^{-1}, \qquad \Theta^2\chi = \chi\varphi^2\Theta^3, \qquad \Theta^3\chi = \varphi^2\Theta, \qquad \Theta^4\chi = \varphi\chi\Theta^2.$$

These permutations are all even; hence the permutation group is the alternating group on five letters. Consequently the icosahedral group is simple, and coincides with the group provisionally called the icosahedral group in the earlier abstract discussion.

§60. The Basic Forms of the Icosahedral Group

We have already found one of the three basic forms of the icosahedral group, namely

$$f = x_1 x_2 (x_1^{10} + 11 x_1^5 x_2^5 - x_2^{10}). \tag{149}$$

The two missing forms have degrees twenty and thirty.

The basic form of degree twenty is the Hessian covariant of f. We put

$$H = \frac{1}{121} \left(f''(x_1, x_1) f''(x_2, x_2) - f''(x_1, x_2)^2 \right). \tag{150}$$

A straightforward calculation gives

$$H = -(x_1^{20} + x_2^{20}) + 228(x_1^{15} x_2^5 - x_1^5 x_2^{15}) - 494 x_1^{10} x_2^{10}. \tag{151}$$

The basic form of degree thirty may be defined as the functional determinant of H and f. Set

$$T = \frac{1}{20} \left(f'(x_1) H'(x_2) - f'(x_2) H'(x_1) \right). \tag{152}$$

Then

$$T = (x_1^{30} + x_2^{30}) + 522(x_1^{25} x_2^5 - x_1^5 x_2^{25}) - 10005(x_1^{20} x_2^{10} + x_1^{10} x_2^{20}). \tag{153}$$

These three forms satisfy an identical relation. To find it, put

$$\lambda = x_1^{10} + x_2^{10}, \qquad \mu = x_1^5 x_2^5.$$

Then

$$f^5 = \mu(\lambda + 11\mu)^5,$$

and after extracting the common factor $x_1^{10} + x_2^{10}$ from T and forming T^2, a numerical calculation gives

$$T^2 + H^3 = 1728 f^5. \tag{154}$$

This is the required relation.

The icosahedral group is simple, hence by the preceding theory it has no relative invariants. Thus the forms f, H, T remain absolutely unchanged when subjected to the binary icosahedral substitutions written with determinant one. This can also be checked directly, without using the general theorem.

Relation (154) may be written as

$$\frac{T^2}{f^5} + \frac{H^3}{f^5} = 1728. \tag{155}$$

The two quotients T^2/f^5 and H^3/f^5 may be regarded as functions of the single variable $x = x_1 : x_2$. If an icosahedral substitution is applied to this variable, the quotients can change only by constant factors. If

$$\frac{T^2}{f^5}(y) = h \frac{T^2}{f^5}(x), \qquad \frac{H^3}{f^5}(y) = k \frac{H^3}{f^5}(x),$$

then, because both x and y are independent variables, the identity (155) gives

$$h\frac{T^2}{f^5} + k\frac{H^3}{f^5} = 1728.$$

Since the two quotients are not in a constant ratio, one must have $h = k = 1$. Hence the quotients remain absolutely unchanged. It follows in turn that f, H, T themselves remain absolutely unchanged under homogeneous icosahedral substitutions of determinant one. For if f were changed into cf, then H and T would be changed into c^2H and c^3T; since the quotients are unchanged, $c = 1$.

If we set

$$\frac{H^3}{f^5} = z,$$

then (154) gives

$$H^3 - zf^5 = 0, \tag{156}$$

and equivalently

$$T^2 - (1728 - z)f^5 = 0. \tag{157}$$

These are two different forms of the same equation of degree sixty in x, called the *icosahedral equation.* Its properties and its relation to the general equation of the fifth degree will be treated later.

§61. The Invariants of the Icosahedron

The basic forms of the polyhedral groups exhaust, as we now show, the invariant theory of the corresponding groups. We give the proof only for the icosahedral group. The proofs for the other groups are entirely analogous and may be supplied by the reader. The fact that, in the tetrahedral and octahedral groups, relative invariants occur besides absolute invariants, whereas the icosahedral group, being simple, has only absolute invariants, has no essential influence on the argument.

We prove that every invariant of the icosahedral group can be represented as an integral rational function of the three forms f, H, T.

First:

1. No two of the icosahedral forms f, H, T have a common divisor.

This follows at once from the definition of these functions: the equations $f = 0$, $H = 0$, and $T = 0$ give respectively the fivefold, threefold, and twofold poles, and these poles are all distinct.

2. A double root of the icosahedral equation

$$H^3 - zf^5 = 0 \tag{158}$$

   is necessarily a root of f, H, or T. Hence it can occur only for one of the values $z = 0, \infty, 1728$.

Indeed, if (158) has a double root, then the function and its first derivative vanish together. In homogeneous notation, by Euler's theorem, the two partial derivatives with respect to x_1 and x_2 vanish simultaneously:

$$3H^2H'(x_1) - 5zf^4f'(x_1) = 0,$$
$$3H^2H'(x_2) - 5zf^4f'(x_2) = 0.$$

Since H and f cannot vanish simultaneously, there are only three possibilities: $H = 0$, or $f = 0$, or

$$H'(x_1)f'(x_2) - H'(x_2)f'(x_1) = 0,$$

i.e. $T = 0$, which corresponds by §60 to $z = 1728$.

3. If $J(x_1, x_2)$ is any invariant form of the icosahedral group and ξ is one of its roots, so that $J(\xi, 1) = 0$, then all values obtained from ξ by the fractional icosahedral substitutions are also roots of J.

Consequently, if J has a common divisor with one of the functions f, H, T, then J is divisible by that corresponding form.

Remove from J the highest possible powers of the three basic forms f, H, T, so that the remaining invariant is relatively prime to all three. If ξ is then a root of the remaining form, choose z so that ξ is also a root of (158). By item 2, ξ is a simple root of that equation. By item 3 all of its icosahedral transforms are roots of J, so J is divisible by the left side of (158). The quotient is again an invariant form, and the argument repeats.

We arrive at the theorem:

4. Every invariant of the icosahedron can be represented in the form

$$J(x_1, x_2) = C f^{\alpha} H^{\beta} T^{\gamma} F(H^3, f^5), \tag{159}$$

   where α, β, γ are non-negative integers, C is a constant, and F is a homogeneous integral function.

§62. Polyhedral Groups of the Second Kind. Crystallographic Groups

In §51 we already drew attention to groups of linear ternary substitutions which contain not only proper but also improper orthogonal substitutions. We called them groups of the second kind. We shall now consider the finite ones among them as polyhedral groups of the second kind, or extended polyhedral groups. The substitutions in them with determinant -1 will be called substitutions of the second kind.

According to §50 these groups are isomorphic to groups of binary linear substitutions with determinant $+1$ and -1. For the fractional substitutions which led us to the polyhedral groups, however, the two kinds cannot be distinguished.

The finite groups of the second kind are important, apart from their general group-theoretic interest, because of their application to crystallography. Their complete determination now presents no essential difficulty. By substitutions we shall here simply mean binary linear substitutions of determinant ± 1, remembering that two such substitutions whose coefficients differ only by a common sign, such as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix},$$

are not to be considered distinct.

Every group of linear substitutions contains the identity, whose determinant is $+1$. Therefore every group Q of the second kind contains, besides improper substitutions, proper substitutions; and these proper substitutions themselves form a group P. If φ is any substitution of the second kind contained in Q, then the product of φ with any other substitution of the second kind in Q lies in P. Hence

$$Q = P + P\varphi. \tag{160}$$

The group P must be one of the polyhedral groups considered above. Moreover

$$\varphi^{-1} P \varphi = P,$$

so that P is a normal divisor of Q. Conversely, if P is a polyhedral group and φ is a substitution of the second kind satisfying $\varphi^{-1} P \varphi = P$, then $P + P\varphi$ is a group of the second kind.

As before, two groups which are transformed into one another are not considered essentially different. We must run through the cases.

I. The identity group

If P is the identity group, viewed as the cyclic group of first degree C_1, then P contains only the identity. Hence $\varphi^2 = 1$, which is also sufficient. By transformation φ may be brought into the form

$$\begin{pmatrix} a & b \\ c & -a \end{pmatrix},$$

with the condition $a^2 + bc = 1$. This gives two essentially different forms:

$$\varphi' = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \varphi'' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{161}$$

The first is called inversion, the second reflection. Translated into rectangular coordinates by §50, φ' changes the signs of all three coordinates, whereas φ'' changes x into $-x$. Thus the two groups of the second kind are

$$C_1' = C_1 + C_1\varphi', \qquad C_1'' = C_1 + C_1\varphi''.$$

II. Cyclic groups

Let $P = C_n$, $n > 1$. Put

$$c = \begin{pmatrix} \varepsilon & 0 \\ 0 & \varepsilon^{-1} \end{pmatrix}, \qquad \varepsilon = e^{\pi i/n}. \tag{162}$$

Then, as in §55,

$$C_n = 1, c, c^2, \ldots, c^{n-1}. \tag{163}$$

If

$$\varphi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad ad - bc = -1, \tag{164}$$

then $\varphi^{-1}c\varphi$ and $\varphi^{-1}d\varphi$ must occur in the row (163). This forces either $b = c = 0$ or $a = d = 0$. In the first case one obtains two possible extensions,

$$C_n' = \left\{ \begin{pmatrix} e^{\pi ih/n} & 0 \\ 0 & e^{-\pi ih/n} \end{pmatrix} : h = 0, 1, \ldots, 2n-1 \right\}, \tag{165}$$

and

$$C_n'' = \left\{ \pm \begin{pmatrix} e^{\pi ih/n} & 0 \\ 0 & e^{-\pi ih/n} \end{pmatrix} : h = 0, 1, \ldots, n-1 \right\}. \tag{166}$$

For even n both C_1' and C_1'' occur in C_n'; for odd n, one of them occurs in C_n' and the other in C_n''. In the second case $a = d = 0$, a transformation which leaves P unchanged reduces φ to

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

producing a third cyclic extension.

III. Dihedral groups

Let $P = D_n$, consisting of

$$D_n = 1, c, c^2, \ldots, c^{n-1}, d, cd, c^2d, \ldots, c^{n-1}d, \qquad d = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}. \tag{167}$$

The substitution c has order n, and the substitutions c^hd all have order two. If $n > 2$, the transform $\varphi^{-1}c\varphi$ must again be among the powers of c, so the preceding analysis applies. Since

d does not occur in the cyclic extensions just described, the dihedral groups of the second kind have the forms

$$C_n' + C_n'd, \qquad C_n'' + C_n''d, \qquad C_n''' + C_n'''d. \tag{168}$$

Written out, the first two are the families

$$\begin{pmatrix} e^{\pi ih/n} & 0 \\ 0 & e^{-\pi ih/n} \end{pmatrix}, \qquad \begin{pmatrix} 0 & e^{\pi ih/n} \\ e^{-\pi ih/n} & 0 \end{pmatrix}, \qquad h = 0, 1, \ldots, 2n-1,$$

and the analogous signed family with $h = 0, 1, \ldots, n-1$. The third coincides with one of these forms according as n is odd or even.

The case $n = 2$ gives only an apparent exception. One might have $\varphi^{-1}c\varphi = d$ or cd. Direct calculation gives two groups

$$1, c, d, cd, \varphi_1, \varphi_1 c, \varphi_1 d, \varphi_1 cd, \qquad 1, c, d, cd, \varphi_2, \varphi_2 c, \varphi_2 d, \varphi_2 cd, \tag{169}$$

where

$$\varphi_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \varphi_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} i & i \\ i & -i \end{pmatrix}.$$

The formulae

$$c\varphi_1 = \varphi_1 d, \quad d\varphi_1 = \varphi_1 c, \quad cd\varphi_1 = \varphi_1 dc,$$

and their analogues for φ_2, show that these groups are transformed forms of D_2.

For the remaining polyhedral groups of the second kind, two general remarks are useful. The inversion

$$j = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \tag{170}$$

is a similarity substitution and therefore commutes with every substitution. Thus from every polyhedral group of the first kind one obtains at least one group of the second kind,

$$P + Pj. \tag{171}$$

By Sylow's theorem, every finite group contains a subgroup whose order is the highest power of two dividing the order of the group. The extended tetrahedral, octahedral, and icosahedral groups have orders $24, 48, 120$; hence they must contain divisors of order $8, 16, 8$. In the tetrahedral and icosahedral groups there is a four-group D_2 but no substitution of fourth order; in the octahedral group there is a dihedral group D_4 but no substitution of eighth order. Hence one of the extended dihedral groups just considered must occur.

For the tetrahedral group two extensions occur. Besides

$$T' = T + Tj,$$

one obtains another

$$T'' = T + Tj_1,$$

where j_1 is the second extension of the four-group. No further tetrahedral extension occurs, because the tetrahedral group is completely determined by any one four-group contained in it.

If an extension of the four-group of one of the exceptional forms (169) is chosen, the resulting tetrahedral group is merely a transform of T''.

For the octahedral group, the special substitution defining the second tetrahedral extension does not commute with the octahedral group, and consequently the octahedral group gives no further extension beyond inversion:

$$O' = O + Oj.$$

The same is true of the icosahedral group. One may transform the icosahedral group so that one of its four-groups has the simplest form; the calculation is longer but entirely straightforward, and leads to no additional case. Thus one obtains

$$J' = J + Jj.$$

The following groups of the second kind are therefore obtained:

$$\begin{array}{llll} C_1', C_1'', & C_n', C_n'', C_n''', & D_n', D_n'', D_n''', \\ T', T'', & O', & J'. \end{array}$$

The thirty-two symmetry systems of crystallographers are contained among them. In the crystallographic notation they are the groups

$$\begin{array}{llllll} C_1, & C_1', & C_1'', & D_1, & & \\ C_2, & C_2', & C_2'', & D_2, & D_2', & D_2'' \\ C_3, & C_3', & C_3'', & D_3, & D_3', & D_3'' \\ C_4, & C_4', & C_4'', & D_4, & D_4', & \\ C_6, & C_6', & C_6'', & D_6, & D_6', & \\ T, & T', & T'', & O, & O'. & \end{array}$$

The remaining polyhedral groups are excluded in crystallography by the crystallographic law of rational indices.

§63. Function Congruences

From the linear substitutions whose construction and composition we have studied in the preceding sections one can derive another kind of finite group, namely the congruence groups.

These groups are of many-sided interest. They arise first in the theory of elliptic functions, where they appear as Galois groups of modular equations. They are also important for general group theory, because they furnish a method for constructing whole series of simple groups. This is all the more noteworthy since, as we saw in the fourth section, simple groups are rather rare, at least among the lower degrees.

The theory of these congruence groups is not only greatly generalized, but the governing laws appear much more sharply, if one bases it on an extension of the notion of congruence due to Gauss and subsequently developed and applied, especially by Galois, in problems of group theory.

To obtain a basis for the theory, consider an integral polynomial in one variable t,

$$f(t) = a_0 t^n + a_1 t^{n-1} + \cdots + a_{n-1} t + a_n, \tag{172}$$

with integral coefficients. Choose a prime number p as modulus, and assume that a_0 is not divisible by p.

Two integral functions of t whose corresponding coefficients are congruent modulo p will themselves be called congruent modulo p. The function $f(t)$ is called reducible modulo p if there exist two integral functions $\varphi(t), \psi(t)$, each with at least one term depending on t whose coefficient is not divisible by p, such that

$$f(t) \equiv \varphi(t)\psi(t) \pmod{p}. \tag{173}$$

Terms with coefficients divisible by p may of course be omitted from φ and ψ, and then the degree of their product equals the degree of f. Hence the degrees of φ and ψ are both lower than n. If no such functions exist, $f(t)$ is called irreducible modulo p.

A function irreducible modulo p is certainly irreducible over the field of rational numbers. The converse, however, need not hold. For example, a quadratic function $t^2 - B$ is reducible modulo p when B is a quadratic residue of p, for if $a^2 \equiv B \pmod{p}$, then

$$t^2 - B \equiv (t-a)(t+a) \pmod{p}.$$

On the other hand, $t^2 - N$ is irreducible when N is a non-residue of p, since otherwise $t^2 - N$ would vanish modulo p for some rational integer t.

The functions

$$t^2 + t + 1, \qquad t^3 + t + 1$$

for the modulus two give another example. They are irreducible modulo two. If either were reducible, one factor would have to be linear, and there would have to be an integer which, substituted for t, makes the function even. This is impossible, since for $t = 0$ and $t = 1$ both functions take odd values.

Let

$$P = t^n + p_1 t^{n-1} + p_2 t^{n-2} + \cdots + p_{n-1} t + p_n \tag{174}$$

be irreducible modulo p, and for simplicity take the leading coefficient to be one. From any other integral function $F(t)$ one may derive a remainder $\Phi(t)$ of degree lower than n by writing

$$F(t) = QP + \Phi(t), \tag{175}$$

where Q has integral coefficients. We describe the relation of F to Φ as a congruence modulo P.

Thus two functions $F(t), F_1(t)$ which have the same remainder modulo P are called congruent modulo P, and this is expressed by

$$F(t) \equiv F_1(t) \pmod{P}.$$

Equivalently, $F(t) - F_1(t)$ is divisible by P.

If the two functions do not have the same remainder, but have remainders congruent modulo p, then the functions are called congruent modulo the double modulus P, p:

$$F(t) \equiv F_1(t) \pmod{P, p}. \tag{176}$$

For this kind of congruence one has:

1. The product of two functions cannot be congruent to zero unless one of the factors is congruent to zero.

The proof follows from the algorithm for the greatest common divisor. Suppose F_1 has degree lower than P and is not congruent to zero modulo p. One can determine functions $F_2, F_3, \ldots$, also of lower degree than P, of decreasing degree, and quotients $Q_1, Q_2, \ldots$, such that

$$P = Q_1 F_1 + F_2, \quad F_1 = Q_2 F_2 + F_3, \quad \ldots, \quad F_{r-2} = Q_{r-1} F_{r-1} + F_r \pmod{p}, \tag{177}$$

until the last remainder F_r is a constant. This constant cannot be congruent to zero modulo p; otherwise P would be reducible modulo p. If now $QF_1 \equiv 0 \pmod{P, p}$, multiplication of the equations (177) by Q gives successively $QF_2 \equiv 0, \ldots, QF_r \equiv 0$, hence $Q \equiv 0$. Taking F_1, Q to be the remainders of the two factors proves the assertion.

The remainders of all functions modulo P have the form

$$X = X(t) = x_0 + x_1 t + \cdots + x_{n-1} t^{n-1}. \tag{178}$$

To obtain one representative from each congruence class modulo P, p, it is enough to let each coefficient $x_0, x_1, \ldots, x_{n-1}$ run through the residues $0, 1, \ldots, p-1$. Thus there are exactly p^n incongruent functions modulo P, p.

§64. Congruence Fields

It is only a different mode of expression for the preceding considerations to introduce, with Galois, an imaginary number ε satisfying

$$P(\varepsilon) \equiv 0 \pmod{p}. \tag{179}$$

No ordinary rational integer of this sort exists when $n > 1$, for then $P(t)$ would be divisible by $t - \varepsilon$ and so would be reducible. Nevertheless one may use such a symbol in calculation, just as one introduces the imaginary unit $i = \sqrt{-1}$ in ordinary arithmetic. As far as addition, subtraction, and multiplication are concerned, one calculates with it according to the ordinary rules of algebra, using the equation $P(\varepsilon) = 0$ whenever desired. One may even think of ε as an actual root of $P(x) = 0$ in the usual sense.

All numbers obtained in this way can be written in the form

$$\alpha = a_0 + a_1\varepsilon + \cdots + a_{n-1}\varepsilon^{n-1}. \tag{180}$$

Because the modulus p is held fixed, two numbers congruent modulo p will simply be regarded as equal. Strictly speaking, therefore, the objects of the calculation are not the individual numbers but the classes of numbers congruent to one another. These numbers α are called the Galois imaginaries. The system associated with a fixed ε will be denoted by $\mathfrak{G}$.

We have first:

2. There are exactly p^n different numbers in $\mathfrak{G}$.

From item 1 of the preceding section we also obtain:

3. A product of two or more numbers of $\mathfrak{G}$ is zero if and only if at least one factor is zero.

The whole system $\mathfrak{G}$ is obtained by allowing the rational coefficients $a_0, a_1, \ldots, a_{n-1}$ in (180) to run independently through a complete residue system modulo p. It contains the ordinary residue classes $0, 1, \ldots, p-1$, and for $n = 1$ it is identical with that system.

If $\alpha \neq 0$ is fixed in $\mathfrak{G}$, then $\alpha\xi$ runs through the whole system at the same time as ξ. Indeed, $\alpha\xi = \alpha\xi'$ implies $\xi = \xi'$. Hence:

4. If $\alpha, \beta \in \mathfrak{G}$ and $\alpha \neq 0$, there exists exactly one $\gamma \in \mathfrak{G}$ such that

$$\alpha\gamma = \beta.$$

This proves that division, except by zero, is allowed in $\mathfrak{G}$. We write this γ as β/α.

The system $\mathfrak{G}$ may be called a finite field: it contains only finitely many numbers, and it has the characteristic property of a field, namely the unrestricted performance of the arithmetic operations, except division by zero. We shall call $\mathfrak{G}$ a *congruence field.* The integer n is its degree and p its modulus.

If $\alpha \neq 0$, then $\alpha\xi = \eta$ runs through the non-zero elements as ξ runs through them. Multiplying all such equations gives

$$\alpha^{p^n-1}\Pi = \Pi,$$

where Π denotes the non-zero product of all non-zero elements. Cancelling Π yields Fermat's theorem in this field:

$$\alpha^{p^n-1} = 1. \tag{181}$$

Multiplication by α gives the equivalent form

$$\alpha^{p^n} = \alpha, \tag{182}$$

which also holds for $\alpha = 0$.

If $\alpha \neq 0$, let e be the least positive exponent with

$$\alpha^e = 1. \tag{183}$$

Then the powers $1, \alpha, \alpha^2, \ldots, \alpha^{e-1}$ are distinct, and α is said to belong to the exponent e. The exponent e divides $p^n - 1$. For if $\alpha^m = 1$ and $m = qe + r$ with $0 \leq r < e$, then $\alpha^r = 1$, hence $r = 0$. In particular $e \mid p^n - 1$.

If $p^n - 1 = ef$, then α^h belongs to the exponent e precisely when h is relatively prime to e. Thus, if there is any element with exponent e, there are $\varphi(e)$ of them, where φ is Euler's function. Since

$$\sum_{e|(p^n-1)} \varphi(e) = p^n - 1, \tag{184}$$

and since every non-zero element belongs to exactly one exponent, it follows that, for every divisor e of $p^n - 1$, there are exactly $\varphi(e)$ elements of exponent e. In particular, there are $\varphi(p^n - 1)$ elements γ whose powers

$$1, \gamma, \gamma^2, \ldots, \gamma^{p^n-2} \tag{185}$$

are all distinct. Such elements are called primitive roots of $\mathfrak{G}$.

Among the elements of $\mathfrak{G}$ some are squares, and the others are non-squares. If $p = 2$, then every element is a square, as follows from (182). If p is odd, then one half of the non-zero elements are squares and the other half are non-squares. Indeed, the two opposite elements α and $-\alpha$ are distinct but have the same square, and $\beta^2 = \alpha^2$ implies $\beta = \pm\alpha$. Thus there are exactly $(p^n - 1)/2$ non-zero squares and the same number of non-squares. Counting zero among the squares, one has:

5. If $p = 2$, all elements of $\mathfrak{G}$ are squares. If p is odd, there are $(p^n + 1)/2$ squares and $(p^n - 1)/2$ non-squares.

For odd p, a primitive root is necessarily a non-square, and in the sequence (185) the even powers are the squares and the odd powers the non-squares.

The product and quotient of two squares is again a square. Hence the product of a non-square and a non-zero square is a non-square, and the product of two non-squares is a square. All these assertions also follow immediately from the representation by powers of a primitive root.

We close with a result needed later:

6. Every non-square is the sum of two squares.

Here p is assumed odd, since otherwise there are no non-squares. First, every element β can be represented as a difference of two squares, because

$$\beta = \left(\frac{\beta+1}{2}\right)^2 - \left(\frac{\beta-1}{2}\right)^2.$$

If -1 is a square in $\mathfrak{G}$, this is already a representation of β as a sum of two squares. If -1 is a non-square, choose in the natural sequence $1, 2, \ldots, p-1$ a square immediately followed by a non-square. Thus let $a = \xi^2$ be a square and $\xi^2 + 1$ a non-square. If β is a non-square, then $\beta/(\xi^2+1) = \eta^2$ is a square, and so

$$\beta = (\xi\eta)^2 + \eta^2.$$

§65. Congruence Groups in the Field $\mathfrak{G}$

In every congruence field $\mathfrak{G}$ one can form a finite group of linear substitutions

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \tag{186}$$

by letting $\alpha, \beta, \gamma, \delta$ run through all elements of $\mathfrak{G}$, and by composing two substitutions according to

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} = \begin{pmatrix} \alpha\alpha' + \beta\gamma' & \alpha\beta' + \beta\delta' \\ \gamma\alpha' + \delta\gamma' & \gamma\beta' + \delta\delta' \end{pmatrix}. \tag{187}$$

The degree of this full group is p^{4n} when p is the modulus and n the degree of $\mathfrak{G}$.

It contains a group of smaller degree. Let

$$\det A = \alpha\delta - \beta\gamma.$$

By (187), determinants multiply. Therefore, if the entries are restricted by

$$\alpha\delta - \beta\gamma = 1, \tag{188}$$

the corresponding substitutions form a group. Its degree is the number of solutions of (188). To count them, first choose α, β not both zero; this gives $p^{2n} - 1$ choices. If a particular solution (γ_0, δ_0) is known, all others have the form

$$\gamma = \gamma_0 + \alpha\xi, \qquad \delta = \delta_0 + \beta\xi,$$

with $\xi \in \mathfrak{G}$. Thus there are p^n choices for (γ, δ), and the degree of the determinant-one group is

$$p^n(p^{2n} - 1). \tag{189}$$

For odd p one can derive a still smaller group. If $p = 2$, a number α is not distinct from $-\alpha$. If p is odd, however, the two matrices

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \begin{pmatrix} -\alpha & -\beta \\ -\gamma & -\delta \end{pmatrix} \tag{190}$$

are distinct, and the whole system splits into such pairs. Multiplication of pairs again gives a pair. Hence these pairs can themselves be regarded as the elements of a new group of degree

$$\frac{1}{2} p^n(p^{2n} - 1). \tag{191}$$

This group, of degree (189) for $p = 2$ and (191) for odd p, will be called the congruence group belonging to $\mathfrak{G}$, and denoted by E. In modern notation it is the projective special linear group over the finite field $\mathfrak{G}$.

For further study it is important to know generators. Write the elements of $\mathfrak{G}$ in the form

$$\xi = x_0 + x_1\varepsilon + \cdots + x_{n-1}\varepsilon^{n-1}, \tag{192}$$

where the x_i are rational residues modulo p. A generating system is

$$A_k = \begin{pmatrix} 1 & \varepsilon^k \\ 0 & 1 \end{pmatrix}, \qquad B_k = \begin{pmatrix} 1 & 0 \\ \varepsilon^k & 1 \end{pmatrix}, \qquad k = 0, 1, \dots, n-1. \tag{193}$$

Indeed,

$$\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \alpha+\beta \\ 0 & 1 \end{pmatrix},$$

and repeated use gives every upper unipotent matrix; similarly every lower unipotent matrix is generated. Combining such elements gives matrices of the forms

$$\begin{pmatrix} \alpha & 1 \\ -1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ \gamma & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and hence, by a further multiplication,

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \tag{194}$$

whenever $\delta \neq 0$. The restriction $\delta \neq 0$ may be removed using the last displayed matrix. Thus the elements (193) generate E.

The group E contains a divisor of index $p^n + 1$, consisting of all substitutions

$$\begin{pmatrix} \alpha & \beta \\ 0 & \delta \end{pmatrix}, \qquad \alpha\delta = 1, \tag{195}$$

where β is arbitrary and $\alpha \neq 0$. This divisor yields a permutation representation of E on $p^n + 1$ letters.

The linear expression

$$\eta = \frac{\alpha\xi + \beta}{\gamma\xi + \delta} \tag{196}$$

assigns to every $\xi \in \mathfrak{G}$ a corresponding η, except when $\gamma\xi + \delta = 0$. Conversely, each η determines a ξ, except when $\gamma\eta - \alpha = 0$. To remove the exceptional cases, add to $\mathfrak{G}$ one new element, called infinity and denoted by ∞. It is governed by the usual rules

$$a + \infty = \infty, \qquad a \cdot \infty = \infty \quad (a \neq 0), \qquad \frac{a}{0} = \infty, \qquad \frac{a}{\infty} = 0.$$

Then every substitution of E acts as a permutation of the $p^n + 1$ symbols of the projective line $\mathfrak{G} \cup \{\infty\}$.

§66. Simplicity of the Group E

The first question in a closer investigation of the group E is whether it has a normal divisor. One can prove that, apart from two quite small exceptional cases, none exists.

Assume that G is a normal divisor of E and that it contains at least one substitution different from the identity,

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \tag{197}$$

By normality, for every element $T \in E$ the transform

$$TAT^{-1} \tag{198}$$

also lies in G. We shall show that by such transformations and their products one can derive all elements of E. It is enough to show that the generators A_k, B_k of (193) lie in G.

We proceed step by step.

First, G contains a substitution of the form

$$\begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}. \tag{199}$$

To see this, form the transform of A by $\begin{pmatrix} 1 & \xi \\ 0 & 1 \end{pmatrix}$. If $\beta \neq 0$, choose ξ so that the lower-left entry vanishes. If $\beta = 0$, then not all of γ and $\delta - \alpha$ vanish, since otherwise A would be the identity; choose ξ so that the transformed matrix has non-zero upper-right entry, and apply the preceding argument once more.

Second, G contains a substitution of the form

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{200}$$

Weber obtains this by solving, for λ, μ, ν, ρ, the system

$$\lambda\rho - \mu\nu = 1, \qquad \mu\rho\gamma - \lambda\nu\beta - \lambda\nu\delta = 0, \qquad -\mu^2\gamma + \lambda\mu\beta + \lambda\mu\delta = 1. \tag{201}$$

From the last two equations one derives

$$\lambda\beta + \mu\delta = 0, \qquad \mu\gamma = \nu, \tag{202}$$

and hence the remaining condition

$$\lambda^2\beta + \lambda\mu\delta - \mu^2\gamma = 1. \tag{203}$$

This equation can always be solved. If β is a square, put $\beta = w^2$ and determine λ. This includes the case $p = 2$. If p is odd and β is a non-square, one reduces the problem to the decomposition of a non-square as a sum of two squares, supplied by §64. Thus the desired transformation exists.

Third, except in the case $n = 1, p = 2$, the group G contains

$$A_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \tag{204}$$

Indeed, from the preceding step the group contains

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and hence also the products

$$\begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ -1 & m \end{pmatrix}$$

for every odd integer m. If p is odd, take $m = p$ and obtain A_0. If $p = 2$, a further calculation gives, for every non-zero $q \in \mathfrak{G}$, a matrix of the form

$$\begin{pmatrix} 0 & q \\ q^{-1} & 0 \end{pmatrix},$$

and products of such matrices yield A_0 provided $n > 1$. The only exception is $n = 1, p = 2$, where one obtains a group of degree six with a normal divisor of degree three.

Fourth, except in the case $n = 1, p = 3$, the group G contains every substitution

$$\begin{pmatrix} 1 & \xi \\ 0 & 1 \end{pmatrix} \tag{205}$$

with arbitrary $\xi \in \mathfrak{G}$. For, using A_0, one obtains

$$\begin{pmatrix} 1 & (q^2 - 1)\lambda \\ 0 & 1 \end{pmatrix} \tag{206}$$

for every non-zero q. If one can choose q so that $q^2 - 1 \neq 0$, then λ can be adjusted to give any ξ. With a primitive root γ of $\mathfrak{G}$, put $q = \gamma^h$. Such a choice is always possible except in the small cases $n = 1, p = 3$ and $n = 1, p = 5$. The latter is removed by an explicit calculation using the matrix

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -2 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix},$$

which again yields all elements (205).

Thus in all cases except $n = 1, p = 2$ and $n = 1, p = 3$, all generators of E lie in G, hence $G = E$. The group E is therefore simple.

For the exceptional case $n = 1, p = 3$, the group E has degree twelve and contains the normal divisor of degree four

$$G = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \right\}. \tag{207}$$

The remaining cosets verify directly that this is normal. For $n = 1, p = 2$, the group is isomorphic to the symmetric group on three letters and has its usual normal subgroup of order three.

For the first non-exceptional cases one obtains the degrees

$$n = 1, p = 5 : \quad g = 60,$$

and then the higher degrees supplied by $\frac{1}{2}p^n(p^{2n} - 1)$ for odd p, and by $p^n(p^{2n} - 1)$ for $p = 2$.

§67. Congruence Fields of Degree Two

Every congruence field $\mathfrak{G}_{n,p}$ of degree n and modulus p contains, as a subfield, the field $\mathfrak{G}_{1,p}$ consisting of rational residues modulo p. Consequently the group of linear substitutions $E_{n,p}$ contains as a divisor the group of all substitutions

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \tag{208}$$

where a, b, c, d are rational residues modulo p and $ad - bc = 1$, again with opposite matrices identified when p is odd.

This group is the group which appears in the theory of elliptic functions as the group of modular equations. It is the real goal of the present investigation. For the study of this group, especially for the search for its divisors, it is most advantageous not to treat it by itself, but to regard it as a divisor of a group $E_{2,p}$.

Let N be a fixed quadratic non-residue modulo p. The polynomial $t^2 - N$ is then irreducible modulo p. We obtain a congruence field of second degree by adjoining a Galois imaginary ε satisfying

$$\varepsilon^2 = N. \tag{209}$$

In what follows, lower-case Latin letters denote rational residues modulo p, which we shall also call real numbers; Greek letters denote elements of the field $\mathfrak{G}_{2,p}$.

The field $\mathfrak{G}_{2,p}$ has the important property that every real number is a square in it. If a is already a quadratic residue modulo p, write $a = x^2$. If b is a non-residue, then $b = aN = (x\varepsilon)^2$ after multiplying by a suitable square a.

A number of the form $a\varepsilon$ will be called purely imaginary. Two numbers $a + b\varepsilon$ and $a - b\varepsilon$ will be called conjugate imaginary. This transfer of terminology from the ordinary complex numbers is justified by the agreement of the calculation rules and cannot cause confusion, since no ordinary complex numbers are involved here.

Since $N^{(p-1)/2} \equiv -1 \pmod{p}$, equation (209) gives

$$\varepsilon^p = -\varepsilon. \tag{210}$$

For $\alpha = a + b\varepsilon$, the binomial theorem gives, modulo p,

$$\alpha^p = a^p + b^p\varepsilon^p = a - b\varepsilon. \tag{211}$$

Therefore

$$\alpha^{p+1} = a^2 - Nb^2, \tag{212}$$

which is always real.

If γ is a primitive root of $\mathfrak{G}_{2,p}$, then γ^r is real if and only if r is divisible by $p + 1$. In particular γ^{p+1} is a primitive root of the prime field in the ordinary sense.

The necessary and sufficient condition that $\alpha \in \mathfrak{G}_{2,p}$ be real is

$$\alpha^p = \alpha. \tag{213}$$

Moreover every real number is a $(p+1)$-st power in $\mathfrak{G}_{2,p}$. Conversely, a power γ^r is real only when r is divisible by $p+1$, because every non-zero real number satisfies Fermat's congruence $x^{p-1} = 1$.

The notion of roots of unity also carries over. If m divides $p^2 - 1$ and

$$\rho = \gamma^{(p^2-1)/m},$$

then $\rho^m = 1$, and no lower positive power is one. Thus ρ is a primitive m-th root of unity in $\mathfrak{G}_{2,p}$. All m-th roots of unity are powers ρ^s, and the primitive ones are precisely those with s prime to m. Every non-zero element of $\mathfrak{G}_{2,p}$ is a root of unity of degree $p^2 - 1$.

§68. The Real Linear Congruence Group L_p

Equipped with these tools, we now investigate the real congruence group L_p, consisting of all substitutions

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad ad - bc = 1, \tag{214}$$

where a, b, c, d are real residues modulo p. In the earlier notation this group is $E_{1,p}$. Excluding $p = 2$, its degree is

$$\frac{1}{2} p(p^2 - 1). \tag{215}$$

If A, U are elements of a group, we call

$$U^{-1} A U = A'$$

the element obtained from A by transformation with U. Elements obtained from one another in this way have the same order. If all elements A are taken from a group G, the transformed elements form an isomorphic group $U^{-1} G U$.

Let $A \in L_p$, and let

$$U = \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix}$$

be an element of $E_{2,p}$. We first try to transform A into the normal form

$$S = \begin{pmatrix} \theta & 0 \\ 0 & \theta^{-1} \end{pmatrix}. \tag{216}$$

From the equation

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix} = \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix} \begin{pmatrix} \theta & 0 \\ 0 & \theta^{-1} \end{pmatrix}$$

one obtains

$$\begin{aligned} (a - \theta)\lambda + b\nu &= 0, \qquad & (a - \theta^{-1})\mu + b\rho &= 0, \\ c\lambda + (d - \theta)\nu &= 0, \qquad & c\mu + (d - \theta^{-1})\rho &= 0. \end{aligned} \tag{217}$$

Thus θ and θ^{-1} must be the roots of

$$\theta^2 - \theta(a + d) + 1 = 0. \tag{218}$$

Once θ and θ^{-1} are known, (217) determines the ratios $\nu : \lambda$ and $\rho : \mu$. The scalar factors are still to be chosen so that $\lambda\rho - \mu\nu = 1$. This is possible unless $a + d = \pm 2$, for otherwise the two eigenvectors would be proportional.

The equation (218) is always solvable in $\mathfrak{G}_{2,p}$, since every real number is a square there. Explicitly,

$$\theta = \frac{a + d}{2} \pm \sqrt{\left(\frac{a + d}{2}\right)^2 - 1}. \tag{219}$$

We therefore have:

1. A substitution A is transformable into the normal form S whenever $a+d \neq \pm 2$. The number θ is real or imaginary according as $(a+d)^2-4$ is a quadratic residue or a non-residue modulo p.

In the remaining case $a+d=\pm 2$, the normal form S cannot be produced. Instead A can be transformed into

$$T=\begin{pmatrix}1 & t\\ 0 & 1\end{pmatrix}, \tag{220}$$

with real t. It is enough to take $a+d=2$, since the case -2 is obtained by changing all signs. Then

$$a-1=-(d-1), \qquad (a-1)(d-1)=bc. \tag{221}$$

If $b=0$ or $c=0$, the transformation is immediate; if both are non-zero, the displayed identities give an explicit transforming matrix. Hence:

2. A substitution A with $a+d=\pm 2$ can always be transformed into the normal form T.

The orders of elements in L_p are now easily determined. There are three cases.

First kind. If $a+d=\pm 2$, every non-identity substitution is transformed into T. Since

$$T^r=\begin{pmatrix}1 & rt\\ 0 & 1\end{pmatrix},$$

such a substitution has order p. Counting these elements gives p^2 substitutions of the first kind, including the identity.

Second kind. If $(a+d)^2-4$ is a quadratic residue modulo p, the substitution is transformed into S with real θ. Let γ be a primitive root of $\mathfrak{G}_{2,p}$. Then the real elements may be written as powers $\gamma^{r(p+1)}$, and

$$a+d=\gamma^{r(p+1)}+\gamma^{-r(p+1)}. \tag{222}$$

Distinct values, different from ± 2, are obtained for

$$r=1,2,\ldots,\frac{p-3}{2}. \tag{223}$$

The order of S, and hence of the substitution, is a divisor of $(p-1)/2$. It is exactly $(p-1)/2$ when r is prime to $(p-1)/2$. The number of substitutions of the second kind is

$$\frac{1}{2}p(p-3)(p+1). \tag{224}$$

Third kind. If $(a+d)^2-4$ is a quadratic non-residue, then θ is not real but is conjugate to its reciprocal. Therefore $\theta^{p+1}=1$, and the order is a divisor of $(p+1)/2$. Write

$$a+d=\gamma^{r(p-1)}+\gamma^{-r(p-1)}, \qquad r=1,2,\ldots,\frac{p-1}{2}. \tag{225}$$

Here neither $b = 0$ nor $c = 0$ can occur; otherwise a and d would be real eigenvalues. Counting gives

$$\frac{1}{2}p(p-1)(p-1) \tag{226}$$

substitutions of the third kind. Together with the numbers from the first and second kinds, this gives the total degree $\frac{1}{2}p(p^2-1)$, as required.

§69. Imaginary Form of the Group L_p

The substitutions of the first and second kinds in L_p have the property that their corresponding normal forms T or S lie again in L_p, and that they can be transformed into these forms by real substitutions, i.e. by substitutions lying in L_p itself. This does not hold for the substitutions of the third kind.

We shall transform the whole group L_p into another group A_p, which is an isomorphic divisor of $E_{2,p}$, in such a way that the normal forms of the transformations of the third kind are contained in A_p, and each substitution of the third kind in A_p can be transformed into normal form by substitutions of A_p itself.

Consider

$$S = \begin{pmatrix} \theta & 0 \\ 0 & \theta^{-1} \end{pmatrix}, \tag{227}$$

where θ and θ^{-1} are conjugate imaginary. We seek a substitution

$$R = \begin{pmatrix} \lambda & \mu \\ \nu & \rho \end{pmatrix} \tag{228}$$

so that

$$RSR^{-1} \tag{229}$$

is real. Direct calculation shows that this is achieved when $\lambda\mu$ and $\rho\nu$ are purely imaginary and $\lambda\rho$, $-\mu\nu$ are conjugate imaginary pairs. A particularly simple choice, with determinant one, is

$$R = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ \varepsilon^{-1} & -\varepsilon^{-1} \end{pmatrix}. \tag{230}$$

If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

is any real substitution in L_p, then, using $\varepsilon^2 = N$, one obtains

$$R^{-1}AR = \begin{pmatrix} \alpha & \beta \\ N\beta' & \alpha' \end{pmatrix}, \tag{231}$$

where α, α' and β, β' are conjugate imaginary pairs; equivalently $\alpha' = \alpha^p$, $\beta' = \beta^p$. This may also be written in the form

$$\begin{pmatrix} \alpha & \beta \\ N\beta^p & \alpha^p \end{pmatrix}. \tag{232}$$

Conversely, from every matrix of the form (232) one obtains by the inverse transformation a real substitution in L_p. Thus, as A runs through L_p, the transformed matrices form an isomorphic group, which we denote by A_p.

In A_p the normal form S for substitutions of the third kind is present. It remains to check that any substitution A of the third kind in A_p can be brought to this normal form by transformation with a substitution $U \in A_p$. Write

$$U = \begin{pmatrix} \lambda & \mu \\ N\mu' & \lambda' \end{pmatrix}, \qquad \lambda\lambda' + N\mu\mu' = 1. \tag{233}$$

The equations corresponding to §68 become

$$(\theta - \alpha)(\theta - \alpha') + N\beta\beta' = 0, \tag{234}$$

and

$$\frac{N\lambda\mu'}{\theta - \alpha} = \frac{\lambda'\mu}{\theta - \alpha'} = \frac{\beta}{\beta'}. \tag{235}$$

The second equation follows from the first by passing to conjugate imaginaries. Hence one can choose

$$N\lambda\mu' = k(\alpha - \theta), \qquad N\lambda'\mu = k'(\alpha' - \theta^{-1}),$$

where k, k' are conjugate imaginary, and then determine k, k' so that $\lambda\lambda' + N\mu\mu' = 1$. This proves the claim.

Let γ be a primitive root of $\mathfrak{G}_{2,p}$. For the normal forms of the second and third kinds we may write

$$S_1 = \begin{pmatrix} \gamma^{p+1} & 0 \\ 0 & \gamma^{-(p+1)} \end{pmatrix}, \qquad S_2 = \begin{pmatrix} \gamma^{p-1} & 0 \\ 0 & \gamma^{-(p-1)} \end{pmatrix}. \tag{236}$$

If A is a substitution of the second kind in L_p, and $\bar{A}$ a substitution of the third kind in A_p, then there are substitutions $U \in L_p$ and $V \in A_p$ such that

$$A = US_1^r U^{-1}, \qquad \bar{A} = VS_2^r V^{-1}.$$

Writing

$$T_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

the same applies to substitutions of the first kind. Hence:

The substitutions of the first, second, and third kinds in L_p or A_p, together with the identity, can be arranged in cycles of respectively

$$p, \qquad \frac{p-1}{2}, \qquad \frac{p+1}{2}$$

elements.

Two such cycles have no element in common except the identity. For first-kind substitutions this is immediate, since a whole cycle is generated by any non-identity element in it. For the second and third kinds, if two cycles have an element in common, transform one of them so that the common element is in normal form:

$$1,\ USU^{-1},\ US^2U^{-1}, \ldots; \qquad 1, S, S^2, \ldots. \tag{237}$$

If for two non-zero exponents

$$US^rU^{-1} = S^s,$$

then calculation forces either U to be diagonal, giving $USU^{-1} = S$, or anti-diagonal, giving $USU^{-1} = S^{-1}$. In both cases the two cycles coincide.

From the enumeration in §68, the numbers of cycles of the first, second, and third kinds are therefore

$$p+1, \qquad \frac{1}{2}p(p+1), \qquad \frac{1}{2}p(p-1). \tag{238}$$

It is useful, besides L_p and A_p, to consider a third group in $E_{2,p}$, isomorphic to them but not transformable into them within the same field. Since N is real, choose $v \in \mathfrak{G}_{2,p}$ with

$$N = v^{p+1}.$$

To the substitution

$$A = \begin{pmatrix} \alpha & \beta \\ N\beta^p & \alpha^p \end{pmatrix}$$

of A_p let correspond

$$B = \begin{pmatrix} \alpha & v\beta \\ -v^p\beta^p & \alpha^p \end{pmatrix}, \qquad \alpha\alpha^p + N\beta\beta^p = 1. \tag{239}$$

The product of two such matrices again has the same form and corresponds to the product of the corresponding A's. Thus the totality of matrices B forms a group isomorphic to A_p, denoted by F_p.

Replacing $v\beta$ by β gives the simpler form

$$B = \begin{pmatrix} \alpha & \beta \\ -\beta^p & \alpha^p \end{pmatrix}, \qquad \alpha\alpha^p + \beta\beta^p = 1. \tag{240}$$

To pass from such a B to the corresponding real substitution A, first form

$$\bar{A} = \begin{pmatrix} \alpha & v^{-1}\beta \\ N(v^{-1}\beta)^p & \alpha^p \end{pmatrix}, \tag{241}$$

and then transform back by

$$A = R\bar{A}R^{-1}. \tag{242}$$

Although F_p is isomorphic to L_p and A_p, it cannot be transformed into either by substitutions whose entries belong to $\mathfrak{G}_{2,p}$. To perform such a transformation one would have to pass to a higher field in which all elements of $\mathfrak{G}_{2,p}$ become squares.

§70. Divisors of the Group L_p Whose Degree Is Divisible by p

It is a problem of the greatest interest to determine all divisors of the group L_p. On the solution of this problem depends the way in which the group L_p can be represented by permutations of letters, and therefore the algebraic equations in which these groups can occur.

The cyclic groups contained in L_p have already been considered in the preceding paragraphs. We have seen that the degree of a cyclic group is either equal to p, or is a divisor of

$$\frac{1}{2}(p-1) \qquad \text{or of} \qquad \frac{1}{2}(p+1),$$

and every divisor of these numbers also occurs among the degrees of the cyclic groups. The index of a cyclic divisor can never be smaller than

$$\frac{1}{2}(p^2-1), \qquad p(p+1), \qquad p(p-1),$$

respectively.

Suppose now that the degree of a divisor G of L_p is divisible by p. Then G contains a cyclic group of degree p, and by §68 we may transform G so that this cyclic group consists of the substitutions

$$T_t: \quad \xi \longmapsto \xi + t, \qquad t = 0, 1, 2, \ldots, p-1.$$

If G contains, besides these, a substitution

$$\xi \longmapsto \frac{a\xi + b}{c\xi + d}$$

in which $c \neq 0$, then by composing it with suitable translations it also contains a substitution interchanging 0 and ∞. From this it follows further that G contains a substitution of the form

$$\xi \longmapsto -\frac{1}{c^2\xi},$$

and hence, by multiplying with the translations, all substitutions needed to generate the full group L_p. More explicitly, the substitution of order two obtained in this way, together with the translation T_1, gives a system of generating elements of L_p as in §65.

Consequently, if G is not the whole group L_p, it can contain only substitutions of the form

$$\xi \longmapsto a^2\xi + ab,$$

or, in matrix notation,

$$\begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix},$$

with the usual identification of proportional matrices. Conversely, all these substitutions form a group of degree

$$\frac{1}{2}\,p(p-1),$$

and therefore a divisor of L_p of index $p+1$.

Among these groups are contained certain further divisors of greater index. They are obtained by not allowing a to run through all admissible values, but only through values of the form a^s, where s is a divisor of $\frac{1}{2}(p-1)$. The degree of such a group is

$$\frac{\frac{1}{2}\,p(p-1)}{s}.$$

If the group L_p is regarded as the group of the linear fractional substitutions

$$\eta = \frac{a\xi + b}{c\xi + d}$$

on the $p+1$ letters

$$\infty, 0, 1, \ldots, p-1,$$

then the group just found is the group of the integral linear substitutions

$$\eta = a^2\xi + ab.$$

These are precisely the substitutions leaving the letter ∞ fixed; they therefore give a permutation group on only p letters. These are the same special metacyclic groups which were treated in the seventeenth section of the first volume.

The various groups conjugate to G leave one of the other $p+1$ letters fixed. Hence the total number of such groups is $p+1$.

§71. Divisors of the Group L_p Whose Degree Is Not Divisible by p

To find the remaining divisors of L_p, whose degree is not divisible by p, we may proceed step by step along the same path which led us, in the seventh and eighth sections, to the determination of the polyhedral groups. It is enough here to emphasize the principal points of the derivation, and for the proofs to refer back to that discussion, where the same conclusions were drawn.

For this purpose we again regard L_p, as in §65, as a group of linear fractional substitutions, and let the variable ξ assume all p^2+1 values of the congruence field $\mathbb{F}_{p^2}$, including ∞. The advantage thereby gained is that the fixed-point equation

$$\xi = \frac{a\xi + b}{c\xi + d}$$

always has two roots. These two roots are distinct if we exclude substitutions of p-th order; such substitutions cannot occur in a group whose degree is not divisible by p, and, by §68, they are exactly those for which the two roots coincide.

We call these two roots the *poles* of the substitution. Let G be a group of degree n, not divisible by p, which is to be a divisor of L_p. If in G the substitutions

$$\Theta_1, \Theta_2, \ldots, \Theta_{\nu-1},$$

and no others, have the same pole α, then we call α a ν-fold pole.

Let S be any substitution of the group over $\mathbb{F}_{p^2}$. If L_p is transformed by S^{-1}, then G is transformed into an isomorphic group. Although the substitutions of the latter group need not have real coefficients, each has two poles, obtained by applying S to the poles of the original substitution; for if α is a pole of Θ, then

$$S\Theta S^{-1}S(\alpha) = S\Theta(\alpha) = S(\alpha).$$

Hence, as in §51, one may transform G so that any chosen substitution has the poles 0 and ∞, and so becomes multiplicative.

We now list, in the order used before, the propositions from §52 which are needed here. Additional remarks are added only where the altered hypotheses require them.

If α is a ν-fold pole, then the substitutions

$$1, \Theta_1, \Theta_2, \ldots, \Theta_{\nu-1}$$

form a cyclic group. Its substitutions may all be transformed into the form

$$\Theta(\xi) = \gamma^{\mu\frac{p^2-1}{\nu}}\xi, \qquad \mu = 0, 1, \ldots, \nu - 1,$$

where γ is a primitive element of $\mathbb{F}_{p^2}$; thus $\gamma^{(p^2-1)/\nu}$ is a primitive ν-th root of unity in this field.

The two poles of a substitution are of the same multiplicity. If Θ denotes the cyclic group generated by such a substitution, and if $n = \nu a$, then the poles may be arranged in systems of conjugate poles

$$\alpha,\ V_1(\alpha),\ V_2(\alpha), \ldots, V_{a-1}(\alpha),$$

where the V_i run through suitable representatives of the group. Hence all poles of G split into systems of conjugate poles, and we obtain the same indeterminate equation as in the theory of the polyhedral groups:

$$2n - 2 = a(\nu - 1) + a'(\nu' - 1) + a''(\nu'' - 1) + \cdots = n\lambda - a - a' - a'' - \cdots ,$$

where λ is the number of systems of conjugate poles. In §52 it was shown that this equation permits only a finite list of solutions.

To obtain the corresponding divisors of the congruence groups, one may put into the formulae for the polyhedral groups, obtained in the eighth section, the roots of unity of the field $\mathbb{F}_{p^2}$ in place of the complex roots of unity which occur there. These roots of unity may of course have only degrees dividing $p^2 - 1$. In each case it remains to decide whether the substitutions thus obtained are contained in L_p, or in a group isomorphic with it.

We thus obtain the following cases. In all of them γ denotes a primitive element of $\mathbb{F}_{p^2}$.

First, there are the cyclic groups C_n of degree n:

$$C_n : \quad \xi \longmapsto \gamma^{\mu \frac{p^2-1}{n}} \xi, \qquad \mu = 0, 1, \ldots, n-1.$$

This group is real when n divides $\frac{1}{2}(p-1)$, and only then; for precisely in this case the exponents of γ are divisible by $p+1$. The group is contained in L_p, and at the same time in the real form A_p, when the corresponding conjugacy condition is satisfied, that is, precisely when n divides $\frac{1}{2}(p+1)$. This agrees with §68: no cyclic groups other than those whose degree is p, or a divisor of $\frac{1}{2}(p-1)$ or of $\frac{1}{2}(p+1)$, occur.

In all remaining polyhedral groups one has three systems of exceptional poles. Thus, secondly, there are the dihedral groups of degree $2m$. Using the second representation of the dihedral group from §55, the group is written more conveniently for the present purpose. It is contained in L_p if

$$p \equiv 1 \pmod{2m},$$

and in the imaginary form F_p if

$$p \equiv -1 \pmod{2m}.$$

The distinction is exactly the same as in the cyclic case.

Thirdly, we obtain tetrahedral groups. The data of the pole multiplicities are

$$\nu = 2, \qquad \nu' = 3, \qquad \nu'' = 3,$$

and the degree is 12. In $\mathbb{F}_{p^2}$ an eighth root of unity always exists. The formulae of §56 therefore give three generating substitutions for the tetrahedral group. If

$$p \equiv 1 \pmod{4},$$

the necessary fourth roots are real, and the group itself is real. If

$$p \equiv 3 \pmod 4,$$

the two conjugate fourth roots lie in the imaginary part of $\mathbb{F}_{p^2}$; a comparison according as $p \equiv 3$ or $7 \pmod 8$ shows that the three generators nevertheless belong to the appropriate real or imaginary form. Thus in every case the group L_p contains a tetrahedral group.

For example, when $p = 5$, since -2 is a quadratic non-residue modulo 5, one may put

$$\varepsilon = \sqrt{-2}.$$

Choosing the primitive roots as in §56 gives explicitly a tetrahedral group in L_5. Written as fractional substitutions, its elements form a divisor of L_5 of index 5.

Fourthly, there are octahedral groups. Since an octahedral group contains a cyclic group of degree 4, it can occur only when one of the numbers

$$\frac{1}{2}(p-1), \qquad \frac{1}{2}(p+1)$$

is divisible by 4, that is,

$$p \equiv \pm 1 \pmod 8.$$

Under this condition the formulae of §57 give an octahedral group. This group is real when $p \equiv 1 \pmod 8$, and it is imaginary but contained in F_p when $p \equiv -1 \pmod 8$.

The first example is $p = 7$. In this case -1 is a quadratic non-residue, and one may put $\varepsilon = \sqrt{-1} = i$. A convenient primitive element is $\gamma = 2 + i$, with conjugate $\gamma^p = 2 - i$. Since 5 is a real primitive root modulo 7, every non-zero element of $\mathbb{F}_{7^2}$ is obtained in the form

$$\gamma^\mu 5^\nu,$$

with the indicated ranges of μ and ν. The generators produced by the formulae of §57 may then be transformed, by the method of §69, into real form. The octahedral group so obtained is a divisor of L_7 of index 7.

Finally, the icosahedral group. For $p = 5$ the group L_p itself is an icosahedral group. For other values of p, an icosahedral group can be contained in L_p only if $p^2 - 1$ is divisible by 5, i.e.

$$p \equiv \pm 1 \pmod 5.$$

Then the formulae of §58 yield the generators of such a group. If $p \equiv 1 \pmod 5$, the required fifth roots are real, and the whole group is real. If $p \equiv -1 \pmod 5$, the corresponding roots are conjugate, and the group first obtained lies in the imaginary form F_p; to obtain the icosahedral group in L_p one has to perform the transformation described in §69.

For $p = 11$ one obtains directly a real icosahedral group of index 11. We may choose, for the relevant root, any primitive root of 11, for example 8; the formulae of §58 then give the desired substitutions.

We have thus determined which kinds of divisors can occur in L_p, and we have also shown that all of them actually occur. We have not pursued here the question how, for each type, all possible divisors are to be obtained. We add only the following remarks.

If a group G already found is transformed by a substitution of L_p, one obtains a conjugate divisor. If, however, one transforms by an imaginary substitution of the group F_p, the transformed group may nevertheless be real and need not be conjugate with G inside L_p, although both are conjugate as divisors of F_p. In this way the octahedral and icosahedral groups generally give a second non-conjugate group. For the tetrahedral group this gives a new group only in the appropriate congruence case. The examples $p = 5, 7, 11$ exhibit these alternatives explicitly.

§72. Constitution of the Group L_7 of Degree 168

For later use we must give the group L_7 a particularly explicit form. In §71 it was shown that L_7 contains octahedral divisors of degree 24, hence of index 7. It was also shown earlier that L_7 contains divisors of degree 21, hence of index 8; these are especially important for our purpose.

Among the conjugate divisors of index 8, the simplest and most convenient for calculation is the one consisting of substitutions of the form

$$\xi \longmapsto a^2\xi + ab.$$

We arrange the preceding transformation so that this subgroup occurs in precisely this form. After transforming the generators of the octahedral group found in §71, one obtains generators which we shall denote by

$$\chi, \quad \omega, \quad \Theta,$$

of orders $3, 2, 4$, respectively. They generate an octahedral subgroup of L_7. In Weber's displayed table the twenty-four elements of this subgroup are arranged in six rows, each row containing four elements. Symbolically they may be written in the form

$$\chi^\lambda \omega^\mu \Theta^\nu, \qquad \lambda = 0, 1, 2, \quad \mu = 0, 1, \quad \nu = 0, 1, 2, 3,$$

with the customary relations of the octahedral group.

To obtain the full group L_7 we adjoin an element τ of seventh order. Thus the elements of L_7 are represented in the form

$$\tau^q \chi^\lambda \omega^\mu \Theta^\nu, \qquad q = 0, 1, \ldots, 6, \quad \lambda = 0, 1, 2, \quad \mu = 0, 1, \quad \nu = 0, 1, 2, 3.$$

No two of these 168 expressions represent the same element. Indeed, no power of τ whose exponent is not divisible by 7 can lie in the octahedral subgroup, since the octahedral subgroup has no element of seventh order.

The subgroup of degree 21 contained in L_7 is represented in this notation by

$$\tau^q \chi^\lambda.$$

In order to describe the composition of the elements it is enough to express, for each q, the products

$$\Theta\tau^q, \qquad \omega\tau^q, \qquad \chi\tau^q$$

again in the normal form just given. Together with the multiplication table inside the octahedral subgroup this determines the product of any two elements of L_7.

One relation is especially simple:

$$\chi\tau = \tau^4\chi.$$

From it the analogous relations for the powers of χ follow. The remaining products are obtained by straightforward calculation from the octahedral table. Weber records twelve such relations;

they may be reduced to four fundamental relations, from which the others follow with the aid of the octahedral relations and the preceding formula.

The result may be summarized as follows. Suppose four elements

$$\tau, \quad \chi, \quad \omega, \quad \Theta$$

of orders

$$7, \quad 3, \quad 2, \quad 4$$

are given, and suppose that their compositions satisfy the octahedral relations among χ, ω, Θ, together with the relations obtained by bringing $\Theta\tau^q, \omega\tau^q, \chi\tau^q$ into the above normal form. Then the 168 elements

$$\tau^q \chi^\lambda \omega^\mu \Theta^\nu$$

form a simple group of degree 168. The two elements ω and τ may be taken as generating elements of this group.

Among the divisors of this group one should especially note the octahedral group

$$\chi^\lambda \omega^\mu \Theta^\nu$$

of index 7, the group

$$\tau^q \chi^\lambda$$

of degree 21 and index 8, and the group of degree 8 generated by suitable involutions. The whole group can also be written with the factors in other orders, for example in one of the forms

$$\chi^\lambda \omega^\mu \Theta^\nu \tau^q, \qquad \omega^\mu \Theta^\nu \chi^\lambda \tau^q, \qquad \tau^q \omega^\mu \Theta^\nu \chi^\lambda.$$

Third Book. Applications of Group Theory

§73. The Resolvents of the Composition Series

The theorems of general group theory treated in the first book of this volume yield important algebraic consequences when applied to the Galois group of an equation.

Let Ω be an arbitrary field taken as the domain of rationality. Let

$$f(x) = 0$$

be an equation in Ω, without multiple roots, and let P be its Galois group. Suppose P has a normal divisor Q. Let n denote the degree of P, and let j be the index of Q, so that n/j is the degree of Q.

In the first volume it was shown that the group of $f(x)$ is reduced from P to Q by adjoining a root of an irreducible normal equation of degree j. This auxiliary equation of degree j was called the partial resolvent.

The Galois group of this partial resolvent is obtained as follows. Let ψ be a function of the roots of $f(x)$ belonging to the group Q, and decompose P into right cosets

$$P = Q + Qa + Qb + Qc + \cdots ,$$

where $a, b, c, \ldots$ are certain permutations in P. Then all elements of the coset Qa carry ψ into one and the same function ψ_a, and the Galois group of the resolvent consists of the substitutions

$$(\psi, \psi_a), \quad (\psi, \psi_b), \quad (\psi, \psi_c), \ldots .$$

If two such substitutions are composed, one must note that

$$Qa \cdot Qb = Qab,$$

and hence

$$(\psi, \psi_a)(\psi, \psi_b) = (\psi, \psi_{ab}).$$

Thus these substitutions compose in exactly the same way as the cosets compose. It follows that:

> The Galois group of the partial resolvent belonging to Q is isomorphic to the quotient group P/Q.

Now take any composition series of P, with the corresponding series of indices:

$$P,\ P_1,\ P_2, \ldots, P_{\mu-1},\ 1, \qquad j_1, j_2, \ldots, j_\mu.$$

By what has just been said, the group of the equation $f = 0$ is reduced from P to P_1 by adjoining a root of a normal equation of degree j_1. It is then reduced from P_1 to P_2 by adjoining a root of a normal equation of degree j_2, and so on; finally the equation is completely solved by adjoining a root of a normal equation of degree j_μ.

The theorem on the invariance of the index series therefore receives a new algebraic meaning:

> To solve the equation $f = 0$, one has to adjoin successively one root of a normal equation of each of the degrees
>
> $$j_1, j_2, \dots, j_\mu.$$
>
> The order in which these degrees occur may be changed, but the total collection of them cannot be changed.

The numbers $j_1, j_2, \dots, j_\mu$ are therefore connected more deeply with the nature of an equation, or more generally with the fields defined by it, than the degree of the equation itself. The degree may be altered in many ways by transforming the equation and taking functions of the roots as new unknowns; but the numbers

$$j_1, j_2, \dots, j_\mu$$

are always retained. They are to be regarded as genuine invariants of the normal field defined by the equation. The degree n of the group P itself is among these invariants only as their product:

$$n = j_1 j_2 \cdots j_{\mu-1} j_\mu.$$

In general, in a composition series of P, each term is a normal divisor only of the term immediately preceding it. It may happen, however, that certain terms are normal divisors of earlier terms as well. Particularly important are those terms which are normal divisors of P itself, and consequently of all preceding terms in the composition series.

Let P_ν be a term of the series which is also a normal divisor of P. The index of P_ν in P is

$$j_1 j_2 \cdots j_\nu,$$

and this product is the degree of the partial resolvent which reduces the group P to P_ν. Denote this resolvent by

$$\chi(y) = 0.$$

The group of this resolvent, which is a normal equation, is P/P_ν. A composition series for this quotient is obtained from the original one:

$$P/P_\nu, \quad P_1/P_\nu, \quad P_2/P_\nu, \dots, \quad P_{\nu-1}/P_\nu, \quad 1,$$

with the index series

$$j_1, j_2, \dots, j_{\nu-1}, j_\nu.$$

We shall need one further consequence. If Q is any normal divisor of P, then, by §7, a composition series of P may be chosen in which Q occurs. If R is another normal divisor of P, and at the same time a divisor of Q, the composition series may be arranged so that both Q and R occur:

$$P,\ Q',\ Q'', \dots, Q, \dots, R.$$

From this one obtains the quotient series

$$P/Q, \quad Q'/Q, \quad Q''/Q, \ldots, 1$$

and

$$P/R, \quad Q'/R, \quad Q''/R, \ldots, Q/R, \ldots.$$

The index of Q'/Q in P/Q is the same as the index of Q' in P, and hence the same as the index of Q'/R in P/R; the same holds for the following terms. Therefore:

> If Q and R are normal divisors of P, and R is a divisor of Q, then the index series of P/Q is a part of the index series of P/R.

If the indices $j_1, j_2, \ldots, j_\nu$ are all prime numbers, the solution of the resolvent

$$\chi(y) = 0$$

is reducible to the solution of a chain of cyclic equations of these prime degrees. Such a resolvent is therefore metacyclic in the sense established in the seventeenth section of the first volume, where metacyclic equations were identified with the equations otherwise called algebraically solvable.

An irreducible equation $f(x) = 0$ is metacyclic if the index series of its group consists entirely of prime numbers. In the first volume the conditions for metacyclic equations of prime degree were studied. We now have to carry out this investigation in the general case, where the degree of the equation may be arbitrarily composite.

§74. Metacyclic Equations

Let $f(x) = 0$ be an irreducible equation of degree m, and let P be its Galois group. Then P is transitive. Let a composition series with index series be

$$P,\ P_1,\ P_2, \ldots, P_{\mu-1},\ 1.$$

Since the last group 1 is intransitive, an intransitive group must occur for the first time somewhere in the series. Let P_λ be this first intransitive group. Then all following groups

$$P_{\lambda+1}, P_{\lambda+2}, \ldots$$

are also intransitive, since they are divisors of P_λ.

Recall the theorems of the first volume, §158. Since P_λ is a normal divisor of $P_{\lambda-1}$, those theorems imply that $P_{\lambda-1}$ is imprimitive. After the necessary adjunctions have reduced the group of $f(x) = 0$ to $P_{\lambda-1}$, a further adjunction of a root of a normal equation of degree j_λ causes $f(x)$ to split into several irreducible factors

$$f(x) = f_1(x) f_2(x) \cdots f_h(x),$$

all of equal degree. The number h of these factors is a divisor of j_λ.

Let m be the degree of $f(x)$, and m_1 the degree of $f_1(x)$. Then

$$m = hm_1.$$

The equations

$$f_1 = 0,\ f_2 = 0, \ldots, f_h = 0$$

all have the same group; denote it by Q. If

$$\alpha, \alpha_1, \ldots, \alpha_{m_1-1}$$

are the roots of $f_1 = 0$, then Q is obtained by taking the permutations of these α's induced by P_λ.

Several distinct permutations in P_λ may induce the same permutation of the roots α. If π_1, π_2 are two such permutations, then

$$\pi_2 \pi_1^{-1}$$

leaves the α's fixed. Conversely, multiplying any one of them by a permutation of P_λ which fixes the α's gives another permutation inducing the same element of Q. Thus every permutation of the α's arises the same number of times from P_λ. If this number is p_0, and if q is the degree of Q, then

$$p_\lambda = p_0 q.$$

Thus the degree of P_λ is a multiple of the degree of Q.

But Q is transitive, since $f_1(x)$ is irreducible. Therefore, by the theorem of the first volume that the degree of a transitive permutation group is divisible by the number of letters it permutes, q

is divisible by m_1. Consequently p_λ is divisible by m_1. Since P_λ is the first intransitive group, $P_{\lambda-1}$ is transitive, and its degree is divisible by $m = hm_1$. Thus the index

$$j_\lambda = \frac{|P_{\lambda-1}|}{|P_\lambda|}$$

is divisible by h, in agreement with the preceding conclusion that h divides j_λ.

Suppose now that, besides j_λ, the number m has another prime divisor p. Then one of the subsequent indices

$$j_\lambda, j_{\lambda+1}, \ldots, j_\mu$$

must be divisible by p, and hence they cannot all be equal to j_λ. It follows from the theorem of §7 that the composition series of P_λ may be chosen in such a way that among

$$P_\lambda, P_{\lambda+1}, \ldots, P_{\mu-1}$$

one of the groups, say P_ν, is a normal divisor of P. This P_ν, being a divisor of the intransitive group P_λ, is itself intransitive. Hence P itself must be imprimitive.

We have proved:

> If more than one distinct prime divides the degree of an irreducible equation, then this equation can become reducible by successive adjunctions of roots of cyclic equations only if it is imprimitive.

Assuming such a reduction possible, we can say a little more about its form. Let P_ν be the first group in the series after P_λ which is a normal divisor of P. Then, with a suitable arrangement of the composition series, the relevant indices

$$j_\lambda, j_{\lambda+1}, \ldots, j_\nu$$

are all equal to the same prime. The resolvent which reduces P to P_ν has therefore a metacyclic group.

Let

$$A, B, \ldots, S$$

be the systems of intransitivity of P_ν, and let their number be s. Put

$$m = rs.$$

After adjoining a root of the corresponding resolvent, $f(x)$ splits into s factors of degree r. These systems are systems of imprimitivity of P.

Let Q be the totality of all permutations in P which leave the individual systems $A, B, \ldots, S$ in place, permuting only the elements inside each system. As was shown in the first volume, Q is a normal divisor of P. By adjoining the roots of an auxiliary equation

$$\varphi(y) = 0$$

of degree s, the equation $f(x)$ splits into s factors of degree r, each associated with one of the systems $A, B, \ldots, S$:

$$f(x) = f(x, y_1)f(x, y_2) \cdots f(x, y_s).$$

Since this auxiliary equation reduces the group P to Q, its group is isomorphic to P/Q. But the intransitive group P_ν is a divisor of Q, because the systems are not moved by P_ν. Applying the theorem of §73 to the three groups P, Q, P_ν, and using that the index series of P/P_ν consists of prime numbers, we see that the same is true of P/Q. Hence the auxiliary equation $\varphi(y) = 0$ is metacyclic.

Therefore:

> If an irreducible equation whose degree contains more than one prime factor splits into factors by successive adjunctions of radicals, then a splitting into s factors of degree r is brought about by adjoining the roots of a metacyclic equation of degree s.

As a special case this contains Abel's theorem: an irreducible equation whose degree m is not a power of a prime can be solvable by radicals only if, by adjoining the roots of a solvable equation of lower degree dividing m, it first decomposes into factors.

This greatly simplifies the question of solving or even reducing equations by radicals. Further investigation may be restricted to equations whose degree is a prime or a prime power, since all other cases are brought back to these by repeated application of the theorem.

For example, the question of all irreducible equations of degree 6 solvable by radicals has an immediate answer: to obtain all metacyclic equations of degree 6 over a field Ω, adjoin a square root to Ω and form all cubic equations in the enlarged field; or adjoin the root of a cubic equation and form all quadratic equations in the enlarged field.

As an example one may cite the equation treated by Hesse, on which the circular sections of a quadric surface not referred to its principal axes depend. This problem is solved by square roots after the principal axes have first been determined by the solution of a cubic equation.

§75. Metacyclic Equations Whose Degree Is a Power of a Prime

After the preceding theorems, the interest of the further investigation of metacyclic equations is concentrated chiefly on the case in which the degree of the equation is a power

$$p^k$$

of a prime number p. We suppose the equation irreducible, and restrict ourselves to primitive equations. The solution of an imprimitive equation reduces to the successive solution of two or more primitive equations whose degrees are again powers of p, and which, if the original equation is metacyclic, must themselves be metacyclic.

Let P be the group of an irreducible primitive metacyclic equation

$$f(x) = 0$$

of degree p^k. By the theorem of the first volume, not only P itself, but every normal divisor of P different from the identity group, must be transitive. We now apply the theorem of §8: P possesses a normal divisor Q all of whose elements commute. If several such divisors exist, choose Q of the smallest possible degree greater than 1.

This group Q is still transitive. It cannot have a proper divisor, different from the identity, which is at the same time a normal divisor of P; otherwise that divisor could have been chosen in place of Q.

After the appropriate adjunctions reduce the group of our equation from P to Q, the equation $f(x) = 0$ has become an abelian equation over the enlarged rationality domain. Since it has remained irreducible, the theorem of the first volume implies that the degree of the equation is equal to the degree of its group. Thus Q is an abelian group of degree

$$p^k.$$

The orders of all elements of Q are therefore powers of p.

We now show that, apart from the identity, only elements of order p can occur in Q. Suppose that p^λ is the highest order occurring among the elements of Q, with $\lambda > 1$. Then all elements of Q whose orders divide $p^{\lambda-1}$ form a divisor Q' of Q, distinct both from Q and from the identity group. This divisor must be a normal divisor of P. Indeed, if $\pi \in Q'$ and $\rho \in P$, then

$$\rho^{-1}\pi\rho$$

has the same order as π, and since Q is normal in P, it again belongs to Q; hence it belongs to Q'. This contradicts the minimal choice of Q. Therefore $\lambda = 1$.

Thus Q is an elementary abelian group: every non-identity element has order p.

§76. Representation of the Abelian Group Q

The preceding paragraph has shown that in a metacyclic group P, when P occurs as the Galois group of a primitive irreducible equation $f(x) = 0$ of degree

$$n = p^k,$$

there must be contained, as a normal divisor, an abelian group Q of degree p^k, and that Q has no elements except the identity and elements of order p.

By §9 this group Q may be represented by a basis consisting of k elements

$$A_1, A_2, \dots, A_k$$

of order p, so that every element of Q is uniquely of the form

$$A_1^{z_1} A_2^{z_2} \cdots A_k^{z_k},$$

where $z_1, z_2, \dots, z_k$ independently run through complete systems of residues modulo p.

We now pass to the corresponding permutation group. Choose an arbitrary root x of $f(x)$, and denote by

$$[z_1, z_2, \dots, z_k]$$

the root into which x is carried by the permutation

$$A_1^{z_1} A_2^{z_2} \cdots A_k^{z_k}.$$

This notation is to remain unchanged if the z_i are altered by multiples of p. The root from which we started is then

$$[0, 0, \dots, 0],$$

and every root of $f(x)$ is represented once and only once by such a symbol.

If x is carried by a permutation A' into x', and x' by A'' into x'', then x is carried by $A'A''$ into x''. It follows that the permutation

$$A = A_1^{a_1} A_2^{a_2} \cdots A_k^{a_k}$$

carries

$$[z_1, z_2, \dots, z_k]$$

into

$$[z_1 + a_1,\ z_2 + a_2, \dots, z_k + a_k],$$

all congruences being taken modulo p. In this way the whole group Q is obtained by allowing

$$a_1, a_2, \dots, a_k$$

to run independently through complete residue systems modulo p.

The group Q, whose construction has now been fixed, must be a normal divisor of P. From this normality an explicit general form can be derived for all permutations of P. For this we need first a lemma on the analytic representation of permutations.

§77. Analytic Representation of the Permutations

Lemma. Suppose an arbitrary permutation of the symbols

$$[z_1, z_2, \ldots, z_k]$$

carries $z_1, z_2, \ldots, z_k$ into

$$z_1', z_2', \ldots, z_k'.$$

Then one may always write

$$z_i' = \varphi_i(z_1, z_2, \ldots, z_k) \pmod p \qquad (i = 1, \ldots, k),$$

where each φ_i is an integral rational polynomial in the variables

$$z_1, z_2, \ldots, z_k$$

with integral coefficients, and where no variable occurs to a degree greater than $p - 1$.

This is a generalization of the corresponding theorem in §180 of the first volume, where the case $k = 1$ was proved. The general proof rests on complete induction. We first note the following.

In the formulae above the arguments z_i have to run through all combinations of k residue systems modulo p, hence through p^k systems. Each of the functions φ has p^k undetermined coefficients. If the values z_i' are prescribed, we therefore obtain kp^k linear congruences for the same number of unknown coefficients. It remains to show that these congruences are independent.

It is enough to consider each function φ separately. Suppose

$$z' = \varphi(z_1, z_2, \ldots, z_k) \pmod p.$$

For each of the p^k arguments the corresponding value of z' is known. We have to determine the coefficients of φ, assuming that the assertion has already been proved for functions of fewer variables.

Arrange φ according to the powers of z_1:

$$z' = \chi_0 z_1^{p-1} + \chi_1 z_1^{p-2} + \cdots + \chi_{p-1} \pmod p,$$

where the coefficients $\chi_0, \chi_1, \ldots, \chi_{p-1}$ are themselves polynomials of the same kind in

$$z_2, \ldots, z_k.$$

Now keep one combination of $z_2, \ldots, z_k$ fixed and let z_1 run through $0, 1, \ldots, p-1$. We then obtain a system of p linear congruences whose determinant is not divisible by p. As in the one-variable case of the first volume, this system determines the values of

$$\chi_0, \chi_1, \ldots, \chi_{p-1}$$

for the fixed combination of $z_2, \ldots, z_k$. Hence, for every combination of the remaining variables, the values of each χ_i are known. By the induction hypothesis the coefficients of the functions χ_i may therefore be determined. The lemma is proved.

Consequently every permutation of the roots may be written

$$[z_1, \ldots, z_k] \longmapsto [\varphi_1(z_1, \ldots, z_k), \ldots, \varphi_k(z_1, \ldots, z_k)].$$

If the new variables z_i' are themselves written as polynomial functions of the old variables, the same symbol may be rewritten with the z_i' in place of the z_i. This gives the rule for composing permutations: substitution of one system of polynomial functions into the other.

§78. Representation of the Metacyclic Group P

The group Q is to be a normal divisor of P. Thus, if A is a permutation in Q and B a permutation in P, a second permutation A' in Q must be determined so that

$$AB = BA'.$$

Using the abbreviated notation of the preceding paragraph, write

$$A : z \mapsto z + a, \qquad A' : z \mapsto z + a', \qquad B : z \mapsto \varphi(z).$$

Then

$$AB : z \mapsto \varphi(z + a), \qquad BA' : z \mapsto \varphi(z) + a',$$

and the condition becomes

$$\varphi(z + a) \equiv \varphi(z) + a' \pmod{p}.$$

Written without abbreviation this is the system

$$\begin{gathered} \varphi_1(z_1 + a_1, z_2 + a_2, \ldots) \equiv \varphi_1(z_1, z_2, \ldots) + a_1', \\ \varphi_2(z_1 + a_1, z_2 + a_2, \ldots) \equiv \varphi_2(z_1, z_2, \ldots) + a_2', \\ \vdots \end{gathered}$$

for all choices of z and a. These congruences show that the differences of the functions φ_i are constant under all translations. It follows that each φ_i is a linear function plus a constant:

$$z_i' = a_{i1}z_1 + a_{i2}z_2 + \cdots + a_{ik}z_k + b_i \pmod{p}.$$

The determinant (a_{ij}) cannot vanish, since B is a permutation of all p^k symbols.

Thus every group P of the kind under consideration is contained in the full linear congruence group

$$R : \qquad z' = Az + b$$

over the field of residues modulo p. Conversely, this affine group contains the translation group Q as a normal divisor.

Let S denote the group of homogeneous linear substitutions

$$z_i' = a_{i1}z_1 + a_{i2}z_2 + \cdots + a_{ik}z_k, \qquad \det(a_{ij}) \neq 0,$$

taken modulo p. The group R is then the product

$$R = SQ,$$

and every element of R may be written in one and only one of the two forms

$$\pi = \delta\gamma = \gamma'\delta,$$

where $\delta \in S$ and $\gamma, \gamma' \in Q$. The translation γ' is obtained from γ by the homogeneous substitution δ.

The group Q is a normal divisor of R. If R_1 is a divisor of R containing Q, then

$$R_1 = S_1 Q$$

for a divisor S_1 of S, and the degree of R_1 is the product of the degrees of S_1 and Q. If R_2 is a normal divisor of R_1, and if R_2 also contains Q, then

$$R_2 = S_2 Q$$

where S_2 is a normal divisor of S_1. Conversely, if S_2 is a normal divisor of S_1, then $S_2 Q$ is a normal divisor of $S_1 Q$.

We therefore obtain:

> If P is the Galois group of a primitive irreducible metacyclic equation of degree p^k, then P is representable in the form
>
> $$P = TQ,$$
>
> where T is a metacyclic group contained in the homogeneous congruence group $S = \mathrm{GL}(k, p)$.

The problem of finding all these metacyclic equations is thus reduced to that of finding all metacyclic divisors of the homogeneous congruence group S.

As an example take $p = 3,\ k = 2$, and ask for metacyclic equations of degree 9. The group S is the group of linear substitutions

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

modulo 3. In §66 the corresponding projective group was considered. There we identified opposite matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix},$$

and also restricted first to determinant 1. In the present group $S = \mathrm{GL}(2, 3)$ the two matrices are distinct, and the matrices of determinant -1 must be included. Thus the degree of the group is 48.

The group denoted previously by G is enlarged by changing all signs into a group of degree 8; the group denoted by E is enlarged into a group of degree 24, in which the enlarged G is a normal divisor. The whole group S is metacyclic. A composition and index series is

$$S,\ S_1,\ S_2,\ S_3,\ S_4, \qquad 2,\ 3,\ 2,\ 2.$$

Accordingly, all equations of degree 9 whose group is the full linear congruence group are metacyclic.

§79. The Ternary Linear Congruence Group for the Modulus 2

Many interesting relations appear if one considers the ternary linear congruence group R for the modulus 2. It may be regarded as a transitive permutation group on eight elements and must contain the groups of metacyclic equations of degree 8. According to what has been proved above, the first object is the homogeneous congruence group S, consisting of the matrices

$$\delta = \begin{pmatrix} a & b & c \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{pmatrix}$$

whose entries are 0 or 1 modulo 2, and whose determinant is non-zero, hence equal to 1 modulo 2.

The degree of S is obtained as follows. The first row can have any of the seven non-zero forms

$$(1,0,0),\ (0,1,0),\ (0,0,1),\ (0,1,1),\ (1,0,1),\ (1,1,0),\ (1,1,1).$$

For the first choice $(1,0,0)$, the determinant is reduced to

$$b_1c_2 - c_1b_2,$$

and this can be made equal to 1 in six ways. The remaining entries in the first column can then be chosen in four ways. Thus there are 24 matrices with first row $(1,0,0)$, and the same number for each first row. Therefore

$$|S| = 7 \cdot 24 = 168.$$

The number 168 has already occurred as the degree of the group L_7. This suggests that S and L_7 are isomorphic. If this conjecture is correct, the investigation of S is greatly simplified, because the divisors of L_7 are already known.

To verify it, we choose generating elements $\tau, \omega, \Theta, \chi$ of S so that they satisfy the characteristic relations of Theorem I in §72. This can be done in several ways. A systematic procedure is lacking, but the choice is readily found by trial.

One first chooses an element τ of seventh order and forms its period. A convenient matrix is

$$\tau = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix},$$

for which the successive powers give the desired cycle of length 7. Next one selects, among the 21 elements of second order, an element suitable for ω, for instance

$$\omega = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

The formulae of §72 then determine the elements Θ and χ. A direct calculation verifies the relations of Theorem I in §72. Hence S is isomorphic to L_7.

It follows that the group $S = \mathrm{GL}(3,2)$, like L_7, is simple.

§80. Resolvents of the Equation of Eighth Degree

We now apply this to equations of degree 8. Let Q again denote the elementary abelian group of degree 8, consisting of the translations

$$(z_1, z_2, z_3) \longmapsto (z_1 + a_1, z_2 + a_2, z_3 + a_3) \qquad (\mathrm{mod} 2),$$

and let S be the ternary congruence group of §79. Put

$$R = SQ,$$

so that R is the full ternary affine congruence group modulo 2.

Consider a system of eight independent quantities

$$X_{z_1,z_2,z_3},$$

where the indices z_1, z_2, z_3 are taken modulo 2. From them form the seven Lagrange resolvents

$$\Psi_{\xi_1,\xi_2,\xi_3} = \sum_{z_1,z_2,z_3} (-1)^{\xi_1 z_1 + \xi_2 z_2 + \xi_3 z_3} X_{z_1,z_2,z_3},$$

where (ξ_1, ξ_2, ξ_3) runs through the seven non-zero triples modulo 2.

If a translation in Q is applied to the indices of the X's, then

$$X_z \longmapsto X_{z+a}.$$

Consequently

$$\Psi_\xi \longmapsto (-1)^{\xi \cdot a} \Psi_\xi.$$

Thus the functions Ψ are changed by the permutations of Q only in sign, and their squares remain unchanged under Q.

It remains to determine the effect of a homogeneous substitution $\delta \in S$ on the functions Ψ. Let q be the transpose of δ. If

$$z' = \delta z,$$

then the scalar product transforms by

$$\xi \cdot z' = (q\xi) \cdot z.$$

Thus applying the permutation δ to the indices of the X's is equivalent, in the functions Ψ, to applying q^{-1} to the indices ξ. Therefore:

> The application of a permutation $\delta \in S$ to the quantities X induces on the seven resolvents Ψ_ξ the inverse transposed substitution on the index triples ξ.

§81. The Triple Systems of the Resolvents

We now write Ψ_ξ simply as Ψ, when the index is clear. Besides the squares of the seven functions, certain products of them are also invariant under the translation group Q. The first is the product of all seven:

$$A = \prod_{\xi \neq 0} \Psi_\xi.$$

There are also products of three functions

$$\Psi_\xi \Psi_\eta \Psi_\zeta$$

and their reciprocal companions, provided the indices satisfy

$$\xi + \eta + \zeta = 0 \pmod 2.$$

We shall call such a system of three functions a *triple*; likewise, the three non-zero index systems ξ, η, ζ satisfying this relation will be called a triple.

Altogether there are exactly seven such triples. No two of the three systems in a triple can be equal, and any two determine the third. Thus, if the first two systems are chosen among the seven possibilities, every triple is counted six times; the number is therefore $7 \cdot 6/6 = 7$.

To characterize the triples, introduce a non-zero triple

$$\alpha = (\alpha_1, \alpha_2, \alpha_3)$$

of residues modulo 2. The congruence

$$\alpha_1\xi_1 + \alpha_2\xi_2 + \alpha_3\xi_3 \equiv 0 \pmod 2$$

is satisfied by exactly three non-zero systems ξ. These three form a triple. Thus the triple belonging to α is characterized by

$$\alpha \cdot \xi = 0, \qquad \alpha \cdot \eta = 0, \qquad \alpha \cdot \zeta = 0.$$

It follows that the function attached to the triple may be denoted unambiguously by v_α.

As functions of the eight quantities X, the v_α admit all permutations of Q. If a substitution $\delta \in S$ is applied to the indices of the X's, then the indices of the Ψ's undergo the inverse transposed substitution. From the defining equations of the triple it follows that the index α itself undergoes the direct substitution δ. Hence:

> The application of a substitution $\delta \in S$ to the indices of X has the same substitution as its effect on the indices α of the functions v_α.

Now take a fixed non-zero system α. The three ξ's satisfying

$$\alpha \cdot \xi = 0$$

form a triple; the four remaining non-zero systems, together with the complementary parity condition, form a quadrupel. Each triple determines a quadrupel and conversely; hence there are seven quadrupels as well.

If $\beta \neq \alpha$, then among the four sums

$$\beta \cdot \xi$$

as ξ runs through the quadrupel belonging to α, two are even and two are odd. This is immediate by comparing with the three elements of the complementary triple. This counting also yields Weber's product relation among A, the four functions v belonging to a quadrupel, and the corresponding resolvent factor. In the notation where the quadrupel belonging to h is

$$\alpha, \beta, \gamma, \delta,$$

one obtains

$$\Psi_h^4 = A^2 v_\alpha v_\beta v_\gamma v_\delta.$$

For practical use the seven triples and the seven complementary quadrupels are most conveniently tabulated. Number the seven non-zero triples

$$100,\ 010,\ 001,\ 011,\ 101,\ 110,\ 111$$

by $1, 2, 3, 4, 5, 6, 7$. Then each number determines the three numbers lying in its associated triple, and the complementary four numbers are the associated quadrupel. This is the combinatorial table used in the following application.

§82. Application to Equations of Eighth Degree

We now put, in the formulae just derived, the roots of an equation of eighth degree for the eight variables X_z. Suppose its Galois group is

$$P = TQ$$

over the fixed rationality domain, and that this group is contained in the affine congruence group

$$R = SQ,$$

so that T is a divisor of S. All functions which admit the permutations of P are rational.

First among these is the sum of the roots:

$$B = \sum_z X_z.$$

Then there is the product A defined in §81. Further, the symmetric functions of the seven squares

$$\Psi_\xi^2$$

are rational. If, for simplicity, we assume for the moment that none of the seven quantities Ψ_ξ vanishes, then the functions v_α are also defined. Not merely the symmetric functions of the v_α's, but every function of them which admits the permutations of the group T, is rational.

The seven quantities v_α are therefore the roots of a rational equation of degree 7, with group T. By using the equation for the sum of the roots and the seven equations expressing the Ψ's, one obtains, after applying the product relations of the preceding paragraph, an explicit expression for a root such as $X_{0,0,0}$. In Weber's notation it has the form

$$8X_{0,0,0} = B + \sqrt{v_\alpha v_\beta v_\gamma v_\delta} + \sqrt{v_{\alpha'} v_{\beta'} v_{\gamma'} v_{\delta'}} + \cdots ,$$

where the displayed square roots run through the quadrupels appearing in the table of §81.

There are fifteen sign changes of the seven square roots under which this expression remains unchanged: the simultaneous change of all seven signs, and the changes of the signs belonging to a triple or to a quadrupel. Therefore the expression assumes only eight distinct values, however the seven square roots occurring in it are chosen. These eight values are precisely the eight roots.

§83. Metacyclic Equations of Eighth Degree

We can now determine the primitive metacyclic equations of eighth degree. First one has to find the metacyclic divisors of $S = \mathrm{GL}(3, 2)$. The group S itself, as already observed, is not metacyclic: it is the simple group of degree 168. Its proper divisors were considered in §§70–72, and all proper divisors of S are metacyclic.

Among them are divisors of degrees 21 and 7, which we denote by

$$T_{21}, \qquad T_7,$$

and also divisors of degree 24, together with other divisors whose degrees divide 24, which we collect under the symbol T_{24}.

The groups T_{24} cannot connect the seven non-zero systems

$$\xi = (\xi_1, \xi_2, \xi_3)$$

transitively, because the degree of a transitive permutation group on seven elements must be divisible by 7. We shall show that the permutation groups corresponding to $T_{24}Q$ are imprimitive.

Adjoin to the seven systems ξ the eighth system

$$(0, 0, 0),$$

and denote the eight quantities X_z by

$$0, 1, 2, 3, 4, 5, 6, 7.$$

There are two cases.

In the first case the substitutions of T_{24} leave two of these eight elements, say $0, 1$, fixed. In Q there is exactly one substitution γ_0 which interchanges 0 and 1. The two substitutions $1, \gamma_0$ form a group Q_0. If $\gamma_1 \in Q$ is not in Q_0, then it carries $0, 1$ into two different elements, say $2, 3$. Similarly Q is decomposed into four cosets

$$Q = Q_0 + Q_0\gamma_1 + Q_0\gamma_2 + Q_0\gamma_3,$$

and the eight elements are split into four pairs

$$0, 1; \qquad 2, 3; \qquad 4, 5; \qquad 6, 7.$$

Each coset carries the pair $0, 1$ into one of these four pairs.

Now let h be any substitution of the group $T_{24}Q$. Suitable elements of T_{24} and of Q may be chosen so that, on multiplying by the representatives above, each of the four pairs is carried into one of the same four pairs. Thus the group $T_{24}Q$ is imprimitive, with four systems of imprimitivity. The two roots $0, 1$ satisfy a quadratic equation whose coefficients depend on the roots of a biquadratic equation.

In the second case the elements of a triple

$$1, 2, 3$$

and the elements of its complementary quadrupel

$$4, 5, 6, 7$$

are each connected transitively by T_{24}. Then the eight quantities split into the two quadrupels

$$0, 1, 2, 3; \qquad 4, 5, 6, 7.$$

The translations in Q either permute the elements inside one of these quadrupels, or carry the whole quadrupel into the other.

Indeed, write $1, 2, 3$ as the three systems

$$\xi, \eta, \zeta.$$

Since they form a triple, there is a non-zero system α such that

$$\alpha \cdot \xi = \alpha \cdot \eta = \alpha \cdot \zeta = 0 \pmod 2.$$

After adding a translation μ to the indices, the three translated systems form the same triple if μ belongs to it; otherwise, together with μ, they form the complementary quadrupel. Hence any element of $T_{24}Q$ carries one of the two quadrupels into one of the two quadrupels. The group is again imprimitive. After adjoining a square root to the rationality domain, the four quantities

$$0, 1, 2, 3$$

are the roots of a biquadratic equation.

These considerations also exclude, for primitive equations of degree 8, the possibility that one of the seven quantities Ψ_ξ vanish. The functions Ψ_ξ are rational functions of the roots X_z. If one of them vanished, then the equation $\Psi_\xi = 0$ would be preserved by every permutation of the group of the equation. By the theorem of §80, either all seven Ψ_ξ would have to vanish, in which case the roots X_z themselves would be rational, or the induced group on the seven non-zero index systems would be intransitive. Then the degree of T would not be divisible by 7, and the corresponding group would fall under the imprimitive cases just discussed.

Thus in the primitive metacyclic case one is led to the divisors whose action on the seven non-zero indices has degree divisible by 7, namely the groups T_7 and T_{21}. The preceding resolvent construction gives the associated equations of degree 8: a rational equation of seventh degree in the quantities v_α, followed by the extraction of square roots in the formula for the eight X_z's.

§84. Biquadratic Equations

The case of biquadratic equations is the same construction in two variables and is sufficiently simple to display explicitly. We have four roots

$$X_{0,0}, \quad X_{1,0}, \quad X_{0,1}, \quad X_{1,1},$$

and the translation group Q of order 4 acts by adding a pair of residues modulo 2 to the indices. The homogeneous group $S = \mathrm{GL}(2,2)$ has degree 6, and acts by the corresponding linear substitutions on the non-zero index pairs.

The three Lagrange resolvents are

$$\begin{aligned}
\Psi_{1,0} &= X_{0,0} - X_{1,0} + X_{0,1} - X_{1,1}, \\
\Psi_{0,1} &= X_{0,0} + X_{1,0} - X_{0,1} - X_{1,1}, \\
\Psi_{1,1} &= X_{0,0} - X_{1,0} - X_{0,1} + X_{1,1}.
\end{aligned}$$

Their squares are invariant under Q, and the group S permutes them.

Let the three quantities formed from the products of these resolvents be denoted

$$v_{1,0}, \qquad v_{0,1}, \qquad v_{1,1}.$$

They remain unchanged under Q, and under two generating substitutions of S they are permuted cyclically and by transposition. Consequently the elementary symmetric functions of

$$v_{1,0}, \; v_{0,1}, \; v_{1,1}$$

are rational. These three quantities are therefore the roots of a cubic equation. This is the cubic resolvent of the general biquadratic equation.

If we put

$$a = X_{0,0} + X_{1,0} + X_{0,1} + X_{1,1},$$

then the roots are recovered from the formula

$$4X_{0,0} = a + \sqrt{v_{1,0}}\sqrt{v_{0,1}} + \sqrt{v_{1,0}}\sqrt{v_{1,1}} + \sqrt{v_{0,1}}\sqrt{v_{1,1}}.$$

By changing the signs of the three square roots in all possible ways this expression gives only four distinct values, and these four values are the four roots of the biquadratic equation. Conversely, if $v_{1,0}, v_{0,1}, v_{1,1}$ are the roots of any cubic equation, the four quantities obtained from this formula satisfy a biquadratic equation with rational coefficients.

Eleventh Section. The Inflection Points of a Curve of the Third Order

§85. Ternary Forms and Algebraic Curves

In order, in the further course of this exposition, to exhibit some of the many and interesting applications of algebra to geometrical problems without referring the reader to other aids, we shall derive briefly the geometrical foundations required for these applications. We restrict ourselves to plane geometry, and even there leave aside everything not directly needed for our purpose.

We shall use homogeneous triangular coordinates. The position of a point x is first described by its three distances from the sides of a fixed coordinate triangle, these distances being measured in arbitrary units, or, what is the same thing, multiplied by three arbitrary constant factors. These three quantities are called the coordinates of x, and are denoted by

$$x_1, \quad x_2, \quad x_3.$$

In the elementary form of the definition they are subject to one non-homogeneous linear relation, obtained by equating the area of the coordinate triangle $(1, 2, 3)$ with the sum of the areas of the triangles $(x, 2, 3)$, $(x, 3, 1)$, and $(x, 1, 2)$. A coordinate changes sign whenever the point passes through the corresponding side of the triangle.

By means of this relation every non-homogeneous expression can be made homogeneous. We shall therefore enlarge the meaning of the coordinates and regard x_1, x_2, x_3 as independent variables, with the convention that only homogeneous equations are used and that the point is determined not by the three numbers themselves, but by their ratios

$$x_1 : x_2 : x_3.$$

The triple $0, 0, 0$ is excluded, and multiplying all three coordinates by the same non-zero factor gives the same point.

A linear substitution

$$x_i = \sum_{j=1}^{3} a_{ij} y_j \qquad (i = 1, 2, 3),$$

whose determinant

$$r = \det(a_{ij})$$

is different from zero, may be interpreted in two ways. It may be regarded as a projective transformation carrying the point y to the point x, or as a change of coordinates for one and the same point. For the present we hold to the latter interpretation.

A ternary form of degree n,

$$f(x_1, x_2, x_3),$$

when equated to zero, represents a curve of order n. We shall call it simply the curve f. Under the substitution above the form goes over into a form of the same degree,

$$f(x_1, x_2, x_3) = \varphi(y_1, y_2, y_3),$$

and the equation $\varphi = 0$ represents the same curve in the new coordinates. Differentiating this identity gives the transformation formulae for the first partial derivatives:

$$\frac{\partial \varphi}{\partial y_j} = \sum_i a_{ij} \frac{\partial f}{\partial x_i}.$$

Thus properties expressed by the simultaneous vanishing of such derivatives are invariant under projective change of coordinates.

§86. Singular Points. Inflection Points. Bitangents

If at a point x the three equations

$$f_{x_1} = 0, \qquad f_{x_2} = 0, \qquad f_{x_3} = 0$$

are simultaneously satisfied, then by Euler's theorem for homogeneous functions,

$$nf = x_1 f_{x_1} + x_2 f_{x_2} + x_3 f_{x_3},$$

the point lies on the curve $f = 0$. The equations just written are transformed into equations of the same kind under every linear change of coordinates. A point so characterized is therefore defined by an invariant condition. Such points are called singular points of the curve.

Not every curve of order n has singular points. In general their existence imposes a condition on the coefficients of f. Eliminating x_1, x_2, x_3 from the three first-derivative equations gives a homogeneous rational function of the coefficients, defined up to a numerical factor, and this function is called the discriminant of the form f. Its explicit calculation is possible only in the simplest cases.

If infinitely many singular points occur, then the partial derivatives have a common factor. Conversely, a common factor of the first partial derivatives gives a component along which all these derivatives vanish. Thus multiple components and higher singularities are detected algebraically by common divisors of the derivative forms.

Let P be a non-singular point of the curve. Choose coordinates so that P is the vertex I_3. Then, expanding f according to the powers of x_3, one has

$$f = f_0 x_3^n + f_1 x_3^{n-1} + f_2 x_3^{n-2} + \cdots,$$

where f_i is a binary form in x_1, x_2. Since P lies on the curve, $f_0 = 0$, and since P is non-singular, $f_1 = 0$ is the tangent at P. If this tangent is chosen to be $x_1 = 0$, then the equation begins in the form

$$f = x_1 x_3^{n-1} + x_3^{n-2} f_2 + \cdots.$$

If the quadratic form f_2 is divisible by x_1, then the tangent $x_1 = 0$ meets the curve at P with multiplicity at least three. The point P is then called an inflection point, and the tangent is called the inflection tangent.

If P is an inflection point and $x_1 = 0$ its inflection tangent, the form f may be written

$$f = x_1 \varphi + x_3^3 \psi,$$

where φ has degree $n-1$ and ψ degree $n-3$. Conversely, any form of this shape has P as an inflection point and $x_1 = 0$ as its inflection tangent.

A line touching the curve in two distinct points is called a bitangent. If $x_3 = 0$ is a bitangent and its two points of contact are the two vertices lying on this line, then, on putting $x_3 = 0$, the remaining binary form must have $x_1^2 x_2^2$ as a factor. Thus f has the form

$$f = x_3 \varphi + x_1^2 x_2^2 \psi,$$

with φ of degree $n-1$ and ψ of degree $n-4$. More generally, if $x_3 = 0$ is a bitangent, the restriction of f to that line is the square of a quadratic factor; the two points of contact are the intersections of $x_3 = 0$ with the corresponding conic.

§87. Fundamental Covariants of a Ternary Form

We recall the notion of invariants and covariants. Suppose the ternary form $f(x)$, with coefficients a, is transformed by a linear substitution into $\varphi(y)$, with coefficients b, and let r be the determinant of the substitution. A form

$$\Phi(x, a)$$

homogeneous in the variables and in the coefficients is a covariant if it satisfies a transformation law

$$\Phi(y, b) = r^{\lambda} \Phi(x, a).$$

The integer λ is called the weight. If Φ is independent of the variables, it is an invariant.

For every form of degree greater than one there is a covariant of weight 2, the Hessian determinant:

$$H = \det \left(\frac{\partial^2 f}{\partial x_i \, \partial x_j} \right)_{1 \leq i,j \leq 3}.$$

It is of degree $3(n-2)$ in the variables and of degree 3 in the coefficients.

Starting from the Hessian H, one obtains further fundamental covariants by forming bordered determinants with f, H, and their first derivatives. In Weber's notation these are denoted J and K. They are covariants of the original ternary form, of successively higher order in the variables and higher degree in the coefficients. For a ternary form of degree n their orders and coefficient-degrees are, respectively,

covariant	order in x	degree in coefficients
H	$3n - 6$	3
J	$6n - 18$	8
K	$12n - 21$	12.

Only the first of these, the Hessian curve, is immediately needed for the elementary geometry of inflection points; the later ones enter in the reduction of the cubic form to its canonical form.

§88. The Hessian Curve

The covariant H has a simple geometrical meaning. Since H is invariantly attached to f, no generality is lost by choosing the coordinate triangle conveniently. Put a vertex I_3 at a non-singular point of the curve and choose the tangent there as the line $x_1 = 0$. Then the equation of the curve has the local form

$$f = hx_1x_3^{n-1} + x_3^{n-2}(ax_1^2 + 2bx_1x_2 + cx_2^2) + \cdots ,$$

with $h \neq 0$. Calculating the Hessian in descending powers of x_3, the constant term at I_3 is proportional to c. Consequently the Hessian curve passes through I_3 precisely when $c = 0$, that is, precisely when I_3 is an inflection point of f.

Furthermore, the inflection tangent is also tangent to the Hessian exactly when the contact is of order four. For irreducible curves of the third order this exceptional case cannot occur unless the curve has a factor; hence in the non-singular irreducible cubic case no such coalescence disturbs the count.

The Hessian curve also passes through the singular points of f. If we restrict attention to a curve without singular points, the intersection points of $f = 0$ with its Hessian $H = 0$ are exactly the inflection points of f. By Bézout's theorem a curve of order n without singular points has

$$n \cdot 3(n-2) = 3n(n-2)$$

inflection points, counted with multiplicity. In particular:

A non-singular curve of the third order has nine distinct inflection points.

§89. Inflection Points of a Curve of the Third Order

Let $f = 0$ now be a non-singular cubic curve. Place two vertices of the coordinate triangle at two of its nine inflection points and choose the corresponding inflection tangents as coordinate sides. Then, after a suitable normalization, the cubic may be brought to the form

$$h(x_1^3 + x_2^3 + x_3^3) + 6mx_1x_2x_3 = 0,$$

or, after a linear change of notation,

$$y_1^3 + y_2^3 + y_3^3 + 6my_1y_2y_3 = 0.$$

This is the canonical form of the ternary cubic. Each coordinate side contains three inflection points. Conversely, the existence of one such triple of inflection points on a line allows the form to be arranged in this way.

For a non-singular cubic the nine inflection points have a highly regular configuration. Given any two of them, there is a third such that the three lie on a line. Thus the inflection points may be arranged into triples. There are twelve such triples in all: every one of the nine points occurs in four triples, and counting incidences gives

$$\frac{9 \cdot 4}{3} = 12.$$

This incidence structure will become the basis of the "triple equations" below.

In the canonical form the nine inflection points can be written explicitly. If ε is a primitive cube root of unity, they are

$$\begin{array}{lll} (0,1,-1), & (0,1,-\varepsilon), & (0,1,-\varepsilon^2), \\ (-1,0,1), & (-\varepsilon,0,1), & (-\varepsilon^2,0,1), \\ (1,-1,0), & (1,-\varepsilon,0), & (1,-\varepsilon^2,0). \end{array}$$

Thus three of them are real, while the other six are imaginary, in conjugate pairs, when the ground field is real and the canonical parameter is real.

The twelve lines through triples can be divided in four ways into three lines so that each of the nine inflection points occurs exactly once in each division. A convenient notation is to label the nine points by pairs

$$(\xi, \eta), \qquad \xi, \eta \in \mathbb{Z}/3\mathbb{Z},$$

with conjugation interchanging (ξ, η) and (η, ξ). Then three labelled points lie on one of the twelve lines exactly when

$$\xi_1 + \xi_2 + \xi_3 \equiv 0, \qquad \eta_1 + \eta_2 + \eta_3 \equiv 0 \pmod 3.$$

This arithmetical notation makes the incidence system transparent.

§90. Transformation of the Cubic Form to the Canonical Form

To reduce a ternary cubic to the canonical form, one must express the required substitution algebraically. For this purpose Weber computes the fundamental covariants of the canonical cubic

$$\varphi(y) = h(y_1^3 + y_2^3 + y_3^3) + 6my_1y_2y_3.$$

The Hessian covariant, after multiplication by a convenient numerical factor, is again a cubic of the same shape:

$$\Delta(y) = -hm^2(y_1^3 + y_2^3 + y_3^3) + (h^3 + 2m^3)y_1y_2y_3.$$

A second covariant J, formed from a bordered determinant involving f, Δ, and their derivatives, is likewise evaluated explicitly. From f, Δ, J one can express the elementary symmetric functions

$$y_1y_2y_3, \qquad y_1^3 + y_2^3 + y_3^3, \qquad y_1^3y_2^3 + y_2^3y_3^3 + y_3^3y_1^3$$

in terms of the corresponding covariants of the original cubic.

The decisive equations may be summarized as follows. If r is the determinant of the transformation and the coefficient of $y_1^3 + y_2^3 + y_3^3$ has been normalized to $h = 1$, then

$$(1 + 8m^3)y_1y_2y_3 = m^2 f + r^2\Delta,$$

and a second formula gives $y_1^3 + y_2^3 + y_3^3$ in terms of f and Δ. A third covariant K gives

$$r^9K = (1 + 8m^3)^2(y_1^3 - y_2^3)(y_2^3 - y_3^3)(y_3^3 - y_1^3).$$

The factor $1 + 8m^3$ cannot vanish when the cubic is non-singular; indeed in that case the three first partial derivatives would have a common zero.

Thus, once m and r have been determined, the quantities y_1^3, y_2^3, y_3^3 are the roots of a cubic equation

$$u^3 - Qu^2 + Pu - R^3 = 0,$$

whose coefficients are rational functions of f, Δ, J and of the invariants m, r. The extraction of cube roots from its roots then gives the linear forms y_1, y_2, y_3, and hence the desired substitution. The reduction of the original cubic to canonical form is therefore converted into the solution of certain algebraic equations attached to its covariants and invariants.

§91. The Invariants of the Curve of the Third Order and the Biquadratic Equation

It remains to find the invariants on which the canonical parameter m depends. Consider the pencil of cubic covariants

$$\Delta_{\lambda,\mu} = \lambda f + \mu\Delta.$$

The Hessian of this pencil can again be expressed as a linear combination of f and Δ:

$$H(\Delta_{\lambda,\mu}) = L(\lambda,\mu)f + M(\lambda,\mu)\Delta.$$

The coefficients of the binary forms L and M are invariants of the original cubic. They may be found by comparing the coefficients of the ten monomials in a ternary cubic. If

$$a_i, \quad A_i, \quad B_i$$

are the corresponding coefficients of $f, \Delta, H(\Delta_{\lambda,\mu})$, then

$$La_i + MA_i = B_i,$$

and hence, for distinct indices i, k,

$$L(a_iA_k - a_kA_i) = B_iA_k - B_kA_i, \qquad M(a_iA_k - a_kA_i) = -B_ia_k + B_ka_i.$$

This also shows that L and M are integral functions of the coefficients of f; a denominator would have to divide all determinants $a_iA_k - a_kA_i$, which is impossible, as is seen from a special cubic.

For the canonical form the calculation gives two fundamental invariants, of degrees 4 and 6, denoted by S and T. With $h = 1$ they satisfy

$$m(1 - m^3) = r^4 S,$$

and

$$1 - 20m^3 - 8m^6 = r^6 T.$$

Eliminating the scale factor r yields a biquadratic equation for the appropriate expression in m. This equation is the algebraic obstacle in the reduction of the cubic to the canonical form; after adjoining one of its roots the remaining steps consist of the cubic equation for y_1^3, y_2^3, y_3^3 and the extraction of cube roots.

The inflection points are also visible at this stage. In the canonical coordinates they are obtained by putting successively one of y_1, y_2, y_3 equal to zero in

$$y_1^3 + y_2^3 + y_3^3 + 6my_1y_2y_3 = 0.$$

With $\varepsilon^3 = 1$, $\varepsilon \neq 1$, the nine points are precisely the nine triples displayed above. Their twelve lines are governed by the congruence rule

$$\xi_1 + \xi_2 + \xi_3 \equiv \eta_1 + \eta_2 + \eta_3 \equiv 0 \pmod 3.$$

The real form of this configuration is as follows. The three points

$$(0,0), \quad (1,1), \quad (2,2)$$

are real, and the remaining six are divided into three conjugate pairs. Among the twelve lines, some are real and some occur in conjugate imaginary pairs, exactly as dictated by the table of triples.

§92. Triple Equations

The nine inflection points of a non-singular cubic give rise to a purely algebraic structure. Let

$$x_1, \ldots, x_9$$

be nine quantities arranged in twelve triples, with the property that any two quantities determine exactly one triple and every quantity lies in four triples. If, whenever x_1, x_2, x_3 are the members of a triple, there is a rational relation

$$x_1 = \Phi(x_2, x_3),$$

or equivalently each member of a triple is rationally determined by the other two, then an equation whose nine roots are so arranged is called a triple equation.

The incidence system of the nine inflection points shows that such a triple equation is naturally labelled by pairs

$$(\xi, \eta) \in (\mathbb{Z}/3\mathbb{Z})^2.$$

The triples are exactly the sets

$$(\xi_1, \eta_1), \quad (\xi_2, \eta_2), \quad (\xi_3, \eta_3)$$

for which

$$\xi_1 + \xi_2 + \xi_3 \equiv 0, \qquad \eta_1 + \eta_2 + \eta_3 \equiv 0 \pmod 3.$$

Conversely, beginning with any one triple and one fourth point, this rule reconstructs the entire system of twelve triples. Thus the geometry of the flexes of a cubic is transferred to the algebra of an equation of degree 9.

§93. The Group of the Triple Equations

We now determine the Galois group P of a triple equation. A permutation in P must carry every triple into a triple. Passing to the notation

$$x = (\xi, \eta),$$

this means that a permutation must preserve the rule

$$\xi_1 + \xi_2 + \xi_3 \equiv 0, \qquad \eta_1 + \eta_2 + \eta_3 \equiv 0 \pmod 3.$$

Every permutation of the nine pairs can be represented by functions

$$\xi' = \phi(\xi, \eta), \qquad \eta' = \psi(\xi, \eta) \pmod 3,$$

where the degree in each variable is at most 2. Holding ξ fixed and applying the permutation to the triple

$$(\xi, 0), \quad (\xi, 1), \quad (\xi, 2),$$

one sees that the quadratic terms in η must vanish. The same argument with the variables interchanged removes the quadratic terms in ξ. Applying the condition to the diagonal triple

$$(0, 0), \quad (1, 1), \quad (2, 2)$$

then removes the mixed quadratic term. Consequently every substitution in P has the affine linear form

$$\begin{aligned} \xi' &= a\xi + b\eta + c, \\ \eta' &= a'\xi + b'\eta + c' \end{aligned} \qquad (\mathrm{mod} 3),$$

with non-zero determinant

$$ab' - a'b \not\equiv 0 \pmod 3.$$

Thus the group of a triple equation is contained in the full affine linear congruence group over $\mathbb{F}_3$.

Conversely, every substitution of this affine linear form carries triples into triples, since it preserves the vanishing of the sums of the two coordinates. Hence a degree-nine equation whose Galois group is a doubly transitive subgroup of this affine group is a triple equation. The affine group decomposes as

$$P = SQ,$$

where Q is the translation group

$$\xi' = \xi + \alpha, \qquad \eta' = \eta + \beta$$

and $S = \mathrm{GL}(2, 3)$ is the homogeneous part.

The translation group Q is simply transitive: if one root is fixed by a translation, all are fixed. The stabilizer of $(0, 0)$ in P is the intersection of P with S. Using the composition series of S discussed earlier, Weber shows that the smaller subgroups S_4Q and its immediate enlargements are still not doubly transitive; true triple equations begin only at the subgroup S_2Q, and therefore also at S_1Q and SQ. Certain other affine subgroups, though transitive, are not groups of genuine triple equations.

§94. Reality Relations of the Triple Equations

Assume now that the rationality field is real. If a triple contains two real roots, the relation defining the third root shows that the third is also real. If an imaginary root occurs, then in every one of the four triples containing it there must be at least one further imaginary root. Hence the number of imaginary roots is at least 5, and therefore, since imaginary roots occur in conjugate pairs, at least 6.

A triple containing two real roots, or two conjugate imaginary roots, will be called a real triple. If a triple contains imaginary roots, then the conjugate roots form the conjugate triple.

There are therefore only three possible reality patterns:

1. all nine roots real,
2. one real root and eight imaginary roots,
3. three real roots and six imaginary roots.

In the second case the four conjugate pairs must form four triples, all having the same real third root. If this real root is labelled $(0,0)$, then the conjugate pairs are

$$(1,2)(2,1), \quad (1,1)(2,2), \quad (1,0)(2,0), \quad (0,1)(0,2).$$

In the third case the three real roots themselves form a triple. We may take them to be

$$(0,0), \quad (1,1), \quad (2,2).$$

The conjugate imaginary pairs cannot lie in the same row of the remaining arrangement, since otherwise the third root of that row would be real. A short check of the twelve triples shows that each of the three real roots occurs, besides the real triple just displayed, in one and only one further real triple whose other two roots are conjugate imaginary. This gives exactly the same arrangement of real and imaginary points as in the inflection points of a real cubic curve.

The first and second cases also occur. Indeed, from a general equation of the ninth degree one obtains a triple equation by adjoining a function belonging to the affine linear group SQ. Weber exhibits such a function as the sum of the products of the three roots over the twelve triples. This function is real in all three reality cases, and hence the enlarged rationality field in which the equation becomes a triple equation is itself real.

Twelfth Section. The Bitangents of a Curve of the Fourth Order

§95. Number of Bitangents of a Curve of the Fourth Order

A geometrical problem of particular importance, both because of its relations to other fields and because it presents algebraic phenomena analogous to those arising from the inflection points of cubic curves, is the problem of the bitangents of a plane quartic. A bitangent is a line touching the curve at two distinct points. For curves of order less than four there are no bitangents; for a quartic a bitangent has no further intersection with the curve. In special cases the two points of contact may coincide, producing a tangent of fourfold contact.

Let

$$f(x_1, x_2, x_3) = 0$$

be the equation of a non-singular curve of the fourth order. If x and y are two fixed points, the points on the line joining them have coordinates $x + ty$. Substituting gives

$$f(x + ty) = 0,$$

a quartic equation in t. Written according to powers of t, it has the form

$$f(x + ty) = P_0 + 4tP_1 + 6t^2P_2 + 4t^3P_3 + t^4P_4,$$

where $P_\nu(x, y)$ are the polars of f.

If x lies on the curve, $P_0 = 0$. If y lies on the tangent at x, then $P_1 = 0$. The two remaining intersections of the tangent line with the quartic are determined by

$$6P_2 + 4tP_3 + t^2P_4 = 0.$$

The tangent is a bitangent precisely when this quadratic has a double root. Its discriminant must vanish; in Weber's notation the condition is

$$R(x, y) = 2P_3^2 - 3P_2P_4 = 0.$$

Thus $R(x, y) = 0$, with x on the curve and y on the tangent at x, is the necessary and sufficient condition for x to be a point of contact of a bitangent.

Weber then eliminates y from this condition and the tangent equation. The result is a curve

$$\chi(x) = 0$$

of degree 14 which passes through the contact points of the bitangents and through no other point of the quartic. Since the quartic $f = 0$ and the curve $\chi = 0$ meet in

$$4 \cdot 14 = 56$$

points, and since every bitangent contributes two contact points, the number of bitangents is

$$56/2 = 28.$$

Possible exceptional contacts are excluded by specializing to a quartic for which the contacts are distinct and then using the fact that the number cannot change while the curve remains non-singular.

Therefore a non-singular plane quartic has twenty-eight bitangents.

§96. The Steiner Complexes

Take two bitangents $x_1 = 0$, $y_1 = 0$. Since their four points of contact lie on a conic, the equation of the quartic may be written in the form

$$f = x_1 y_1 v - u^2,$$

where $u = 0$ is a conic and $v = 0$ another conic. If a second representation

$$f = x_2 y_2 v_1 - u_1^2$$

is possible, then comparison gives

$$x_1 y_1 (v - v_1) = (u - u_1)(u + u_1).$$

One of the two factors $u - u_1$, $u + u_1$ must be divisible by $x_1 y_1$. After changing the sign of u_1 if necessary,

$$u_1 = u + \lambda x_1 y_1,$$

and hence

$$f = x_1 y_1 (v + 2\lambda u + \lambda^2 x_1 y_1) - (u + \lambda x_1 y_1)^2.$$

Whenever the quadratic form

$$v + 2\lambda u + \lambda^2 x_1 y_1$$

splits into two linear factors $x_2 y_2$, the two lines $x_2 = 0$, $y_2 = 0$ are two further bitangents, and the eight points of contact of

$$x_1,\ y_1,\ x_2,\ y_2$$

lie on a conic.

The condition that this conic split is the vanishing of its determinant. It is an equation of the fifth degree in λ, and therefore yields five further pairs of bitangents. Hence:

To every pair of bitangents x_1, y_1 belong five further pairs x_i, y_i such that the eight contact points of x_1, y_1, x_i, y_i lie on a conic.

The six pairs

$$x_1 y_1,\ x_2 y_2,\ x_3 y_3,\ x_4 y_4,\ x_5 y_5,\ x_6 y_6$$

so determined form what Weber calls a Steiner complex.

Two bitangents in a complex form a syzygetic pair; three bitangents are called syzygetic if their six contact points lie on a conic, and azygetic otherwise. Weber proves that three bitangents of one Steiner complex, no two of which form one of the distinguished pairs x_i, y_i, are always azygetic. This follows directly from the equation

$$f = x_1 y_1 x_2 y_2 - u_1^2$$

and from the impossibility, for a non-singular quartic, that the relevant three lines meet in a point or that a conic pass through the wrong six contact points.

§97. Pairs and Triples of Complexes

The preceding results provide an effective calculus for the twenty-eight bitangents. Begin with one Steiner complex

$$x_1y_1,\ x_2y_2,\ \dots,\ x_6y_6.$$

The form of the equation shows that the complex determined by the pair x_1, x_2 contains the pair y_1, y_2, while the complex determined by x_1, y_2 contains x_2, y_1. These two new complexes, together with the original one, contain all twenty-eight bitangents.

Consequently every syzygetic triple of bitangents is completed by one uniquely determined fourth bitangent to a syzygetic quadruple; equivalently, a conic through the six contact points of three such bitangents cuts the quartic again in the two contact points of a fourth bitangent.

Counting pairs gives the number of complexes. There are

$$\binom{28}{2} = 14 \cdot 27$$

pairs of bitangents. Each Steiner complex contains six distinguished pairs. Since every pair determines one complex, the total number of Steiner complexes is

$$\frac{14 \cdot 27}{6} = 63.$$

Two complexes are called syzygetic if they have four pairs of bitangents in common, and azygetic if they have six azygetic elements in common. A syzygetic pair of complexes is completed in one and only one way to a syzygetic triple of complexes; an azygetic pair is similarly completed to an azygetic triple. In a syzygetic triple all twenty-eight bitangents occur, while in an azygetic triple only eighteen occur.

§98. The Aronhold Seven-Systems

Especially important are azygetic systems of seven bitangents, first considered by Aronhold. Weber calls them Aronhold seven-systems, complete seven-systems, or simply complete systems.

Such systems exist. For instance, from an azygetic pair of complexes one obtains a set of seven bitangents no three of which form a syzygetic triple. Conversely, no azygetic system can contain more than seven elements.

The key theorem is:

Any six elements of a complete seven-system occur in one and only one Steiner complex.

Uniqueness is proved by contradiction: if the six elements belonged to two different complexes, the corresponding syzygetic triple of complexes would force two of the six to be paired inside a third complex, which would create a forbidden syzygetic triple. Existence is obtained by choosing two of the six, forming a complex through them, and using the decomposition of the associated syzygetic complex triple to show that the other four must enter the same complex.

This theorem makes it possible to introduce a notation for all twenty-eight bitangents. Let a complete system be

$$x_1, x_2, \ldots, x_7.$$

Introduce an eighth symbol 8, and denote the seven elements by

$$[18], [28], \ldots, [78].$$

Then all twenty-eight bitangents are denoted uniformly by

$$[\mu\nu], \qquad 1 \leq \mu < \nu \leq 8.$$

This is Cayley's notation.

§99. The Hesse–Cayley Algorithm for Denoting the Bitangents

The notation $[\mu\nu]$ is best visualized as a line segment joining two of eight labelled points. Two symbols with no common index, such as [12] and [34], are represented by disjoint segments; two symbols with one common index, such as [12] and [13], by two segments issuing from one point.

In this notation the distinction between syzygetic and azygetic triples becomes combinatorial. Weber checks the possible types of three segments. Triples shaped like a star, a path, or a triangle of certain types are azygetic; the remaining types are syzygetic. The check is made by comparing them with the three complexes obtained from an azygetic complex triple.

The complete seven-systems have only two apparent graphical shapes:

a seven-rayed star, a triangle together with a four-rayed star.

There are eight systems of the first type, one for each possible centre. For the second type, the triangle can be chosen in

$$\frac{8\cdot 7\cdot 6}{2\cdot 3} = 56$$

ways, and then the centre of the four-rayed star in five ways, giving 280. Thus the total number of Aronhold systems is

$$8 + 280 = 288.$$

The two graphical shapes do not represent two intrinsically different species; the difference lies only in the notation.

In the same notation the Steiner complexes have two standard forms. One is

$$[12][34],\ [13][24],\ [14][23],\ [56][78],\ [57][68],\ [58][67],$$

and the other is

$$[17][18],\ [27][28],\ [37][38],\ [47][48],\ [57][58],\ [67][68],$$

with the obvious relabellings.

§100. Rational Determination of the Curve from a Complete Seven-System

The importance of Aronhold systems is expressed in two facts.

First, if a complete seven-system of bitangents of a quartic is known, then all the remaining bitangents can be determined rationally from it.

Second, given seven general lines in the plane, one can, provided certain rational functions of their coefficients do not vanish, determine rationally the equation of a non-singular quartic for which these seven lines form an Aronhold system.

To prove the first assertion, it suffices by symmetry to determine three new bitangents at once. Let

$$x_1, \dots, x_7$$

be the given complete system. Form the three Steiner complexes in which six of the seven occur, excluding in turn x_1, x_2, x_3. The newly appearing bitangents are denoted $\ell_1, \ell_2, \ell_3, \dots$. By choosing suitable constant factors one may write the quartic in the form

$$\sqrt{x_1\ell_1} + \sqrt{x_2\ell_2} + \sqrt{x_3\ell_3} = 0,$$

or, equivalently,

$$f = 4x_2\ell_2x_3\ell_3 - u_1^2 = 4x_3\ell_3x_1\ell_1 - u_2^2 = 4x_1\ell_1x_2\ell_2 - u_3^2,$$

with

$$u_1 = -x_1\ell_1 + x_2\ell_2 + x_3\ell_3$$

and the two analogous formulae obtained cyclically. The syzygetic relations among the complexes yield linear equations from which ℓ_1, ℓ_2, ℓ_3 are rationally determined. Repeating the same argument determines all remaining bitangents.

The second assertion is obtained by reversing the construction. From seven general lines one forms the rational functions which play the role of the missing ℓ's, and then the displayed equation gives the quartic. The exceptional cases occur exactly when the determinants used in solving the linear systems vanish, or when the resulting quartic becomes singular.

§101. The Galois Group of the Bitangent Problem

The bitangents are roots of an equation of degree 28. Choose a line L not tangent to the quartic and not passing through any intersection point of two bitangents. In Cartesian coordinates with L as the x-axis, a bitangent can be written in the form

$$y = \theta(x - \xi).$$

If $F(x, y) = 0$ is the equation of the quartic, the condition that

$$F(x, \theta(x - \xi))$$

be a perfect square quartic in x gives two equations in ξ, θ. Eliminating θ gives the bitangent equation of degree 28.

The roots of this equation inherit all the arrangements of the corresponding bitangents: Steiner complexes, Aronhold systems, syzygetic and azygetic triples. Denote the roots, as before, by symbols $[ik]$, $1 \leq i < k \leq 8$.

Given two roots p, q, one can rationally construct the conic through their four contact points and therefore form

$$f = pqv - u^2.$$

The five conics in the pencil

$$v + 2\lambda u + \lambda^2 pq = 0$$

which split into pairs of lines give the ten other bitangents in the Steiner complex determined by p, q. Hence there is a rational function

$$X(\xi_1, \xi_2, \xi_3)$$

which vanishes exactly when ξ_1, ξ_2, ξ_3 form a syzygetic triple, and does not vanish for an azygetic triple.

Similarly, by adjoining a complete seven-system

$$\xi_1, \ldots, \xi_7$$

all other roots are rationally determined. Therefore every permutation in the Galois group must preserve the whole incidence structure of syzygetic triples and complete seven-systems. Conversely, every permutation preserving this structure is realized in the group. Thus the Galois group of the bitangent equation is the full group of permutations of the twenty-eight symbols $[ik]$ induced by the transformations described in the next section.

Counting the complete systems gives the order of the group. There are 288 choices of a complete seven-system, and each such system can be arranged in 7! ways. Thus

$$|P| = 288 \cdot 7! = 8! \cdot 36.$$

§102. Representation of the Group

The notation $[ik]$ immediately shows that every permutation of the eight indices $1, \ldots, 8$ gives a permutation of the twenty-eight roots. These permutations form a subgroup

$$S \cong S_8$$

of index 36 in the full group P.

The missing cosets are obtained by replacing the original Aronhold system

$$[18], [28], \ldots, [78]$$

by a system of the other type, for instance

$$[23], [31], [12], [48], [58], [68], [78].$$

The induced permutation is denoted

$$\Pi_{1,2,3,8} = \Pi_{4,5,6,7},$$

where the subscript may be replaced by any set of four indices or by its complement. There are 35 such permutations.

The whole group is therefore

$$P = S + \sum S\Pi_{\alpha_1\alpha_2\alpha_3\alpha_4},$$

where the sum extends over the 35 partitions of the eight indices into two complementary sets of four.

The action of $\Pi_{\alpha_1\alpha_2\alpha_3\alpha_4}$ is simple. If the two indices of $[\mu\nu]$ both lie in the same half of the partition, $[\mu\nu]$ is changed to the complementary pair in that half; if one index lies in each half, $[\mu\nu]$ remains fixed. Each Π has order two.

The composition laws are consequently explicit. If $\sigma \in S_8$, then

$$\Pi_{1234}\sigma = \sigma\Pi_{\sigma(1)\sigma(2)\sigma(3)\sigma(4)},$$

and

$$\Pi_{1234}^2 = 1.$$

For overlapping quadruples one obtains formulae such as

$$\Pi_{1234}\Pi_{1235} = (45)\Pi_{1234},$$

and

$$\Pi_{1234}\Pi_{1256} = (12)(34)(56)(78)\Pi_{1278}.$$

These relations completely determine the group P.

§103. Simplicity of the Group of the Bitangent Problem

The representation just obtained gives a direct proof that P is simple. Write

$$P = S + \sum S\Pi_{\alpha_1\alpha_2\alpha_3\alpha_4},$$

where $S \cong S_8$. Let S' be the alternating group on the eight letters. Every non-trivial normal subgroup of S_8 contains S'.

Let Q be a non-trivial normal divisor of P. If $Q \cap S$ contains a non-identity element, then it contains S'. But if Q contains S', conjugating by the Π's and using the relations of §102 gives elements in all cosets $S\Pi_{\alpha_1\alpha_2\alpha_3\alpha_4}$. Thus $Q = P$.

It remains to exclude the case in which Q contains no non-trivial element of S. If Q contains an element $\sigma\Pi_{1234}$, conjugate it by a transposition $(\alpha\beta)$ chosen so that the conjugated transposition differs from the original one. Multiplying the two conjugates produces a non-identity element of S, contradicting the assumption unless $Q = P$. Therefore the only normal divisors of P are 1 and P.

Consequently the group of the bitangent equation is simple.

An immediate consequence is that the group acts by even permutations on the twenty-eight roots. Hence the discriminant of the bitangent equation is a square. The stabilizer of one root is transitive on the other 27, so P is doubly transitive, but it is not triply transitive: after two roots are fixed, a root syzygetic with them cannot be carried to an azygetic one. Adjoining two roots splits off a factor of degree 10, whose roots are the bitangents completing the Steiner complex with the two given roots.

The subgroup $S \cong S_8$, of index 36, is also distinguished. By adjoining a root of a degree-36 equation one reduces the bitangent problem to a general equation of degree 8. In this sense the bitangent problem is governed by an 8-letter symmetric group sitting inside the simple group P of order $8! \cdot 36$.

§104. Reality of the Bitangents

Let the quartic curve be real and non-singular. If a bitangent l is imaginary, its conjugate l' is also a bitangent. Every rational relation among the roots of the bitangent equation remains valid after replacing all imaginary roots by their conjugates. Hence conjugation carries syzygetic systems to syzygetic systems and azygetic systems to azygetic systems.

A Steiner complex C is therefore carried by conjugation into a Steiner complex C'. If $C = C'$, the complex is called real; otherwise C, C' form a conjugate pair of complexes. In a real complex a pair is either a pair of real bitangents, a conjugate pair l, l', or part of a conjugate double-pair

$$(l, m), \quad (l', m').$$

A real bitangent can never be paired in a real complex with a non-real bitangent, since then the conjugate pair would share the same real element, which is impossible inside a Steiner complex.

Similarly, an Aronhold seven-system is carried by conjugation into another Aronhold system. It is real if it is identical with its conjugate; such a system contains, together with every imaginary bitangent, its conjugate, and therefore contains an odd number of real bitangents.

Weber derives the following restrictions.

First, not all twenty-eight bitangents can be imaginary. Starting with a conjugate pair and a third imaginary bitangent, one would obtain conjugate Steiner complexes arranged in a way that forces the existence of a real pair. Thus at least one real pair occurs, and then further incidence arguments force at least four real bitangents.

Second, there are always four real bitangents forming two pairs in a real Steiner complex.

Third, if one further real bitangent exists beyond these four, then there are eight real bitangents, grouped as four pairs in one Steiner complex.

Fourth, if more than eight real bitangents occur, then there are sixteen real bitangents arranged in a syzygetic triple of Steiner complexes, each complex containing four real pairs.

Repeating the argument shows that the only possible numbers and arrangements of real bitangents of a real non-singular quartic are:

1. 4 real syzygetic bitangents,
2. 8 real bitangents, forming four pairs of a Steiner complex,
3. 16 real bitangents, arranged in a syzygetic complex triple,
4. 28 real bitangents.

§105. Proof of the Existence of the Four Cases

The previous section proved only that no other cases are possible. To show that all four really occur, Weber uses the theorem of §100: from seven general lines one can construct rationally a non-singular quartic for which those seven lines are an Aronhold system, and from this system all bitangents are rationally determined.

The seven given lines may include imaginary conjugate pairs. The resulting curve is real whenever the conjugate of the seven-system is again an Aronhold system of the same curve. This is certainly the case when the seven-system itself is real, i.e. when every imaginary line occurs together with its conjugate.

A real Aronhold system of seven lines can contain:

1. 1 real line and 3 conjugate imaginary pairs,
2. 3 real lines and 2 conjugate imaginary pairs,
3. 5 real lines and 1 conjugate imaginary pair,
4. 7 real lines.

Using Cayley's notation for the bitangents, the remaining twenty-one bitangents are computed from the seven by the Hesse–Cayley rules. In the first case, if the given lines are

$$0,\ 1, 1',\ 2, 2',\ 3, 3',$$

then the real bitangents are exactly

$$0,\ [11'],\ [22'],\ [33'],$$

and all others are imaginary. This gives the case of four real bitangents.

In the second case, with

$$0,\ 1,\ 2,\ 3, 3',\ 4, 4',$$

the real bitangents are

$$0,\ 1,\ 2,\ [12],\ [20],\ [01],\ [33'],\ [44'],$$

and the remaining bitangents occur in conjugate pairs. These eight real bitangents form four pairs of a Steiner complex.

In the third case, with

$$0,\ 1,\ 2,\ 3,\ 4,\ 5, 5',$$

the construction gives sixteen real bitangents. They are arranged in a syzygetic triple of complexes, each containing four real pairs, and the remaining twelve are imaginary in conjugate pairs.

In the fourth case, when all seven lines of the Aronhold system are real, every bitangent determined rationally from them is real, and all twenty-eight bitangents are real.

This proves the existence of the four possible real cases. Weber adds that the reality of a bitangent does not imply the reality of its two points of contact, since those contact points depend on a quadratic equation. Thus the real bitangents themselves split further into bitangents with real contact points and bitangents with imaginary contact points; this finer classification is not pursued here.

Thirteenth Section. General Theory of the Equation of the Fifth Degree

§106. Statement of the Problem

In §44 we saw that, beyond the question of the Galois group of an algebraic equation, there is a still more far-reaching problem: to ask for a linear substitution group of as small a number of dimensions as possible, to whose problem of forms the given equation can be reduced. In metacyclic equations, and in particular in the general equations of the third and fourth degrees, this number of dimensions is equal to one. This is precisely another way of saying that these equations are solvable by radicals. The next equations to be investigated are therefore the equations of the fifth degree; and it will appear that the solution of the general equation of the fifth degree leads to a binary linear substitution group, namely to the icosahedral problem.

Connected with this is another question. The general equation of the fifth degree contains five coefficients, which may be regarded as independent variables; hence a root of such an equation is an algebraic function of five variables. But the number of these variables may already be reduced by simple linear substitutions. By a Tschirnhausen transformation, for example to Jerrard's form, the equation can even be made to depend on only one variable coefficient, that is, on one parameter. Thus, by mediation of equations which do not reach the fifth degree, one may make the root of a general equation of the fifth degree depend on an algebraic function of one variable.

The question to which we are led is therefore this: how can the solution of an algebraic equation, whose coefficients depend on a certain number of variables, be reduced to an equation with as small a number of parameters as possible; and, in particular, under what circumstances and by what auxiliary means can it be reduced to equations with only one parameter?

We accordingly pose the following question: under what hypotheses do resolvents with only one parameter exist for a general equation of degree n?

Let

$$x_0, x_1, \ldots, x_{n-1}$$

be a system of n independent variables, and let

$$u = \psi(x_0, x_1, \ldots, x_{n-1}) \tag{243}$$

be a rational function of these variables. Under the substitutions of the alternating group on the n letters x, suppose that u is changed into the mutually distinct functions

$$u, u_1, u_2, \ldots, u_{\nu-1}, \tag{244}$$

where ν may be equal to or smaller than the degree of the alternating group. Let z denote a function of the variables x which remains unchanged under the alternating group. We ask when there exists a rational equation of degree ν in u,

$$F(u, z) = 0, \tag{245}$$

which is identically satisfied by the ν functions just written. The coefficients in this equation must be independent of the x and hence must be pure numbers. For the time being we impose no restriction on the numerical field of rationality.

If $\nu > 1$, the equation $F(u,z) = 0$ is a resolvent of the equation of degree n whose roots are the x's, and it depends on the single parameter z. We exclude the case $\nu = 1$.

§107. Luroth's Theorem

We first need the following auxiliary theorem.

1. Let

$$u, u_1, u_2, \ldots, u_{\nu-1}$$

be a system of rational functions of one variable t. Then there is a rational function ϑ of the $u, u_1, \ldots, u_{\nu-1}$ such that $u, u_1, \ldots, u_{\nu-1}$ can be expressed rationally by ϑ alone.

For the proof, write

$$u = \frac{\varphi(t)}{\psi(t)}, \qquad u_1 = \frac{\varphi_1(t)}{\psi_1(t)}, \quad \ldots, \quad u_{\nu-1} = \frac{\varphi_{\nu-1}(t)}{\psi_{\nu-1}(t)}, \tag{246}$$

where $\varphi(t), \psi(t)$ are integral functions of t without a common divisor, and similarly for the pairs $\varphi_i(t), \psi_i(t)$. We may also suppose that none of the functions $u, u_1, \ldots, u_{\nu-1}$ is constant. Introduce, beside t, a second variable τ and put

$$\varphi(t)\psi(\tau) - \varphi(\tau)\psi(t) = \Phi(t,\tau) = -\Phi(\tau,t),$$

$$\varphi_i(t)\psi_i(\tau) - \varphi_i(\tau)\psi_i(t) = \Phi_i(t,\tau) = -\Phi_i(\tau,t).$$

As functions of t all these expressions vanish for $t = \tau$; consequently they have a greatest common divisor, of positive degree in t. Let this common divisor be

$$R(t,\tau). \tag{247}$$

This function is rational also in τ. It may be normalized so that τ occurs neither in a denominator nor in a superfluous factor. Then, upon interchanging t and τ, $R(t,\tau)$ can change at most by sign. For the expressions above are alternating in the two variables and are divisible by $R(t,\tau)$, and also by $R(\tau,t)$. Hence the two functions divide one another and differ only by a constant factor, which must be ± 1.

It remains to observe that, for undetermined τ, none of the functions $\Phi_i(t,\tau)$, considered as functions of t, contains a repeated factor. Suppose, for example, that

$$\Phi(t,\tau) = \varphi(t)\psi(\tau) - \varphi(\tau)\psi(t),$$

$$\Phi'(t,\tau) = \varphi'(t)\psi(\tau) - \varphi(\tau)\psi'(t)$$

had a common divisor P as functions of t. Then P would divide

$$\psi(t)\Phi'(t,\tau) - \psi'(t)\Phi(t,\tau) = (\varphi'(t)\psi(t) - \varphi(t)\psi'(t))\varphi(\tau),$$

and therefore P could be assumed independent of τ. If now two values τ_1, τ_2 are chosen so that $\Phi(\tau_2,\tau_1)$ is non-zero, the identity obtained by eliminating the parameter shows that both $\varphi(t)$ and $\psi(t)$ would be divisible by P. This contradicts the relative primality of φ and ψ.

Thus $R(t,\tau) = -R(\tau,t)$ up to the chosen sign. To form $R(t,\tau)$ one may take the greatest common divisor of the functions

$$\varphi_h(t)\psi_h(\tau) - \varphi_h(\tau)\psi_h(t).$$

It follows that, apart from a factor independent of t, $R(t,\tau)$ is rationally expressible by $u, u_1, \ldots, u_{\nu-1}$. If a, b are any fixed numerical values, put

$$\vartheta = \frac{R(b,t)}{R(a,t)}. \tag{248}$$

Then ϑ is a rational function of $u, u_1, \ldots, u_{\nu-1}$.

For a new variable ξ the equation

$$R(a,\xi) - \vartheta R(b,\xi) = 0 \tag{249}$$

is of degree equal to the degree of R in t. If τ is an arbitrary value and if the roots of

$$R(t,\tau) = 0$$

are

$$t = \tau, \tau', \tau'', \ldots,$$

then for every index h the functions

$$\Phi_h(t,\tau), \quad \Phi_h(t,\tau'), \quad \Phi_h(t,\tau''), \ldots$$

have the same common divisor. The equations

$$R(t,\tau) = 0, \quad R(t,\tau') = 0, \quad R(t,\tau'') = 0, \ldots$$

therefore have the same roots. Hence the roots of the last displayed equation in ξ are precisely

$$\xi = \tau, au', \tau'', \ldots.$$

On the other hand, from $\Phi_h(\tau', \tau) = 0$ one obtains

$$\frac{\varphi_h(\tau)}{\psi_h(\tau)} = \frac{\varphi_h(\tau')}{\psi_h(\tau')} = \frac{\varphi_h(\tau'')}{\psi_h(\tau'')} = \cdots .$$

Therefore each u_h is a symmetric function of the roots of the equation above and is rationally expressible by its coefficients, hence rationally expressible by ϑ. On interchanging again t and τ, we obtain formulae of the form

$$u = f(\vartheta), \quad u_1 = f_1(\vartheta), \quad \ldots, \quad u_{\nu-1} = f_{\nu-1}(\vartheta), \tag{250}$$

where $f, f_1, \ldots, f_{\nu-1}$ are rational functions. This proves the theorem.

§108. Resolvents with One Parameter

We return to the hypotheses of §106 and denote by

$$u, u_1, \ldots, u_{\nu-1}$$

a system of rational functions of $x_0, x_1, \ldots, x_{n-1}$ which arise from one of them by the permutations of the alternating group and which are roots of the equation

$$F(u, z) = 0, \tag{251}$$

where z is a rational function of the x's remaining unchanged under the alternating group.

We prove a second auxiliary theorem.

2. If the rational function

$$\Psi(u, u_1, \ldots, u_{\nu-1})$$

vanishes identically after substituting for $x_0, x_1, \ldots, x_{n-1}$ certain rational functions of one variable t, and if under this substitution z does not become independent of t, then Ψ also vanishes identically as a function of the independent variables $x_0, x_1, \ldots, x_{n-1}$.

The Galois group of the equation $F(u, z) = 0$ over the field of rational functions in z consists of certain permutations of the roots $u, u_1, \ldots, u_{\nu-1}$. Perform these permutations in the function Ψ and form the product

$$\prod \Psi(u, u_1, \ldots, u_{\nu-1}) = f(z) \tag{252}$$

over all functions so obtained. This gives a rational function $f(z)$ of z. If, after the substitution in the variable t, one factor of this product vanishes identically and z is not independent of t, then $f(z)$ must be identically zero. Consequently one factor of the product must vanish identically as a function of the x's. But if one factor vanishes identically, then all factors vanish identically, since in every rational equation among the u's one may apply all substitutions of the Galois group. This proves the assertion.

Together with Luroth's theorem, this gives the following result.

3. Let $u, u_1, \ldots, u_{\nu-1}$ be the roots of an equation depending on one parameter z, and let $u, u_1, \ldots, u_{\nu-1}, z$ be rational functions of the independent variables $x_0, x_1, \ldots, x_{n-1}$. Then one may put

$$u = f(\vartheta), \quad u_1 = f_1(\vartheta), \quad \ldots, \quad u_{\nu-1} = f_{\nu-1}(\vartheta), \tag{253}$$

where $f, f_1, \ldots, f_{\nu-1}$ are rational functions of ϑ and

$$\vartheta = \chi(u, u_1, \ldots, u_{\nu-1}) = \psi(x_0, x_1, \ldots, x_{n-1}) \tag{254}$$

is a rational function of the u's, equivalently of the x's.

Indeed, after replacing $x_0, x_1, \ldots, x_{n-1}$ by rational functions of one variable t, in such a way that z does not become independent of t, Luroth's theorem first gives the representation just stated. The relations

$$u_h = f_h(\vartheta) = f_h(\chi(u, u_1, \ldots, u_{\nu-1}))$$

are then identities in t, and by the preceding theorem are identities in the variables x.

4. If two variables ϑ, ϑ_1 are each rational functions of the other, then they are linear fractional functions of one another.

For suppose

$$\vartheta_1 = \frac{\varphi(\vartheta)}{\psi(\vartheta)}, \qquad \vartheta = \frac{\varphi_1(\vartheta_1)}{\psi_1(\vartheta_1)},$$

where numerator and denominator in each fraction are relatively prime integral functions. If the second relation is arranged as a polynomial equation in ϑ, and if the first relation is substituted, the denominator ψ must divide the leading coefficient and hence must be constant or linear. Similarly φ must be constant or linear. Since not both can be constant, the assertion follows.

Apply this to the function ϑ of theorem 3. If the variables x are subjected to any permutation of the alternating group, then the functions $u, u_1, \ldots, u_{\nu-1}$ are also permuted. Hence ϑ is changed into another function ϑ_1 which, by the representation above, is rationally expressible by ϑ. Conversely ϑ is rationally expressible by ϑ_1. Therefore ϑ and ϑ_1 are linear fractional functions of one another.

This proves:

5. If for a general equation of degree n there exists a resolvent with one parameter, then there is a rational function ϑ of the n variables x which, under every permutation of the alternating group, is changed into a linear fractional function of itself,

$$\vartheta \longmapsto \frac{a\vartheta + b}{c\vartheta + d}, \tag{255}$$

where a, b, c, d are constants, that is, numbers.

§109. Group of the Resolvents with One Parameter

The function ϑ in theorem 5 of the preceding paragraph is itself the root of a resolvent with one parameter,

$$\Phi(\vartheta, z) = 0.$$

Since each root is rationally expressible by every other root, this equation is normal, and its Galois group is isomorphic to the group of the linear fractional substitutions $\chi(\vartheta)$ occurring in theorem 5. In the identity $\Phi(\vartheta, z) = 0$ all permutations of the alternating group $\mathfrak{A}$ of the variables x may be performed; z remains unchanged, while ϑ is carried into every other root of Φ. Hence the group of the linear substitutions $\chi(\vartheta)$, that is, the Galois group of Φ, is isomorphic, in one or more stages, with the alternating permutation group on n letters. We denote this linear group by L. Since it is finite, it must be one of the polyhedral groups considered in the seventh section.

For simplification we use the theorem of §51, according to which the group L can be transformed so that any chosen non-identity substitution of L becomes a multiplication. Such a transformation of L is simply the replacement of ϑ by a linear fractional function of ϑ, which has the same property as the original function.

Let ϑ_π denote the function obtained from ϑ by applying the permutation π to the variables $x_0, x_1, \ldots, x_{n-1}$. We may therefore choose ϑ in such a way that for a fixed but arbitrary permutation π one has

$$\vartheta_\pi = \varepsilon\vartheta.$$

If π is repeated, and if p is its order, then ε is a p-th root of unity.

First let $n = 3$. The alternating group consists of the powers of the cyclic permutation

$$\gamma = (0, 1, 2).$$

We may assume that the corresponding substitution is multiplicative, say $\vartheta_\gamma = \varepsilon\vartheta$, where ε is a third root of unity. Thus ϑ^3 is an alternating, or symmetric, function. As a resolvent with one parameter we obtain the pure equation

$$\vartheta^3 = z.$$

For example, one may take for ϑ the Lagrange resolvent

$$x_0 + \varepsilon x_1 + \varepsilon^2 x_2,$$

which gives the familiar reduction of the cubic equation to a pure equation.

Next consider $n = 4$. The alternating group $\mathfrak{A}$ contains the four-group

$$1, \quad \alpha_1 = (0, 1)(2, 3), \quad \alpha_2 = (0, 2)(1, 3), \quad \alpha_3 = (0, 3)(1, 2),$$

and also eight cyclic permutations of three letters. If

$$\gamma = (0, 1, 2), \qquad \alpha_1 = (0, 1)(2, 3),$$

then γ and α_1 generate $\mathfrak{A}$. Put

$$\gamma\alpha_1\gamma = / = (0,3,1),$$

so that γ and the indicated generator may also be regarded as generators. Moreover

$$\gamma^2/ = \alpha_1. \tag{256}$$

Choose ϑ so that γ acts by multiplication:

$$\vartheta_\gamma = \varepsilon\vartheta,$$

where ε is a third root of unity. Write

$$\vartheta = \frac{\varphi(x_0, x_1, x_2, x_3)}{\psi(x_0, x_1, x_2, x_3)} \tag{257}$$

with φ, ψ integral forms without common factor. From the identity $\vartheta_\gamma = \varepsilon\vartheta$ it follows that

$$\varphi_\gamma = \varepsilon_1\varphi, \qquad \psi_\gamma = \varepsilon_2\psi,$$

where $\varepsilon_1, \varepsilon_2$ are again third roots of unity. Applying any of the α's gives a second linear fractional expression in ϑ. By comparing numerator and denominator and by eliminating, one obtains a relation of the form

$$h\varphi + h_1\varphi_1 + h_2\varphi_2 = 0.$$

Repeating this relation under the substitutions α_1, α_2 and adding suitable multiples, one concludes that one of the relations

$$\varphi = \pm\varphi_1, \quad \varphi = \pm\varphi_2, \quad \varphi = \pm\varphi_3$$

must hold. Since we may equally well use γ, α_1, or γ, α_2, or γ, α_3 as generators, it follows:

1. The function φ is changed by every permutation of the alternating group only by multiplication by a constant factor.

The same proof applies to ψ, and hence also to their quotient ϑ. Thus for every permutation π of the alternating group

$$\vartheta_\pi = \varepsilon\vartheta. \tag{258}$$

If π is a cyclic permutation of three letters, ε is a third root of unity. Since every permutation of $\mathfrak{A}$ is compounded from such cycles, ε is always a third root of unity. If π belongs to the four-group, ε is at the same time a second root of unity and must therefore be 1. Hence:

2. The function ϑ is unchanged by the permutations of the four-group.

The third power ϑ^3 is invariant under the alternating group, and ϑ is therefore the root of a pure cubic equation

$$\vartheta^3 = z.$$

This is the resolvent $\Phi(\vartheta, z) = 0$ in this case; it is a partial resolvent.

The purpose of these considerations is the following theorem.

3. A general equation of degree higher than the fourth has no resolvents with one parameter.

For if $n > 4$, we may regard ϑ as a function of any four of the variables x and apply the preceding result. It follows that ϑ remains unchanged when any pair of transpositions is performed among the variables. But the alternating group is generated by such pairs of transpositions; hence ϑ is invariant under the whole alternating group, that is, it is alternating or symmetric. The equation $\Phi(\vartheta, z) = 0$ then reduces to the identity $\vartheta = z$ and is not a resolvent. If the symmetric group is used instead of the alternating group, the theorem is true a fortiori.

This theorem was first stated by Kronecker, who did not publish a proof. The first proof was given by F. Klein and was then simplified by Gordan; the argument above follows Gordan in its main lines.

§110. The Icosahedral Equation

The preceding result is at first negative for the general equation of degree greater than four: it is not possible, by forming rational resolvents, to reduce the general equation of the fifth degree to an equation with one parameter. If nevertheless one does not give up this aim, there is only one remaining course: the variables

$$x_0, x_1, \dots, x_{n-1}$$

must no longer be regarded as entirely independent, but certain equations among them must be allowed. We then no longer have the general equation of the degree in question, but a more special one, in which the coefficients are not independent. A second question arises: how, and by what irrational functions, can the general equation be reduced to this special one? If this can be done by equations of lower degree, for example by square roots, then a real reduction of the problem has nevertheless been achieved.

We now suppose that the variables $x_0, x_1, \dots, x_{n-1}$ are connected by an arbitrary number of relations

$$A(x_0, x_1, \dots, x_{n-1}) = 0, \quad B(x_0, x_1, \dots, x_{n-1}) = 0, \quad \dots, \tag{259}$$

where $A, B, \dots$ are rational functions of the x's. These relations are assumed to be so constituted, in number and mutual dependence, that the values of the x's are not fixed numerically but still vary. Moreover, let a permutation group $\mathfrak{R}$ of the x's be given. We regard as rational precisely those functions of the x's which admit the substitutions of $\mathfrak{R}$, so that $\mathfrak{R}$ is the Galois group of the equation whose roots are the $x_0, x_1, \dots, x_{n-1}$. The relations must then be such that they too admit the substitutions of $\mathfrak{R}$.

Under these hypotheses we ask whether two rational functions u, z of the x's can be found such that an equation

$$F(u, z) = 0 \tag{260}$$

is satisfied for all values of the x's satisfying the relations, and in which z remains invariant, with respect to the relations, under the substitutions of $\mathfrak{R}$. The function u is to be changed thereby into ν different functions

$$u, u_1, \dots, u_{\nu-1},$$

which are the roots of the equation; if $\nu > 1$, this equation is a resolvent of the equation for the x's.

In this generality we have no point of attack. We add one further restricting hypothesis: it must be possible to put

$$x_h = \varphi_h(t) \tag{261}$$

with rational functions of a variable t, so that all relations are identically satisfied, and so that at the same time z does not become independent of t.

As an example, which will occupy us later in detail, take five variables x_0, x_1, x_2, x_3, x_4 satisfying

$$\sum x = 0, \qquad \sum x^2 = 0. \tag{262}$$

They are then the roots of a principal equation of the fifth degree. The parametrization given in Volume I, §74, shows that our hypothesis is satisfied in this case. Geometrically, if the x and the u are regarded as coordinates of points X and U in spaces of n and ν dimensions, the relations determine for X a manifold of lower dimension. To every X corresponds a U, and by (2) the totality of the U forms a one-dimensional manifold, a curve. The hypothesis says that this curve is rational, or unicursal, or in the terminology of function theory, of genus zero.

Under this hypothesis the preceding arguments remain valid. Thus, if an equation whose roots are the x's has a resolvent

$$F(u, z) = 0 \tag{263}$$

with one parameter z invariant under $\mathfrak{R}$, then there must be a function ϑ of the variables x which, under every substitution π of $\mathfrak{R}$ and with respect to the relations, undergoes a linear fractional substitution

$$\vartheta_\pi = \frac{a\vartheta + b}{c\vartheta + d}. \tag{264}$$

The roots of the resolvent are rationally expressible by ϑ, and ϑ itself is the root of an equation with one parameter

$$\Phi(\vartheta, z) = 0. \tag{265}$$

If the resolvent is a total resolvent, which always happens when the group is simple, the x's themselves are rationally expressible by ϑ and the functions belonging to the group $\mathfrak{R}$.

Up to this point there is complete agreement with the results of §108. The restriction encountered in §109 need not occur now, since from the equality of two fractional functions one can no longer infer, as in the case of completely independent variables, the equality of numerator and denominator.

The linear substitutions in ϑ form a group P isomorphic, in one or more stages, with $\mathfrak{R}$. This group must be one of the polyhedral groups. We shall treat in detail only the most interesting case, namely that in which P is the icosahedral group, and we may take it without loss of generality in the normal form of §58.

For this group the three fundamental forms were found in §60. Denote their variables by y_1, y_2. They are the homogeneous forms of degrees $12, 20, 30$:

$$\begin{aligned}
f(y) &= y_1 y_2 (y_1^{10} + 11 y_1^5 y_2^5 - y_2^{10}), \\
H(y) &= -(y_1^{20} + y_2^{20}) + 228(y_1^{15} y_2^5 - y_2^{15} y_1^5) - 494 y_1^{10} y_2^{10}, \\
T(y) &= y_1^{30} + y_2^{30} + 522(y_1^{25} y_2^5 - y_1^5 y_2^{25}) - 10005(y_1^{20} y_2^{10} + y_2^{20} y_1^{10}),
\end{aligned} \tag{266}$$

with the relation

$$T^2 + H^3 = 1728 f^5. \tag{267}$$

If in the quotient $T^2 : f^5$ we put for $y_1 : y_2$ one of the sixty values ϑ_α obtained from ϑ by icosahedral substitutions, then this quotient has always the same value. It is therefore rationally expressible by the coefficients of the equation for ϑ, hence rationally by z. We set it equal to z and obtain the resolvent

$$T^2 - z f^5 = 0, \tag{268}$$

or, equivalently,

$$H^3 - (1728 - z)f^5 = 0. \tag{269}$$

This equation is homogeneous in y_1, y_2 and of degree 60. If $y_2 = 1$, its sixty roots $y_1 = \vartheta_\alpha$ form the icosahedral equation. Written explicitly in terms of ϑ it is

$$\left(\vartheta^{30} + 522\vartheta^{25} - 10005\vartheta^{20} - 10005\vartheta^{10} - 522\vartheta^5 + 1\right)^2 = z\,\vartheta^5(\vartheta^{10} + 11\vartheta^5 - 1)^5. \tag{270}$$

The problem, in the form now reached, is therefore reduced to the question:

Which equations can be solved by the icosahedral equation?

§111. The Resolvents of the Icosahedral Equation

The formulation at the end of the preceding paragraph leads us to the various resolvents of the icosahedral equation. To each divisor of the icosahedral group there corresponds such a resolvent; its degree is the index of the divisor and hence always a divisor of 60. The icosahedral group contains:

1. tetrahedral groups, resolvents of degree 5,
2. dihedral groups D_5, resolvents of degree 6,
3. dihedral groups D_3, resolvents of degree 10,
4. cyclic groups C_5, resolvents of degree 12,
5. four-groups D_2, resolvents of degree 15,
6. cyclic groups C_3, resolvents of degree 20,
7. cyclic groups C_2, resolvents of degree 30.

Since the icosahedral group is simple, all these equations are total resolvents of one another. We restrict ourselves to the most important among them.

The resolvents of lowest degree, the quintic resolvents, belong to the tetrahedral group. To find them we must seek a function of ϑ which remains unchanged under the substitutions of a tetrahedral group but is five-valued in the icosahedral group.

Take the icosahedral group in the normal form of §58 and choose the tetrahedral group Q contained in it. It is generated over the four-group by

$$1, \quad \psi, \quad \chi, \quad \psi\chi, \tag{271}$$

where, for a fifth root of unity ε,

$$\psi(x) = -\frac{1}{x}, \qquad \chi(x) = \frac{\omega x + 1}{x - \omega}, \qquad \omega = \varepsilon + \varepsilon^{-1}. \tag{272}$$

Putting $\Theta(x) = \varepsilon x$, the whole icosahedral group decomposes as

$$P = Q + Q\Theta + Q\Theta^2 + Q\Theta^3 + Q\Theta^4, \tag{273}$$

and

$$Q_h = \Theta^{-h} Q \Theta^h \tag{274}$$

are the conjugate tetrahedral groups.

In §56 we met two forms of degrees 6 and 8, say t and ω, invariant under the tetrahedral substitutions when the latter are taken homogeneously with determinant one. In the present normal form their first member is obtained from the equation for the double poles of the tetrahedral group:

$$x = \psi(x), \qquad x = \chi(x), \qquad x = \psi\chi(x),$$

that is,

$$x^2 + 1 = 0, \quad x^2 - 2\omega x - 1 = 0, \quad \omega x^2 + 2x - \omega = 0,$$

and, using $\omega^2 + \omega = 1$,

$$(x^2 + 1)(x^4 + 2x^3 - 6x^2 - 2x + 1) = 0. \tag{275}$$

Thus the required tetrahedral form of degree 6 in homogeneous variables x_1, x_2 is

$$t(x_1, x_2) = x_1^6 + 2x_1^5 x_2 - 5x_1^4 x_2^2 - 5x_1^2 x_2^4 - 2x_1 x_2^5 + x_2^6. \tag{276}$$

The second tetrahedral form is its Hessian determinant. If

$$\Omega(x_1, x_2) = \frac{1}{5^2 \cdot 4^2} \left(t''_{11} t''_{22} - (t''_{12})^2\right), \tag{277}$$

then a simple calculation gives

$$\begin{aligned}\Omega(x_1, x_2) = &- (x_1^8 + x_2^8) + (x_1^7 x_2 - x_1 x_2^7) - 7(x_1^6 x_2^2 + x_1^2 x_2^6) \\ &- 7(x_1^5 x_2^3 - x_1^3 x_2^5).\end{aligned} \tag{278}$$

The three icosahedral forms are the forms f, H, T displayed in the preceding paragraph.

From these invariant forms of the tetrahedron one obtains functions of $x_1 : x_2$. Setting $\vartheta = x_1 : x_2$, they are functions unchanged under the fractional tetrahedral substitutions and hence are roots of quintic resolvents of the icosahedral equation. The icosahedral equation itself assumes the two forms

$$H^3 = zf^5, \qquad T^2 = z_1 f^5, \qquad z + z_1 = 1728. \tag{279}$$

If $\varphi(x_1, x_2)$ is any absolute tetrahedral invariant form, then under the substitutions $Q\Theta$ it gives the five forms

$$\varphi_\varepsilon = \varphi(\varepsilon^{-2} x_1, \varepsilon^2 x_2). \tag{280}$$

Every symmetric function of these five forms is an invariant of the icosahedral group, and therefore, by §61, can be expressed by the fundamental forms f, H, T. This gives a relatively simple calculation of the resolvents.

The simplest functions to use as roots are

$$r = \frac{t^2}{f}, \qquad u = \frac{f^2 t}{T}, \qquad v = \frac{f\Omega}{H}. \tag{281}$$

The symmetric fundamental functions of the five functions t_ε are icosahedral invariants of degrees $6, 12, 18, 30$. Since degrees 6 and 18 do not occur among the fundamental invariants in §61, one obtains an equation of the form

$$t^5 + aft^3 + bf^2 t + cT = 0, \tag{282}$$

where a, b, c are numerical coefficients. Determining them from Newton's formulae gives, in the notation above, the quintic equations

$$r(r^2 - 10r + 45)^2 = rz, \tag{283}$$

for r, and

$$u^5 - 10u^3 + 45u - z = 0 \tag{284}$$

for u, after the corresponding normalization. By the substitution

$$y = -\frac{u}{3}, \qquad z = 27Z,$$

the latter becomes

$$y^5 + 15y - 10y^3 + 3Z = 0, \tag{285}$$

which has already been encountered in the first volume as a normal form of a quintic equation.

§112. The Principal Resolvent of the Fifth Degree

If the two functions u, v are used simultaneously, one obtains a family of resolvents having the form of the principal equation of the fifth degree. They may be brought directly into agreement with any given principal quintic equation.

Since the icosahedron has no invariants of degrees $8, 14, 16, 22, 28$, the sums

$$S(\Omega), \quad S(t\Omega), \quad S(\Omega^2), \quad S(t\Omega^2), \quad S(t^2\Omega^2)$$

vanish. Thus, if

$$Y = \alpha\Omega^2 + \beta t\Omega, \tag{286}$$

then the functions $S(Y)$ and $S(Y^2)$ vanish for arbitrary α, β. This yields an identity of the form

$$Y^5 + 5aY^2 + 5bY + c = 0, \tag{287}$$

where a, b, c are homogeneous functions of degrees $3, 4, 5$ in α, β, their coefficients being icosahedral invariants.

The last coefficient follows at once from the product formula:

$$c = -\Pi(\Omega)\,\Pi(\alpha + \beta t). \tag{288}$$

Using the formula from the preceding paragraph for $\Pi(\lambda - t)$, one obtains

$$c = H^2(\alpha^5 - 10f\alpha^3\beta^2 + 45f^2\alpha\beta^4 + T\beta^5). \tag{289}$$

The two coefficients a, b are obtained most simply from Newton's formulae. The calculation gives

$$\begin{aligned} a &= 8f^2\alpha^3 + T\alpha^2\beta + 12f^3\alpha\beta^2 + fT\beta^3, \\ b &= -fH\alpha^4 + 18f^2H\alpha^2\beta^2 + HT\alpha\beta^3 + 27f^3H\beta^4. \end{aligned} \tag{290}$$

To obtain from this the resolvent of the icosahedral equation, introduce the functions u, v of the preceding paragraph. Put

$$Y = \lambda v + \mu u. \tag{291}$$

Then, if

$$z_1 = 1728 - z, \tag{292}$$

the preceding formulae assume the corresponding normalized form

$$Y^5 + 5aY^2 + 5bY + c = 0. \tag{293}$$

This is a principal equation of the fifth degree. It can represent every principal equation if a, b, c may be taken arbitrarily. Thus, for arbitrary prescribed a, b, c, the three quantities λ, μ, z_1 are to be determined. A remarkable property of this system is that these unknowns are rationally expressible by a, b, c and by the discriminant of the quintic.

Weber derives this by eliminating the parameters. From the formulae one obtains first

$$z_1(7b + c) = \lambda^2 a, \tag{294}$$

and another linear relation between a, b, c and the remaining parameter. Squaring, subtracting, and using $z + z_1 = 1728$, one is led to the quadratic equation

$$\lambda^2(a^4 + abc - b^3) - \lambda(11a^2b - ac^2 + 2b^2c) - 27a^2c + 64ab^2 - bc^3 = 0. \tag{295}$$

Its discriminant is

$$\Delta = a^2(108a^3c - 135a^2b^2 + 90abhc^2 - 320ab^2c + 256b^3 + c^4), \tag{296}$$

which is, up to the factor displayed in Weber's normalization, the discriminant D of the principal quintic. Thus

$$\Delta = a^2 D. \tag{297}$$

Consequently λ is obtained with the square root of the discriminant, and after λ is known the ratio of the remaining parameters follows rationally; the last unknown is then obtained rationally as well.

Thus, besides the square root of the discriminant, which is rationally expressible by the roots, a further occurrence of $\sqrt{5}$ appears.

§113. Resolvents of the Sixth Degree

To find the resolvents of degree six of the icosahedral equation, begin with one of the dihedral groups D_5 contained in the icosahedral group. The representation of the icosahedral group in §58 gives the decomposition into cosets directly. If

$$Q = \Theta^r \psi, \qquad r = 0, 1, 2, 3, 4, \tag{298}$$

is the dihedral group with which we begin, then the full icosahedral group may be written

$$P = Q + Q\chi + Q\chi\Theta + Q\chi\Theta^2 + Q\chi\Theta^3 + Q\chi\Theta^4. \tag{299}$$

If U_∞ is a function invariant under the substitutions of Q, and if it is changed by

$$\chi, \quad \chi\Theta, \quad \chi\Theta^2, \quad \chi\Theta^3, \quad \chi\Theta^4$$

into

$$U_0, U_1, U_2, U_3, U_4,$$

then

$$U_\infty, U_0, U_1, U_2, U_3, U_4$$

are the roots of a resolvent of the sixth degree.

One may use the invariant forms of the dihedral group and divide by suitable powers of the icosahedral invariants f, H, T. It is enough here to carry through the simplest case. If the substitution χ is applied to the quadratic form in the notation of §58, and the five conjugates are written down, we obtain six forms w; the first may be written

$$w_\infty = 5x_1^3 x_2^3, \tag{300}$$

and the others in the shape

$$w_r = (\varepsilon^r x_1 + \varepsilon^{-r} x_2)^6 \tag{301}$$

with the appropriate signs and normalizing factors used in §58.

The elementary symmetric functions of these six forms are icosahedral invariants. From comparison of degrees and of one coefficient each Weber obtains the power sums, whence

$$\Pi(w) = 5f^2. \tag{302}$$

Newton's formulae then give the equation

$$w^6 - 10fw^3 + Hw + 5f^2 = 0. \tag{303}$$

If

$$U = \frac{Tw}{fH}, \tag{304}$$

then one obtains the sextic resolvent of the icosahedral equation

$$U^6 - 10zU^3 + z^2U + 5 = z^2. \tag{305}$$

The six roots satisfy the Jacobi relations. If they are denoted by

$$U_\infty, U_0, U_1, U_2, U_3, U_4,$$

then the linear combinations with coefficients $1, \varepsilon, \varepsilon^2, \varepsilon^3, \varepsilon^4$ have the form displayed by Jacobi, and the equations of degree six whose roots satisfy these relations are called Jacobi equations of the sixth degree.

§114. On the Solution of the Icosahedral Equation by Transcendental Functions

We finally indicate, without proof and without detailed development, the relations between the preceding algebraic investigations and the theory of certain transcendent functions, especially elliptic functions and hypergeometric series. From these relations the solution of the icosahedral equation, and therefore of all its resolvents, by transcendent functions follows. This lies outside algebra proper, but stands to it in roughly the same relation as the use of logarithms in the computation of roots, or the trisection of angles in the solution of cubic equations.

Let ω be a variable whose imaginary part is positive, so that

$$q = e^{2\pi i\omega} \tag{306}$$

has absolute value less than one. Weber introduces three functions of ω, denoted in the source by combinations of infinite products and series, from which the modular invariant is formed. In modern notation these are the theta-constant expressions which transform into one another under the substitutions of the modular group. From them one constructs a function $J(\omega)$ and a sextic equation whose Galois group is the same as that of the Jacobi sextic resolvent.

The roots of this sextic may be put in correspondence with the quantities $v_\infty, v_0, \ldots, v_4$ occurring in the previous section. By choosing the fifth root of unity suitably, Weber obtains

$$v_\xi = -\eta_\xi(\omega), \qquad \xi = \infty, 0, 1, 2, 3, 4, \tag{307}$$

where the η_ξ are explicit theta functions. Thus, once ω has been determined from the transcendent equation giving the prescribed value of z, the formulae of the preceding paragraph give the complete solution of the sextic icosahedral resolvent.

In particular, for $\xi = 0, 1, 2, 3, 4$ Weber writes

$$\eta_\xi(\omega) = \sum (-1)^n e^{\pi i\omega(6n+1)^2} \tag{308}$$

with the congruence classes of n divided according to their residues modulo 5. From the decomposition of this series one recognizes directly that the five quantities satisfy Jacobi's relations. The root $x = x_1 : x_2$ of the icosahedral equation is then represented as a quotient of two such theta series; one of Weber's forms is

$$x = \frac{\sum (-1)^n e^{\pi i\omega(6n+1)^2}}{\sum (-1)^n e^{\pi i\omega(6n-1)^2}}, \tag{309}$$

with the summation over the appropriate residue classes.

The remaining roots are obtained by applying the icosahedral substitutions to x. They may also be obtained by replacing the variable ω by an equivalent value.

Weber also records the solution by hypergeometric series. For Gauss' series

$$F(\alpha, \beta, \gamma; \lambda) = 1 + \frac{\alpha\beta}{\gamma}\lambda + \frac{\alpha(\alpha+1)\beta(\beta+1)}{1 \cdot 2\gamma(\gamma+1)}\lambda^2 + \cdots,$$

put

$$z = 1728\lambda.$$

Then a root of the icosahedral equation is given by a quotient of two hypergeometric series, one of the forms being

$$\vartheta = \frac{F\left(\frac{1}{60}, \frac{19}{60}, \frac{2}{3}; \lambda\right)}{F\left(\frac{31}{60}, \frac{49}{60}, \frac{4}{3}; \lambda\right)} \tag{310}$$

up to the conventional normalizing factor used in the icosahedral equation. These series converge when $|\lambda| < 1$; for other values they must be replaced by the corresponding analytic continuations.

Fourteenth Section. Groups of Linear Ternary Substitutions

§115. A Ternary Linear Substitution Group of Degree 168

The problem of all possible finite groups of linear substitutions in several variables, especially in three and four dimensions, was treated by C. Jordan. As in the binary case, Jordan found a finite number of possible types. We shall not follow these general investigations here; we restrict ourselves to one example of a ternary group.

In the section on congruence groups we met a series of simple groups. The first, of order 60, proved to be isomorphic with the icosahedral group. We now consider the next one, of order 168, which was denoted by L_7, and we shall construct a ternary linear substitution group isomorphic with it.

We are led to this by the theorem of §72. We seek four elements

$$\tau, \chi, \omega, \Theta$$

which satisfy the composition relations stated there. First take a substitution τ in normal form

$$\tau = \begin{pmatrix} \varepsilon & 0 & 0 \\ 0 & \varepsilon^2 & 0 \\ 0 & 0 & \varepsilon^4 \end{pmatrix}, \tag{311}$$

where ε is a primitive seventh root of unity. The product of the three diagonal factors is one.

We next determine χ from the condition

$$\chi\tau\chi^{-1} = \tau^2. \tag{312}$$

This leads to a cyclic permutation of the three variables. Up to a simultaneous transformation of the whole group, we may put

$$\chi = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \chi^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}. \tag{313}$$

The substitutions $\tau^r\chi^s$ then form a group of order 21.

To determine ω, use the relation

$$\omega\chi\omega = \chi^{-1}, \tag{314}$$

or, equivalently, the relation given in §72. If

$$\omega = \begin{pmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{pmatrix}, \tag{315}$$

then the condition that ω have order two gives

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 &= 1, \\ \alpha\beta + \beta\gamma + \gamma\alpha &= 0. \end{aligned} \tag{316}$$

Together with the determinant condition one obtains

$$\alpha + \beta + \gamma = -1, \tag{317}$$

and hence

$$\alpha = \beta\gamma - \alpha^2, \quad \beta = \gamma\alpha - \beta^2, \quad \gamma = \alpha\beta - \gamma^2. \tag{318}$$

The coefficients are expressed by the seventh root ε as

$$\alpha = h(\varepsilon^3 - \varepsilon^{-3}), \quad \beta = h(\varepsilon^2 - \varepsilon^{-2}), \quad \gamma = h(\varepsilon - \varepsilon^{-1}), \tag{319}$$

where

$$h = \frac{1}{\sqrt{-7}} \tag{320}$$

with the sign chosen according to the chosen primitive seventh root. Equivalently,

$$\alpha = -\frac{2\sin(6\pi/7)}{\sqrt{7}}, \quad \beta = -\frac{2\sin(4\pi/7)}{\sqrt{7}}, \quad \gamma = -\frac{2\sin(2\pi/7)}{\sqrt{7}}. \tag{321}$$

The useful relations

$$\alpha\varepsilon + \beta\varepsilon^{-2} + \gamma\varepsilon^2 = 1, \quad \alpha\varepsilon^{-1} + \beta\varepsilon^{-4} + \gamma\varepsilon^{-2} = 1 \tag{322}$$

follow from these formulae.

The remaining generator Θ is obtained from the relations in §72. Weber gives it in the form

$$\Theta = \omega\chi\omega\chi^2 \tag{323}$$

after the substitutions are multiplied out. It has order four, and with τ, χ, ω it generates a group of degree 168. The defining relations are exactly those of the congruence group L_7; consequently the ternary group so obtained is isomorphic with that simple group.

For the following paragraphs it is not the abstract presentation but the explicit ternary substitutions which will be used: the diagonal element τ, the cyclic permutation χ, the symmetric involution ω with the coefficients α, β, γ, and the fourth-order element Θ generated from them.

§116. Poles and Axes of Ternary Groups

A ternary linear substitution

$$y_i = \sum_j a_{ij} x_j \tag{324}$$

may be regarded as a projective collineation of the plane. To a point with homogeneous coordinates (x_1, x_2, x_3) it assigns a point (y_1, y_2, y_3); to every line it assigns a line. If two points are joined by a line, then the images of the two points are joined by the image of that line.

If a second substitution S is used to transform A into

$$A' = S^{-1}AS,$$

this has the same effect as changing the system of coordinates. Composition is preserved by such transformations, and every group is thereby transformed into an isomorphic group.

The most important points for a substitution A are those left fixed by it. If

$$A = \begin{pmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{pmatrix}, \tag{325}$$

then a fixed point is determined by the equations

$$\begin{aligned} \lambda x_1 &= \alpha x_1 + \beta x_2 + \gamma x_3, \\ \lambda x_2 &= \alpha' x_1 + \beta' x_2 + \gamma' x_3, \\ \lambda x_3 &= \alpha'' x_1 + \beta'' x_2 + \gamma'' x_3. \end{aligned} \tag{326}$$

The possible values of λ are the roots of the characteristic equation. For each root one obtains, unless exceptional coincidences occur, a fixed point. Such fixed points Weber calls the *poles* of the substitution.

Correspondingly, lines may be left invariant. In the ordinary case of three distinct poles, the sides of the triangle formed by the poles are invariant lines. Weber calls these invariant lines the *axes* of the substitution.

Special cases occur. Two or all three poles may coincide. More important is the case in which infinitely many points are fixed. If the substitution is not merely a scalar multiplication, these fixed points lie on a line. Such a line is called a *principal axis.* A substitution with a principal axis has, besides it, one pole, which in exceptional cases may lie on the principal axis; in a finite group this exceptional case does not occur except for scalar multiplication.

If the principal axis is the line $x_1 = 0$, then the substitution has the form

$$y_1 = \alpha x_1, \quad y_2 = \beta x_2, \quad y_3 = \beta x_3$$

after a suitable choice of coordinates, with $\alpha \neq \beta$. A line is then invariant either if it is the principal axis itself or if it passes through the remaining pole. Thus:

If a substitution has a pole and a principal axis, then, besides the principal axis, the invariant lines are exactly the lines passing through the pole.

Collecting the possibilities, ternary linear substitutions may have:

1. three poles,
2. two poles,
3. one pole,
4. one principal axis and one pole,
5. one principal axis only,

with scalar multiplication as the limiting case in which every point is fixed. The type is invariant under transformation.

If a point is fixed by a substitution A, then it is fixed by all powers of A. Conversely, a power of A may have a principal axis even when A has three poles: then two poles of A lie on the principal axis of the power, and the third pole is the isolated pole of that power. If A has finite order μ, its eigenvalues are μ-th roots of unity; the first power of A which has a principal axis corresponds to the first coincidence among suitable powers of two eigenvalues.

§117. Application to the Group G_{168}. Sevenfold Poles

We apply the preceding generalities to the ternary group constructed in §115. It is convenient first to arrange the elements of the abstract group according to their periods. The elements of order seven, together with the identity, are divided into eight cyclic groups of seven elements each. The elements of order three are divided into twenty-eight cyclic groups of three elements each. The elements of orders four and two are divided into twenty-one cyclic groups of four elements; each of these contains one element of order two and two elements of order four.

Thus the group contains

48 elements of order 7, 56 elements of order 3, 42 elements of order 4, 21 elements of order 2.

If a point of the projective plane is fixed by v substitutions of the group, the identity being included, we shall call it a v-fold pole. This is the same terminology as in the discussion of binary groups, but now the poles are points in the projective plane.

For the determination of all poles and axes it is enough to examine one representative of each cyclic class, since all the others arise by transformation with elements of the group. We begin with the substitution of seventh order, denoted in §115 by r. Since, in the chosen coordinates, r is diagonal, the vertices of the coordinate triangle are its three distinct poles. All non-trivial powers of r have the same three poles.

The cyclic permutation χ carries these three vertices cyclically into one another. On the other hand, the substitutions denoted by $\omega, \omega^2, \Theta, \omega\Theta, \omega^2\Theta, \omega\Theta^2, \omega^2\Theta^2$, when applied to the coordinate triangle, produce triangles which are wholly different from it. For example, under one of these transformations the triangle with vertices

$$(1:0:0), \qquad (0:1:0), \qquad (0:0:1)$$

passes into a triangle whose vertices have coordinates proportional to

$$(\alpha:\beta:\gamma), \qquad (\beta:\gamma:\alpha), \qquad (\gamma:\alpha:\beta).$$

It follows that no one of these poles can have a stabilizer larger than the cyclic group of order seven which has already been found.

There are therefore eight triples of such sevenfold poles, corresponding to the eight cyclic subgroups of order seven. These twenty-four points are all distinct. Moreover any one of them can be transformed into any other by a substitution of the full group G_{168}. We have therefore the theorem:

> There are twenty-four sevenfold poles. They divide into eight triples, corresponding to the eight cyclic groups of order seven, and the full group G_{168} is transitive on the twenty-four points.

Each triple may be regarded as a triangle. The coordinate triangle is only the first of the eight; the remaining seven are its transforms. Later these same triangles will appear as triangles of inflection points of the fundamental quartic. The passage from the algebraic substitution r to its three projective poles is therefore not only a calculation of fixed points; it is the first step in the construction of the whole geometric configuration belonging to G_{168}.

The dual lines of these triangles are obtained from the contragredient group. Whenever a substitution has three distinct poles, it also has three axes, namely the lines joining the poles pairwise in the dual description. The systems of poles and axes will be treated together in the following paragraphs.

§118. The Principal Axes

We pass next to the substitutions of fourth and second order. As representatives we choose Θ and Θ^2. The poles of a ternary substitution are found by solving the characteristic equation and substituting the eigenvalues in the equations

$$Ax = \lambda x.$$

For Θ, after inserting the formulae of §115 and using the relations among α, β, γ, the characteristic equation reduces to

$$\lambda^3 - \lambda^2 + \lambda - 1 = 0.$$

Its roots are

$$\lambda = 1, \qquad \lambda = i, \qquad \lambda = -i.$$

Thus the substitution Θ has three distinct poles.

For Θ^2 the corresponding cubic has the form

$$(\lambda + 1)^2(\lambda - 1) = 0.$$

Here the double root $\lambda = -1$ gives not a single pole but a whole line of fixed points: this line is a principal axis. The simple root $\lambda = 1$ gives an isolated pole not lying on that line.

Putting $\lambda = -1$ into the equations for Θ^2, Weber obtains a line which, in his notation, is of the form

$$A = \alpha\gamma x_1 + \beta\gamma\epsilon^m x_2 + \alpha\beta\epsilon^n x_3 = 0,$$

with definite exponents depending on the representative chosen. Applying the group to this line gives all principal axes. Since the group has order 168, and the stabilizer of a principal axis has order 8, there are exactly

$$\frac{168}{8} = 21$$

principal axes.

Let A be one of these axes. Weber determines its stabilizer directly. The line is fixed by Θ^2, because it is the principal axis belonging to the double root; it is also fixed by a second substitution, denoted ω, for substitution in the equation of A changes its left-hand side only by a factor -1. Consequently the whole stabilizer of A is a group of order eight, which Weber denotes by G_A. In symbolic form it is generated by the substitutions corresponding to ω and to powers of Θ.

Among these eight substitutions only two, the identity and Θ^2, leave every point of A fixed. The others have only two fixed points on A. Thus the same line is the principal axis for exactly one involution in the stabilizer, but it also contains poles belonging to substitutions of fourth order.

The isolated pole of Θ^2, obtained from the simple root $\lambda = 1$, has coordinates proportional to

$$x_1 : x_2 : x_3 = \alpha\gamma\epsilon^{-3} : \beta\gamma\epsilon : \alpha\beta\epsilon^2.$$

Call this point temporarily a. It is fixed not only by Θ^2 but by the whole group G_A of order eight; conversely its orbit under G_{168} has twenty-one elements. Hence a is exactly an eightfold pole. We therefore obtain twenty-one eightfold poles.

The four substitutions of fourth order in G_A have their poles on the principal axis A. Their isolated poles, transformed from a, are again eightfold poles. On each principal axis lie four eightfold poles, and through each eightfold pole pass four principal axes. This follows equally from the stabilizer calculation and from the incidence count

$$21 \cdot 4 = 21 \cdot 4.$$

The two remaining fixed points of Θ on the principal axis are not eightfold. They are poles of substitutions of order four and give the system of fourfold poles. There are forty-two of these points; each principal axis contains two of them. Thus the principal axes already account for the systems

$$21P_8, \qquad 42P_4.$$

The notation P_v indicates a v-fold pole.

The result of this paragraph is the following concise statement. The group has twenty-one principal axes. Each principal axis contains four eightfold poles and two fourfold poles. Through each eightfold pole pass four principal axes. The principal axes and the eightfold poles are thus paired in a symmetric incidence system, while the fourfold poles appear in pairs on each axis.

§119. The Threefold and Sixfold Poles

It remains to examine the substitutions of third order and the related sixfold poles. A representative is the cyclic substitution χ, or a conjugate of it. Its characteristic equation has the cube roots of unity as roots. Accordingly the corresponding eigenvectors give three fixed points in the projective plane.

Two of these points form a pair belonging to the substitution of third order. Their coordinates may be written, in a suitable normalization, in the form

$$x_1 = \rho x_2 = \rho^2 x_3,$$

or in the two conjugate forms obtained by replacing ρ by ρ^2, where ρ is a primitive cube root of unity. These points are fixed by the cyclic group generated by the substitution, but by no larger subgroup. Hence they are threefold poles.

Weber proves this by excluding the other possibilities. If a larger group fixed one of these two points, the additional substitutions would have to be of orders seven, four, or two. But the coordinates of the sevenfold and fourfold poles are rational functions of the seventh root ϵ, while the coordinates here contain the independent cubic irrationality ρ. A substitution of second order would put the point on a principal axis, and the explicit equations of the principal axes show that this cannot occur. Therefore the stabilizer has exactly three elements.

Thus there are

$$\frac{168}{3} = 56$$

threefold poles. They are paired, two by two, according to the cyclic subgroup of order three to which they belong. Equivalently, the twenty-eight cyclic subgroups of order three give twenty-eight pairs of threefold poles. The full group is transitive on the fifty-six points.

Let

$$T = x_1 + x_2 + x_3.$$

The equation $T = 0$ is the line joining one such pair of threefold poles. The line remains fixed under the substitution χ and changes sign under a suitable involution. Under the full group it gives twenty-eight distinct lines, which Weber denotes collectively by the lines T.

On the line $T = 0$ lie three of the eightfold poles described in the preceding paragraph. They are obtained from the representative eightfold pole by the substitutions belonging to the same normalizer. Thus:

> Each line T contains three eightfold poles and two threefold poles.

Since the group acts transitively on the T-lines, this statement holds for all twenty-eight of them.

The third pole of the substitution χ, call it t, is of a different kind. It is fixed by six substitutions, namely by the identity, by the two powers of χ, and by three further substitutions which, together with these, form a group of order six. To show that the stabilizer is no larger, Weber again uses the list of possible orders and the already determined axes. The point t lies on three principal axes, those corresponding to the three involutions in the stabilizer. If another involution fixed t, it would have to add a fourth principal axis through t, but the explicit equations show that no such fourth axis passes through it.

Therefore the point t is exactly sixfold. There are

$$\frac{168}{6} = 28$$

sixfold poles. They form a transitive system. Through each sixfold pole pass exactly three principal axes. Conversely, since there are twenty-one principal axes, each containing the same number of sixfold poles, the incidence count gives

$$28 \cdot 3 = 21 \cdot 4,$$

so that each principal axis contains four sixfold poles.

The twenty-eight lines T and the twenty-eight sixfold poles t are thus linked to the same order-three subgroups, but the sixfold pole does not lie on the corresponding line T. This will be important in the configuration below.

§120. The Configuration of the Group G_{168}

We now collect the results in a single configuration. No new calculation is involved; the purpose is only to make the statements of the last paragraphs more transparent.

The configuration consists first of

$$21 \text{ lines } A \quad \text{(the principal axes)},$$

$$21 \text{ points } a \quad \text{(the eightfold poles)},$$

$$28 \text{ lines } T, \qquad 28 \text{ points } t \quad \text{(the sixfold poles)}.$$

The whole system is transformed into itself by the 168 substitutions of the group.

The principal incidence statements are:

on each line A lie four points a, through each point a pass four lines A,

on each line T lie three points a, through each point t pass three lines A.

Moreover the points a and t together form the complete system of intersections of the twenty-one lines A. Indeed, twenty-one lines have

$$\frac{21 \cdot 20}{2} = 210$$

pairwise intersections when multiplicities are counted. At a point a four lines meet; this contributes $\binom{4}{2} = 6$ pairwise intersections, and at a point t three lines meet; this contributes $\binom{3}{2} = 3$. Since

$$21 \cdot 6 + 28 \cdot 3 = 126 + 84 = 210,$$

all intersections are accounted for.

From the same count one obtains the remaining incidences. Since through every point t pass three lines A, and since all lines A are conjugate, the number of points t lying on a fixed line A is

$$\frac{28 \cdot 3}{21} = 4.$$

Similarly every point a lies on four of the lines T. Finally no point t lies on a line T; otherwise the incidence counts and the stabilizers would be incompatible.

If the v-fold poles are denoted by P_v, the systems of poles are as follows:

system	number of points
P_8	21
P_6	28
P_4	42
P_3	56
P_7	24
P_2	infinitely many

The twofold poles are all points on the principal axes, except for the special points already counted as P_4, P_6, P_8. On each principal axis lie two P_4's; on each line T lie two P_3's. The sevenfold poles do not lie on the principal axes, and it will follow from the curve theory that they do not lie on the lines T either.

The corresponding system of axes is analogous. Every line through an eightfold pole is an axis; among them are the principal axes, four of the lines T, and two further lines joining that eightfold pole to suitable fourfold poles. It is natural, therefore, to regard the eightfold poles as the principal poles. There are also twenty-eight pairs of connecting lines from a sixfold pole to the two associated threefold poles; and there are twenty-four lines joining paired sevenfold poles, represented by the sides of the coordinate triangle.

This configuration is the projective image of the abstract structure of G_{168}. The incidence relations determine the later invariant curves almost without further calculation: if an invariant curve contains one point of any of these systems, it contains the whole system; if a line is tangent or bitangent in one place, all its conjugates have the corresponding contact.

§121. Invariant Curves of the Group G_{168}

Let $\varphi(x_1, x_2, x_3)$ be a homogeneous ternary form of degree m. Under the substitutions of G_{168} it generally passes into 168 different forms. For special forms this number may be smaller. In that case the substitutions by which φ is unchanged form a subgroup G' of G_{168}; the number of distinct forms obtained from φ is then the index of G'. The form φ is an absolute invariant of G', or at least a relative invariant if it is multiplied by a constant factor.

The equation

$$\varphi = 0$$

represents a curve in the projective plane whose coordinates are x_1, x_2, x_3. The substitutions of the group map this curve to its conjugate curves. If not all the images are distinct, then the curve is left invariant by a subgroup. In particular, if φ is invariant under the whole group, the curve is mapped onto itself by all 168 collineations.

Projective properties are preserved by these collineations. If a line is a tangent, an inflection tangent, or a bitangent of a curve, then every image of the line has the same relation to the corresponding image of the curve. Likewise double points, inflection points, cusps, points of contact of bitangents, and all similar projective properties pass to their images.

We are chiefly concerned with forms which remain unchanged under the whole group. Since the group is simple, the invariants that occur here are absolute. If $\Phi(x_1, x_2, x_3)$ is such an invariant, the curve

$$\Phi = 0$$

will be called an invariant curve of the group. A point on an invariant curve carries with it all its images. In general a point has 168 distinct images. The number is smaller only when the point is one of the poles; then the possible orbit sizes are

$$24 \text{ for } P_7, \qquad 21 \text{ for } P_8, \qquad 28 \text{ for } P_6, \qquad 42 \text{ for } P_4, \qquad 56 \text{ for } P_3,$$

and, for a connected system of twofold poles, 84. Thus, if one pole of a given system lies on an invariant curve, all poles of that system lie on it.

The subsequent determination of the invariants is therefore guided by the configuration. The degree of a curve and Bezout's theorem control which systems of poles it can contain. Conversely, an invariant curve of small degree is often forced once its intersections with the fundamental configuration are known.

§122. The First Invariant of the Group G_{168} and the Fundamental Curve

We now form the invariants explicitly. It is enough to require invariance under the three generators r, χ, ω introduced in §115, since the whole group is generated by them. The substitution χ cyclically permutes the variables, while r is diagonal:

$$x_1, x_2, x_3 \longmapsto \epsilon x_1, \epsilon^2 x_2, \epsilon^4 x_3.$$

If a monomial

$$x_1^{h_1} x_2^{h_2} x_3^{h_3}$$

occurs in an invariant, invariance under r requires

$$h_1 + 2h_2 + 4h_3 \equiv 0 \pmod{7}.$$

Invariance under χ requires that, together with this monomial, its two cyclic transforms occur with corresponding coefficients, unless $h_1 = h_2 = h_3$.

No invariant of degree smaller than three can satisfy the congruence. In degree three the only possibility is $x_1 x_2 x_3$, but this product is not left invariant by ω. Thus the least possible degree is four. Solving

$$h_1 + h_2 + h_3 = 4, \qquad h_1 + 2h_2 + 4h_3 \equiv 0 \pmod{7}$$

in non-negative integers gives only the three cyclic possibilities

$$(h_1, h_2, h_3) = (3, 1, 0), \quad (0, 3, 1), \quad (1, 0, 3).$$

Hence the only form of fourth degree invariant under r and χ, apart from a constant factor, is

$$f(x_1, x_2, x_3) = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_1.$$

It remains to verify invariance under ω. Weber substitutes

$$\begin{aligned} x_1 &= \alpha y_1 + \beta y_2 + \gamma y_3, \\ x_2 &= \beta y_1 + \gamma y_2 + \alpha y_3, \\ x_3 &= \gamma y_1 + \alpha y_2 + \beta y_3, \end{aligned}$$

where α, β, γ have the values fixed in §115. If

$$F(y_1, y_2, y_3) = f(x_1, x_2, x_3),$$

then the second derivatives of F, computed with the relations among α, β, γ, reduce to exactly the second derivatives of

$$y_1^3 y_2 + y_2^3 y_3 + y_3^3 y_1.$$

By Euler's theorem for homogeneous functions it follows that

$$F(y_1, y_2, y_3) = y_1^3 y_2 + y_2^3 y_3 + y_3^3 y_1.$$

Thus f is indeed an invariant of the group. Apart from multiplication by a constant, it is the unique invariant of degree four.

The curve

$$f = 0$$

is the fundamental curve of the group. It is a plane quartic. The side $x_1 = 0$ of the coordinate triangle cuts the curve in three coincident points at one vertex and in a fourth point at another vertex. Thus the vertices of the coordinate triangle are inflection points, and the sides of the coordinate triangle are inflection tangents. With the usual notation, side 1 is the inflection tangent at vertex 2, side 2 at vertex 3, and side 3 at vertex 1.

Since the vertices of the coordinate triangle are sevenfold poles, all the sevenfold poles lie on the fundamental curve. They are all inflection points, and they form eight inflection triangles, corresponding to the eight cyclic groups of order seven.

The threefold poles also lie on the fundamental curve. If ρ is a primitive cube root of unity and one substitutes

$$x_2 = \rho x_1, \qquad x_3 = \rho^2 x_1$$

(or the conjugate relation), the equation $f = 0$ is satisfied because

$$1 + \rho + \rho^2 = 0.$$

Their connecting line

$$T = x_1 + x_2 + x_3 = 0$$

therefore meets the quartic at the two threefold poles. In fact the four intersections coalesce two by two; the line T is a bitangent of the fundamental curve. Consequently all twenty-eight lines T are bitangents, and their points of contact are precisely the fifty-six threefold poles.

The sixfold and eightfold poles do not lie on the fundamental curve. For the sixfold pole one may use coordinates proportional to $(1 : 1 : 1)$, which gives $f = 3$. For the eightfold pole Weber substitutes the coordinates found in §118 and uses the relations among α, β, γ; the resulting value is non-zero.

It remains to determine the intersections of the principal axes with the quartic. Since all intersections of principal axes with one another are the points P_8 and P_6, and neither of these systems lies on the curve, no two principal axes meet on the curve. If a fourfold pole on a principal axis lay on the quartic, its conjugate fourfold pole on the same axis would also lie there; then the axis would have to be a bitangent. But the twenty-eight lines T already give all bitangents of a nonsingular quartic. Hence the fourfold poles do not occur there. The intersections of a principal axis with the quartic are therefore twofold poles.

We denote these special twofold poles by P_2. The distinguished point systems are now:

$$24P_7, \qquad 56P_3, \qquad 84P_2 \quad \text{on the curve } f = 0,$$

and

$$21P_8, \qquad 42P_4, \qquad 28P_6 \quad \text{not on the curve.}$$

All other points of the curve are transformed into 168 distinct points.

This also proves that the fundamental curve has no double point and cannot decompose into curves of lower degree. For if it had a double point, all its images would be double points; a quartic cannot have so many double points unless it contains a repeated component. But then f would have to be the square of a quadratic invariant, and no quadratic invariant exists. Directly, the partial derivative equations force

$$x_1 = x_2 = x_3 = 0,$$

which is not a point of the projective plane. Hence the quartic is nonsingular.

For brevity Weber calls a system of points, any one of which can be transformed into any other by substitutions of the group, a system of connected points.

§123. The Higher Invariants

Further invariants are obtained from the quartic by forming covariants. The first is the Hessian covariant. With Weber's normalization it is

$$\Delta = \frac{1}{54} \det \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right),$$

an invariant of degree six. Its developed expression is again invariant under the cyclic permutations and under the diagonal substitution. The curve $\Delta = 0$ contains the vertices of the coordinate triangle; consequently, by the group action, it contains all sevenfold poles. Since the intersections of $f = 0$ and $\Delta = 0$ have total number

$$4 \cdot 6 = 24,$$

these sevenfold poles form the complete system of intersections of the fundamental quartic with its Hessian curve.

A second covariant, of degree fourteen, is obtained from the determinant described in the general theory of covariants. Weber denotes it by C. In determinant form it is the bordered determinant built from the second derivatives of f and the first derivatives of Δ. Written symbolically,

$$C = \det \begin{pmatrix} f_{11} & f_{12} & f_{13} & \Delta_1 \\ f_{21} & f_{22} & f_{23} & \Delta_2 \\ f_{31} & f_{32} & f_{33} & \Delta_3 \\ \Delta_1 & \Delta_2 & \Delta_3 & 0 \end{pmatrix},$$

up to the constant factor chosen by Weber. When developed, it is a cyclic-sum expression in the variables x_1, x_2, x_3; Weber writes the expansion explicitly and uses the notation Σ for the sum of the three cyclic terms obtained from the first term by cyclic permutation of the variables.

For the subsequent geometry only two features of this expression are immediately needed. First, from the leading term one sees that the curve $C = 0$ does not pass through the vertices of the coordinate triangle. Second, being an invariant curve, if it contains any one point of a connected pole system, it contains the entire system. It will meet the fundamental curve in the system of threefold poles.

A fourth invariant, of degree twenty-one, is obtained by taking the functional determinant of the three forms f, Δ, C:

$$K = \frac{1}{14} \det \begin{pmatrix} f_1 & \Delta_1 & C_1 \\ f_2 & \Delta_2 & C_2 \\ f_3 & \Delta_3 & C_3 \end{pmatrix}.$$

Only the first terms are needed in the present paragraph to see that this curve also does not pass through the vertices of the coordinate triangle. The full expression is lengthy; Weber refers to Gordan's computation of the complete formula.

The four forms

$$f, \qquad \Delta, \qquad C, \qquad K$$

have degrees

$$4, \qquad 6, \qquad 14, \qquad 21.$$

They will now be shown to generate all invariants of the group. Geometrically, $f = 0$ is the fundamental quartic, $\Delta = 0$ is its Hessian, $C = 0$ is the next distinguished invariant curve, and $K = 0$ is the product of the twenty-one principal axes.

The last assertion is already suggested by the degree. The twenty-one principal axes form an invariant curve of degree twenty-one. In §124 Weber proves that there is, up to a constant factor, only one invariant of degree twenty-one. Hence the invariant K must split into the product of the twenty-one linear forms defining the principal axes.

§124. The Complete System of Invariants

We now prove that the four forms

$$f, \quad \Delta, \quad C, \quad K$$

constitute a complete system of invariants for the group. This means that every invariant of G_{168} can be represented as an integral rational function of these four forms. The proof uses Bezout's theorem: two curves of degrees m and n which have more than mn points in common must have a common component; points of contact are counted twice or with the corresponding higher multiplicity.

Let Φ be an invariant of degree m which does not contain f as a factor. Consider the intersections of the curve $\Phi = 0$ with the fundamental quartic $f = 0$. Among them let the systems P_7, P_3, P_2 occur with multiplicities h_1, h_2, h_3, and let there also be h_4 arbitrary connected systems of 168 points. Since the intersections with the quartic are counted with multiplicity, one obtains

$$4m = 24h_1 + 56h_2 + 84h_3 + 168h_4,$$

and therefore

$$m = 6h_1 + 14h_2 + 21h_3 + 42h_4. \tag{1}$$

The integers h_1, h_2, h_3, h_4 are non-negative and cannot all vanish.

The smallest value of m is therefore 6. Every invariant of degree six must pass through the twenty-four points P_7, just as Δ does. If Δ' is another invariant of degree six, choose a constant a so that $\Delta' - a\Delta$ passes through one further point of the quartic. Then, by Bezout's theorem, $\Delta' - a\Delta$ must be divisible by f. The quotient would be an invariant of degree two, which does not exist. Hence Δ' is a constant multiple of Δ.

The same reasoning gives the invariants of degrees twelve and eighteen. If an invariant of degree twelve has the same intersection multiplicities with the fundamental quartic as Δ^2, then after subtracting a suitable multiple of Δ^2 it must be divisible by f; the quotient is an invariant of degree eight and is therefore a multiple of f^2. Thus every invariant of degree twelve has the form

$$a\Delta^2 + bf^3.$$

Similarly every invariant of degree eighteen has the form

$$a\Delta^3 + bf^3\Delta.$$

Consequently all invariants of degrees $6, 12, 18$ are functions of f and Δ.

The next possible degree is 14, corresponding to $h_2 = 1$. An invariant of degree fourteen must pass through the fifty-six points P_3, and these are precisely the complete intersection of the curve $C = 0$ with the fundamental quartic. If C' is another invariant of degree fourteen, choose a so that $C' - aC$ is divisible by f. The quotient is an invariant of degree ten. Since there is no invariant of degree ten except the product $f\Delta$, it follows that every invariant of degree fourteen has the form

$$aC + bf\Delta.$$

Conversely every expression of this form is plainly an invariant of degree fourteen.

For degree twenty, equation (1) allows the system consisting of P_7 and P_3. The curve must therefore pass through the eighty intersections of $f = 0$ with $\Delta C = 0$. By the same Bezout argument it follows that an invariant of degree twenty is of the form

$$f(a\Delta^2 + bf^3) + c\Delta C,$$

with arbitrary constants a, b, c. Thus there is no new independent invariant of degree twenty.

Consider now invariants of degree twenty-one. They must pass through the eighty-four twofold poles P_2 on the fundamental quartic. If K_1 and K are two invariants of degree twenty-one, a constant a can be chosen so that $K_1 - aK$ is divisible by f. The quotient would be an invariant of degree seventeen, which cannot occur by (1). Therefore, up to a constant factor, there is only one invariant of degree twenty-one.

But the product of the twenty-one linear equations of the principal axes is an invariant of degree twenty-one. Hence this product must be proportional to K. Thus:

> The invariant K splits into twenty-one linear factors, and these factors, when set equal to zero, are the principal axes of the group.

We next need a relation. For any constant k the expression

$$Q = K^2 - kC^3$$

is an invariant of degree forty-two. The constant may be chosen so that the curve $Q = 0$ passes through a prescribed connected system of points on the fundamental quartic. Hence, given any invariant Φ not divisible by f, one forms products of such curves Q so as to remove all connected systems occurring in the intersection with $f = 0$, and one subtracts a suitable product of powers of Δ, C, K. The resulting invariant still contains all intersections with $f = 0$, and after one additional choice of a constant it has an extra intersection with the quartic. Bezout's theorem then forces it to be divisible by f. This lowers the degree and permits induction.

This proves:

> Every invariant of the group is an integral rational function of the four fundamental invariants f, Δ, C, K.

Since K^2 is of even degree, it must itself be expressible in terms of f, Δ, C. Gordan computed the relation explicitly. In Weber's notation it is

$$\begin{aligned} K^2 = C^3 &- 88f^2\Delta C^2 + 16\cdot 63 f\Delta^4 C + 17\cdot 64 f^4\Delta^2 C - 256 f^7 C \\ &+ 27\cdot 64\Delta^7 - 128\cdot 469 f^3\Delta^5 + 43\cdot 512 f^6\Delta^3 - 2048 f^9\Delta. \end{aligned}$$

Every term has weighted degree forty-two, when the weights of f, Δ, C are $4, 6, 14$.

This relation allows all powers of K except the first to be eliminated from any expression in the invariants. Consequently an invariant of even degree can be expressed rationally through

$$f, \quad \Delta, \quad C,$$

whereas an invariant of odd degree is the product of K and an invariant of even degree. The invariant theory of the ternary group G_{168} is thereby complete.

The next section begins the application of this invariant system to the form problem of the group G_{168} and to the theory of equations of the seventh degree.

Fifteenth Section. The Form Problem of the Group G_{168} and the Theory of Equations of the Seventh Degree

§125. The Resolvents of the Form Problem

The form problem for the group G_{168}, in the sense of §43, gives first of all, by the general theory, an equation of degree 168, whose coefficients are invariants. Every divisor of the group, however, gives a resolvent of lower degree, namely of degree equal to the index of that divisor.

We shall consider here only the two most interesting cases of such divisors. The first is the octahedral group

$$G_{24} = \langle \chi, \omega, \theta \rangle,$$

and the second is the subgroup of order 21,

$$G_{21} = \langle \tau, \chi \rangle,$$

which was already met in §72. They lead respectively to resolvents of the seventh and eighth degrees.

We begin with the resolvents of degree seven. In their construction one may start from the principal axes of the group. A principal axis is left fixed by the group $\chi^a \theta^b$, and under the whole group G_{168} it goes into three different lines. Thus there must be seven triples of principal axes, and these triples are determined by an equation of the seventh degree.

Put, in the notation of §118,

$$\begin{aligned} A_1 &= \alpha\gamma x_1 + \gamma\beta\varepsilon^2 x_2 + \alpha\beta\varepsilon^4 x_3, \\ A_2 &= \beta\gamma\varepsilon^2 x_1 + \alpha\beta\varepsilon^4 x_2 + \alpha\gamma x_3, \\ A_3 &= \alpha\beta\varepsilon^4 x_1 + \alpha\gamma x_2 + \beta\gamma\varepsilon^2 x_3. \end{aligned} \tag{1}$$

Then $A_1 = 0, A_2 = 0, A_3 = 0$ are the equations of three of these axes. The substitution χ cyclically permutes A_1, A_2, A_3. The substitution θ, when actually carried out and reduced by the formulae of §115, produces the corresponding interchange among the same three functions. Since ω is expressible by θ and χ, it follows that the quantities A_1, A_2, A_3 are permuted among themselves by the octahedral subgroup. Their symmetric functions therefore are roots of resolvents of the seventh degree.

The simplest symmetric function of these three quantities is

$$A_1^2 + A_2^2 + A_3^2.$$

After multiplication by a convenient numerical factor, it will be used as the unknown of the resolvent. If the sum is arranged according to $x_1^2 + x_2^2 + x_3^2$ and $x_1x_2 + x_2x_3 + x_3x_1$, one obtains an expression of the form

$$\lambda(x_1^2 + x_2^2 + x_3^2) + \mu(x_2x_3 + x_3x_1 + x_1x_2).$$

By substituting the values of α, β, γ and simplifying with the relations among the seventh roots of unity, this assumes the form

$$z = x_1^2 + x_2^2 + x_3^2 - \frac{1 - \sqrt{-7}}{2}(x_2x_3 + x_3x_1 + x_1x_2). \tag{2}$$

This function is taken as one root of the resolvent of the seventh degree.

The remaining six roots are obtained from it by the substitutions τ^r. They may all be written in a common form

$$z_r = \varepsilon^r x_1^2 + \varepsilon^{2r} x_2^2 + \varepsilon^{4r} x_3^2 - \frac{1-\sqrt{-7}}{2}(\varepsilon^{3r} x_2 x_3 + \varepsilon^{5r} x_3 x_1 + \varepsilon^{6r} x_1 x_2), \quad r = 0, 1, \ldots, 6. \quad (3)$$

The coefficients of the equation whose roots are the seven quantities z_r are invariants of the group. Their degrees are easily determined. If a_ν is the coefficient of the term of degree $7-\nu$ in the unknown, after the leading coefficient has been made equal to one, then a_ν is an invariant of degree 2ν. Since no quadratic invariant exists, one must have $a_1 = 0$. The other coefficients are expressed linearly and homogeneously by the fundamental invariants:

$$a_2 \text{ by } f, \quad a_3 \text{ by } \Delta, \quad a_4 \text{ by } f^2, \quad a_5 \text{ by } f\Delta,$$

$$a_6 \text{ by } \Delta^2, f^3, \quad a_7 \text{ by } C, K.$$

Thus only a finite number of numerical constants has to be computed. They are found by comparing a few terms in the expressions of the elementary symmetric functions of the z_r with the invariant expressions already obtained in §123. Apart from rational numbers these constants can contain only $\sqrt{-7}$.

We first perform the calculation under the simplifying hypothesis $f = 0$. This case is especially important because, like the icosahedral equation, it leads to transformation equations in the theory of elliptic functions. It is enough to compute the first terms in the power sums $\sum z_r^3$ and $\sum z_r^6$, and then to apply Newton's formulae. The last coefficient is obtained from the term $x_1^7 x_2^7 x_3^7$ in the product of the seven z_r. The resulting special resolvent is

$$u^7 - 7u^5 g - 7\frac{1+\sqrt{-7}}{2} u^3 g^2 - g^3 = 0. \quad (4)$$

Here the variable has been normalized so that only the parameter g remains.

If the same substitution is made in the general resolvent, in which f is no longer set equal to zero, then one has to introduce a second parameter, say h. The coefficients of the resolvent, apart from numerical factors, occur successively in the forms

$$h, quadf, \; h, \quad hg, \quad g^2, \; h^2, \quad g^2, \; h^2 g,$$

more precisely as homogeneous linear expressions in the indicated products. The computation is carried out exactly as in the special case. For simplification one may put one of the variables equal to zero and still obtains sufficiently many equations for all coefficients.

The two functions g, h are fractional invariants depending only on the ratios $x_1 : x_2 : x_3$. Conversely every other invariant of the same kind is rationally expressible by g and h. For, if such an invariant is written as a quotient of two forms of equal degree without common divisor, then numerator and denominator must be integral invariants of the same degree, determined individually up to constant factors. They cannot be of odd degree, for otherwise they would have the common factor K. Hence they are rational functions of f, Δ, C. A monomial

$$f^a \Delta^b C^c$$

of degree m satisfies

$$4a + 6b + 14c = m,$$

and after expressing f and Δ through the normalized parameters g, h, the common power of C cancels in numerator and denominator. Everything is therefore rational in g, h.

In the same way the function u depends only on the ratios $x_1 : x_2 : x_3$. Any other function with the same property and admitting the substitutions of the subgroup G_{24} is rationally expressible by u, g, h. Indeed, by the general theorems on the form problem it is first a rational function of u and of the invariants, and in a unique way it may be written as an integral polynomial in u of degree at most six. Since the function depends only on ratios, its coefficients do so as well, and hence are rational in g, h.

§126. Reduction of the General Seventh-Degree Resolvent to the Special One

In the preceding section we obtained two forms of the seventh-degree resolvent of the form problem. One of them, the special resolvent, belongs to the case $f = 0$, that is, to the case in which the required point lies on the fundamental curve. The other belongs to an arbitrary position of the point.

We shall denote by

$$u_x, \qquad g_x, \qquad h_x$$

the quantities u, g, h when they are formed from the point (x). The general resolvent is then written

$$R(u_x, g_x, h_x) = 0. \tag{1}$$

The special resolvent is obtained from this by putting $h_x = 0$. The quantities u_x, g_x, h_x depend only on the ratios of the coordinates of the point (x).

Klein observed that the solution of the general resolvent can be reduced to the solution of the special one after adjoining the root of a biquadratic equation. To see this, introduce beside the point (x) a second point (y), and form the polar of f,

$$f_1(x, y) = y_1 f_{x_1}(x) + y_2 f_{x_2}(x) + y_3 f_{x_3}(x). \tag{2}$$

If (x) and (y) are transformed simultaneously by the same substitution of G_{168}, this polar remains unchanged.

Let the point (x) be arbitrary, but require the point (y) to lie both on the fundamental curve and on the polar of (x). Thus

$$f(y_1, y_2, y_3) = 0, \qquad f_1(x, y) = 0. \tag{3}$$

For each point (x) there are four corresponding points (y). If (x) is replaced by a connected point under the group, then these four points are likewise replaced by connected points. For the point (y) the quantity h_y is zero, and g_y becomes a four-valued function of (x). The elementary symmetric functions of these four values remain invariant under G_{168}. Consequently g_y is the root of a biquadratic equation whose coefficients are rational functions of g_x, h_x. The root of this biquadratic equation has to be adjoined.

The corresponding u_y is a root of the special resolvent

$$R(u_y, g_y, 0) = 0. \tag{4}$$

Let the four points belonging to the same (x) be denoted by

$$y, \quad y', \quad y'', \quad y'''.$$

For an indeterminate t form

$$(t - u_y)(t - u_{y'})(t - u_{y''})(t - u_{y'''}) = \Phi(t). \tag{5}$$

This expression is left fixed by all substitutions of the subgroup G_{24}, and is therefore rationally expressible by u_x, g_x, h_x.

Since the line represented by the polar equation can be made arbitrary by a suitable choice of (x), the point (x) may be chosen in such a way that, among the four points y, y', y'', y''', no two are connected by the group. Putting $t = u_y$ in (5), one obtains a rational equation

$$\Phi(u_y; u_x, g_x, h_x) = 0. \tag{6}$$

This equation is not satisfied after a substitution from G_{168} which changes u_y into a value different from the four values associated with the same (x). Hence, considered as an equation in u_x, it has exactly one root in common with the general resolvent $R(u_x, g_x, h_x) = 0$. If one takes the greatest common divisor of these two equations, one obtains u_x rationally through u_y, g_x, h_x. This is the desired reduction of the general resolvent to the special one.

The group G_{168} also contains the subgroup G_{21} generated by χ and τ. It gives rise to a resolvent of degree eight. A root of this resolvent may be taken to be the product $x_1 x_2 x_3$. Its coefficients are invariants, of degrees rising as high as the ninety-fourth. We shall not pursue this resolvent further here.

§127. A Permutation Group on Seven Letters of Degree 168

The ternary substitution group of order 168 has a resolvent of degree seven. The Galois resolvent of this equation of the seventh degree is therefore of degree 168. It follows that the full permutation group on seven letters, whose order is

$$1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 = 5040 = 168 \cdot 30,$$

contains a subgroup of order 168. Kronecker first recognized the existence of this subgroup; its closer investigation is of great importance for the general theory of equations of the seventh degree.

We obtain this permutation group by examining the permutations induced by the substitutions of G_{168} among the seven quantities

$$z_0, z_1, \ldots, z_6$$

of §125. If two linear substitutions s_1, s_2 induce the permutations π_1, π_2, then their product must be read in the reverse order: the composed permutation $\pi_1\pi_2$ is induced by $s_2 s_1$. This is merely the convention that the substitution acts first on the variables, whereas the displayed permutation acts first on the list of symbols. In this manner we obtain, corresponding to G_{168}, a permutation group of order 168, denoted by P_{168}. The two groups are isomorphic.

Since τ and ω generate G_{168}, it is enough to determine the corresponding permutations. The substitution τ gives the cyclic permutation

$$\tau = (0, 1, 2, 3, 4, 5, 6). \tag{1}$$

For ω one uses the fact that z_0 is left unchanged by the octahedral subgroup. Thus it is enough to know the effect of $\omega\tau^r$ on z_0. The formulae of §72 give the required values. Consequently one obtains a second generating permutation, together with the cycle (1), for the group P_{168}.

The same group may also be represented as the ternary linear congruence group modulo 2, previously studied in §79. Let the seven letters be denoted by the non-zero triples of residues modulo 2, for instance

$$\begin{aligned} &z_0 = (1,0,0), \qquad && z_1 = (1,1,0), \qquad && z_2 = (1,1,1), \\ &z_3 = (0,1,1), && z_4 = (1,0,1), && z_5 = (0,1,0), \qquad && z_6 = (0,0,1). \end{aligned}$$

Then the substitutions corresponding to the two generators are ternary linear substitutions modulo 2. In particular one may choose a matrix which leaves z_0, z_3, z_6 fixed and interchanges z_1, z_2 and z_4, z_5. Together with the cyclic generator it yields precisely the congruence group of order 168.

Both generating permutations are even. Hence P_{168} is a subgroup of the alternating group on seven letters.

§128. Equations of the Seventh Degree with a Group of Order 168

We now turn to the general theory of those special equations of the seventh degree whose Galois group is reduced to P_{168}.

Let

$$A_0, A_1, A_2, A_3, A_4, A_5, A_6$$

be arbitrary quantities, and let

$$\tau = (A_0, A_1, A_2, A_3, A_4, A_5, A_6) \tag{1}$$

be the cyclic permutation of order seven. If we add the transposition pair

$$\omega = (A_1, A_2)(A_3, A_5), \tag{2}$$

then the compositions and powers of these permutations generate the whole group P_{168}. The permutations corresponding to the ternary generators of G_{168} are readily derived from these.

The element A_0 remains fixed by the subgroup P_{24} contained in P_{168}. It is easy to form a function of the seven quantities A_i which belongs to the group P_{168}. Take, for example, the product of the three letters left fixed by a suitable octahedral subgroup, and add the products obtained by applying the cyclic permutation. Thus one obtains

$$\begin{aligned} v = A_0A_4A_6 &+ A_1A_5A_0 + A_2A_6A_1 + A_3A_0A_2 \\ &+ A_4A_1A_3 + A_5A_2A_4 + A_6A_3A_5. \end{aligned} \tag{3}$$

Application of the generators ω and τ shows that v is unchanged by all permutations of P_{168}.

Since P_{168} has index 30 in the full symmetric group and index 15 in the alternating group, v is a root of an equation of degree 30 whose coefficients are symmetric functions of the A_i, and of an equation of degree 15 whose coefficients also may contain the product of differences of the A_i. If the A_i are the roots of a seventh-degree equation without affect, then v is the root of a resolvent of degree 30. After adjoining the square root of the discriminant, this resolvent splits into two factors of degree 15.

Suppose now that, besides the symmetric functions of the A_i, the quantity v belongs to the rationality field. This may hold from the beginning or may be achieved by adjoining v. Then the A_i are the roots of a special equation of the seventh degree whose Galois group is contained in P_{168}.

The solution of such an equation will be obtained from the form problem of G_{168}. It remains to find functions

$$X_1, \quad X_2, \quad X_3$$

of the roots A_i which are transformed by the permutations of P_{168} in the same way as the variables x_1, x_2, x_3 are transformed by the corresponding ternary substitutions of G_{168}. If these functions have been found, then substitution of them into the invariants of G_{168} gives quantities fixed by P_{168}, and hence belonging to the rationality field. The determination of X_1, X_2, X_3 from these invariant values is precisely the form problem for G_{168}.

Any function of X_1, X_2, X_3 changed by the substitutions of G_{168} is then a resolvent of the given equation of the seventh degree; and since both G_{168} and P_{168} are simple, it is a total resolvent.

Thus, if a non-vanishing system X_1, X_2, X_3 is available, the form problem of G_{168} gives the solution of the equation of the seventh degree with group P_{168}. To reduce this solution further to the special form problem occurring in the theory of elliptic functions, namely to the case $f = 0$, one still has to solve an accessory equation of the fourth degree. This is analogous to the square root needed to reduce the general equation of the fifth degree to the icosahedral equation.

Everything now depends on constructing the functions X_1, X_2, X_3 of the roots A_i. For this purpose we shall first need a few elementary propositions from general invariant theory.

§129. Contragredient Groups

We have already introduced in §37 the notion of contragredient transformations. Let

$$y = A(x), \qquad \xi = A^{-T}(\eta)$$

be two transposed substitutions. The two rows of variables (x_1, x_2, x_3) and (ξ_1, ξ_2, ξ_3), and likewise their transformed rows, are called contragredient.

Under the substitution $y = A(x)$ every function $\Phi(x_1, x_2, x_3)$ goes over into a function of the y's. Differentiating gives the first fundamental fact:

> The row of variables (x_1, x_2, x_3) and the row of differential operators
>
> $$\left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \frac{\partial}{\partial x_3}\right)$$
>
> are contragredient.

By repeated application of this assertion, higher differentials transform in the corresponding symmetric powers. In practical terms, to express the m-th derivatives with respect to x by the derivatives with respect to y, one expands the relevant products in the variables and then replaces products of the contragredient variables by the corresponding differential operators. The coefficients which occur are integral rational functions of the coefficients of the substitution.

This rule may be put in a more general form. Suppose a form $\varphi(x)$ is carried by the substitution $y = A(x)$ into $\Phi(y)$, and a second form $\psi(\xi)$ is carried by the transposed substitution into $\Psi(\eta)$. Then a new transformation is obtained by replacing in ψ and Ψ the powers and products of the variables by the corresponding differentiations of φ and Φ. In other words, one may replace symbolic products in the contragredient variables by differential operators acting on the original form.

From the composition law of linear substitutions follows immediately:

> If A runs through a group G, then the transposed inverse substitution A^{-T} runs through a group G_1, called contragredient to G. If $A, B \in G$ correspond to $A_1, B_1 \in G_1$, then the order of multiplication is reversed. The two groups are isomorphic if to A one assigns not A_1, but A_1^{-1}.

The invariants of G_1 are called the contravariants of G. Conversely, the invariants of G are the contravariants of G_1.

The differential rule just stated gives two useful principles.

First, if in a contravariant of G the powers and products of the variables are replaced by the corresponding derivatives of an invariant of G, the result is again an invariant of G.

Second, if in an invariant of G the powers and products of the variables are replaced by the corresponding derivatives of a contravariant, then the result is again a contravariant.

These elementary rules are the mechanism by which the required functions for the seventh-degree equation will be formed.

§130. Solution of the Seventh-Degree Equation with Group P_{168} by the Form Problem of G_{168}

The ternary linear substitution group G_{168} has the remarkable property of being contragredient to itself. Its generating substitutions τ and ω are unchanged by transposition; hence the transpose of any substitution of the group is again a substitution occurring in the group. Thus the contravariants of G_{168}, apart from the names of the variables, coincide with its invariants.

The octahedral subgroup contained in G_{168}, however, is not contragredient to itself as a subgroup. For example, the transpose of one of its substitutions is a substitution of G_{168} which need not lie in the octahedral subgroup. This distinction is important for what follows.

Let η_0 be any function of the seven quantities

$$A_0, A_1, \ldots, A_6$$

belonging to the subgroup P_{24}; the root A_0 itself may be taken as an example. By the cyclic permutations τ^r it goes into

$$\eta_0, \eta_1, \ldots, \eta_6.$$

The sums

$$\omega_r = \sum_{s=0}^{6} \varepsilon^{rs} \eta_s, \qquad r = 0, 1, \ldots, 6, \tag{1}$$

give a system of functions which are linearly, but not ternarily, transformed by the permutations of P_{168}. This is because the η_s may be expressed linearly in terms of the ω_r.

A ternarily transforming system of three functions is obtained as follows. Take three functions $\eta_0, \eta_0', \eta_0''$, each belonging to the subgroup P_{24}, and form from them the corresponding Fourier sums

$$\omega_r, \quad \omega_r', \quad \omega_r''.$$

Using the contragredient invariants and the differential rule of §129, one constructs a trilinear expression in these three rows which remains invariant when a permutation from P_{168} is applied to the roots and the corresponding contragredient substitution is applied to the variables.

Let $f(\xi_1, \xi_2, \xi_3)$ denote the biquadratic contravariant

$$f(\xi_1, \xi_2, \xi_3) = \xi_1^3\xi_2 + \xi_2^3\xi_3 + \xi_3^3\xi_1, \tag{2}$$

written in variables contragredient to (x_1, x_2, x_3). If in this contravariant the products of the variables are replaced by the corresponding derivatives of a suitable invariant expression in the three rows $\omega, \omega', \omega''$, the result is a linear form

$$L = X_1\xi_1 + X_2\xi_2 + X_3\xi_3. \tag{3}$$

The coefficients X_1, X_2, X_3 are functions of the roots A_i. The construction is such that L remains invariant if the roots A_i are subjected to a permutation π of P_{168} and simultaneously the variables ξ are subjected to the corresponding contragredient substitution.

If π carries X_1, X_2, X_3 into Y_1, Y_2, Y_3, and if

$$A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$$

is the corresponding ternary substitution, invariance of L gives

$$Y_1\eta_1 + Y_2\eta_2 + Y_3\eta_3 = X_1\xi_1 + X_2\xi_2 + X_3\xi_3. \tag{4}$$

Expanding this identity yields

$$\begin{aligned} Y_1 &= a_1X_1 + b_1X_2 + c_1X_3, \\ Y_2 &= a_2X_1 + b_2X_2 + c_2X_3, \\ Y_3 &= a_3X_1 + b_3X_2 + c_3X_3. \end{aligned} \tag{5}$$

Thus application of the permutation π to the functions X_1, X_2, X_3 has exactly the same effect as the corresponding linear substitution of G_{168}. These are therefore precisely the functions required at the end of §128.

§131. Possibility of Determining the Functions X_1, X_2, X_3

To make the reduction of the equation of the seventh degree with group P_{168} to the form problem of G_{168} complete, one point remains to be checked. We have to show that the initial functions $\eta_0, \eta_0', \eta_0''$ can be chosen in such a way that the functions X_1, X_2, X_3 do not vanish identically.

For this purpose the construction of the quantities X_i has to be examined a little more closely. When the substitutions described in the preceding section are inserted into the symbolic formulae, the coefficients can be written as trilinear forms in the three rows

$$\eta_r, \quad \eta_r', \quad \eta_r''.$$

Their coefficients are rational functions of the seventh root of unity ε. In a convenient abbreviated determinant notation the three functions take the form

$$\begin{aligned} X_1 &= (q_3, p_2, q_1) + (p_1, q_1, q_2) + (p_1, q_3, p_3), \\ X_2 &= (q_2, q_1, p_3) + (q_3, p_2, q_2) + (q_2, p_2, q_1), \\ X_3 &= (q_2, q_3, q_2) + (q_3, q_1, p_3) + (q_2, p_2, p_3). \end{aligned} \tag{1}$$

The precise letters are less important than the consequence: these are not identically zero trilinear forms.

Indeed the variables may be linearly replaced by Fourier coordinates. Put

$$\eta_r = \sum_{s=0}^{6} a_s \varepsilon^{rs}, \qquad \eta_r' = \sum_{s=0}^{6} a_s' \varepsilon^{rs}, \qquad \eta_r'' = \sum_{s=0}^{6} a_s'' \varepsilon^{rs}. \tag{2}$$

The determinant of this transformation is, up to sign, the product of all differences $\varepsilon^r - \varepsilon^s$, and is not zero. Hence the transformation is invertible. If X_1, X_2, X_3 were identically zero in the η's, they would also be identically zero in the a's. But the explicit formulae show that suitable values of the a's make the three functions non-zero.

Since the coefficients of these forms lie in the cyclotomic field of seventh roots of unity, one may choose rational numerical values of the auxiliary variables so that the functions X_1, X_2, X_3 assume prescribed non-zero values. The desired system of functions therefore exists. This completes the proof that the solution of the special seventh-degree equations with group P_{168} is obtained by the form problem of G_{168}, with only the auxiliary equation of degree four required in passing from the general to the special form of that problem.

Fourth Book. Algebraic Numbers

Sixteenth Section. Numbers and Functionals of an Algebraic Field

§132. Definition of Algebraic Numbers

A number θ is called an algebraic number if it satisfies an algebraic equation with rational coefficients,

$$f(\theta) = 0. \tag{1}$$

Among all such equations there is one of least degree. Let it be

$$f(x) = x^n + a_1 x^{n-1} + \cdots + a_n = 0, \tag{2}$$

chosen monic. This equation is irreducible over the field of rational numbers. For if $f(x)$ decomposed into two rational factors of smaller degree, then θ would satisfy one of those factors, contrary to the minimal choice of n.

If n is the degree of the rational equation of least degree satisfied by θ, then θ is called an algebraic number of degree n.

§133. Integral Algebraic Numbers

An algebraic number θ is called an integral algebraic number, or simply an integer in the algebraic sense, if it satisfies an equation

$$\theta^m + A_1\theta^{m-1} + \cdots + A_m = 0 \tag{1}$$

whose coefficients A_i are rational integers.

It is enough that among the infinitely many equations of this form satisfied by θ there be one with integral coefficients. The integral algebraic numbers contain, as a special case, the ordinary rational integers. Positive rational integers will also be called natural numbers. When we speak of integers without qualification in this part, we mean algebraic integers, rational or irrational.

First, if an algebraic integer is rational, it is necessarily a rational integer. For let $\theta = P/Q$, with P, Q rational integers without common divisor and $Q > 0$. Substitution in (1) gives

$$P^m + A_1P^{m-1}Q + \cdots + A_mQ^m = 0.$$

Every prime divisor of Q would then have to divide P, which is impossible unless $Q = 1$.

Second, the sum, difference, and product of two algebraic integers are again algebraic integers. Let α, β be algebraic integers of degrees μ, ν. If

$$\omega = \alpha + \beta, \quad \alpha - \beta, \quad \alpha\beta,$$

then the $\mu\nu$ quantities obtained from the conjugates of α and β may be used as a finite system closed under multiplication by ω, with integral coefficients after reduction by the equations of α and β. Eliminating this system gives a monic equation with integral coefficients for ω. Hence ω is an algebraic integer.

Third, if $f(x)$ is irreducible over the rational numbers and one root of $f(x) = 0$ is an algebraic integer, then all roots are algebraic integers. Indeed any rational polynomial vanishing at one root is divisible by the irreducible polynomial f, and therefore vanishes at every conjugate root. Applying an integral equation for the first root gives the same assertion for all conjugates. The elementary symmetric functions of the roots then show that the coefficients of the monic irreducible equation are rational integers.

Thus, if θ is an algebraic integer, then its equation of least degree, written monically, has integral coefficients.

Finally, every algebraic number can be made integral by multiplication by a natural number. If

$$\theta^m + A_1\theta^{m-1} + \cdots + A_m = 0$$

and the rational coefficients have common denominator a, then multiplication by a^m gives a monic equation with integral coefficients for $a\theta$. Thus $a\theta$ is an algebraic integer.

§134. Algebraic Fields

In the thirteenth section of the first volume we saw how, from a root θ of an equation of degree n irreducible over a field, one obtains an algebraic field $\mathfrak{R}(\theta)$ over that field. The fields derived from the various roots of the irreducible equation are called conjugate fields; some of them, or even all of them, may coincide.

Let R denote the rational field. Every algebraic number θ of degree n gives rise to an algebraic number field

$$R(\theta),$$

which will be called briefly an algebraic number field of degree n. It has been shown earlier that one can always find an algebraic number field containing any finite number of prescribed algebraic numbers. Hence no generality is lost, in any discussion involving finitely many algebraic numbers, by supposing that they all lie in one algebraic number field.

Every number ω of such a field can be represented as an integral function $\varphi(\theta)$ of θ with rational coefficients. To ω there correspond in the conjugate fields certain conjugate numbers. These may coincide in part; accordingly one distinguishes primitive and imprimitive numbers of the field. A symmetric function of the conjugates is rational.

Among the symmetric functions of the conjugates two are especially important: their sum and their product. The first is called the trace, the second the norm of ω. We write

$$S(\omega), \qquad N(\omega).$$

Repeated conjugates are counted with their multiplicities.

Since the four fundamental operations can be carried out in every algebraic number field as in the rational field, one asks how far the elementary laws of arithmetic familiar from rational integers remain valid in an arbitrary algebraic number field. The first question concerns factorization of integers into prime factors. In general this factorization fails if one uses only the numbers of the algebraic field. The computational material must be enlarged in order to restore simple laws.

Kummer first solved this great problem for numbers built from roots of unity, the cyclotomic numbers, by creating ideal numbers. Dedekind gave another general solution, without exceptions, by introducing the ideals named after him. Kronecker gave a third approach through an arithmetical theory of algebraic quantities. Weber will use here a method close to these theories but adapted to the applications that follow.

§135. Integral Functions in an Algebraic Field

Let Ω be an algebraic number field. We shall consider integral functions in any number of independent variables whose coefficients are numbers of Ω. Such a function is a finite sum of terms

$$\alpha x^r y^s z^t \cdots ,$$

where the exponents are non-negative integers and $\alpha \in \Omega$. We write

$$\varphi(x, y, z, \ldots) = \sum \alpha x^r y^s z^t \cdots . \tag{1}$$

The expression is always supposed to be collected, so that the same combination of exponents does not occur twice. Two such functions are equal only when they contain the same monomials with the same coefficients. An integral function is zero only when all its coefficients are zero.

Sums, differences, and products of integral functions are again integral functions. Moreover, rational numerical values may be assigned to the variables so that any prescribed finite set of non-zero such functions remains non-zero.

Integral functions over the rational field occur as a special case:

$$\Phi(x, y, z, \ldots) = \sum a x^r y^s z^t \cdots ,$$

with rational coefficients. If the coefficients are rational integers without a common divisor, the function is called primitive. The product of two primitive integral functions is again primitive. More generally, the divisor or content of a product is the product of the contents of its factors.

If the coefficients of an integral function in Ω depend on an algebraic number θ, the conjugate functions are obtained by replacing θ by its conjugates. A function belongs to the rational field exactly when all of its conjugates are equal. Its norm is the product of all conjugate functions,

$$N(\varphi) = \varphi\varphi'\varphi'' \cdots ,$$

and is an integral function over the rational field. It is divisible by φ, and its quotient is the product of the remaining conjugates.

These elementary facts will be used repeatedly when functions are treated as arithmetic objects.

§136. The Functionals of an Algebraic Field Ω and the Extended Field Ω^*

The variables occurring in the theory of algebraic numbers do not have the meaning of variables ranging over independently varying numerical systems, as in function theory. They are only symbols of calculation. What matters is the system of coefficients; the variables serve to make the familiar rules of literal calculation available for that system. It is not excluded that one sometimes substitutes numerical values for them, but this is not their primary role.

We therefore introduce integral and fractional functions in any number of variables, with coefficients in the algebraic field Ω, and agree to calculate with them according to the ordinary rules of algebra.

Every such expression ω can be written as a quotient of two integral functions in Ω,

$$\omega = \frac{\varphi}{\psi}, \qquad \psi \neq 0.$$

Two quotients are equal if $\varphi\psi_1 = \varphi_1\psi$. If numerator and denominator have no common divisor, the quotient is irreducible. Among all representations there is one in irreducible form; in it numerator and denominator are determined by ω up to a common factor belonging to Ω.

Such a fractional expression will be called a *functional* of the field Ω. Integral functions and numbers themselves are special cases. The four fundamental operations are applicable to the functionals in the same extent as to numbers. Thus the totality of all functionals of Ω is again a field, denoted by Ω^*, the functional field of Ω. The number field Ω is contained in Ω^*.

A functional whose coefficients are rational numbers is a rational functional. The rational functionals form the functional field R^* of the rational field R, and R^* is contained in every algebraic functional field Ω^*.

Every rational functional can be represented as a quotient of two integral rational functions with integral coefficients. If a rational integral function has integral coefficients with greatest common divisor a, it can be written

$$aE,$$

where E is primitive. For a rational functional A write

$$A = aE. \tag{1}$$

The natural number a is called the absolute value of A; the primitive factor E is called a rational unit. In the special case in which A is a number, a is the ordinary absolute value and $E = \pm 1$ is its sign. The general definition may look unfamiliar at first, but it is the right analogue for the calculations to follow.

The absolute value of a product of rational functionals is the product of their absolute values. If ω is a functional in Ω^*, then its norm $N(\omega)$ is rational. The absolute value of this rational functional is called the absolute norm of ω, and is denoted by

$$N(\omega) = N_a(\omega)E(\omega). \tag{2}$$

Here $N_a(\omega)$ is a positive rational number and $E(\omega)$ a rational unit. The absolute norm is multiplicative:

$$N_a(\alpha\beta) = N_a(\alpha)N_a(\beta). \tag{3}$$

Replacing all coefficients of a functional by the corresponding numbers in a conjugate field gives a conjugate functional. Every identity between functionals remains valid after this simultaneous replacement. If t is a new variable not occurring in ω, the function

$$\Phi(t) = N(t-\omega) = t^m + A_1 t^{m-1} + \cdots + A_m \tag{4}$$

has rational functional coefficients and vanishes for $t = \omega$. More generally, there may be equations of lower degree with rational functional coefficients that vanish at ω.

A functional ω is called integral if it satisfies a monic equation

$$\omega^m + A_1\omega^{m-1} + \cdots + A_m = 0 \tag{5}$$

with coefficients A_i integral rational functionals. This definition extends the definition of algebraic integers from numbers to functionals.

§137. Integral Functionals

The integral functionals enjoy the same elementary closure properties as the algebraic integers.

First, if two functionals are integral, then their sum, difference, and product are integral. This follows by the same determinant argument used for algebraic integers: the products of a suitable finite system by the new functional are expressible linearly in that system, with integral rational functional coefficients, and elimination gives a monic integral equation.

Second, if ω is integral and $\Phi(t)$ is a monic function with integral rational functional coefficients which vanishes for $t = \omega$, then every rational divisor of $\Phi(t)$ has integral rational functional coefficients. In particular the irreducible equation of ω over the rational functional field has integral rational functional coefficients. This proves that a functional which is fractional in Ω^* cannot become integral merely by considering it in a higher field.

Third, every functional ω can be made integral by multiplication by a rational integer. If ω satisfies a monic equation with rational functional coefficients, multiply by a common denominator of the absolute values of the coefficients. Then a suitable multiple $a\omega$ satisfies a monic integral equation. Hence every functional is a quotient of two integral functionals, and the denominator may be chosen to be a natural number.

A useful criterion is the following. Suppose there are m quantities $\omega_1, \ldots, \omega_m$, not all zero, such that all products $\omega\omega_i$ can be expressed as

$$\omega\omega_i = A_{1i}\omega_1 + A_{2i}\omega_2 + \cdots + A_{mi}\omega_m, \tag{1}$$

where the A_{ji} are integral rational functionals. Then ω is integral. For the determinant of this linear system gives a monic equation for ω with integral rational functional coefficients.

Consequently any integral function in integral functionals is integral. In particular, by addition, subtraction, and multiplication one obtains only integral functionals from integral functionals. Conjugates of an integral functional are integral. The absolute norm of an integral functional is a natural number.

Finally, if a functional satisfies a monic equation whose coefficients are integral functionals of Ω^*, then the functional itself is integral. To prove this, form the norm of the polynomial in a new variable t. The norm has integral rational functional coefficients and vanishes at the same functional; the preceding criterion applies.

If an integral functional is in fact a number, then it is an algebraic integer in the earlier sense. Thus the algebraic integers are special cases of integral functionals.

§138. Divisibility, Associated Functionals, Units

The integral functionals obey laws of divisibility analogous to those of rational integers. We begin with the definition.

An integral functional α is called divisible by another non-zero integral functional β if the quotient

$$\gamma = \frac{\alpha}{\beta}$$

is again an integral functional. We then write $\alpha = \beta\gamma$.

Two integral functionals α, β are called associated if each is divisible by the other. Equivalently,

$$\alpha = \beta\varepsilon,$$

where ε is an integral functional whose reciprocal is also integral. Such a functional ε is called a unit.

The absolute norm of a unit is equal to one. Conversely, if an integral functional has absolute norm one, then it is a unit. Indeed, its reciprocal is, up to a rational unit, the product of its conjugates divided by its norm and is integral.

Every rational unit is, in the present sense, a unit. An integral rational functional is associated with its absolute value. Thus the sign of an ordinary integer is here generalized to the rational unit factor of a rational functional.

For any integral functional α, the norm $N(\alpha) = \alpha\alpha' \cdots$ shows that the natural integer $N_a(\alpha)$ is divisible by α. Hence there are infinitely many natural integers divisible by α. Among them is a least one, say a. Every rational integer divisible by α is then divisible by a; for the remainder upon division by a would again be a rational integer divisible by α, smaller than a, and so must be zero.

§139. Greatest Common Divisor

Let $\alpha, \beta, \ldots$ be a finite number of non-zero integral functionals in Ω^*, and let $x, y, \ldots$ be variables not occurring in them. Form

$$\delta = \alpha x + \beta y + \cdots . \tag{1}$$

This is an integral functional. Its norm is a homogeneous integral rational function of the variables. If D denotes the absolute norm of δ, one may write

$$N(\delta) = DE(x, y, \ldots), \tag{2}$$

where E is a rational unit.

Now substitute arbitrary rational integers $x_0, y_0, \ldots$ for the variables and put

$$\delta_0 = \alpha x_0 + \beta y_0 + \cdots .$$

The quotient δ_0/D is an integral functional. This is seen by considering

$$\delta t - \delta_0 = \alpha(xt - x_0) + \beta(yt - y_0) + \cdots$$

and its norm, ordered according to powers of t. The resulting monic equation has integral rational functional coefficients and is satisfied by δ_0/D. Thus δ_0 is divisible by D.

Since the integers $x_0, y_0, \ldots$ are arbitrary, it follows in particular that $\alpha, \beta, \ldots$ are all divisible by D. Conversely, every common divisor of $\alpha, \beta, \ldots$ divides every expression of the form (1), and hence divides D. Therefore D has the characteristic property of a greatest common divisor.

Thus every finite system of integral functionals has a greatest common divisor, determined up to multiplication by a unit. It may be represented by a linear expression in the functionals with integral functional coefficients, in the same way as for rational integers.

§140. Prime Functionals in the Field Ω

An integral functional π, different from zero and not a unit, is called a prime functional if from a divisibility

$$\pi \mid \alpha\beta$$

it follows that $\pi \mid \alpha$ or $\pi \mid \beta$. This is the direct analogue of a prime number.

If π is a prime functional, there are natural integers divisible by π. Let p be the least among them. It cannot be composite. For if $p = p_1 p_2$ with both factors greater than one, then, since π is prime, one of the two factors would be divisible by π, contradicting the minimality of p. Hence p is an ordinary rational prime.

Every natural integer divisible by π is divisible by this same prime p. Since

$$p = \pi\alpha \tag{1}$$

with integral α, taking absolute norms gives

$$p^n = N_a(\pi)N_a(\alpha), \tag{2}$$

where n is the degree of the number field. Therefore $N_a(\pi)$ is a power of p. Write

$$N_a(\pi) = p^f. \tag{3}$$

The integer f, with $1 \leq f \leq n$, is called the degree of the prime functional π.

§141. Decomposition of Integral and Fractional Functionals into Prime Factors

Every non-zero integral functional which is not a unit is divisible by a prime functional. If the functional itself is prime, there is nothing to prove. Otherwise it has a proper integral divisor ω_1 which is neither a unit nor associated with the original functional:

$$\omega = \omega_1 \alpha.$$

Since the absolute norm of α is greater than one, the absolute norm of ω_1 is strictly smaller than that of ω. If ω_1 is still not prime, the same argument may be applied again. The absolute norms form a strictly decreasing sequence of natural numbers, and therefore the process stops at a prime functional.

It follows that every integral functional different from zero and from the units can be decomposed into finitely many prime factors:

$$\omega = \pi_1 \pi_2 \cdots \pi_r. \tag{1}$$

This decomposition is unique, up to associated prime factors and the order of the factors. The proof is the usual one. If

$$\pi_1 \pi_2 \cdots \pi_r \quad \text{and} \quad \rho_1 \rho_2 \cdots \rho_s$$

are two decompositions into prime functionals, then π_1 divides the product on the right and hence divides one of the ρ_j. Since both are prime, they are associated. Cancelling associated factors and repeating the argument gives equality of the decompositions.

The same result applies to fractional functionals. Since every functional is a quotient of two integral functionals, and both numerator and denominator can be decomposed into prime factors, the functional is represented uniquely as a product of prime powers with integral exponents, again up to units.

This restores, in the enlarged domain of functionals, the law of unique factorization that may fail for the algebraic integers themselves.

§142. Integral Functions in the Field Ω^*

The proof of unique factorization also rests on the following analogue of Gauss's lemma. Let

$$\varphi = a_0 t^h + a_1 t^{h-1} + \cdots + a_h, \qquad \psi = b_0 t^k + b_1 t^{k-1} + \cdots + b_k$$

be integral functions of one variable t, whose coefficients are integral functionals not involving t. If all coefficients of the product $\varphi\psi$ have a common prime divisor π, then π divides all coefficients of φ or all coefficients of ψ.

From this one draws an important conclusion. A functional

$$\Phi(t) = \varphi_0 t^h + \varphi_1 t^{h-1} + \cdots + \varphi_h \tag{1}$$

with coefficients in Ω^* is integral only if all its coefficients φ_i are integral functionals.

Indeed, suppose not all coefficients are integral. Let μ be a common denominator, so that $\mu\varphi_i = \alpha_i$ are integral and not all divisible by a prime factor π of μ. If Φ itself were integral, then it would satisfy a monic equation with integral rational functional coefficients. After multiplying by a sufficiently high power of μ, one obtains an identity between integral functions in which the right-hand side has all coefficients divisible by π, whereas the left-hand side does not, contradicting the lemma.

Conversely, a polynomial whose coefficients are integral functionals is integral. Hence integral functions in Ω^* are precisely those whose coefficients are integral functionals.

Every functional can be multiplied by a primitive rational function so as to become an integral function of the variables with coefficients in Ω. If the functional is already integral, it is associated with such an integral function whose coefficients are integral numbers of Ω. Thus integral functionals may be represented by ordinary integral functions, up to multiplication by rational units.

§143. The Prime Factors of the Numbers of the Field Ω

Since the numbers of Ω are among the functionals of Ω^*, the preceding theory shows that integral numbers of Ω can be decomposed into prime factors. These prime factors need not themselves be numbers; in general they are functionals. In special cases prime functionals may be numbers, and then they are called prime numbers of the field.

All equations between functionals are ultimately identities between integral rational functions. Hence they remain valid when the variables are replaced by other symbols. It follows that an integral functional, a unit, and a prime functional remain respectively integral, a unit, and prime after such a replacement.

Consequently the prime factors of an integral functional may be represented so that they do not contain the variables on which the functional itself depends. For every integral functional divides a number, in particular its absolute norm. Its prime factors are therefore among the prime factors of such a number, and may be chosen with variables different from those occurring in the original functional.

Let

$$E\omega = \varphi(x, y, \ldots) \tag{1}$$

where E is a unit and φ an integral function in Ω. If ω is decomposed into prime factors chosen free of the variables $x, y, \ldots$, their product gives a number ω_1 associated with ω. The quotient φ/ω_1 is then an integral function. Hence all coefficients of φ are divisible by ω_1, and therefore by ω. Conversely, if all coefficients of φ are divisible by a factor, then so is φ.

Thus the divisibility of an integral function by a number or by a prime functional is equivalent to the divisibility of all its coefficients. This observation is the bridge between factorization of functionals and arithmetic of the numbers of Ω.

Seventeenth Section. Theory of Algebraic Number Fields

§144. Basis of an Algebraic Number Field; Discriminants

Let $\Omega = R(\theta)$ be an algebraic number field of degree n, where θ satisfies the irreducible equation

$$f(t) = t^n + a_1 t^{n-1} + \cdots + a_n = 0. \tag{1}$$

Every number of Ω is uniquely expressible in the form

$$\omega = c_1\omega_1 + c_2\omega_2 + \cdots + c_n\omega_n, \tag{2}$$

with rational coefficients, after a suitable system $\omega_1, \ldots, \omega_n$ has been chosen. Such a system is called a basis of the field. The powers

$$1, \theta, \theta^2, \ldots, \theta^{n-1} \tag{3}$$

form one basis.

Let $\omega_{r,1}, \omega_{r,2}, \ldots, \omega_{r,n}$ denote the conjugates of ω_r. The square of the determinant

$$\Delta(\omega_1, \ldots, \omega_n) = (\det(\omega_{r,s}))^2 \tag{4}$$

is a symmetric function of the conjugates and hence a rational number. It is called the discriminant of the system $\omega_1, \ldots, \omega_n$.

For the power basis one obtains the discriminant of the equation (1), equivalently

$$\Delta(1, heta, \ldots, \theta^{n-1}) = (-1)^{n(n-1)/2} N(f'(\theta)). \tag{5}$$

This number is non-zero. If another system $\alpha_1, \ldots, \alpha_n$ is obtained from a basis by a rational linear substitution with determinant H, then

$$\Delta(\alpha_1, \ldots, \alpha_n) = H^2 \Delta(\omega_1, \ldots, \omega_n). \tag{6}$$

Thus the discriminants of different bases differ by a rational square and have the same sign. Conversely, any system of n numbers whose determinant is non-zero is a basis.

§145. The Minimal Basis and the Field Discriminant

A basis remains a basis if each of its elements is multiplied by a non-zero rational number. Since every algebraic number can be made integral by multiplication by a rational integer, there exist bases all of whose elements are algebraic integers. The discriminant of such an integral basis is a non-zero rational integer, and all such discriminants have the same sign.

Among their absolute values choose the least. A basis of algebraic integers whose discriminant has this least absolute value is called a minimal basis. The corresponding discriminant is called the discriminant of the field or the ground number of the field.

Let

$$\omega_1, \ldots, \omega_n$$

be a minimal basis. Any algebraic integer $\alpha \in \Omega$ has a rational representation

$$\alpha = u_1\omega_1 + \cdots + u_n\omega_n.$$

If some coefficient had a denominator, multiplying by a suitable rational integer and comparing discriminants would yield an integral basis with smaller absolute discriminant, which is impossible. Hence all u_i are rational integers.

Thus the minimal basis has the fundamental property that every algebraic integer of Ω is represented uniquely as

$$\alpha = x_1\omega_1 + x_2\omega_2 + \cdots + x_n\omega_n, \qquad x_i \in \mathbb{Z}. \tag{1}$$

Conversely, every expression of this form is an algebraic integer. The minimal basis therefore gives the integral lattice of the field.

The same reasoning applies to integral functionals. If $\omega_1, \ldots, \omega_n$ is a minimal basis, then a functional

$$\alpha = u_1\omega_1 + \cdots + u_n\omega_n$$

is integral precisely when the coefficients u_i are integral rational functionals. This will be used in the construction of bases of functionals.

§146. The Bases of Functionals

Let μ be an integral functional of Ω^*. A basis of μ is a system of n algebraic integers

$$\alpha_1, \alpha_2, \ldots, \alpha_n \tag{1}$$

which is a basis of the field Ω, and which has the following property: all algebraic integers of Ω divisible by μ, and no others, are obtained in the form

$$\alpha = x_1\alpha_1 + x_2\alpha_2 + \cdots + x_n\alpha_n, \qquad x_i \in \mathbb{Z}. \tag{2}$$

Every integral functional has such a basis. To construct one, begin with a minimal basis

$$\omega_1, \omega_2, \ldots, \omega_n.$$

There are positive rational integers a such that $a\omega_1$ is divisible by μ. Let a_{11} be the least positive such integer and put

$$\alpha_1 = a_{11}\omega_1.$$

Next choose a_{22} least positive such that integers a_{12} exist with

$$\alpha_2 = a_{12}\omega_1 + a_{22}\omega_2$$

divisible by μ. Continuing in this way one obtains a triangular system

$$\begin{aligned} \alpha_1 &= a_{11}\omega_1, \\ \alpha_2 &= a_{12}\omega_1 + a_{22}\omega_2, \\ &\vdots \\ \alpha_n &= a_{1n}\omega_1 + a_{2n}\omega_2 + \cdots + a_{nn}\omega_n, \end{aligned} \tag{3}$$

where each diagonal coefficient a_{rr} is chosen minimal positive subject to divisibility by μ.

The system (3) is a basis of the functional μ. Every integral combination of the α_r is divisible by μ by construction. Conversely, if an algebraic integer divisible by μ is expressed in the minimal basis, the triangular equations determine successively residues modulo $a_{nn}, a_{n-1,n-1}, \ldots, a_{11}$. The minimality conditions force the required representation in the α_r.

A basis form of μ is the linear form

$$\alpha = x_1\alpha_1 + \cdots + x_n\alpha_n. \tag{4}$$

Substituting rational integers for the variables gives precisely all algebraic integers divisible by μ.

If $\mu = 1$, a basis of μ is a minimal basis of the field. For arbitrary μ, all other bases of μ are obtained from one by an integral linear substitution of determinant ± 1.

§147. The Absolute Norms of Functionals

Let

$$\alpha_1, \ldots, \alpha_n$$

be a basis of the functional μ. These algebraic integers may be expressed integrally in terms of a minimal basis $\omega_1, \ldots, \omega_n$:

$$(\alpha_1, \alpha_2, \ldots, \alpha_n) = A(\omega_1, \omega_2, \ldots, \omega_n), \tag{1}$$

where A is an integral matrix. In the triangular basis constructed above, the determinant of A is

$$a_{11}a_{22}\cdots a_{nn}.$$

Let

$$\lambda = t_1\alpha_1 + \cdots + t_n\alpha_n \tag{2}$$

be the basis form of μ. Since the products $\omega_r\alpha_s$ are all divisible by μ, they can be expressed in the basis $\alpha_1, \ldots, \alpha_n$ with rational integral coefficients:

$$\omega_r\alpha_s = \sum_u g_{rsu}\alpha_u. \tag{3}$$

Multiplying the basis form and eliminating the α_u, one obtains the equation for λ. The product of its roots, that is, $N(\lambda)$, factors as

$$N(\lambda) = A\,T(t_1, \ldots, t_n), \tag{4}$$

where $A = \det(a_{rs})$ and T is an integral rational function of the variables t_i.

It remains to show that T is primitive. If a rational prime p divided all its coefficients, then by elementary determinant theory one could choose integral forms in the variables, not all divisible by p, making all corresponding linear combinations divisible by p. This would produce an algebraic integer divisible by μ whose coordinates in the basis are not all divisible by p, contradicting the defining property of the basis.

Therefore T is primitive. Consequently the absolute value of the determinant A is the absolute norm of μ. In particular, for the triangular basis (3) of §146,

$$N_a(\mu) = a_{11}a_{22}\cdots a_{nn}. \tag{5}$$

If one applies to the variables of a basis form an integral linear substitution of determinant ± 1, one obtains a new basis form of the same functional. Equivalently, the transposed substitution acts on the basis elements. The discriminant of a basis of μ is therefore

$$\Delta(\alpha_1, \ldots, \alpha_n) = N_a(\mu)^2\, d_\Omega, \tag{6}$$

where d_Ω is the discriminant of the field. Conversely, if a system of integral numbers divisible by μ has discriminant equal to this number, then it is a basis of μ.

§148. Complete Residue System Modulo a Functional

Two algebraic integers ξ, η are called congruent modulo a non-zero integral functional μ if the difference $\xi - \eta$ is divisible by μ. We write, in Gauss's notation,

$$\xi \equiv \eta \pmod{\mu}. \tag{1}$$

The modulus may be a functional, not merely a number. Multiplying the modulus by a unit does not change the congruence.

If

$$\xi \equiv \xi' \pmod{\mu}, \qquad \eta \equiv \eta' \pmod{\mu},$$

then every expression formed from the numbers by addition, subtraction, and multiplication remains congruent to the expression obtained by replacing each number by its congruent counterpart. The usual rules of congruence therefore hold.

Let

$$\alpha_1, \ldots, \alpha_n \tag{2}$$

be a basis of the modulus μ, written in triangular form relative to a minimal basis $\omega_1, \ldots, \omega_n$:

$$\begin{aligned} \alpha_1 &= a_{11}\omega_1, \\ \alpha_2 &= a_{12}\omega_1 + a_{22}\omega_2, \\ &\vdots \\ \alpha_n &= a_{1n}\omega_1 + \cdots + a_{nn}\omega_n. \end{aligned} \tag{3}$$

Every algebraic integer η of the field can be written

$$\eta = y_1\omega_1 + y_2\omega_2 + \cdots + y_n\omega_n, \qquad y_i \in \mathbb{Z}. \tag{4}$$

Now form the system

$$\xi = x_1\omega_1 + x_2\omega_2 + \cdots + x_n\omega_n, \tag{5}$$

where x_1 runs through a complete residue system modulo a_{11}, x_2 modulo a_{22}, and so on up to x_n modulo a_{nn}. The number of such ξ is

$$a_{11}a_{22}\cdots a_{nn} = N_a(\mu).$$

We prove that every integer η is congruent modulo μ to exactly one of these ξ. The condition that $\eta - \xi$ be divisible by μ means that it can be written as an integral combination of the basis elements α_i. Comparing coefficients in the minimal basis gives the triangular system

$$\begin{aligned} y_1 &= x_1 + h_1a_{11} + h_2a_{12} + \cdots + h_na_{1n}, \\ y_2 &= x_2 + h_2a_{22} + \cdots + h_na_{2n}, \\ &\vdots \\ y_n &= x_n + h_na_{nn}. \end{aligned}$$

The last equation determines x_n and h_n uniquely; then the preceding equation determines x_{n-1} and h_{n-1}, and so on. Thus all x_i are determined uniquely.

The system (5), or any system congruent to it, is therefore called a complete residue system modulo μ. The number of individuals in a complete residue system modulo μ is the absolute norm $N_a(\mu)$.

§149. Congruences

From the existence, proved in the preceding section, of a complete residue system modulo a functional μ, there follows a series of propositions exactly analogous to the elementary propositions on congruences in rational number theory.

Let α be an arbitrary integer of the field Ω, and let μ be an integral functional serving as modulus. If α is relatively prime to μ, then two products $\alpha\xi$ and $\alpha\xi'$ are congruent modulo μ only when ξ and ξ' are themselves congruent modulo μ. Hence, when ξ runs through a complete residue system modulo μ, the products $\alpha\xi$ also run through such a complete residue system. Thus:

> If α is an integer of Ω relatively prime to the modulus μ, and if γ is any integer of Ω, the congruence
> $$\alpha\xi \equiv \gamma \pmod{\mu} \tag{1}$$
> is soluble in an integer ξ, and has exactly one solution when ξ is required to lie in a prescribed complete residue system modulo μ.

If the modulus itself is represented by a number β, this gives the following form. If α, β, γ are integers in Ω and if α, β are relatively prime, then two integers ξ, η can be chosen in Ω so that

$$\alpha\xi + \beta\eta = \gamma. \tag{2}$$

In particular, for any two relatively prime integers α, β, the equation

$$\alpha\xi + \beta\eta = 1 \tag{3}$$

can be satisfied by integers ξ, η of the field.

The same proposition extends to any finite system of integers. If

$$\alpha, \beta, \gamma, ldots$$

have no common divisor, then integers $\xi, \eta, \zeta, \ldots$ may be found such that

$$\alpha\xi + \beta\eta + \gamma\zeta + \cdots = 1. \tag{4}$$

Indeed one first chooses coefficients in such a way that a certain linear combination of $\beta, \gamma, \ldots$ is relatively prime to α, and then applies the preceding two-term result. Conversely, if such an equation exists, no prime divisor can divide all of $\alpha, \beta, \gamma, \ldots$.

It follows more generally that if a system of equations of the form

$$\varepsilon_i = \alpha\xi_i + \beta\eta_i + \gamma\zeta_i + \cdots \tag{5}$$

is obtained by expanding a unit ε in a basis, then the coefficients ε_i have no common divisor. Multiplying by suitable integers and adding gives (4). This is the form of Bezout's theorem which will be needed below.

§150. Fermat's Theorem

The theory of congruences now gives consequences exactly corresponding to the theorems derived in rational arithmetic from Fermat's theorem. Let π be a prime functional of degree f, and let p be the rational prime divisible by π. The norm of π is then

$$N(\pi) = p^f.$$

For any integer α not divisible by π, multiplication by α permutes the non-zero residue classes modulo π. Multiplying all non-zero residues and cancelling their product, which is not divisible by π, gives

$$\alpha^{p^f-1} \equiv 1 \pmod{\pi}. \tag{1}$$

Multiplication by α yields the equivalent form

$$\alpha^{p^f} \equiv \alpha \pmod{\pi}, \tag{2}$$

which remains valid even when α is divisible by π. These formulae are the analogue, for algebraic number fields, of Fermat's theorem.

A polynomial congruence modulo a prime functional has no more roots than its degree. If

$$f(t) \equiv 0 \pmod{\pi} \tag{3}$$

where $f(t)$ is an integral polynomial of degree m, then there are at most m incongruent integers t modulo π satisfying it. For if a is one root, then

$$f(t) = (t-a)f_1(t) + f(a),$$

and all other roots either are congruent to a or satisfy a congruence of degree $m-1$. Induction proves the assertion.

Every integer not divisible by π satisfies

$$\omega^{p^f-1} \equiv 1 \pmod{\pi}. \tag{4}$$

If a is the least positive exponent for which

$$\omega^a \equiv 1 \pmod{\pi},$$

then every exponent l with $\omega^l \equiv 1 \pmod{\pi}$ is a multiple of a; hence $a \mid (p^f-1)$. We say that ω belongs to the exponent a. If ω belongs to a, then

$$1, \omega, \omega^2, \ldots, \omega^{a-1}$$

are incongruent and give all roots of $t^a \equiv 1 \pmod{\pi}$. Among them, ω^l belongs to the exponent a precisely when l is relatively prime to a. Therefore the number of elements belonging to exponent a is $\varphi(a)$, whenever such elements exist.

As in rational arithmetic one proves that for every divisor a of $p^f - 1$ there are elements belonging to exponent a. The elements belonging to the greatest exponent $p^f - 1$ are called primitive roots of π. If γ is such a primitive root, then

$$1, \gamma, \gamma^2, \ldots, \gamma^{p^f-2}$$

form a complete system of the non-zero residues modulo π.

Finally, if an integer α is congruent modulo π to a rational integer, then by the ordinary Fermat theorem it satisfies

$$\alpha^p \equiv \alpha \pmod{\pi}. \tag{5}$$

Conversely, the congruence (5) characterizes precisely those residue classes which are represented by rational integers. For a polynomial $\psi(x, y, \ldots)$ with integral algebraic coefficients, the congruence

$$\psi(x, y, \ldots)^p \equiv \psi(x^p, y^p, \ldots) \pmod{\pi} \tag{6}$$

is therefore the necessary and sufficient condition that all coefficients of ψ be congruent modulo π to rational integers.

§151. Dedekind Ideals

Dedekind based the theory of algebraic numbers on the concept of the ideal. We now show that the theory of ideals is, in essence, the same as the theory of functionals, and describe the passage from one language to the other.

Let $\mathfrak{o}$ denote, as before, the system of all integers of the algebraic number field Ω. A system $\mathfrak{a}$ of numbers contained in $\mathfrak{o}$ is called an ideal if it satisfies the following two conditions:

I. The sum and difference of any two numbers of $\mathfrak{a}$ again belong to $\mathfrak{a}$.

II. The product of any number of $\mathfrak{a}$ by any number of $\mathfrak{o}$ again belongs to $\mathfrak{a}$.

The system consisting only of zero satisfies these conditions, but for simplicity is not counted as an ideal here. The system $\mathfrak{o}$ itself is an ideal. Also, the set of all multiples in $\mathfrak{o}$ of a fixed integer α is an ideal; such an ideal is called a principal ideal.

The product $\mathfrak{ab}$ of two ideals is the set of all finite sums of products $\sum \alpha\beta$, with $\alpha \in \mathfrak{a}$, $\beta \in \mathfrak{b}$. This is again an ideal. With respect to this multiplication, $\mathfrak{o}$ is the identity.

Ideals and functionals correspond as follows. To each integral functional φ there belongs the ideal consisting of all integers divisible by φ. Associated functionals give the same ideal. Conversely, to each ideal there correspond infinitely many integral functionals, but they are all associated. Thus the ideal records the divisor common to all its numbers.

If the functional φ corresponds to $\mathfrak{a}$ and ψ to $\mathfrak{b}$, then the product $\varphi\psi$ corresponds to the ideal product $\mathfrak{ab}$. One inclusion is immediate, since every product of an element divisible by φ and an element divisible by ψ is divisible by $\varphi\psi$. For the converse, represent φ and ψ as greatest common divisors of pairs of integers; expanding the products shows that every integer divisible by $\varphi\psi$ belongs to $\mathfrak{ab}$.

If a number or functional is divisible by the functionals belonging to an ideal, we say that it is divisible by the ideal. Thus a factorization of a functional can also be written in the form

$$\varphi = \mathfrak{ab}\cdots, \tag{1}$$

where unit factors are suppressed.

A basis of the functional is also a basis of the corresponding ideal. The absolute norm of the representing functional is the number which Dedekind calls the norm of the ideal. We therefore write, if φ belongs to $\mathfrak{a}$,

$$N_a(\varphi) = N(\mathfrak{a}). \tag{2}$$

The norm of an ideal is always a natural number. If $\mathfrak{p}$ is a prime ideal and p the rational prime divisible by it, then

$$N(\mathfrak{p}) = p^f, \tag{3}$$

and f is called the degree of the prime ideal. In congruences the corresponding ideals may be used as moduli in place of the functionals.

Kummer's ideal numbers may be understood through the same correspondence. If α, β have greatest common divisor χ, and if $\beta = \chi\varphi$, the congruence $\alpha x \equiv 0 \pmod{\beta}$ is governed precisely by the functional φ. What Kummer called an ideal number is thereby represented by the functional or, equivalently, by the ideal attached to it. This removes the ambiguity from the manipulation of ideal factors.

§152. Equivalence

We now call two functionals associated, even when fractional, if they differ only by a unit factor. We make the following definition.

> Two integral or fractional functionals φ, ψ in the field Ω are called equivalent if their quotient $\varphi : \psi$ is associated with a number.

Equivalently, there are a unit ε and a number α of Ω such that

$$\varphi = \alpha\varepsilon\psi. \tag{1}$$

If two functionals are each equivalent to a third, they are equivalent to each other. The functionals of Ω therefore split into classes: two functionals lie in the same class exactly when they are equivalent. Each class is determined by any one of its representatives.

Associated functionals are equivalent, hence occur in the same class. A class contains not merely single functionals, but all ideals determined by those functionals. For this reason the classes are called functional classes, or more commonly ideal classes.

The collection of all integral and fractional numbers of the field, together with all units and all products of numbers by units, forms a single class. This is called the principal class. As an ideal class it contains the ideal $\mathfrak{o}$; in what follows it is denoted by 0.

Every ideal class contains integral functionals. For if φ is any functional and α is a number chosen so that $\alpha\varphi$ is integral, then $\alpha\varphi$ is equivalent to φ.

More strongly, from every ideal class one can choose a representative which is integral and relatively prime to any prescribed integral functional. Let φ be an integral representative of a class C, choose a number α divisible by φ, and write

$$\varphi\chi = \alpha. \tag{2}$$

Choose a number β, divisible by χ, so that $\beta : \chi$ is relatively prime to the prescribed functional. Then

$$\psi = \beta : \chi \tag{3}$$

is equivalent to φ and is a representative of C relatively prime to the given functional.

§153. The Class Number of the Field Ω

We now prove the important theorem:

> The number of ideal classes of a field Ω is finite.

It is equivalent to prove that in every class there are integral functionals whose absolute norm does not exceed a fixed finite number depending only on the field. Indeed every integral functional divides its absolute norm, and any fixed natural number has only finitely many non-associated divisors in the field. Hence a uniform bound for the norm gives only finitely many classes.

Let

$$\omega_1, \omega_2, \ldots, \omega_n$$

be a minimal basis of Ω. Every integer has the form

$$\omega = x_1\omega_1 + x_2\omega_2 + \cdots + x_n\omega_n, \qquad x_i \in \mathbb{Z}. \tag{1}$$

Choose a positive integer k and restrict the coefficients x_i to lie in the interval from 0 to k. Then the absolute value of each conjugate of ω is bounded by a constant times k, and the absolute norm is bounded by

$$|N_a(\omega)| < Bk^n, \tag{2}$$

where B depends only on the basis and the field.

Let μ be an integral functional and choose k so that

$$k^n < N_a(\mu) < (k+1)^n. \tag{3}$$

The $(k+1)^n$ numbers obtained from (1) by taking $x_i = 0, 1, \ldots, k$ are more numerous than the residue classes modulo μ, whose number is $N_a(\mu)$. Therefore two of them are congruent modulo μ. Their difference

$$\alpha = a_1\omega_1 + \cdots + a_n\omega_n \tag{4}$$

is non-zero, divisible by μ, and has coefficients $|a_i| \le k$. From (2) and (3) it follows that

$$|N_a(\alpha)| < Bk^n < BN_a(\mu). \tag{5}$$

Since α is divisible by μ, write

$$\alpha = \mu\varphi, \qquad N_a(\alpha) = N_a(\mu)N_a(\varphi). \tag{6}$$

Then

$$|N_a(\varphi)| < B. \tag{7}$$

If ψ is any representative of a given class C, choose μ so that $\mu\psi$ is a number. Applying the preceding construction and writing $\alpha = \mu\varphi$, one obtains a functional φ equivalent to ψ, hence still representing C, and satisfying (7). Thus every class has a representative whose absolute norm is bounded by the same constant. The number of classes is therefore finite. This finite number is called the class number of Ω.

In the simplest case, where the class number is one, every functional is associated with a number. Then every integer of the field admits a factorization into prime number factors, and the arithmetic of the field essentially agrees with rational number theory. The field of rational numbers, the Gaussian imaginary numbers, and the field generated by the third roots of unity are among such examples.

§154. The Group of Ideal Classes

The ideal classes may be composed. If $\varphi, \psi, \varphi_1, \psi_1$ are functionals, with φ equivalent to φ_1 and ψ equivalent to ψ_1, then $\varphi\psi$ is equivalent to $\varphi_1\psi_1$. This follows immediately from the definition, since the quotient

$$\frac{\varphi\psi}{\varphi_1\psi_1}$$

is associated with a number.

Conversely, if φ is equivalent to φ_1 and $\varphi\psi$ is equivalent to $\varphi_1\psi_1$, then ψ is equivalent to ψ_1.

Now let A and B be two ideal classes, not necessarily distinct. Choose a representative φ of A and a representative ψ of B. The class C containing the product $\varphi\psi$ is independent of the choice of representatives. It is called the composition of A and B, and is written

$$C = AB = BA. \tag{1}$$

Because this composition is derived from ordinary multiplication, it obeys the commutative and associative laws.

The identity element is the principal class 0. To define the inverse of a class A, take a representative φ and choose a number α divisible by φ. If

$$\alpha = \varphi\chi,$$

then χ represents a class denoted A^{-1}, and

$$AA^{-1} = 0. \tag{2}$$

Thus the ideal classes form an abelian group, the ideal class group of the field.

Since the number of ideal classes is finite, this group is finite. If h is its order, then for every class A

$$A^h = 0. \tag{3}$$

If h_A is the least positive integer for which $A^{h_A} = 0$, then h_A divides h. Hence every functional φ belongs to a definite exponent dividing the class number: some power φ^{h_A} is associated with a number of the field.

Eighteenth Section. Discriminants

§155. The Minimum of a Quadratic Form

The finiteness theorem for the classes of ideals, which was obtained earlier by a purely arithmetical argument, may be reached in another and very fruitful way. The method, due to Minkowski, depends first on a theorem about positive quadratic forms. We state and prove it in the form in which it is needed for algebraic number fields.

Let

$$f(x_1, \ldots, x_n) = \sum_{i,k} a_{ik} x_i x_k$$

be a real quadratic form which is positive for every non-zero system of real values x_i. It is then possible to write it as a sum of n squares of real linear forms,

$$f = \xi_1^2 + \xi_2^2 + \cdots + \xi_n^2, \qquad \xi_i = \alpha_{1i} x_1 + \alpha_{2i} x_2 + \cdots + \alpha_{ni} x_n.$$

If A denotes the determinant of the coefficients α_{ij}, and if D denotes the determinant of the quadratic form f, then

$$D = A^2.$$

The inverse linear substitution has the form

$$x_i = \beta_{1i}\xi_1 + \beta_{2i}\xi_2 + \cdots + \beta_{ni}\xi_n \qquad (i = 1, \ldots, n).$$

When the x_i range over all rational integral systems, the corresponding points

$$(\xi_1, \ldots, \xi_n)$$

form a lattice in n-dimensional space. If the system $x_1, \ldots, x_n$ is not identically zero, then f assumes positive values; let M be the least of these values. Thus the least distance between two distinct lattice-points, measured in the ordinary Euclidean metric of the ξ-space, is $\sqrt{M}$.

Consider in the ξ-space an arbitrary bounded region G_ξ, and let its volume be

$$V_\xi = \int \cdots \int d\xi_1 \cdots d\xi_n.$$

Under the linear change of variables this volume is changed in the ratio $|A|$, so that

$$V_\xi = \sqrt{D}\, V_x.$$

In particular, if K_n is the volume of the unit sphere

$$\xi_1^2 + \cdots + \xi_n^2 \leq 1,$$

then a sphere of radius r has volume $K_n r^n$.

Around every lattice point draw a sphere of radius $\frac{1}{2}\sqrt{M}$. No two of these spheres have an interior point in common; for otherwise two lattice-points would have distance smaller than $\sqrt{M}$. Now take a large region similar to a fixed region G_ξ, and count the lattice-points in it. The quotient of the number of lattice-points by the volume tends, in the limit, to the reciprocal of the fundamental volume $\sqrt{D}$. Hence the spheres just constructed cannot, in the limiting sense, occupy more than the whole space. Therefore

$$\frac{K_n\left(\frac{1}{2}\sqrt{M}\right)^n}{\sqrt{D}} \le 1,$$

and consequently

$$M \le 4\left(\frac{D}{K_n^2}\right)^{1/n}.$$

For the applications the strict inequality is sufficient:

$$M < 4\left(\frac{D}{K_n^2}\right)^{1/n}.$$

A rough but simple estimate of K_n already gives a useful form of the theorem. The sphere of radius 1 contains the cube

$$|\xi_i| \le \frac{1}{\sqrt{n}} \qquad (i = 1, \dots, n),$$

whose volume is $(2/\sqrt{n})^n$. Hence

$$K_n > \left(\frac{2}{\sqrt{n}}\right)^n,$$

and from the preceding inequality follows

$$M < n\sqrt[n]{D}.$$

For $n > 1$ the inequality is strict:

$$M < n\sqrt[n]{D}. \tag{1}$$

A sharper estimate, which will be useful below, is obtained recursively. Cut the unit sphere by a cylinder with axis in the direction of the first coordinate. If $0 < a < 1$, the slice

$$|\xi_1| \le a$$

contains, for each ξ_1 in that interval, an $(n-1)$-sphere of radius at least $\sqrt{1-a^2}$. Hence

$$K_n > 2a(1-a^2)^{(n-1)/2}K_{n-1}.$$

Choosing a so as to make the right hand side as large as possible gives a decreasing sequence of constants θ_n for which the preceding estimate may be written

$$M < \theta_n n \sqrt[n]{D}.$$

In dimension 2, since $K_2 = \pi$, one obtains

$$\theta_2 = \frac{2}{\pi} < 0.636, \qquad \theta_2^2 < 0.404.$$

Since the numbers θ_n decrease, we have, for every $n > 1$,

$$M < n\sqrt[n]{0.404\,D}. \tag{2}$$

Thus every positive quadratic form in n variables with determinant D assumes, for some non-zero integral system $x_1, \dots, x_n$, a value smaller than the right hand side of (2). This is the form of Minkowski's theorem that will be applied to algebraic integers.

§156. Application to Algebraic Fields

We first record the elementary inequality that will be used in passing from sums of conjugate absolute values to norms. If $a_1, \ldots, a_n$ are positive real numbers, then

$$a_1 + \cdots + a_n \geq n(a_1 a_2 \cdots a_n)^{1/n}. \tag{1}$$

The proof is by induction. For $n = 2$ it is the identity

$$a + b - 2\sqrt{ab} = (\sqrt{a} - \sqrt{b})^2 \geq 0.$$

The general case follows by applying the assertion first to $n-1$ of the numbers and then to the two quantities so obtained.

Let Ω be an algebraic field of degree n, with minimal basis

$$\omega_1, \ldots, \omega_n$$

and field discriminant Δ. Let φ be a whole functional in the ring of integers o of Ω. Choose a basis of φ,

$$\beta_i = h_{1i}\omega_1 + h_{2i}\omega_2 + \cdots + h_{ni}\omega_n \qquad (i = 1, \ldots, n),$$

so that the determinant $h = (h_{ij})$ has absolute value equal to the absolute norm of the functional:

$$N_a(\varphi) = |\det(h_{ij})|.$$

Every whole number divisible by φ is then of the form

$$\eta = \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n,$$

where $x_1, \ldots, x_n$ are rational integers.

Let

$$\eta_1, \ldots, \eta_n$$

be the conjugates of η. We form the positive quadratic expression

$$f(x_1, \ldots, x_n) = |\eta_1|^2 + |\eta_2|^2 + \cdots + |\eta_n|^2. \tag{2}$$

For a real conjugate this is simply the square of a real linear form. For a pair of conjugate imaginary fields the product $\eta_i \eta_{i'}$ is the sum of two squares of real linear forms. Therefore (2) is a positive quadratic form in n real variables.

The determinant of this form is, apart from the usual sign of the field discriminant,

$$|D| = |\,\Delta\,|\, N_a(\varphi)^2.$$

Applying the theorem of the preceding section, we can choose the rational integers x_i, not all zero, so that

$$\sum_{i=1}^{n} |\eta_i|^2 < \theta_n\, n \sqrt[n]{|\Delta|\, N_a(\varphi)^2}, \tag{3}$$

where $\theta_n^n < 0.404$ for $n > 1$.

By (1),

$$n\,|N(\eta)|^{2/n} \leq |\eta_1|^2 + \cdots + |\eta_n|^2.$$

Combining this with (3) gives

$$|N(\eta)|^2 < \theta_n^n\,|\Delta|\,N_a(\varphi)^2. \tag{4}$$

Since η is divisible by φ, we may write, in the language of functionals,

$$\eta = \varphi\psi$$

up to a unit factor; consequently

$$|N(\eta)| = N_a(\varphi)\,N_a(\psi).$$

It follows from (4) that

$$N_a(\psi) < \sqrt{\theta_n^n|\Delta|} < \sqrt{0.404\,|\Delta|}. \tag{5}$$

Thus we obtain Minkowski's fundamental theorem:

For every ideal class in an algebraic field Ω of degree $n > 1$ there is an integral ideal belonging to that class whose absolute norm is smaller than

$$\sqrt{|\,\Delta\,|}. \tag{6}$$

Indeed, take φ in the inverse class. The construction gives ψ in the desired class, and (5) is stronger than (6).

A first consequence is the following. No algebraic field of degree $n > 1$ can have discriminant $\Delta = \pm 1$. For if $|\Delta| = 1$, the theorem would give, in every ideal class and in particular in the principal class, an integral ideal of norm smaller than 1, which is impossible. Hence the rational field is the only algebraic number field with fundamental discriminant ± 1.

This is the point at which the geometry of numbers enters the ideal theory. The earlier finiteness theorem for classes is sharpened: not only are there finitely many ideal classes, but each class contains an ideal of bounded norm, with a bound expressed solely in terms of the discriminant of the field.

§157. Prime Factors of the Natural Prime Numbers

We now determine how an ordinary rational prime number p decomposes into prime factors in the field Ω. The following treatment follows Hensel's method.

Let $\mathfrak{p}$ be a prime functional, or prime ideal, of degree f. Thus the number of residue classes modulo $\mathfrak{p}$ is p^f, where p is the rational prime lying below $\mathfrak{p}$. The smallest positive rational integer divisible by $\mathfrak{p}$ is then p. Congruences modulo $\mathfrak{p}$ may be applied not only to numbers of the field, but also to forms with rational integral coefficients in auxiliary variables.

Let

$$\tau = \omega_1 t_1 + \omega_2 t_2 + \cdots + \omega_n t_n$$

be the basis form of the field, and let

$$F(z) = N(z - \tau)$$

be the equation satisfied by τ; its coefficients are rational integral forms in $t_1, \dots, t_n$. Since $F(\tau) = 0$, the congruence $F(\tau) \equiv 0 \pmod{\mathfrak{p}}$ is immediate.

If a rational integral form $\Phi(z)$ satisfies

$$\Phi(\tau) \equiv 0 \pmod{\mathfrak{p}}, \tag{1}$$

then τ may be regarded as a root of the congruence $\Phi(z) \equiv 0 \pmod{\mathfrak{p}}$. In a residue field with p^f elements the Frobenius operation gives new roots. Thus, setting

$$\tau_r = \omega_1^{p^r} t_1 + \omega_2^{p^r} t_2 + \cdots + \omega_n^{p^r} t_n,$$

we have, together with τ,

$$\tau, \tau_1, \dots, \tau_{f-1}$$

as roots. Moreover $\tau_r \equiv \tau_s \pmod{\mathfrak{p}}$ precisely when $r \equiv s \pmod{f}$, so that these f roots are distinct modulo $\mathfrak{p}$.

It follows that the product

$$P(z) = (z - \tau)(z - \tau_1) \cdots (z - \tau_{f-1}) \tag{2}$$

is congruent modulo $\mathfrak{p}$ to a rational integral form of degree f in $z, t_1, \dots, t_n$; its coefficients are determined modulo p. We shall denote this form again by $P(z)$.

The form $P(z)$ is irreducible in the following sense. If $\Phi(z)$ is a rational integral form for which $\Phi(\tau)$ is divisible by $\mathfrak{p}$, then

$$\Phi(z) \equiv P(z)\Phi_0(z) \pmod{p} \tag{3}$$

with a rational integral form $\Phi_0(z)$. Thus P has the smallest possible degree in z among all congruences over the rational residue field that are satisfied by τ modulo $\mathfrak{p}$.

We next recover the prime ideal $\mathfrak{p}$ itself from P. The greatest common divisor of the rational prime p and the number $P(\tau)$ is precisely $\mathfrak{p}$. Indeed, $P(\tau)$ is divisible by $\mathfrak{p}$, by construction.

If another prime divisor $\mathfrak{q}$ of p also divided $P(\tau)$, then P would vanish at the corresponding residue value of τ, contrary to the irreducibility just established for the factor attached to $\mathfrak{p}$, unless $\mathfrak{q} = \mathfrak{p}$.

Now let

$$p = \mathfrak{p}_1^{e_1} \mathfrak{p}_2^{e_2} \cdots \mathfrak{p}_r^{e_r} \tag{4}$$

be the decomposition of p in Ω, and let f_i be the degree of $\mathfrak{p}_i$. To each $\mathfrak{p}_i$ corresponds a form $P_i(z)$ of degree f_i. Since $F(z) = N(z - \tau)$ is divisible modulo $\mathfrak{p}_i$ by $P_i(z)$, and since every factor over p is obtained in this way, we have the congruence

$$F(z) \equiv P_1(z)^{e_1} P_2(z)^{e_2} \cdots P_r(z)^{e_r} \pmod{p}. \tag{5}$$

The degrees give the familiar relation

$$n = e_1 f_1 + e_2 f_2 + \cdots + e_r f_r. \tag{6}$$

This is Dedekind's decomposition law in the form required here: the decomposition of the minimal congruence $F(z)$ modulo p determines the prime ideal factors over p, their residue degrees f_i, and their exponents e_i.

§158. Dedekind's Theorem on the Field Discriminant

Let again

$$\tau = \omega_1 t_1 + \cdots + \omega_n t_n$$

be the basis form of Ω, and put

$$F(z) = N(z - \tau).$$

The powers

$$1, \tau, \tau^2, \ldots, \tau^{n-1}$$

may be expressed linearly in the basis $\omega_1, \ldots, \omega_n$. Let U be the determinant of these coefficients. Then the discriminant of the system $1, \tau, \ldots, \tau^{n-1}$ is

$$\Delta(1, \tau, \ldots, \tau^{n-1}) = (-1)^{n(n-1)/2} N(F'(\tau)). \tag{1}$$

On the other hand it is also equal to

$$U^2 \Delta, \tag{2}$$

where Δ is the field discriminant.

It remains to see that U is a unit. If a rational prime p divided U, then the powers $1, \tau, \ldots, \tau^{n-1}$ would be linearly dependent modulo p. There would therefore be a congruence of degree $< n$, with rational integral coefficients, satisfied by τ modulo a prime ideal above p. This contradicts the minimality of the factors P_i established in the preceding section. Hence $U = \pm 1$.

Thus

$$|N(F'(\tau))| = |\Delta|. \tag{3}$$

The number $F'(\tau)$, or more precisely the functional which it determines, is therefore the fundamental functional of the field. Changing the basis form τ only replaces $F'(\tau)$ by an associated functional, so that the ideal so defined is independent of the auxiliary choice.

We now compare this with the decomposition of a rational prime. Suppose that p has the factorization

$$p = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$$

and that the factor $P_i(z)$ of $F(z)$ belonging to $\mathfrak{p}_i$ occurs with exponent e_i modulo p. From

$$F(z) \equiv P_i(z)^{e_i} \Phi_i(z) \pmod{p}$$

we get, by differentiation,

$$F'(z) \equiv e_i P_i(z)^{e_i - 1} P_i'(z) \Phi_i(z) + P_i(z)^{e_i} \Phi_i'(z) \pmod{p}.$$

Putting $z = \tau$, the factor $P_i(\tau)$ is divisible by $\mathfrak{p}_i$ but not by a higher power than is forced by the exponent already visible in the congruence. Hence the fundamental functional $F'(\tau)$ is divisible by $\mathfrak{p}_i^{e_i - 1}$. If e_i is not divisible by p, it is not divisible by $\mathfrak{p}_i^{e_i}$. If e_i is divisible by p, a higher divisibility may occur.

The principal conclusion is the theorem of Dedekind:

A rational prime p divides the field discriminant Δ if and only if, in the field Ω, it is divisible by the square of at least one prime ideal factor. Equivalently, p is ramified in Ω if and only if $p \mid \Delta$.

The formula also gives a useful basis for the inverse different. If

$$F(z) = z^n + a_1 z^{n-1} + \cdots + a_n,$$

and if

$$F_\nu(z) = z^\nu + a_1 z^{\nu-1} + \cdots + a_\nu \qquad (\nu = 0, \ldots, n-1),$$

with $F_0 = 1$, then

$$\frac{F_\nu(\tau)}{F'(\tau)}$$

has trace 0 or 1 according to the usual duality rule. More explicitly, the expansion

$$\frac{F(z)}{(z-\tau)F'(\tau)}$$

has trace 1. Hence the fractions

$$\frac{F_0(\tau)}{F'(\tau)}, \ldots, \frac{F_{n-1}(\tau)}{F'(\tau)}$$

form a system dual to the powers $1, \tau, \ldots, \tau^{n-1}$ with respect to the trace. Therefore, if ω is any whole number of Ω, then

$$\mathrm{Tr}\left(\frac{\omega}{F'(\tau)}\right)$$

is a rational integer. This is the trace-theoretic form of the fundamental ideal.

Nineteenth Section. The Relation of a Field to Its Divisors

§159. Relative Norms

The preceding theory was developed over the rational field. We now let the ground field itself be an algebraic field.

Let R be a field of degree m over the rational field, and let

$$\Omega = R(\Theta)$$

be an algebraic extension of relative degree n. The absolute degree of Ω is then mn. If ρ is a primitive element of R, then the products

$$\rho^i \Theta^j$$

may be used, in the usual range of exponents, to represent the elements of Ω. Every number of Ω is expressible as a polynomial in Θ of degree $< n$, with coefficients in R.

Let

$$\Omega_1, \dots, \Omega_n$$

be the conjugate fields of Ω over the fixed field R. If $\omega \in \Omega$, its *relative norm* is

$$N_{\Omega/R}(\omega) = \omega_1 \omega_2 \cdots \omega_n, \tag{1}$$

where ω_i denotes the image of ω in Ω_i. This is a number of R. The absolute norm is then obtained by taking the norm in R:

$$N_{\Omega/\mathbb{Q}}(\omega) = N_{R/\mathbb{Q}}\big(N_{\Omega/R}(\omega)\big). \tag{2}$$

In exactly the same way one defines the relative norm of a functional or an ideal in Ω; it is then a functional or ideal in R.

Similarly, the relative trace is

$$S_{\Omega/R}(\omega) = \omega_1 + \cdots + \omega_n, \tag{3}$$

and the absolute trace is

$$S_{\Omega/\mathbb{Q}}(\omega) = S_{R/\mathbb{Q}}\big(S_{\Omega/R}(\omega)\big). \tag{4}$$

The fundamental ideal is compatible with passage through intermediate fields. Let τ be a basis form of Ω over the rational field, and let

$$F(z) = N_{\Omega/\mathbb{Q}}(z - \tau).$$

Then $F'(\tau)$ represents the absolute fundamental ideal of Ω. Let θ be a basis form of R, with

$$\psi(z) = N_{R/\mathbb{Q}}(z - \theta),$$

and let

$$f(z) = N_{\Omega/R}(z - \tau)$$

be the relative equation of τ over R. The derivative $f'(\tau)$ defines the relative fundamental ideal of Ω over R.

From the chain rule for norms and from the trace-duality interpretation of the derivative one obtains

$$F'(\tau) \sim f'(\tau)\,\psi'(\theta), \tag{5}$$

where $\sim$ denotes association by a unit. Thus

$$\mathfrak{D}_{\Omega/\mathbb{Q}} \sim \mathfrak{D}_{\Omega/R}\,\mathfrak{D}_{R/\mathbb{Q}}. \tag{6}$$

More generally, for a tower of fields the corresponding fundamental ideals multiply transitively.

It is important that the relative fundamental ideal may be a unit even when the absolute fundamental ideal is not. The ramification which is visible over the rational field can disappear after the ground field has been enlarged; the relative ideal records only the ramification still present over R.

We shall call the extension Ω/R normal if every conjugate of Ω over R is again equal to Ω. In that case the substitutions carrying Ω into itself over R form a group, the Galois group of Ω over R. If this group is commutative, the relative field is Abelian.

§160. Prime Ideals in the Relatively Normal Field

Assume now that Ω is normal over R, and denote its group over R by $\mathcal{O}$. If $\varphi \in \mathcal{O}$, and if $\omega \in \Omega$, write

$$\omega^\varphi$$

for the conjugate of ω obtained by the substitution φ. The same notation will be used for ideals:

$$\mathfrak{a}^\varphi = \{\alpha^\varphi : \alpha \in \mathfrak{a}\}.$$

Prime ideals go into prime ideals under the substitutions of the group.

Let $\mathfrak{p}$ be a prime ideal of Ω. There is a unique prime ideal $\mathfrak{P}$ of R below it. Every number of R that is divisible by $\mathfrak{p}$ is divisible by $\mathfrak{P}$, and conversely the ideal $\mathfrak{P}$ in Ω is divisible by all conjugates of $\mathfrak{p}$. If

$$N_{\Omega/R}(\mathfrak{p}) = \mathfrak{P}^f, \tag{1}$$

the exponent f is called the degree of $\mathfrak{p}$ over R.

Let the distinct conjugates of $\mathfrak{p}$ under the group be

$$\mathfrak{p}_1, \mathfrak{p}_2, \dots, \mathfrak{p}_g.$$

Suppose that the highest power of each of them dividing $\mathfrak{P}$ is the e-th power. Then

$$\mathfrak{P} = (\mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_g)^e. \tag{2}$$

Taking relative norms gives the fundamental relation

$$n = efg, \tag{3}$$

where $n = [\Omega : R]$.

The substitutions that carry $\mathfrak{p}$ into itself form a subgroup Ψ of $\mathcal{O}$. This is the decomposition group of $\mathfrak{p}$. Its cosets describe the conjugate prime ideals. The substitutions in Ψ act on the residue field $o_\Omega/\mathfrak{p}$, and their action fixes the residue field of R modulo $\mathfrak{P}$. The quotient by the subgroup acting trivially will therefore measure the residue degree.

The preceding factorization is the normal-field form of the decomposition law. In a non-normal field one must pass to a normal overfield, or else use the double-coset formula described later. In the normal case the symmetry supplied by $\mathcal{O}$ ensures that all prime ideals over a fixed $\mathfrak{P}$ have the same ramification exponent e and the same residue degree f.

§161. Primitive Roots of Prime Ideals

Let $\mathfrak{P}$ be a prime ideal of R, and let $\mathfrak{p}$ be a prime ideal of Ω above it, of degree f. Denote by $P = N_{R/\mathbb{Q}}(\mathfrak{P})$ the number of residue classes modulo $\mathfrak{P}$ in R. Then the number of residue classes modulo $\mathfrak{p}$ in Ω is

$$P^f.$$

Consequently every number α of Ω satisfies

$$\alpha^{P^f} \equiv \alpha \pmod{\mathfrak{p}}. \tag{1}$$

If α is not divisible by $\mathfrak{p}$, then

$$\alpha^{P^f-1} \equiv 1 \pmod{\mathfrak{p}}.$$

As in the rational case, the non-zero residue classes modulo $\mathfrak{p}$ form a cyclic group. There is therefore a residue class γ, represented by a whole number of Ω, such that

$$\gamma^{P^f-1} \equiv 1 \pmod{\mathfrak{p}}$$

and no smaller positive exponent gives 1. The powers

$$1, \gamma, \gamma^2, \ldots, \gamma^{P^f-2}$$

then represent all non-zero residue classes modulo $\mathfrak{p}$. Such a γ will be called a primitive root of the prime ideal $\mathfrak{p}$.

Let θ be any whole number of Ω. Among all congruences over R satisfied by θ modulo $\mathfrak{p}$, choose one of least degree:

$$G(\theta) \equiv 0 \pmod{\mathfrak{p}}. \tag{2}$$

Division by powers of γ shows that every residue class modulo $\mathfrak{p}$ is representable by a polynomial in γ with coefficients taken modulo $\mathfrak{P}$. Hence the degree of G cannot be less than f.

On the other hand, the f elements

$$\gamma, \gamma^P, \gamma^{P^2}, \ldots, \gamma^{P^{f-1}}$$

are distinct modulo $\mathfrak{p}$, and all are roots of the same congruence over R. Therefore

$$(z-\gamma)(z-\gamma^P)\cdots(z-\gamma^{P^{f-1}}) \tag{3}$$

is congruent modulo $\mathfrak{p}$ to a whole polynomial $G(z)$ of degree f with coefficients in R. This polynomial is the relative irreducible congruence belonging to the prime ideal $\mathfrak{p}$. It plays, over R, the same role as the polynomial $P(z)$ of §157 played over the rational field.

The construction also shows that the residue field modulo $\mathfrak{p}$ is obtained from the residue field modulo $\mathfrak{P}$ by adjoining the residue class of γ. Its degree is exactly f.

§162. The Partial Fundamental Ideal

Let τ be a basis form of Ω over R, and let $\mathfrak{p}$ be a prime ideal of Ω above $\mathfrak{P}$. Among the substitutions in the decomposition group Ψ of $\mathfrak{p}$, consider first those for which

$$\tau^{\chi} \equiv \tau \pmod{\mathfrak{p}}. \tag{1}$$

They form a subgroup X of Ψ. This subgroup is the inertia group. Its elements act trivially on the residue field.

There is also a distinguished coset in Ψ/X, represented by a substitution ψ_0, characterized by

$$\omega^{\psi_0} \equiv \omega^{P} \pmod{\mathfrak{p}} \tag{2}$$

for every whole number ω of Ω. In other words, ψ_0 induces the Frobenius substitution on the residue field. The decomposition group splits into cosets

$$\Psi = X + X\psi_0 + \cdots + X\psi_0^{f-1}. \tag{3}$$

The quotient Ψ/X is cyclic of order f.

The subgroup X may itself have a filtration by higher congruence conditions. Let $X^{(1)}$ consist of those substitutions $\chi \in X$ for which

$$\tau^{\chi} \equiv \tau \pmod{\mathfrak{p}^2};$$

let $X^{(2)}$ consist of those for which the congruence holds modulo $\mathfrak{p}^3$, and so on. We obtain a chain

$$X \supset X^{(1)} \supset X^{(2)} \supset \cdots, \tag{4}$$

which eventually reaches the identity. Hilbert's terminology is: Ψ is the decomposition group, X the inertia group, and $X^{(1)}$ the ramification group.

We now compute the contribution of $\mathfrak{p}$ to the relative fundamental ideal. If

$$f(z) = N_{\Omega/R}(z - \tau),$$

then the relative fundamental ideal is represented by

$$f'(\tau) = \prod_{\varphi \neq 1} (\tau - \tau^{\varphi}). \tag{5}$$

The factor $\tau - \tau^{\varphi}$ is divisible by $\mathfrak{p}$ precisely when $\varphi \in X$, except for the identity factor which is omitted. It is divisible by $\mathfrak{p}^2$ precisely when $\varphi \in X^{(1)}$, by $\mathfrak{p}^3$ precisely when $\varphi \in X^{(2)}$, and so forth. If

$$|X| = e, \quad |X^{(1)}| = e_1, \quad |X^{(2)}| = e_2, \ldots,$$

then the exponent with which $\mathfrak{p}$ enters the relative fundamental ideal is

$$(e-1) + (e_1 - 1) + (e_2 - 1) + \cdots. \tag{6}$$

The same exponent occurs for every conjugate of $\mathfrak{p}$.

Consequently, if $\mathfrak{p}_1, \ldots, \mathfrak{p}_g$ are the prime ideals over $\mathfrak{P}$, the relative fundamental ideal contains the factor

$$(\mathfrak{p}_1 \mathfrak{p}_2 \cdots \mathfrak{p}_g)^{(e-1)+(e_1-1)+(e_2-1)+\cdots}. \tag{7}$$

Taking the norm down to R gives the relative discriminant exponent. For $R = \mathbb{Q}$ this formula becomes the usual discriminant formula for the field Ω.

§163. The Ideals in the Divisor Fields of Ω

Let Ω/R be normal with group $\mathcal{O}$, and let $\mathcal{O}'$ be a subgroup. The elements fixed by $\mathcal{O}'$ form a divisor field Ω' of Ω. If

$$|\mathcal{O}'| = n',$$

and if $|\mathcal{O}| = n$, then

$$[\Omega : \Omega'] = n', \qquad [\Omega' : R] = \frac{n}{n'}.$$

The extension Ω/Ω' is normal, though Ω'/R need not be normal.

We describe the prime ideals of Ω' lying over a given prime ideal $\mathfrak{P}$ of R. Let $\mathfrak{p}$ be a prime ideal of Ω over $\mathfrak{P}$, with decomposition group Ψ. The prime ideals in Ω' are obtained by grouping the conjugates of $\mathfrak{p}$ under the action of $\mathcal{O}'$. This grouping is governed by double cosets:

$$\mathcal{O} = \Psi\varphi_1\mathcal{O}' + \Psi\varphi_2\mathcal{O}' + \cdots + \Psi\varphi_s\mathcal{O}'. \tag{1}$$

Each double coset gives one prime ideal $\mathfrak{p}'_r$ of Ω' over $\mathfrak{P}$.

Let $\Psi_r = \varphi_r^{-1}\Psi\varphi_r$, and let

$$H_r = \Psi_r \cap \mathcal{O}'.$$

This group is the decomposition group, inside Ω/Ω', of a prime ideal of Ω lying over $\mathfrak{p}'_r$. Its inertia subgroup and residue degree determine how $\mathfrak{p}'_r$ decomposes further in Ω.

Thus the factorization of $\mathfrak{P}$ in Ω' has the form

$$\mathfrak{P} = (\mathfrak{p}'_1)^{a_1}(\mathfrak{p}'_2)^{a_2}\cdots(\mathfrak{p}'_s)^{a_s}. \tag{2}$$

The exponents a_r are obtained from the inertia indices supplied by the intersections H_r. Their sum, with residue degrees counted, gives the degree

$$[\Omega' : R].$$

When $\mathfrak{p}'_r$ is further decomposed in Ω, say

$$\mathfrak{p}'_r = (\mathfrak{p}_{r1}\mathfrak{p}_{r2}\cdots\mathfrak{p}_{r,e_r})^{g_r}, \tag{3}$$

the relative norm has the form

$$N_{\Omega/\Omega'}(\mathfrak{p}_{rs}) = (\mathfrak{p}'_r)^{f_r}, \tag{4}$$

and

$$n' = e_r f_r g_r. \tag{5}$$

If f is the degree of $\mathfrak{p}$ over R, and f'_r is the degree of $\mathfrak{p}'_r$ over R, then

$$f = f_r f'_r. \tag{6}$$

This identity expresses the transitivity of residue degrees in the tower

$$\Omega \supset \Omega' \supset R.$$

The complete law of decomposition in an intermediate field is therefore the double-coset law: double cosets of the decomposition group of $\mathfrak{p}$ with the subgroup defining the intermediate field correspond to the prime ideals in that intermediate field, and the intersections determine ramification and residue degree.

Twentieth Section. Quadratic Fields

§164. Basis of a Quadratic Field

Let

$$\Omega = \mathbb{Q}(\sqrt{d})$$

be a quadratic field, where d is a square-free rational integer different from 1. If $d > 0$ the field is real; if $d < 0$ it is imaginary. The field is normal, since the only non-trivial substitution changes $\sqrt{d}$ into $-\sqrt{d}$.

Every number of the field is of the form

$$x + y\sqrt{d}, \qquad x, y \in \mathbb{Q}.$$

If ω is an irrational number of the field, then it satisfies an equation

$$c\omega^2 = a + b\omega \tag{1}$$

with rational integers a, b, c. The discriminant of this equation is

$$D = b^2 + 4ac,$$

and D/d is a square. Conversely the roots are

$$\omega = \frac{b \pm \sqrt{D}}{2c}. \tag{2}$$

A number is integral if its equation may be taken monic. From (2) one obtains the usual distinction according to the residue of d modulo 4. If

$$d \equiv 2, 3 \pmod{4},$$

then the whole numbers of Ω are precisely

$$x + y\sqrt{d}, \qquad x, y \in \mathbb{Z}. \tag{3}$$

If

$$d \equiv 1 \pmod{4},$$

then the whole numbers are

$$\frac{x + y\sqrt{d}}{2},$$

where x and y are both even or both odd. Equivalently they are

$$x + y\theta, \qquad \theta = \frac{1 + \sqrt{d}}{2}, \quad x, y \in \mathbb{Z}. \tag{4}$$

Thus a basis of the ring of integers is

$$1, \sqrt{d} \qquad (d \equiv 2, 3 \pmod{4}), \tag{5a}$$

and

$$1, \frac{1 + \sqrt{d}}{2} \qquad (d \equiv 1 \pmod{4}). \tag{5b}$$

It is convenient to write in both cases

$$1, \theta$$

for the basis, with

$$\theta = \sqrt{d}$$

in the first case and

$$\theta = \frac{1+\sqrt{d}}{2}$$

in the second.

The discriminant, which Weber calls the ground number, is therefore

$$\Delta = 4d \qquad (d \equiv 2, 3 \pmod 4), \tag{6a}$$

and

$$\Delta = d \qquad (d \equiv 1 \pmod 4). \tag{6b}$$

If θ' is the conjugate of θ, then

$$\theta - \theta' = \sqrt{\Delta} \tag{7}$$

with the evident choice of the square-root. This compact formula is useful in the following sections.

The values $\Delta = \pm 1, \pm 2$ cannot occur as ground numbers of quadratic fields. The smallest absolute value of a negative ground number is therefore 3, attained by the field $\mathbb{Q}(\sqrt{-3})$.

§165. Prime Ideals of the First Degree

Let φ be a whole functional in the quadratic field. Its basis may be written in the form

$$\alpha_1 = a_{11}, \qquad \alpha_2 = a_{12} + a_{22}\theta, \tag{1}$$

where a_{11} is the smallest positive rational integer divisible by φ, and a_{22} is the least positive rational integer for which $a_{22}\theta$ is congruent modulo φ to a rational integer. The absolute norm of φ is

$$N_a(\varphi) = a_{11}a_{22}. \tag{2}$$

Apply this to a prime functional π. Since the field has degree 2, the prime ideal π can have degree 1 or 2. If its absolute norm is p, then the rational prime p decomposes into two prime factors. If its absolute norm is p^2, then p remains prime in the quadratic field.

For $\varphi = \pi$ one has at once

$$a_{11} = p.$$

The number a_{22} is determined by the condition that $a_{22}\theta$ be congruent to a rational number modulo π. The necessary and sufficient condition is

$$a_{22}(\theta^p - \theta) \equiv 0 \pmod{\pi}. \tag{3}$$

If

$$\theta^p - \theta \equiv 0 \pmod{\pi},$$

then $a_{22} = 1$, and the prime ideal has degree 1. Otherwise $a_{22} = p$, and the prime ideal has degree 2.

In the first case θ, and therefore every whole number of Ω, is congruent modulo π to a rational number. The p rational residues

$$0, 1, \ldots, p-1$$

then form a complete residue system modulo π. Thus

$$N_a(\pi) = p, \qquad a_{22} = 1. \tag{4a}$$

In the second case

$$N_a(\pi) = p^2, \qquad a_{22} = p, \tag{4b}$$

and the prime functional is principal:

$$\pi = p(u + \theta v). \tag{5}$$

Suppose now that p splits. Then a_{12} is determined by

$$a_{12} + \theta \equiv 0 \pmod{\pi}. \tag{6}$$

Taking the norm, the product

$$(a_{12} + \theta)(a_{12} + \theta')$$

must be divisible by p. Using

$$\theta - \theta' = \sqrt{\Delta},$$

one obtains the following criterion. The rational prime p splits into two prime factors in the quadratic field if and only if the congruence

$$x^2 \equiv \Delta \pmod{4p} \tag{7}$$

is soluble in rational integers x. In Weber's terminology, the ground number Δ must be a quadratic residue modulo $4p$.

If x is chosen so that (7) holds, the two conjugate prime functionals may be written

$$\pi = pu + (a_{12} + \theta)v, \qquad \pi' = pu + (a_{12} + \theta')v. \tag{8}$$

Their product is associated with the rational prime p.

Finally, the two conjugate factors π, π' may themselves be associated. This happens exactly when π', hence also $\pi - \pi'$, is divisible by π. Since

$$\pi - \pi' = (\theta - \theta')v,$$

the condition is that

$$(\theta - \theta')^2 = \Delta$$

be divisible by p. Thus p is associated with the square of a prime functional only when $p \mid \Delta$, in agreement with the general ramification theorem of §158.

§166. Functionals in the Quadratic Field and Quadratic Irrationalities

We now take an arbitrary whole functional φ of the quadratic field and write its basis in the form

$$\alpha_1 = a_{11}, \qquad \alpha_2 = a_{12} + a_{22}\theta. \tag{1}$$

Then

$$N_a(\varphi) = a_{11}a_{22}. \tag{2}$$

Put

$$\varphi = a_{11}t_1 + (a_{12} + a_{22}\theta)t_2. \tag{3}$$

Taking the norm of this linear form gives a binary quadratic form. Its divisor is $a_{11}a_{22}$. We write

$$a_{11} = c\,a_{11}a_{22},$$

and, after dividing the coefficients of the norm form by the common divisor, obtain relatively prime rational integers a, b, c for which the quotient

$$\omega = \frac{-b + \sqrt{\Delta}}{2c} \tag{4}$$

satisfies

$$c\omega^2 = a + b\omega, \tag{5}$$

or equivalently

$$b^2 + 4ac = \Delta. \tag{6}$$

Thus ω is a quadratic irrationality belonging to the discriminant Δ.

The functional φ itself may then be written

$$\varphi = ec(t_1 + \omega t_2), \tag{7}$$

where e is a positive rational integer. Here c is the least positive integer for which $c\omega$ is a whole number of the quadratic field; every integer h with $h\omega$ integral is divisible by c.

Conversely, let

$$\omega = \frac{-b + \sqrt{\Delta}}{2c}$$

be any quadratic irrationality of discriminant Δ, with a, b, c relatively prime and satisfying $b^2 + 4ac = \Delta$. Then for every positive integer e the form

$$ec(t_1 + \omega t_2) \tag{8}$$

is the basis form of a whole functional. Indeed its coefficients ec and $ec\omega$ are whole numbers, and the norm calculation gives

$$N_a(\varphi) = e^2 c. \tag{9}$$

The least rational integer divisible by the functional is ec, and the second basis element is $ec\omega$.

The result may be stated succinctly.

Every whole functional φ in a quadratic field with ground number Δ determines a quadratic irrationality ω of discriminant Δ and a positive integer e such that

$$ec(t_1 + \omega t_2) \tag{10}$$

is a basis form of φ. Conversely, every quadratic irrationality of discriminant Δ, together with a positive integer e, gives in this way a whole functional of the quadratic field.

This is the bridge between the ideal theory of the quadratic field and the classical theory of binary quadratic forms.

§167. Equivalent Functionals and Equivalent Numbers

Let

$$\omega, \omega_1$$

be two quadratic irrationalities belonging to the same discriminant Δ. Their associated basis forms are

$$\varphi = ec(t_1 + \omega t_2), \qquad \varphi_1 = e_1 c_1(t_1 + \omega_1 t_2). \tag{1}$$

Suppose first that the two irrationalities are equivalent in the sense of the theory of binary quadratic forms; that is,

$$\omega_1 = \frac{p\omega + q}{r\omega + s}, \tag{2}$$

where p, q, r, s are rational integers and

$$ps - qr = \pm 1. \tag{3}$$

Substituting (2) into φ_1, and putting

$$u_1 = st_1 + qt_2, \qquad u_2 = rt_1 + pt_2,$$

we obtain

$$(r\omega + s)\varphi_1 = e_1 c_1(u_1 + \omega u_2). \tag{4}$$

The change of variables has determinant ± 1. Hence the form on the right is associated with φ, and therefore the two functionals φ, φ_1 are equivalent.

Conversely, assume that the functionals φ, φ_1 are equivalent. Let ψ be a functional in the reciprocal class to that of φ, so that $\varphi\psi$ is associated with a principal functional. Write

$$\varphi\psi = \varepsilon\xi, \tag{5}$$

where ε is a unit and ξ a number of the ring. Take basis forms

$$\varphi = ec(t_1 + \omega t_2), \qquad \psi = e'c'(t_1 + \eta t_2). \tag{6}$$

Multiplying by the conjugate form of ψ, and comparing discriminants, gives a new basis of ψ'. This basis differs from

$$e'c', \quad e'c'\eta'$$

by an integral unimodular substitution. Therefore there are rational integers p, q, r, s with

$$ps - qr = \pm 1 \tag{7}$$

such that

$$\omega = \frac{p\eta' + q}{r\eta' + s}. \tag{8}$$

Thus ω is equivalent to η'. Replacing φ by an equivalent form φ_1 gives the same ψ, and hence ω_1 is also equivalent to η'. Therefore ω and ω_1 are equivalent.

We have therefore proved the two directions:

If two quadratic irrationalities of discriminant Δ are equivalent, then the corresponding functionals are equivalent; and if two whole functionals are equivalent, then the corresponding quadratic irrationalities are equivalent.

Consequently each ideal class of the quadratic field corresponds to exactly one class of equivalent quadratic irrationalities of discriminant Δ, and conversely. The number of inequivalent quadratic irrationalities of the discriminant Δ is equal to the number of ideal classes of the field.

In the terminology of Gauss one sometimes obtains a finer division into proper and improper equivalence. With the present notion of equivalence of irrationalities, in which this distinction is not made, no such further division is necessary.

§168. Composition of Quadratic Irrationalities

We have seen that to every whole functional φ in the quadratic field there belongs a quadratic irrationality ω of discriminant Δ. We now determine how far ω is fixed by φ, and then define composition.

Let the basis of φ be

$$\alpha_1 = a_{11}, \qquad \alpha_2 = a_{12} + a_{22}\theta. \tag{1}$$

The numbers a_{11} and a_{22} are uniquely determined by φ, since

$$N_a(\varphi) = a_{11}a_{22}. \tag{2}$$

The remaining coefficient a_{12} is determined modulo φ by

$$a_{12} + a_{22}\theta \equiv 0 \pmod{\varphi}. \tag{3}$$

Since a_{12} is rational integral, it is determined only up to a multiple of a_{11}. Replacing

$$\alpha_2$$

by

$$\alpha_2 + h\alpha_1$$

therefore replaces the corresponding irrationality $\omega = \alpha_2/\alpha_1$ by

$$\omega + h.$$

We have the following statement.

To each whole functional of the quadratic field belongs one and only one series of quadratic irrationalities of discriminant Δ. The numbers of this series differ from one another by arbitrary rational integers and are consequently equivalent. Associated functionals lead to the same series.

Now let φ_1, φ_2 be two whole functionals, and put

$$\varphi_3 = \varphi_1\varphi_2. \tag{4}$$

Choose irrationalities $\omega_1, \omega_2, \omega_3$ belonging to $\varphi_1, \varphi_2, \varphi_3$. The class of ω_3 is determined by the classes of ω_1 and ω_2, up to the addition of a rational integer. We call ω_3 the composite of ω_1 and ω_2, and write symbolically

$$\omega_1\omega_2 = \omega_3. \tag{5}$$

If either factor is replaced by an equivalent irrationality, the composite is replaced by an equivalent irrationality. Thus the classes of quadratic irrationalities of the discriminant Δ form an Abelian group, isomorphic to the group of ideal classes of the quadratic field.

It remains to describe the actual computation. Let the bases of the three functionals be

$$\begin{aligned}
\varphi_1 &: \quad \alpha_1 = a_{11}, \qquad \alpha_2 = a_{12} + a_{22}\theta, \\
\varphi_2 &: \quad \beta_1 = b_{11}, \qquad \beta_2 = b_{12} + b_{22}\theta, \\
\varphi_3 &: \quad \gamma_1 = c_{11}, \qquad \gamma_2 = c_{12} + c_{22}\theta.
\end{aligned} \tag{6}$$

Then

$$\omega_1 = \frac{\alpha_2}{\alpha_1}, \qquad \omega_2 = \frac{\beta_2}{\beta_1}, \qquad \omega_3 = \frac{\gamma_2}{\gamma_1}.$$

The number c_{11} is the least positive rational integer divisible by the product $\varphi_1\varphi_2$, and therefore divides $a_{11}b_{11}$. Write

$$a_{11}b_{11} = kc_{11}. \tag{7}$$

Since absolute norms multiply, one has

$$c_{22} = k\, a_{22}b_{22}. \tag{8}$$

The last coefficient c_{12} is determined by

$$c_{12} + c_{22}\theta \equiv 0 \pmod{\varphi_1\varphi_2}. \tag{9}$$

For the purpose of finding a representative of the composed class one may proceed more simply. Keep φ_1 fixed, and replace φ_2 by an equivalent functional chosen so that it is relatively prime to a_{11}. This can always be done. In that situation b_{11} is relatively prime to a_{11}, and therefore

$$c_{11} = a_{11}b_{11}, \qquad c_{22} = a_{22}b_{22}. \tag{10}$$

The congruence (9) splits into the two congruences

$$\begin{aligned} c_{12} &\equiv b_{22}a_{12} \pmod{a_{11}}, \\ c_{12} &\equiv a_{22}b_{12} \pmod{b_{11}}. \end{aligned} \tag{11}$$

These determine c_{12} modulo $c_{11} = a_{11}b_{11}$. The irrationality

$$\omega_3 = \frac{c_{12} + c_{22}\theta}{c_{11}} \tag{12}$$

is then a representative of the composed class.

It remains only to justify the preliminary replacement of φ_2. Since associated forms are equivalent, φ_2 may be replaced by a form

$$c_2(t_1 + \omega_2 t_2).$$

By making a unimodular substitution

$$\omega_2 \mapsto \frac{p\omega_2 + q}{r\omega_2 + s}, \qquad ps - qr = \pm 1,$$

the leading coefficient c_2 is transformed into

$$c_2' = -a_2r^2 + b_2rs + c_2s^2.$$

The integers r, s may be chosen relatively prime and so that c_2' is not divisible by any prescribed finite set of rational primes. In particular it may be chosen prime to a_{11}. This proves the asserted reduction.

The laws of composition have here been derived under the assumption that the discriminant Δ is the ground number of a quadratic field, that is, a fundamental discriminant. With slight modifications the same laws remain valid for arbitrary discriminants, as in Gauss's theory of binary quadratic forms.

Twenty-First Section. Cyclotomic Fields

§169. Decomposition of the Prime q in the Cyclotomic Field Ω_m

We make a second application of the general theory of algebraic numbers to cyclotomic fields. This will supply the means for bringing the theory of Abelian number fields to a particularly transparent conclusion.

Let q be a rational prime, including the case $q = 2$, and let

$$m = q^\lambda \tag{1}$$

be a positive power of q. In the case $q = 2$ we assume $\lambda > 1$. Let ζ be a primitive m-th root of unity, and denote by Ω_m the full cyclotomic field generated by ζ. Its degree is

$$\mu = \varphi(m) = q^{\lambda-1}(q-1). \tag{2}$$

We let n run through a complete system of residues modulo m, with the residues divisible by q omitted. Then the μ numbers

$$\zeta^n$$

are the roots of the irreducible cyclotomic equation $f(x) = 0$, where

$$f(x) = \frac{x^m - 1}{x^{m/q} - 1}. \tag{3}$$

The root ζ is a whole number, and its norm is $+1$; hence it is a unit.

The field Ω_m is normal. The substitutions

$$s_n = (\zeta, \zeta^n) \tag{4}$$

where n is prime to m, form an Abelian group isomorphic to the group of reduced residue classes modulo m.

Putting $x = 1$ in the factorization

$$f(x) = \prod_n (x - \zeta^n),$$

we get

$$q = \prod_n (1 - \zeta^n). \tag{5}$$

Let

$$\delta_n = 1 - \zeta^n, \qquad \delta = 1 - \zeta. \tag{6}$$

If n, n' are prime to m, then $n' \equiv an \pmod{m}$ for some a, and

$$\frac{1 - \zeta^{n'}}{1 - \zeta^n} = 1 + \zeta^n + \zeta^{2n} + \cdots + \zeta^{(a-1)n}$$

is a whole number. The reverse divisibility is obtained by interchanging n and n'. Thus all δ_n are associated. Formula (5) therefore gives

$$q \sim \delta^\mu, \qquad N(\delta) = q. \tag{7}$$

Since any proper divisor of δ would have norm dividing q, it follows that δ itself is a prime factor of first degree.

Thus the rational prime q is associated in Ω_m with the μ-th power of a prime factor of the first degree.

If $m = m_1 m_2$, with m_1, m_2 again powers of q, then applying the factorization of $x^{m_1} - 1$ and putting $x = \zeta^{m_2}$ gives

$$N(\zeta^{m_2} - 1) = q^{\varphi(m_1)}. \tag{8}$$

From this one computes the discriminant of the cyclotomic equation. Since

$$\operatorname{disc} f = (-1)^{\mu(\mu-1)/2} N(f'(\zeta)),$$

the discriminant is, apart from sign, a power of q. More explicitly it is the usual cyclotomic discriminant

$$\Delta_m = \pm\, q^{\,q^{\lambda-1}(\lambda q - \lambda - 1)}. \tag{9}$$

Only the prime q occurs in it, as expected from the complete ramification of q in Ω_m.

§170. Minimal Basis of the Field Ω_m

The preceding results lead to the following theorem.

The numbers

$$1, \zeta, \zeta^2, \ldots, \zeta^{\mu-1} \tag{1}$$

form a basis of the system o of whole numbers of the cyclotomic field Ω_m.

It is enough to show that a number

$$\omega = x_0 + x_1\zeta + \cdots + x_{\mu-1}\zeta^{\mu-1} \tag{2}$$

with rational integral coefficients can be divisible by a rational prime p only if all the x_i are divisible by p. If ω is divisible by p, then every conjugate

$$\omega_n = x_0 + x_1\zeta^n + x_2\zeta^{2n} + \cdots + x_{\mu-1}\zeta^{(\mu-1)n} \tag{3}$$

is divisible by p. The determinant of this system in the unknowns x_i is the square-root of the discriminant. If $p \neq q$, the discriminant is not divisible by p, and hence all x_i are divisible by p.

It remains to consider $p = q$. Put

$$F(t) = x_0 + x_1 t + \cdots + x_{\mu-1}t^{\mu-1}.$$

Since $\zeta = 1 - \delta$, Taylor's expansion gives

$$\omega = F(1-\delta) = F(1) - \delta F'(1) + \frac{\delta^2}{2!}F''(1) - \cdots . \tag{4}$$

If ω is divisible by q, then it is divisible by δ^μ. From (4) it follows successively that

$$F(1), F'(1), F''(1), \ldots$$

are divisible by q. Written out in terms of the x_i, these congruences imply successively that

$$x_{\mu-1}, x_{\mu-2}, \ldots, x_0$$

are all divisible by q. This proves the theorem.

Instead of the sequence (1), one may take any sequence of μ consecutive powers of ζ,

$$\zeta^a, \zeta^{a+1}, \ldots, \zeta^{a+\mu-1}, \tag{5}$$

as a basis. This follows by multiplication with the unit ζ^a. In particular, the symmetric sequence of powers around 1 is often useful.

The ground number of the field is therefore the discriminant of the cyclotomic equation itself:

$$\Delta(\Omega_m) = \operatorname{disc}(1, \zeta, \ldots, \zeta^{\mu-1}). \tag{6}$$

Thus the discriminant computed in the preceding section is the field discriminant.

§171. The Prime Ideals in the Field Ω_m

We have already seen that the prime q belonging to the exponent $m = q^\lambda$ is totally ramified:

$$q \sim (1-\zeta)^\mu.$$

It remains to decompose the rational primes $p \neq q$.

Let p be such a prime. If

$$\omega = x_0 + x_1\zeta + \cdots + x_{\mu-1}\zeta^{\mu-1} \tag{1}$$

is a whole number, then its conjugates are

$$\omega_n = x_0 + x_1\zeta^n + x_2\zeta^{2n} + \cdots + x_{\mu-1}\zeta^{(\mu-1)n}. \tag{2}$$

Two such conjugates can be congruent modulo a prime ideal $\mathfrak{p}$ over p only when the corresponding exponents are congruent modulo m. Hence the inertia group is trivial: no prime $p \neq q$ is divisible by the square of a prime ideal in Ω_m. This is also clear from the discriminant, which is a power of q.

Raising (2) to the p-th power and using Fermat's theorem for the rational coefficients gives

$$\omega^p \equiv \omega_p \pmod{\mathfrak{p}}. \tag{3}$$

Thus the Frobenius substitution belonging to $\mathfrak{p}$ is

$$\zeta \mapsto \zeta^p.$$

Let f be the least positive exponent for which

$$p^f \equiv 1 \pmod{m}. \tag{4}$$

Then the decomposition group is generated by

$$(\zeta, \zeta^p),$$

and has order f. Hence every prime ideal over p has residue degree f, and

$$N(\mathfrak{p}) = p^f. \tag{5}$$

If

$$\mu = \varphi(m) = ef,$$

then there are e distinct conjugate prime ideals over p. Therefore:

If $p \neq q$ belongs to the exponent f modulo m, and if $\varphi(m) = ef$, then p decomposes in Ω_m into e distinct conjugate prime factors, each of degree f.

§172. The Conjugate Prime Ideals

We now describe the conjugate prime ideals over a prime $p \neq q$ more explicitly. Let again f be the least positive exponent with

$$p^f \equiv 1 \pmod{m}, \qquad \varphi(m) = ef. \tag{1}$$

The subgroup of the residue-class group modulo m generated by p is

$$1, p, p^2, \ldots, p^{f-1}. \tag{2}$$

Choose representatives

$$\xi_1, \xi_2, \ldots, \xi_e$$

for the cosets of this subgroup in the group of reduced residue classes modulo m. Then the prime factors over p may be denoted

$$\mathfrak{p}_{\xi_1}, \mathfrak{p}_{\xi_2}, \ldots, \mathfrak{p}_{\xi_e},$$

and one has, up to association,

$$p = \mathfrak{p}_{\xi_1}\mathfrak{p}_{\xi_2}\cdots\mathfrak{p}_{\xi_e}. \tag{3}$$

For an odd $m = q^\lambda$, let g be a primitive root modulo m, and write

$$p \equiv g^\gamma \pmod{m}.$$

Then f is the least positive integer with

$$\gamma f \equiv 0 \pmod{\varphi(m)},$$

and $e = (\gamma, \varphi(m))$. The coset representatives may be taken as

$$1, g, g^2, \ldots, g^{e-1}. \tag{4}$$

If $p-1$ is divisible by q, a part of the splitting already occurs in the smaller cyclotomic field determined by the greatest common divisor of $p-1$ and m. If $p-1$ is not divisible by q, then the series (4) gives the representatives directly.

For $m = 2^\lambda$ the same discussion is made with the customary generators of the odd residue classes; if $p-1$ is divisible by 4, powers of 5 are used, and if only divisibility by 2 occurs, the representatives are correspondingly adjusted. The exceptional small case $m = 4$ is to be separated.

The substance of the result is that the decomposition of p is completely governed by the order of p in the multiplicative group of residues modulo m. The conjugate prime ideals correspond to the cosets of the subgroup generated by p.

§173. Representation of the Prime Factors of p

The representation of the prime factors can now be made quite explicit. Let

$$f(t) = \prod_n (t - \zeta^n)$$

be the cyclotomic polynomial, the product being over the reduced residues n modulo m. For a fixed prime $p \neq q$, with order f modulo m, put

$$F_n(t) = \prod_{h=0}^{f-1} (t - \zeta^{np^h}). \tag{1}$$

This depends only on the coset of n modulo the subgroup generated by p. If $\xi_1, \ldots, \xi_e$ are coset representatives, then

$$f(t) = F_{\xi_1}(t) F_{\xi_2}(t) \cdots F_{\xi_e}(t). \tag{2}$$

By the binomial theorem,

$$(t - \zeta^{np^h})^p \equiv t^p - \zeta^{np^{h+1}} \pmod{p},$$

and the exponents p^{h+1} run through the same cycle as the exponents p^h. Hence

$$F_n(t)^p \equiv F_n(t^p) \pmod{p}. \tag{3}$$

This is exactly the criterion that $F_n(t)$ be congruent modulo p to a polynomial $P_n(t)$ with rational integral coefficients:

$$F_n(t) \equiv P_n(t) \pmod{p}. \tag{4}$$

Accordingly,

$$f(t) \equiv P_{\xi_1}(t) P_{\xi_2}(t) \cdots P_{\xi_e}(t) \pmod{p}. \tag{5}$$

The prime ideal $\mathfrak{p}_1$ corresponding to the coset of 1 is the greatest common divisor of p and

$$P_1(\zeta). \tag{6}$$

More generally, $\mathfrak{p}_n$ is the greatest common divisor of p and $P_1(\zeta^n)$, or, if $nn' \equiv 1 \pmod{m}$, the greatest common divisor of p and $P_{n'}(\zeta)$. Thus all prime factors of p are obtained by taking the greatest common divisors of p with the functions

$$P_{\xi_1}(\zeta), \ldots, P_{\xi_e}(\zeta). \tag{7}$$

In the special case

$$p \equiv 1 \pmod{m}, \tag{8}$$

we have $f = 1$ and $e = \varphi(m)$. Then every factor is of the first degree. The root ζ is congruent modulo each prime ideal $\mathfrak{p}_n$ to a rational residue. If g is a primitive root modulo p, this may be written in the form

$$\zeta \equiv g^{(p-1)n/m} \pmod{\mathfrak{p}_n}, \tag{9}$$

with n prime to m. Consequently $\mathfrak{p}_n$ is the greatest common divisor of p and

$$\zeta - g^{(p-1)n'/m}, \tag{10}$$

where

$$nn' \equiv 1 \pmod{m}. \tag{11}$$

This gives an exact characterization of each prime ideal in the completely split case.

We shall also use the following terminology. In the factorization of any number or functional into prime factors, a prime ideal $\mathfrak{p}$ occurs with a well-defined exponent, positive, negative, or zero. If the exponent is zero, $\mathfrak{p}$ is not contained in the number. Two numbers or functionals are called related if they contain a common prime functional, and relatively prime if they contain none. A number which has no related rational prime is a unit. In a normal field all conjugates of a number are related to the same rational primes.

§174. Kummer's Theorem

We now return to certain decompositions already met in the theory of cyclotomy. They involve factors built from roots of unity, and it is important to relate these factors to the prime ideals of the field Ω_m.

Let p be a rational prime satisfying

$$p \equiv 1 \pmod{m}, \qquad p-1 = mm'. \tag{1}$$

Let g be a primitive root modulo p, and form the periods of m' terms from the p-th roots of unity. If $\eta_0, \eta_1, \ldots, \eta_{m-1}$ denote these periods, and if ε is a primitive $(p-1)$-st root of unity with

$$\varepsilon^{m'} = \zeta, \tag{2}$$

then the resolvent

$$(\zeta, \eta) = \eta_0 + \zeta\eta_1 + \zeta^2\eta_2 + \cdots + \zeta^{m-1}\eta_{m-1} \tag{3}$$

is a whole algebraic number. It satisfies a product formula of the form

$$(\zeta, \eta)(\zeta^{-1}, \eta) = \pm p, \tag{4}$$

and hence is a divisor of p.

There are further cyclotomic factors, denoted in Weber's notation by $\psi_s(\zeta)$. They also divide p, and their products with the corresponding conjugates are again equal to p, whenever the exceptional congruences are excluded:

$$\psi_s(\zeta)\psi_s(\zeta^{-1}) = p. \tag{5}$$

The powers of the resolvent are expressible in terms of these factors. In particular,

$$(\zeta, \eta)^m = \pm p\,\psi_1(\zeta)\psi_2(\zeta)\cdots\psi_{m-2}(\zeta). \tag{6}$$

Let $\mathfrak{p}_n$ be the prime ideals over p described in the preceding section. The congruences for ζ modulo $\mathfrak{p}_n$, together with the congruences for the binomial coefficients used in the construction of the ψ_s, determine exactly which $\mathfrak{p}_n$ divides which $\psi_s(\zeta)$. If n' is defined by

$$nn' \equiv 1 \pmod{m}, \tag{7}$$

then a simple counting of residues gives Kummer's factorization

$$(\zeta, \eta)^m \sim \prod_n \mathfrak{p}_{n'}^{n}, \tag{8}$$

where n runs through the positive integers smaller than m and prime to m. The exponents are taken from the representatives n, and n' is always the inverse of n modulo m.

This is Kummer's theorem in the form needed below. It says that the ideal factorization of the m-th power of the resolvent is completely determined by the inverse residue classes modulo m.

A slightly more general form is useful when $p-1$ is divisible only by a proper divisor of m. Let m_1 be the greatest common divisor of $p-1$ and m, so that

$$m = m_1 m_2.$$

The splitting of p in Ω_m is then the same as in the smaller field Ω_{m_1}, and the prime ideals satisfy

$$\mathfrak{p}_n = \mathfrak{p}_{n+sm_1} \qquad (s = 0, \ldots, m_2 - 1).$$

Applying the preceding formula in Ω_{m_1}, and then lifting back to Ω_m, gives a cyclotomic number Q_n such that

$$Q_n^m \sim \prod_t \mathfrak{p}_{t'}^t, \tag{9}$$

with t running over the reduced residues modulo m, and t' determined by $tt' \equiv 1 \pmod{m}$. The statement applies simultaneously to products of prime factors belonging to rational primes congruent to 1 modulo q, or modulo 4 when $q = 2$.

These formulae are the bridge between the explicit cyclotomic resolvents of the first volume and the ideal factorization in the cyclotomic field.

§175. The Roots of Unity in the Field Ω_m

For the later applications a more exact knowledge of the units of the cyclotomic field is required. We first discuss only the numerical units, that is, the whole numbers of Ω_m whose reciprocals are again whole numbers, or equivalently whose norm is ± 1.

Among the units are certainly the roots of unity contained in Ω_m. We shall show that no others of this kind occur beyond those already forced by ζ. Every root of unity in Ω_m is, up to sign, a power of ζ.

Let ρ be a root of unity contained in Ω_m, and write

$$\rho = \Phi(\zeta) \tag{1}$$

as a rational function of ζ. Let q^a be the highest power of q dividing the order of ρ. Then

$$\rho^{q^a} \tag{2}$$

is a root of unity whose order is prime to q. By the irreducibility of the cyclotomic equation in the extended sense, the substitutions

$$\zeta \mapsto \zeta^n \qquad (n, m) = 1$$

may be applied to the expression in (2) without changing its value. Hence ρ^{q^a} is rational. Being a rational root of unity, it is equal to $+1$ or -1. Therefore ρ, apart from its sign, has order a power of q, and so is a power of the primitive m-th root ζ. Thus the roots of unity in Ω_m are precisely

$$\pm\zeta^k. \tag{3}$$

We shall also need the following theorem of Kronecker. If α is a whole algebraic number and if all conjugates of α have absolute value 1, then α is a root of unity.

Indeed, the absolute value of the norm of a whole number is the product of the absolute values of its conjugates and is a rational integer at least 1, unless the number is zero. Under the present hypothesis the norm is therefore ± 1, so α is a unit. The same is true of every power of α.

It remains to see that only finitely many whole numbers can have all conjugates of absolute value 1. Let

$$\omega_1, \dots, \omega_n$$

be a basis of the ring of integers, and write

$$\alpha = a_1\omega_1 + \cdots + a_n\omega_n, \qquad \beta = b_1\omega_1 + \cdots + b_n\omega_n.$$

If two such numbers α, β are not equal and not negatives of each other, then among their conjugates there is no real quotient equal to ± 1, and hence

$$|\alpha \pm \beta| < |\alpha| + |\beta| = 2$$

for at least one choice of sign in every conjugate. Thus $(\alpha \pm \beta)/2$ cannot be a whole number, since its norm would have absolute value < 1. Consequently two distinct numbers of this type cannot have all their basis coefficients congruent modulo 2, except for the pair α and $-\alpha$. There are only finitely many residue classes modulo 2, and so only finitely many such whole numbers.

Now the infinite sequence

$$1, \alpha, \alpha^2, \alpha^3, \ldots$$

consists entirely of numbers of this finite set. Two terms must coincide; hence

$$\alpha^{h-k} = 1$$

for some $h > k$, and α is a root of unity.

It follows in particular that if a number of a field is a root of unity of degree m, then all of its conjugates are roots of unity of the same degree. For they satisfy the same rational equation

$$x^m - 1 = 0.$$